

TABLE BXII.γ.242. (cont.)

Characteristic	1. <i>P. carotovorum</i>	2. <i>P. cacticida</i>	3. <i>P. chrysanthemi</i>	4. <i>P. cypripedii</i>
Phenylalanine	+	+	+	+
Quinolinic acid	—	—	—	—
Sarcosine	—	—	—	—
Serine	+	+	+	+
Spermidine	—	—	—	—
Spermine	—	—	—	—
Taurine	nd	nd	—	nd
Threonine	nd	nd	+	nd
Trigonelline	—	—	—	—
Trimethylammonium	—	—	—	—
Tryptamine	—	—	nd	nd
Tryptophane	+	+	+	+
Tyrosine	+	+	+	+
Ureum	+	+	+	+
Valine	nd	nd	+	nd
Xanthin	—	—	nd	nd
<i>Sensitivity toward:</i>				
Amoxycillin	+	+	nd	nd
Cephalotine	+	+	nd	nd

^aFor symbols see standard definitions; nd, not determined.

TABLE BXII.γ.243. Differentiation of *Pectobacterium carotovorum* subspecies^a

Characteristic	<i>P. carotovorum</i> subsp. <i>carotovorum</i>	<i>P. carotovorum</i> subsp. <i>atrosepticum</i>	<i>P. carotovorum</i> subsp. <i>betavascularum</i>	<i>P. carotovorum</i> subsp. <i>odoriferum</i>	<i>P. carotovorum</i> subsp. <i>wasabiae</i>
Reducing sugars produced from sucrose	—	+	+	+	—
Methyl red	+	+	—	nd	+
Growth in 5% NaCl	+	+	+	nd	—
Growth at 37°C	+	—	+	+	nd
Utilization of citrate	+	+	—	nd	+
H ₂ S from cysteine	v	v	v	—	+
Acetoin production	+	v	+	+	—
Citrate (Simmons)	+	+	—	+	—
Caseinase	+	—	—	nd	+
β-Galactosidase	+	+	+	+	—
Gelatinase	+	—	v	+	+
<i>Acid production from:</i>					
d-Arabitol	—	—	—	+	nd
Citrate	+	+	—	nd	+
α-Methylglucoside	—	+	v	+	—
Gluconate	—	+	—	nd	nd
myo-Inositol	—	—	+	—	—
Inulin	—	—	+	—	—
Maltose	—	+	+	nd	—
Melibiose	+	+	—	+	—
Palatinose	—	+	+	+	nd
Raffinose	+	+	v	+	—
Sorbitol	—	—	—	+	—
<i>Utilization of carbon sources:</i>					
D-Arabitol	—	—	—	+	nd
Citrate	+	+	—	+	nd
Gluconate	—	+	+	v	nd
Inulin	—	—	+	—	nd
Melibiose	+	+	—	+	nd
Palatinose	—	+	+	+	nd
Sorbitol	—	—	—	+	—
D-Turanose	—	d	+	d	nd

^aFor symbols see standard definitions; nd, not determined.

(2) PCR detection with primers ECA1f (5'-CGGCAT-CATAAAAACACG-3') and ECA2r (5'-GCACACTTCA-TCCAGCGA-3'), annealing at 62°C, amplifying a 690-bp DNA fragment (De Boer and Ward, 1995); (3) hybridization with a nonradioactive labeled DNA probe (Helander and Persson, 1996); (4) PCR detection with prim-

ers derived from sequences of metalloprotease-coding genes: ERWFOR (5'-ACGCATGAAATCGGCCATGC-3') *T_m* 62°C, ATROREV (5'-ATCGATAATTTGATTGTCT-3') *T_m* 52°C, and CHRREV (5'-AGTGCTGCCGTACAGCACGT-3') *T_m* 64°C (Smid et al., 1995); (5) detection with a digoxigenin-labeled DNA probe, selected from an

TABLE BXII.γ.244. Additional characteristics of *Pectobacterium carotovorum* subspecies^a

Characteristic	<i>P. carotovorum</i> subsp. <i>carotovorum</i>	<i>P. carotovorum</i> subsp. <i>atrosepticum</i>	<i>P. carotovorum</i> subsp. <i>betavascularum</i>	<i>P. carotovorum</i> subsp. <i>odoriferum</i>	<i>P. carotovorum</i> subsp. <i>wasabiae</i>
Indole production	—	—	—	—	—
Growth in 7% NaCl	+	nd	nd	nd	nd
Growth at 40°C	—	—	—	nd	—
Production of aminopeptidase	+	+	+	+	nd
Arginine dihydrolase	—	—	—	—	nd
β-Glucuronidase	nd	nd	nd	—	nd
Lecithinase	—	—	—	—	—
Production of pectolytic enzymes	+	+	+	+	+
Phenylalanine deaminase	—	—	—	nd	—
Phosphatase	—	—	—	—	—
<i>Acid production from:</i>					
Amygdalin	+	+	+	+	nd
L-Arabinose	+	+	+	+	nd
Arbutine	+	+	+	+	nd
Cellobiose	+	+	v	+	nd
Dulcitol	—	—	—	—	nd
Erythritol	—	—	—	—	nd
Esculin	+	+	+	+	nd
DL-Fucose	—	—	—	—	nd
Glycerol	+	+	+	+	nd
Glycogen	—	—	—	—	nd
Lactose	+	+	+	+	+
Malonate	—	—	—	nd	—
Mannitol	+	+	+	+	+
L-Rhamnose	+	+	+	+	nd
D-Ribose	+	+	+	+	nd
Salicin	+	+	+	+	nd
L-Sorbose	—	—	—	—	nd
Starch	—	—	—	—	nd
Sucrose	+	+	+	+	nd
D-Tagatose	—	—	—	—	nd
Trehalose	+	+	+	+	+
Xylitol	—	—	—	—	nd
D-Xylose	+	+	+	+	nd
L-Xylose	—	—	—	—	nd
β-Methylxyloside	—	—	—	—	nd
<i>Utilization of carbon sources:</i>					
Aconitate	nd	nd	nd	—	nd
Adonitol	nd	nd	nd	—	nd
Amygdalin	d	d	+	d	nd
D-Arabinose	—	—	—	—	nd
L-Aspartate	nd	nd	nd	+	nd
Cellobiose	+	+	v	+	+
Dulcitol	nd	nd	nd	—	nd
Erythritol	nd	nd	nd	—	nd
Fucose	nd	nd	nd	—	nd
Methyl-α-galactoside	nd	nd	nd	d	nd
Galacturonic acid	nd	nd	—	+	nd
Gentiobiose	+	+	+	+	nd
α-Methylglucoside	v	+	+	+	nd
N-Acetyl-D-glucosamine	nd	nd	nd	+	nd
L-Glutamate	nd	nd	nd	+	nd
D-Glucosamine	nd	nd	nd	+	nd
Ketogluconate	nd	nd	nd	d	nd
Glucuronate	nd	nd	nd	d	nd
Glutarate	nd	nd	nd	—	nd
DL-Glycerate	nd	nd	nd	+	nd
L-Histamine	nd	nd	nd	—	nd
Hypoxanthin	nd	—	nd	nd	nd
myo-Inositol	+	+	+	+	+
α-Ketoglutarate	+	+	+	+	+
DL-Lactate	nd	nd	nd	—	nd
Lactose	+	+	+	+	nd
Lactulose	nd	nd	nd	+	nd

(continued)

TABLE BXII.γ.244. (cont.)

Characteristic	<i>P. carotovorum</i> subsp. <i>carotovorum</i>	<i>P. carotovorum</i> subsp. <i>atrosepticum</i>	<i>P. carotovorum</i> subsp. <i>betavascularum</i>	<i>P. carotovorum</i> subsp. <i>odoriferum</i>	<i>P. carotovorum</i> subsp. <i>wasabiae</i>
Malonate	—	—	—	—	—
Maltitol	nd	nd	nd	d	nd
Maltose	v	d	+	d	nd
Maltotriose	nd	nd	nd	—	nd
Mucate	nd	nd	nd	+	nd
Proline	nd	nd	nd	d	nd
Raffinose	+	+	v	+	+
Saccharate	nd	nd	nd	+	nd
L-Serine	nd	nd	nd	+	nd
D-Tagatose	nd	nd	nd	—	nd
Tartrate	—	—	—	—	—
Trehalose	+	+	+	+	nd
Triacetate	—	—	—	—	—
D-Xylose	nd	nd	nd	+	nd
<i>Utilization of nitrogen sources:</i>					
Hypoxanthin	nd	nd	+	nd	nd
Pyrazinamide	nd	nd	+	nd	nd
<i>Sensitivity toward:</i>					
Erythromycin	—	—	—	nd	—
Hypoxanthin	—	nd	nd	nd	nd

^aFor symbols see standard definitions; nd, not determined.

EcoRI digest of a cloned library (Ward and De Boer, 1994).

Several monoclonal antibodies have been produced against *P. carotovorum* subsp. *atrosepticum* strains that can be used for identification and detection in cell suspensions as well as in plant material by (1) ODD (Alarcon et al., 1995); (2) IIF with a sensitivity of up to 240 cells/ml (De Boer and McNaughton, 1987; Gorris et al., 1994; Alarcon et al., 1995); (3) ELISA with a sensitivity of up to 10³ cells/ml tuber peel extract (Jones et al., 1993; Gorris et al., 1994; Alarcon et al., 1995; Basalp et al., 1995; Hyman et al., 1995; Pérombélou and Hyman, 1995); (4) immunofluorescence (IF) cell staining with a sensitivity of up to 10⁵ cells/ml tuber peel extract (Fraaije et al., 1996b); (5) immunofluorescence colony staining (Gorris et al., 1994; Pérombélou and Hyman, 1995; Schober and Van Vuurde, 1997); and (6) slide agglutination test (McLeod and Pérombélou, 1992).

Causes a vascular and parenchymal disease (blackleg) of potato (*Solanum tuberosum*), cauliflower (*Brassica oleracea* var. *botrytis*), tomato (*Lycopersicon esculentum*), and cabbage (*Brassica oleracea* var. *capitata*), and a storage rot of potato tubers. Bases of stems are blackened, rotted, and slimy. Plants wilt and are stunted or die. The soft rot may affect various parts of the plant, particularly the tubers in storage.

The mol% G + C of the DNA is: 51.3–53.1 (*T_m*, Bd).

Type strain: ATCC 33260, NCPPB 549, LMG 2386.

GenBank accession number (16S rRNA): Z96090.

- c. ***Pectobacterium carotovorum* subsp. *betavascularum*** (Thomson, Hildebrand and Schroth 1981) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 393) (*Erwinia carotovora* subsp. *betavascularum* Thomson, Hildebrand and Schroth 1981, 1040.) *be.ta.vas.cu.lo'rum*. L. n. *beta* beet; L. n. *vasculum* vascular tissue; M.L. pl. gen. n. *betavascularum* of the beet's vascular tissues.

Characteristics distinguishing this subspecies from the other *Pectobacterium carotovorum* subspecies are indicated in Table BXII.γ.243. Strain Ecb168 was found to produce an antibiotic that suppresses growth of *P. carotovorum* subsp. *carotovorum* (Costa and Loper, 1994). Causes soft rot and vascular necrosis of sugar beet (*Beta vulgaris*) and, after artificial inoculation, of potato stems.

The mol% G + C of the DNA is: 54.4–54.7 (*T_m*, Bd).

Type strain: ATCC 43762, NCPPB 2795, LMG 2466.

GenBank accession number (16S rRNA): U80198, Z96091.

- d. ***Pectobacterium carotovorum* subsp. *odoriferum*** (Gallois, Samson, Ageron and Grimont 1992) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 394) (*Erwinia carotovora* subsp. *odorifera* Gallois, Samson, Ageron and Grimont 1992, 586.) *o.do.ri'fe.rum*. L. masc. adj. *odoriferum* bringing odors, fragrant.

Characteristics distinguishing this subspecies from the other *Pectobacterium carotovorum* subspecies are indicated in Table BXII.γ.243. Grow on nutrient agar, producing colonies that are about 2.5 mm in diameter after 48 h at 25°C and also on yeast peptone dextrose (YPDA) medium (3 g/l yeast extract, 5 g/l peptone, 5 g/l dextrose, 15% agar). Colonies are circular, grayish (to yellowish on media containing sucrose or gelatin), slightly domed, and semitranslucent. Causes odorous soft rot of chicory (*Cichorium intybus* L.). The disease typically results in water-soaked, translucent, wet rot of the etiolated leaves. The development of slimy rot in witloof chicories is always associated with the production of a sweet, ripe, banana-like odor.

The mol% G + C of the DNA is: unknown.

Type strain: CFBP 1878, LMG 17566.

GenBank accession number (16S rRNA): AJ223407.

e. ***Pectobacterium carotovorum* subsp. *wasabiae*** (Goto and Matsumoto 1987) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 393) (*Erwinia carotovora* subsp. *wasabiae* Goto and Matsumoto 1987, 132.)

wa.sa' bi.a.e. L. gen. n. *wasabiae* of wasabi (*Eutrema wasabi*), name of host plant.

Characteristics distinguishing this subspecies from the other *Pectobacterium carotovorum* subspecies are indicated in Table BXII.γ.243. Colonies on yeast extract-peptone agar plates at 28°C are white, transparent, circular with entire margins, convex, and 1.0–2.0 mm in diameter after 24 h. Growth on yeast extract-peptone agar slants is moderate, white, and butyrous, with high viability. Abundant growth is obtained on yeast extract-peptone agar slants supplemented with glucose or on potato-glucose agar slants, although viability on these media is poor. Causes soft rot and an internal black discoloration on slices of wasabi rhizomes and, after artificial inoculation, also on potato tubers, carrot and radish roots, midribs of chinese cabbage and intact wasabi, and tomato and tobacco plants.

The mol% G + C of the DNA is: 51.4–51.7 (T_m).

Type strain: ATCC 43316, ICMP 9121, LMG 8444.

GenBank accession number (16S rRNA): U80199, AJ223408.

2. ***Pectobacterium cacticida*** (Alcorn, Orum, Steigerwalt, Foster, Fogleman, and Brenner 1991) Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1^{VP} (Effective publication: Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1998a, 394) (*Erwinia cacticida* (Hori 1911) Dye 1969a, 93; *Erwinia cypripedii* Alcorn, Orum, Steigerwalt, Foster, Fogleman, and Brenner 1991, 210.)

cac.ti.ci' da. Gr. n. *kaktos* prickly plant; Gr. v. *cid* to kill; L. adj. *cacticida* cactus killing.

The characteristics are as described for the genus and as listed in Tables BXII.γ.241 and BXII.γ.242. Colonies grown on PDP (40 g potato dextrose agar (Difco), 5 g agar, 20 g peptone, 1 l distilled water, pH 7.2) for 24 h at 30°C are small, smooth, glistening, circular, entire, slightly convex, ivory, and translucent, and frequently have striations. They produce a distinct odor and become opaque, off-white, and butyrous with age. Causes soft rot of *Opuntia* (cactus) fruits and pads. After artificial inoculation, also causes soft rot of saguaro, organ pipe, senita cacti, tomato fruits, potato slices, and slices of carrot roots.

The mol% G + C of the DNA is: 50.8–51.7 (T_m).

Type strain: ATCC 49481, LMG 17936, ICPB EC186.

GenBank accession number (16S rRNA): AJ223409.

3. ***Pectobacterium chrysanthemi*** (Burkholder, McFadden and Dimock 1953) Brenner, Steigerwalt, Miklos and Fanning 1973b, 205^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Erwinia chrysanthemi* Burkholder, McFadden and Dimock 1953, 526.) *chrys.an'the.mi.* M.L. n. *Chrysanthemum* generic name; M.L. gen. n. *chrysanthemi* of chrysanthemums.

The characteristics are as described for the genus and as listed in Tables BXII.γ.241 and BXII.γ.242.

Colonies on potato-glucose agar (pH 6.5) (ATCC culture medium 97) are characteristically umbonate with undulate to coralloid margins ("fried egg") at 3–6 d.

The chrysobactin-mediated iron uptake is essential in enabling the bacterium to systematically attack plants (Expert et al., 1996). Chrysobactin (*N*-(*N*²-(2,3-dihydroxybenzoyl)-D-lysyl)-L-serine) is a compound with siderophore activity (Persmark et al., 1989). The iron status is involved in *pel* regulation. Iron acquisition involves an inductive process resulting in differential expression of two siderophore-mediated pathways in relation to external accessibility (Mahe et al., 1995).

The general protein secretory pathway was studied by mutagenesis of cellulase EGZ (Py et al., 1993). *P. chrysanthemi* produces four proteases encoded by *prtA*, *prtB*, *prtC*, and *prtG*, and they are secreted by a type I secretion system encoded by *prtD*, *prtE* (inner membrane), and *prtF* (outer membrane) (Barras et al., 1994). PrtG is an extracellular metalloprotease secreted through a signal peptide-independent secretion pathway. The COOH-terminal exposition of the last four amino acids in the secretion of PrtG plays a key role in the protease secretion pathway (Ghigo and Wandersman, 1994). These metalloproteases are secreted independently of the general export pathway encoded by the *sec* genes. They are secreted via a C-terminal secretion signal and by a secretion apparatus composed of the two inner membrane proteins, PrtD and PrtE, and the outer membrane protein PrtF. PrtD exhibits a secretion signal-regulated ATPase activity (Delepelaire, 1994). PrtD is the ATP-binding cassette (ABC) integral membrane component from the metalloprotease secretion system. PrtA and prtB are two tandem metalloprotease-encoding structural genes (Boyd and Keen, 1993). Further, *P. chrysanthemi* produces two cellulases, encoded by *celZ* (the majority) and *celY* (5% of the total activity), and four types of pectinases. The importance of these various isoenzymes in pathogenicity depends on the plant species infected (Barras et al., 1994). The pectinases are subjected to multiple regulations—KdgR, PecT, PecS, PecM—and products of pectic catabolism encoded by *ogl*, *kduI*, *kduD*, *kdgK*, and *kdgA*. Mutants were isolated, containing genetic fusions of the *kdgK*, encoding 2-keto-3-deoxygluconate (KDG) kinase (Hugouvieux-Cotte-Pattat et al., 1994) or *kdgA* genes to the *lacZ* gene of *Escherichia coli* by infection of a *lacZ* mutant of *E. chrysanthemi* with the phage Mu d(Ap lac) (Hugouvieux-Cotte-Pattat and Robert-Baudouy, 1985). A second set of secondary pectate lyases are preferentially expressed in plants (Barras et al., 1994). The *out* genes are required for the translocation across the outer membrane during secretion of pectate lyases, pectin methyl esterases, exopolysaccharuronases, and CelZ. An open reading frame *outT*, located between *outB* and *outC*, has no homology with the *pul* cluster but is involved in secretion. *OutC*, *outD*, and *outE* form an operon, while *outS*, *outB*, and *outT* constitute independent transcription units. *OutT* and the *outCDE* operon are regulated by *kdgRa*, encoding a repressor that negatively regulates the expression of genes involved in pectinolysis and in pectinase secretion. *OutB* and *outS* seem to be expressed constitutively (Ji et al., 1989; Condemine et al., 1992; Nasser et al., 1994).

Mutants defective in secretion of pectinase and cellulase were obtained by chemical and insertion mutagenesis (Andro et al., 1984).

The regulation of pectinolysis was reviewed by Hugouvioux-Cotte-Pattat et al. (1996). Four types of pectinases have been identified: two pectin methyl esterases (PemA, PemB), a polygalacturonase (PehX), nine pectate lyases (PelA, PelB, PelC, PelD, PelE, PelI, PelL, PelZ, PelX), and a pectin acetyl esterase (PaeY) (Shevchik et al., 1997). *PecT*, encoded by *pecT*, represses the expression of pectate lyase genes *pelC*, *pelD*, *pelE*, *pelL*, and *kdgC*, activates *pelB*, and has no effect on the expression of *pelA* or the pectin methyl-esterase genes *pemA* and *pemB*. *PecT* activates its own expression (Surgey et al., 1996). *PecS* encodes a repressor that negatively regulates the expression of virulence factors such as pectinases or cellulases (Reverchon et al., 1994; Praillet et al., 1996, 1997). The oligogalacturonate lyase (*ogl*) gene is involved in oligogalacturonides degradation (Reverchon and Robert-Baudouy, 1987) and the *pem* gene encodes the pectin methyl-esterase (PME) (Laurent et al., 1993). *kduD* mutants altered in pectin degradation were isolated by chemical and Mu d(Ap lac) insertion mutagenesis (Condemine et al., 1986).

Harpins are synthesized in a nutrient-deprived medium and induce plant response, including alkalization of the apoplastic fluid. Harpin from *P. chrysanthemi* contributes to soft-rot pathogenesis (Bauer et al., 1995). Harpins are encoded by the *hrp* genes, which have at least three biochemical functions: gene regulation, protein secretion, and production of HR elicitor proteins.

P. chrysanthemi strains can be genotypically characterized by RFLP of PCR-amplified fragments of *pel* genes (Nassar et al., 1996a) and ribotyping, which correlates well with the established pathovars (Nassar et al., 1994).

A transducing phage Phi EC2 was described by Resibois et al. (1984) and Phi EC2-resistant mutants, lipopolysaccharide-defective, were isolated by Schoonejans et al. (1987).

Several monoclonal antibodies have been produced against *P. chrysanthemi* strains that can be used for detection or identification by double-antibody sandwich (DAS)-ELISA with a sensitivity of 5×10^3 CFU/ml in carnation

stem samples (Nassar et al., 1996b), time-resolved fluoroimmunoassay (TR-FIA) in potato peel extracts with a sensitivity of 10^5 cells/ml (Van De Wolf, 1993).

L-Asparaginase from *P. chrysanthemi* provides an alternative to the enzyme from *E. coli* for the effective treatment of acute lymphoblastic leukemia (Goward et al., 1992). It alters the coagulation system less severely than does *E. coli* asparaginase (Carlsson et al., 1995).

Causes vascular wilts, stunting, soft rots, spotting of leaves or parenchymal necroses of a wide range of plant species and cultivars like *Allium cepa*, *Capsicum anuum*, *Colocasia esculanta*, *Kalanchoe blossfeldiana*, *Nicotiana tabacum*, *Nopalea* sp., *Pelargonium zonale*, and *Primula* sp. The disease symptoms are usually systemic.

The mol% G + C of the DNA is: 55.1–57.1 (T_m , Bd).

Type strain: ATCC 11663, DSM 4610, LMG 2804, ICMP 5703, NCPPB 402.

GenBank accession number (16S rRNA): U80200, Z96093.

4. ***Pectobacterium cypripedii*** (Hori 1911) Brenner, Steigerwalt, Miklos and Fanning 1973b, 205^{AL} emend. Hauben, Moore, Vauterin, Steenackers, Mergaert, Verdonck and Swings 1999a, 1 (*Erwinia carotovora* biovar *cypripedii* (Hori 1911) Dye 1969a, 93; *Erwinia cypripedii* (Hori 1911) Bergey, Harrison, Breed, Hammer and Huntton 1923, 171; *Bacillus cypripedii* Hori 1911, 91.)
cyp.ri.ped' i.i. M.L. n. *Cypripedium* generic name; M.L. gen. n. *cypripedii* of *cypripedium* orchids.

The characteristics are as described for the genus and as listed in Tables BXII.γ.241 and BXII.γ.242. The gene cluster encoding three subunits of membrane-bound glutonate dehydrogenase (GADH) was studied by Yum et al. (1997). Causes a brown rot of *cypripedium* orchids (*Cypripedium* spp.). The disease usually attacks orchids with fleshy leaves. It starts with small water-soaked lesions that enlarge and become slightly sunken, brownish greasy-looking areas. It may spread down the stem and involve the growing point.

The mol% G + C of the DNA is: 54.1–54.6 (T_m , Bd).

Type strain: ATCC 29267, DSM 3873, ICMP 1591, LMG 2657.

GenBank accession number (16S rRNA): U80201, Z96094.

Genus XXV. ***Candidatus Phlomobacter*** Zreik, Bové and Garnier 1998, 260

MONIQUE GARNIER

Phlo.mo.bac.ter. Gr. n. *phlomos* bark; Gr. n. *bakterion* a small rod; M.L. neut. *Phlomobacter* rod in the bark.

Filamentous bacteria occurring in the sieve tubes of phloem tissue of plants (Figs. BXII.γ.202 and BXII.γ.203). The original description is from infected strawberry plants (*Fragaria x ananassa*) showing leaf marginal chlorosis and stunting in France (Nourrisseau et al., 1993). Like most other phloem-restricted bacteria, *Candidatus Phlomobacter fragariae* has resisted *in vitro* cultivation.

The mol% G + C of the DNA is: not determined.

Type species: *Candidatus Phlomobacter fragariae* Zreik, Bové and Garnier 1998, 260.

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic treatment Phylogenetic analysis of the 16S rRNA gene indicates that *Candidatus Phlomobacter* is a member of the *Enterobacteriaceae* in the *Gammaproteobacteria*. Its closest cultured relative is *Arsenophonus nasoniae*, the recently characterized agent of the son-killer trait in the parasitic wasp *Nasonia vitripennis* (92% 16S rDNA identity) (Fig. BXII.γ.204). The complete sequence of *Candidatus Phlomobacter* 16S rRNA gene is deposited in GenBank under the accession number U91515. The oligo-

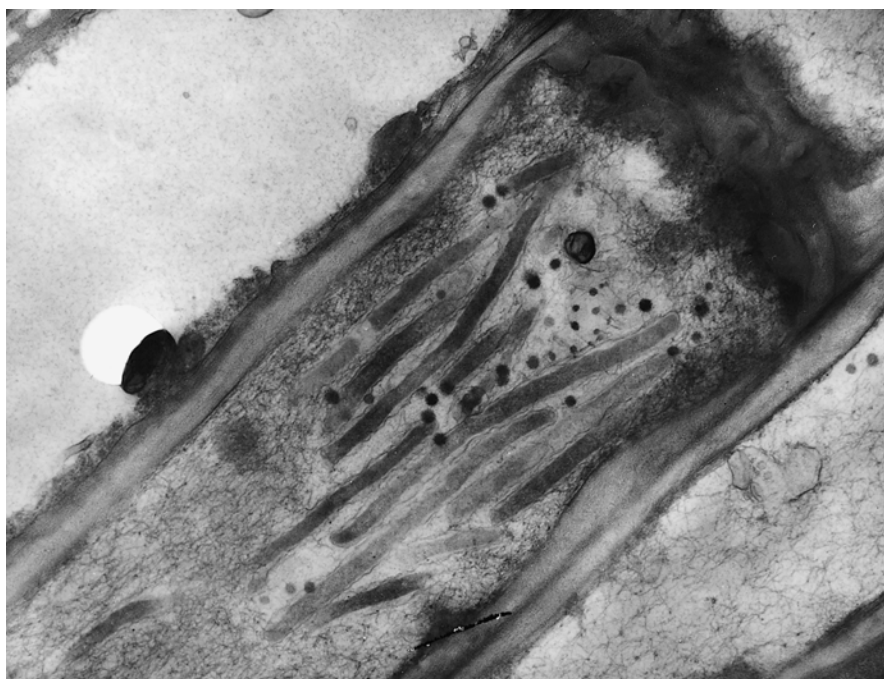


FIGURE BXII.γ.202. *Candidatus Phlomobacter fragariae* cells in a phloem sieve tube of strawberry leaves showing marginal chlorosis ($\times 20,000$).



FIGURE BXII.γ.203. *Candidatus Phlomobacter fragariae* cell in a sieve tube with thickened walls ($\times 29,000$).

nucleotide sequence complementary to unique region of the 16S rRNA is 5'-AGCAATTGACATTAGCGA-3' (Zreik et al., 1998).

Cultivation To date, all attempts to culture *Candidatus Phlomobacter* have failed.

Strain morphology Electron microscopy measurements on thin sections show that *Candidatus Phlomobacter fragariae* cells are 0.2–0.27 μm in diameter and up to 4 μm in length. The cell envelope is 250 nm thick. There is no evidence for flagella or pili (Figs. BXII.γ.202 and BXII.γ.203).

Ecological data, host range *Candidatus Phlomobacter* are phytopathogenic bacteria affecting strawberry (*Fragaria x ananassa*) cultivars in which they induce marginal chlorosis of leaf

blades, stunting, and fruit malformations. At the ultrastructural level, they induce thickening of the sieve tube walls (Fig. BXII.γ.203). To date, they have not been described in other hosts, and trials to transmit the bacterium via dodder (*Cuscuta campestris*) to periwinkle (*Catharanthus roseus*), the experimental plant for phloem-restricted bacteria, have failed. Vector transmission of the bacterium occurs in nature. The insect vector has been identified as *Cixius wagneri* (China). The disease was described in France, and its incidence is highest in strawberry production fields in Southern France. Strawberry nurseries, located in northern France, are not affected. Whether *Candidatus Phlomobacter fragariae* is present in countries other than France has not been investigated.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Specific identification *Candidatus Phlomobacter* can be identified by amplification and sequencing of the 16S rDNA. By sequence comparisons, two primers specific for phlomobacteria have been selected on the 16S rDNA sequence. They are efficient and specific for *Candidatus Phlomobacter* detection in plants. When used for DNA amplification in insects, these primers are not specific, as *Candidatus Phlomobacter* shares strong homologies with insect bacterial symbionts and parasites as well as with enterobacteria. Thus, specific identification from hosts other than plants requires sequencing of the 16S rDNA. Genes other than 16S rDNA have been isolated and are under study.

ACKNOWLEDGMENTS

Monique Garnier-Semancik died suddenly in May, 2003. She was Director of Research at INRA Bordeaux and served as director of a "Joined Research Unit" in which INRA researchers and University teachers work together. Most of her scientific contributions were to the field of phloem- and xylem-restricted plant pathogenic bacteria, which won her respect both in France and internationally.

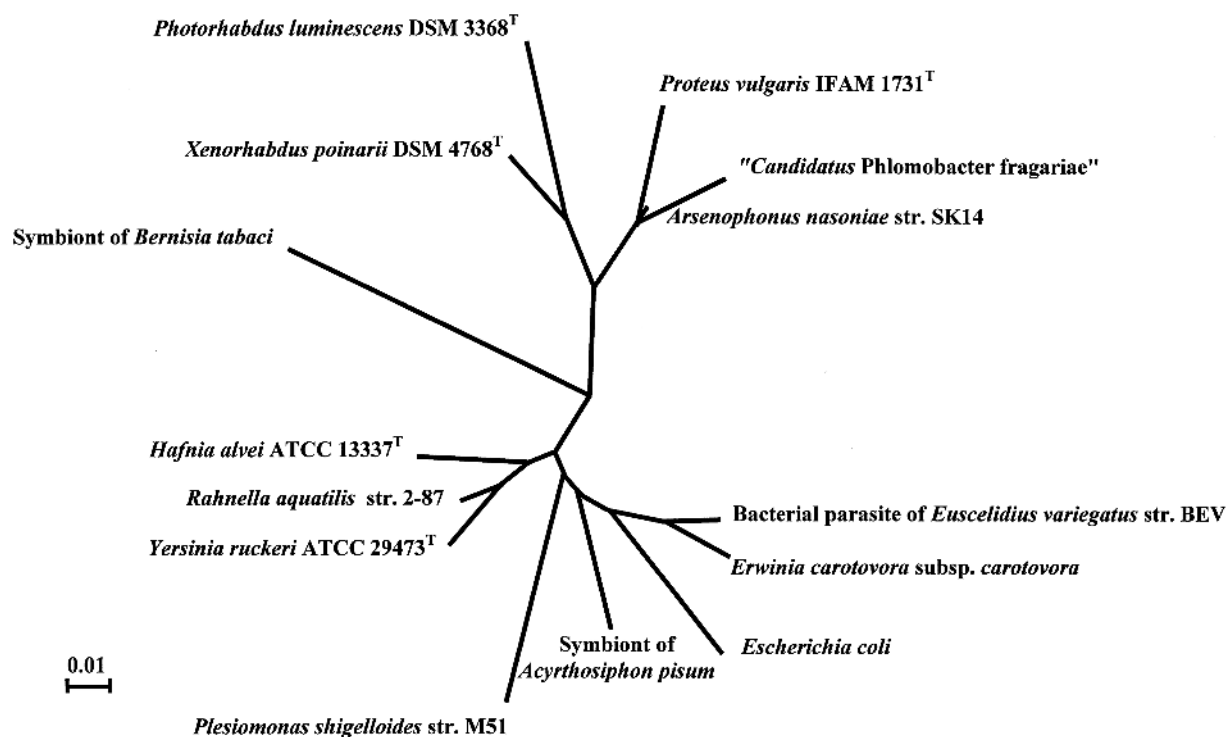


FIGURE BXII.γ.204. Phylogenetic tree constructed with the following 16S rDNA sequences from GenBank using the Suggest Tree Tool of the Ribosomal Database Project (Olsen et al., 1991): *Arsenophonus nasoniae* SK14 (M90801), *Proteus vulgaris* 1731 (X07653), *Hafnia alvei* 13337 (M59155), *Rahnella aquatilis* 2-87 (X79937), *Yersinia ruckeri* 29473 (X75275), *Plesiomonas shigelloides* M61 (M59159), *Acyrthosiphon pisum* symbiont S (M27040), *Escherichia coli* (J01859), parasite of *Euscelidius variegatus* (Z14096), *Erwinia carotovora* (M59149), *Photorhabdus luminescens* 3368 (X82248), *Xenorhabdus poinarii* 4768 (X82253), and symbiont of *Bemisia tabaci* (Z11925). The scale corresponds to 1% substitution. (Reproduced with permission from L. Zreik et al., International Journal of Systematic Bacteriology 48: 257–261, 1998, ©International Union of Microbiological Societies.)

List of species of the genus *Candidatus Phlomobacter*

1. *Candidatus Phlomobacter fragariae* Zreik, Bové and Garnier 1998, 260.

fra.ga'riae. L. gen. n. *fragaria*, *fragariae* of *fragaria*, strawberry.

The description of this organism is identical to that of the genus *Candidatus Phlomobacter*.

The mol% G + C of the DNA is: not determined.
GenBank accession number (16S rRNA): U91515.

Genus XXVI. *Photorhabdus* Boemare, Akhurst and Maurant 1993, 253^{VP}

NOËL E. BOEMARE AND RAYMOND J. AKHURST

Pho.to.rhab'dus. Gr. n. *photo* light; Gr. fem. n. *rhabdus* rod; M.L. masc. n. *Photorhabdus* bioluminescent rod-shaped bacterium.

Asporogenous rod-shaped cells 0.5–2 × 1–10 μm. Cell size is **highly variable** within and between cultures with occasional filaments up to 30 μm long. In the last stage of exponential growth and during stationary growth period, **spheroplasts may occur with an average of 2.6 μm in diameter** (10–20% of cell population), resulting from the partial disintegration of the cell wall. Proteinaceous **protoplasmic inclusions** are synthesized inside a high proportion of cells (50–80%) during the stationary period. Gram negative, motile by means of peritrichous flagella. Facultatively anaerobic, having both a respiratory and a fermentative

type of metabolism. Optimum growth temperature usually ~28°C; some strains grow at 37–38°C. Most strains produce pink, red, orange, yellow, or green **pigmented colonies on nutrient agar, and especially on rich media** (tryptic soy agar, egg yolk agar). Bioluminescent, usually detectable by the dark-adapted eye; intensity varies within and between isolates and may only be detectable by photometer or scintillation counter in some isolates; only very few nonluminescent isolates have been reported. **Spontaneous phase shift occurs in subcultures inducing the appearance of phase II clones** characterized by the loss of neutral

red adsorption on MacConkey agar, of production of antibiotics, and of some other properties usually exhibited by wild clones freshly isolated from the natural environment and named phase I variants. Catalase positive. **Do not reduce nitrate.** Negative for oxidase, o-nitrophenyl- β -D-galactopyranoside (ONPG), Voges-Proskauer, arginine dihydrolase, lysine and ornithine decarboxylase tests. Proteolytic for gelatin. Most strains hemolytic for sheep and/or horse blood, some producing an unusual annular hemolysis on sheep blood at 25°C (Fig. BXII.γ.205). Lipolytic activity on Tween 20; many strains lipolytic for Tweens 40, 60, 80, and/or 85. Acid production from glucose without gas. Acid produced from fructose, D-mannose, maltose, ribose, and N-acetylglucosamine; weak for acid production from glycerol. Fumarate, glucosamine, L-glutamate, L-malate, L-proline, succinate, and L-tyrosine are utilized as sole carbon and energy sources. Biochemical identification of *Phototrhhabdus* within the family *Enterobacteriaceae* is summarized in Table BXII.γ.245. Sequence analyses of 16S rDNA show that all *Phototrhhabdus* strains branch deeply within the radiation of the family *Enterobacteriaceae*, and have a **specific TGAAAG sequence at positions 208–213** (*E. coli* numbering). **The natural habitat for most strains is the intestinal lumen of entomopathogenic nematodes of the genus *Heterorhabditis*** and insects infected by these nematodes. However, some nonsymbiotic strains have been identified as opportunistic pathogens for humans, not nematodes.

The mol% G + C of the DNA is: 43–45.

Type species: ***Phototrhhabdus luminescens*** (Thomas and Poinar

1979) Boemare, Akhurst and Mourant 1993, 254 *Xenorhabdus luminescens* Thomas and Poinar 1979, 354.)

FURTHER DESCRIPTIVE INFORMATION

The strains formerly considered as belonging to *Xenorhabdus luminescens* clearly form a DNA relatedness group that is distinct from all the other *Xenorhabdus* strains (Boemare et al., 1993; Akhurst et al., 1996). These DNA data, together with the significant differences in phenotypic characters between “*X. luminescens*” and the other *Xenorhabdus* species (Akhurst and Boemare, 1988; Boemare and Akhurst, 1988), fatty acids of the whole cellular composition (Janse and Smits, 1990), and chemotaxonomic data (Suzuki et al., 1990), led to the transfer of *X. luminescens* into a new genus, *Phototrhhabdus*, as *Phototrhhabdus luminescens* comb. nov. (Boemare et al., 1993). Table BXII.γ.245 lists the characteristics of *Phototrhhabdus* genus for the three species and four subspecies recognized today (Fischer-Le Saux et al., 1999).

Nevertheless, comparison of 16S rDNA sequences of the type strains of *Phototrhhabdus* and *Xenorhabdus* species indicate the close phylogenetic relationship of these two genera (Rainey et al., 1995). However, all *Xenorhabdus* strains could be clearly distinguished from strains of *Phototrhhabdus* by the occurrence of a TTCCG sequence at positions 208–211 (*E. coli* numbering) of the 16S rDNA, while *Phototrhhabdus* have a TGAAAG sequence (Szallás et al., 1997). The nearest phylogenetic neighbor is *Proteus vulgaris* as demonstrated by PCR-RFLP (Brunel et al., 1997) and sequencing of 16S rDNA (Rainey et al., 1995; Suzuki et al., 1996; Szallás et al., 1997; Fischer-Le Saux et al., 1999).

The major cellular fatty acids of *Phototrhhabdus* are C_{16:0} and C_{18:1}; C_{15:0 iso}, C_{17:0 iso}, and C_{16:1} are major components in some strains (Janse and Smits, 1990; Suzuki et al., 1990). Ubiquinone-8 is the respiratory quinone in all strains (Suzuki et al., 1990).

The main difficulty when subculturing *Phototrhhabdus* strains is the occurrence of a colonial dimorphism that appears suddenly on agar plates. It can be easily detected by two major properties: dye adsorption and antibiotic production (Akhurst, 1980). Each strain occurs as two phase variants that can be described as morphovar, chemovar, and/or biovar. The bacterium isolated from the infective stage (*dauer* larvae) of *Heterorhabditis* was named the phase I variant (Boemare and Akhurst, 1988). Phase I colonies are mucoid and stick to the loop when streaked on plates, produce antibiotic molecules (Akhurst, 1982a), adsorb dyes when incorporated into agar (e.g., the neutral red in MacConkey agar), and are differently pigmented from phase II variants (e.g., red in phase I and yellow in phase II). Among the *Enterobacteriaceae*, cells of some strains of *Phototrhhabdus* may be the largest known (0.5–2 × 1–10 μm). Phase I cells are larger than phase II cells; they are pleomorphic, comprising rods (80–90%) and spheroplasts (10–20%), and harbor protoplasmic inclusions (Boemare et al., 1983) (Fig. BXII.γ.206). Phase II appears spontaneously during stationary growth period from *in vitro* culture and during nematode rearing on artificial diets. Phase II colonies are not mucoid, do not adsorb dye, and do not produce antibiotics. Several intermediate colony forms, possessing at least some phase I properties, have been recorded (Gerritsen et al., 1992); however, it is not certain that these are not mixtures of phase I and phase II. In general, variation in phase-related characters has been reported qualitatively (“+” and “–”). However, for every character that can be quantified (e.g., luminescence, antibiotic production), it is clear that the difference between phases is a matter of magnitude, not presence/absence. It is highly probable that this holds true for all phase-related characters. Table BXII.γ.246

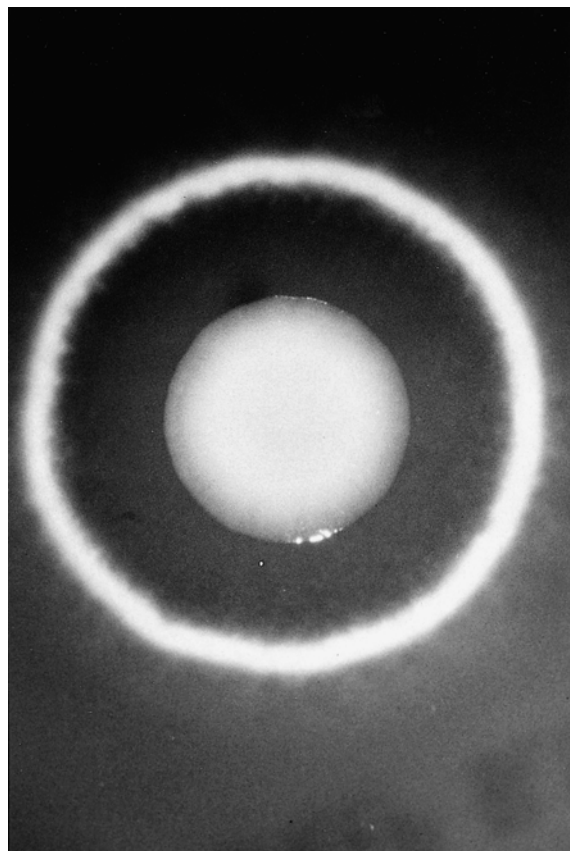


FIGURE BXII.γ.205. Annular hemolysis on sheep blood (10%) agar produced at 25°C by many *Phototrhhabdus* strains.

TABLE BXII.γ.245. Main characteristics of *Photorhabdus* and characteristics differentiating species and subspecies^a

Characteristic	<i>Photorhabdus luminescens</i> ^b	<i>P. luminescens</i> subsp. <i>luminescens</i> ^b	<i>P. luminescens</i> subsp. <i>laumondii</i> ^c	<i>P. luminescens</i> subsp. <i>akhurstii</i> ^d	<i>Photorhabdus asymbiotica</i> ^e	<i>Photorhabdus temperata</i> ^f
DNA–DNA hybridization with <i>E. coli</i>	4%	4%	4%	4%	4%	4%
Mol% G + C of DNA	43–45	43–45	43–45	43–45	43–45	43–45
Maximum growth temperature, °C	35–39	38–39	35–36	38–39	37–38	33–35
Pathogenicity for insects	+	+	+	+	+	+
Motility	+	+	+	+	+	+
Peritrichous flagella	+	+	+	+	+	+
Protoplasmic inclusions	+	+	+	+	–	+
Bioluminescence	+	+	+	+	+	+
Pigmentation	+	+	+	+	+	+
Dye	+	+	+	+	–	+
Antimicrobial production	+	+	+	+	–	+
Oxidation-fermentation	F	F	F	F	F	F
Catalase	+	+	+	+	+	+
Nitrate reduced to nitrite	–	–	–	–	–	–
Oxidase (Kovac's)	–	–	–	–	–	–
Growth in KCN	–	–	–	–	–	–
Indole production	+	+	+	d	–	[–]
Methyl red	–	–	–	–	–	–
Voges–Proskauer	–	–	–	–	–	–
Simmons citrate	d	+	d	d	+ w	d
Hydrogen sulfide production	–	–	–	–	–	–
ONPG (β-galactosidase)	–	–	–	–	–	–
Esculin hydrolysis	+	+	+	+	+	[+]
Urease (Christensen's)	d	–	[+]	d	+	d
Phenylalanine deaminase	[–]	–	d	–	–	[+]
Tryptophan deaminase	[–]	–	d	–	–	[–] w
<i>Amino acid decarboxylases (Moeller's)</i>						
Lysine decarboxylase	–	–	–	–	–	–
Ornithine decarboxylase	–	–	–	–	–	–
Arginine dihydrolase	–	–	–	–	–	–
D-Glucose, acid production	+	+	+	+	+	+
D-Glucose, gas production	–	–	–	–	–	–
<i>Acid production from:</i>						
D-Adonitol	–	–	–	–	–	–
L-Arabinose	–	–	–	–	–	–
Cellobiose	–	–	–	–	–	–
Dulcitol	–	–	–	–	–	–
Fructose	+	+	+	+	+	+
Glycerol	+ w	+ w	+ w	+ w	+ w	+ w
N-acetylglucosamine	+	+	+	+	+	+
myo-Inositol	d	+	[+]	[+]	d w	d w
Lactose	–	–	–	–	–	–
Maltose	+	+	+	+	+	+ w
D-Mannitol	d	d w	–	+	–	[–]
D-Mannose	+	+	+	+	+	+
Melibiose	–	–	–	–	–	–
α-Methyl-D-glucoside	–	–	–	–	–	–
Raffinose	–	–	–	–	–	–
L-Rhamnose	–	–	–	–	–	–
Ribose	+	+	+	+	+	+
Salicin	–	–	–	–	–	–
D-Sorbitol	–	–	–	–	–	–
Sucrose	–	–	–	–	–	–
Trehalose	[+] w	+ w	[+] w	[+] w	[+]	[+]
D-Xylose	–	–	–	–	–	–
<i>Utilization of:</i>						
L-Fucose	d	d	–	[+]	–	d
DL-Glycerate	[–]	d	–	–	d	+
L-Histidine	d	+	[+] w	d	d	[+]

(continued)

TABLE BXII.γ.245. (cont.)

Characteristic	<i>Photorhabdus luminescens</i> ^b	<i>P. luminescens</i> subsp. <i>luminescens</i> ^b	<i>P. luminescens</i> subsp. <i>laumondii</i> ^c	<i>P. luminescens</i> subsp. <i>akhurstii</i> ^d	<i>Photorhabdus asymbiotica</i> ^e	<i>Photorhabdus temperata</i> ^f
myo-Inositol	+	+	+	+	d	[+]
DL-Lactate	[−]	−	−	d w	−	−
D-Mannitol	d	+	−	+	−	[−]
Ribose	+	+	+	+	+	+
L-Tyrosine	+	+	+	+	d	+
Gelatin hydrolysis (Kohn's)	+	+	+	+	+	+
Lecithinase (egg yolk agar)	+	+	+	+	−	+
Lipase (Tween 20)	+	+	+	+	+	+
Lipase (Tween 80)	+	+	+	+	[+]	+
DNase	[−]	−	+	−	−	+
<i>Annular hemolysis at 25°C on:</i>						
Sheep blood agar	d	+	[−]	+	+	+
Horse blood agar	d	+	−	d	+	+

^aSymbols: +, 90–100% of strains are positive; [+], 76–89% are positive; d, 26–75% are positive; [−], 11–25% are positive; −, 0–10% are positive; F, fermentative. The letter w indicates a weak reaction. Data from Akhurst et al. (1996) and Fischer-Le Saux et al. (1999). All tests were done at 28° ± 1°C except annular hemolysis.

^bType strain ATCC 29999 isolated from *Heterorhabditis bacteriophora* group Brecon.

^cType strain CIP 105565 isolated from *Heterorhabditis bacteriophora* group HP88.

^dType strain CIP 105564 isolated from *Heterorhabditis indica*.

^eType strain ATCC 43950 isolated from human blood and/or wounds.

^fType strain CIP 105563 isolated from *Heterorhabditis megidis* Palaearctic group.

summarizes the characters affected by phase variation. However, both phases show a similar entomopathogenic effect and share all the other bacteriological properties of members of the genus.

In natural conditions, cells of *Photorhabdus* are carried in the intestinal lumen of the free-living stage (infective juvenile L3 or *dauer* larva) of *Heterorhabditis*. The bacterial cells are stored and do not multiply in the gut of the *dauer* host, which is a non-feeding stage. When infective juveniles infect an insect, they release their symbionts into the body cavity of the insect prey. *Photorhabdus* cells multiply inducing toxemia and septicemia and the insect dies. The carcass is a sort of monoxenic microcosm where the symbionts eliminate competitive microorganisms by using several antimicrobial barriers, such as antibiotics possessing a wide spectrum of activity (Akhurst, 1982a) and bacteriocins acting against closely related species (Boemare et al., 1992; Baghdiguian et al., 1993). Nematodes reproduce in the insect carcass, feeding on the insect remains metabolized by the symbiotic bacteria and on the bacterial biomass. When the *dauers* escape the insect cadaver to search for new prey, they carry the symbiont in their gut, ensuring the vertical transmission of the mutualistic association. Although the nematode hosts are the natural vectors of their propagation in the insects, in nutritional terms *Photorhabdus* might be considered entomophilic rather than nematophilic microorganisms.

The life cycle of the clinical strains of *Photorhabdus* is much less certain. *Photorhabdus* has been isolated from five clinical sources in the U.S.A. (Farmer et al., 1989) and six recently in Australia (Peel et al., 1999; Gerrard et al., 2003). Isolations were variously made from tissue, blood, and sputum samples; no definite route of infection has been established. Although some

patients may have been immunocompromised, this was definitely not the case for at least two of the Australian patients. These clinical isolates were all easily cultured on standard media at 37°C. They all exhibited the annular hemolysis on sheep blood agar at 25°C. By the time they were identified, all were phase II cultures; it is not known if that is the form in which they were originally isolated.

In vitro subcultures from fresh isolates can be obtained without major difficulties. *Photorhabdus* are mesophilic bacteria able to grow between 15 and 35°C, and some strains at 37–38°C. Subculturing and all biochemical tests should be undertaken around the optimal temperature at 28°C. Usually, nutrient agar or Luria-Bertani agar are sufficient for growth. On minimal media, nicotinic acid, *para*-aminobenzoic acid, proline, tyrosine, and serine are required as growth factors, the mix of growth factors varying between strains (Grimont et al., 1984b). Minimal medium II (BioMérieux) contains all the necessary requirements to test utilization of organic compounds.

Although the shift from phase I to phase II is spontaneous, it is remarkable that the wild *dauer* *Heterorhabditis* almost exclusively harbor phase I *Photorhabdus* (Akhurst and Boemare, 1990). The role of phase II variants is not clarified, and today we have no convincing data to support a good explanation of their occurrence. Phase change occurs during the *in vitro* stationary period in a highly unpredictable manner (Akhurst and Boemare, 1990). The two phases of *Photorhabdus* differ significantly in respiratory activity (Smigielski et al., 1994). After periods of starvation, phase II cells recommenced growth within 2–4 h after the addition of nutrients, compared with 14 h for phase I cells, indicating a more efficient nutrient uptake ability in the former.

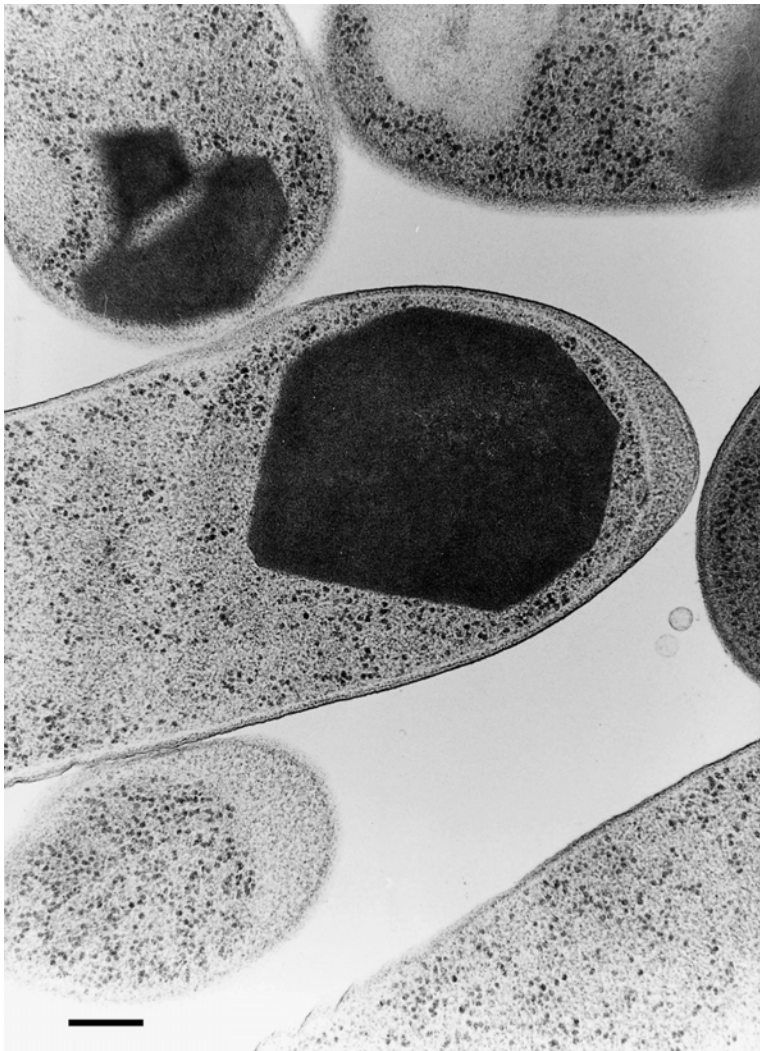


FIGURE BXII.γ.206. Electron micrograph of a fast-freeze fixation and freeze substitution of a culture on nutrient agar in stationary growth of the type strain ATCC 29304 of *Photorhabdus* showing typical inclusions in the protoplasm of the cells. Sections were contrasted with uranyl acetate and lead citrate. Bar = 0.5 μm.

TABLE BXII.γ.246. Differential characteristics of phase variants in *Photorhabdus*^{a,b}

Characteristic	Phase I	Phase II
Colonies	mucoid	smooth
Colonial adhesiveness	+	—
Dye adsorption	+	—
Pigmentation	+	+ ^c
Protoplasmic inclusions	+	[—] w
Bioluminescence	+	+ w
Antibiotic production	+	—
Lecithinase (egg yolk agar)	+	—

^aSymbols: +, 90–100% of strains are positive; [—], 11–25% are positive; —, 0–10% are positive. The letter w indicates a weak reaction. All tests were done at 28° ± 1°C.

^bCharacteristics scored in this table are those that always differ between phases, but there are others that differ for the two phases of an individual strain. For those characteristics that can be quantified (e.g., bioluminescence, pigmentation, and protoplasmic inclusions), the difference between phases is a matter of magnitude, not presence/absence. It is highly probable that this holds true for all phase-related characteristics.

^cPhase II variants of *Photorhabdus* are differently pigmented from phase I variants (e.g., colonies of strain K80 are red in phase I, and yellow in phase II).

Phase II variants may grow a little on complex media previously utilized by phase I variants (Akhurst and Boemare, 1990). However, although reciprocal phase change occurs in *Xenorhabdus*, conversion from phase II *Photorhabdus* to phase I has not been unambiguously demonstrated.

A DNA relatedness study demonstrated unequivocally that phase variation of *Photorhabdus* is a real phenomenon, and not merely an artifact due to contamination, by showing that the two phases of any strain have 100% DNA relatedness. Furthermore, restriction digests and Southern cross analysis of phase I and phase II DNA indicate that the organization of the genome is the same in the two phases (Akhurst et al., 1992).

There are several groups and subgroups within the *Photorhabdus* strains associated with nematodes (*Heterorhabditis* spp.) and human clinical specimens. By using the hydroxyapatite (HA) method to analyze DNA–DNA *Photorhabdus* heteroduplexes, two DNA relatedness groups associated with nematode (*Heterorhabditis* spp.) and another that contains the American human clinical specimens (Farmer et al., 1989) were recognized (Akhurst et al., 1996). More recently, restriction patterns obtained after amplification of the 16S rDNA allowed identification of 12 genotypes

among *Photorhabdus* strains (Fischer-Le Saux et al., 1998). By hybridizing the DNA of some representative strains of each ribosomal genotype, and by using the S1 nuclease method, the previous three genomic groups were confirmed (Fischer-Le Saux et al., 1999). They exhibited between them DNA–DNA hybridization values lower than 42% with ΔT_m higher than 8.7°C. Moreover, the phylogenetic trees inferred from the complete 16S rDNA sequence analysis (neighbor joining, parsimony, and maximum likelihood) delineate the same clusters as both DNA–DNA hybridization methods. Consequently, species and subspecies among *Photorhabdus* were delineated by applying a polyphasic approach, combining 16S rDNA, DNA–DNA hybridization, and phenotypic data (Fischer-Le Saux et al., 1999) (Table BXII.γ.245). Based on DNA–DNA hybridization, 16S rDNA and *gyrB* sequencing, and phenotypic data, the new Australian clinical strains (Peel et al., 1999) will probably constitute a subspecies of *P. asymbiotica* (Akhurst et al., unpublished data). A nonluminescent strain (Akhurst and Boemare, 1986) probably constitutes yet another species (Akhurst et al., 1996).

Within *P. luminescens*, subgroups of strains, which shared very high DNA–DNA hybridization values and ΔT_m lower than 1.5°C and were separated by stable 16S rDNA branching, were identified (Fischer-Le Saux et al., 1999). Consequently, *P. luminescens* was divided into three subspecies (Table BXII.γ.245).

The molecular mechanism of phase variation is uncertain. Lipase and protease are regulated at a posttranslational level in the Irish K122 strain of *P. temperata* (Wang and Dowds, 1993), whereas the *lux* genes are posttranscriptionally regulated in the Hm strain (Hosseini and Nealson, 1995). This contrasts with *X. nematophila* in which the flagellin genes are not transcribed in phase II (Givaudan et al., 1996). These preliminary studies seem to indicate that phase variation in *Photorhabdus* is regulated at a different genetic level than in *Xenorhabdus*.

Several strains of *Photorhabdus* harbor plasmids for which no role or gene has been identified. Heating cultures at 45°C (20 min) and subculturing at 28°C, or mitomycin C treatment, may lead to complete lysis of the cultures, suggesting the occurrence of lysogenic strains (Boemare et al., 1992). Phage tail-like particles, different from those of *Xenorhabdus*, are associated with bacteriocin production in *Photorhabdus* (Baghdiguian et al., 1993).

Antibiograms (Bauer et al., 1966) must be done at 28°C and incubated for 3 days to observe clear zones. *Photorhabdus* have large zones of inhibition around disks impregnated with nalidixic acid, gentamicin, streptomycin, kanamycin, tetracycline, and chloramphenicol, but none around penicillin. Resistance is variable from strain to strain with colistin, ampicillin, carbenicillin, and cephalothin (Farmer, 1984b). It is interesting to note that one patient infected with an isolate sensitive to gentamicin *in vitro* did not respond to gentamicin treatment (Peel et al., 1999).

All strains of *Photorhabdus* have been reported to be entomopathogenic, the LD₅₀ usually being <100 cells when injected into hemocoel of the insect *Galleria mellonella* (Farmer et al., 1989; Akhurst and Boemare, 1990; Akhurst and Dunphy, 1993). *Photorhabdus* has also been isolated from human wounds and blood in the U.S.A. (Farmer et al., 1989) and more recently in Australia (Peel et al., 1999). None of the infections was lethal, but some required weeks of treatment.

Photorhabdus strains are the only terrestrial bioluminescent bacteria known today. Their *lux* genes share a strong identity with the *lux* genes of the marine luminescent *Photobacterium*, and

Vibrio spp. (Frackman et al., 1990). No satisfactory demonstration has yet been provided to explain the role of luminescence in this genus.

Two ecological niches have been identified for *Photorhabdus*: one as a metabolically active form in the insect host, and the other as a quiescent form in the gut of the nonfeeding dauer nematode. The occurrence of nonsymbiotic clinical strains possessing the phase II properties of the nematophilic symbionts and the remarkable differences in respiratory activity between the two phases of the symbiotic strains suggest that there may be a third niche in the soil. However, experiments to test the symbiotic strains ability to grow and survive in external environments indicate that they disappear within a few days (Morgan et al., 1997). *Photorhabdus* strains may enter into a nonculturable but viable survival strategy, as do *Aeromonas*, *Vibrio*, *E. coli*, and *Salmonella* spp.

The specific interaction with the nematode host has been tested by gnotobiological experiments. These assays demonstrate that *Photorhabdus* isolates do not support culture of any *Steinernema* species *in vitro* (Akhurst, 1983b) but in some combinations support culture of nonhost *Heterorhabditis* spp. (Akhurst and Boemare, 1990; Han et al., 1990). Similarly, *Xenorhabdus* spp. do not support culture of *Heterorhabditis* (Akhurst, 1983b). In all the microbial ecological surveys undertaken, a *Xenorhabdus* isolate has never been recovered from *Heterorhabditis* or a *Photorhabdus* isolate from *Steinernema*. Despite the intestinal location of these bacteria, which would allow environmental contamination, the specificity of *Photorhabdus* for *Heterorhabditis* is a remarkable feature of this symbiosis (Boemare et al., 1997a; Forst et al., 1997). When accurate microbial ecology studies are undertaken, using a simple and fast method of PCR-RFLP of 16S rDNA (Brunel et al., 1997), a clear correspondence between *Photorhabdus* isolates and nematode species can be seen. From a total of 75 isolates identified in the Caribbean region (Fischer-Le Saux et al., 1998), two genotypes were associated only with *Heterorhabditis bacteriophora* and another two only with *Heterorhabditis indica*.

Photorhabdus–*Heterorhabditis* and *Xenorhabdus*–*Steinernema* symbioses are widely divergent. Similar patterns of infectivity, life cycle, and mutualism with nematodes have to be considered to be the result of evolutionary convergence. The symbiotic, pathogenic, and phase variation properties that are the conditions for such associations (Boemare and Akhurst, 1994) do not necessarily imply the same physiological mechanisms. As more genes are sequenced, it will be possible to formulate a clearer picture to explain the convergent evolution of *Photorhabdus*–*Heterorhabditis* and *Xenorhabdus*–*Steinernema* symbioses (Boemare and Akhurst, 1994).

ENRICHMENT AND ISOLATION PROCEDURES

Three methods have been used for isolating *Photorhabdus* from nematodes. The “hanging drop” uses a sterile drop of insect hemolymph to which surface-disinfected dauer stage *Heterorhabditis* are added (Poinar and Thomas, 1966). The nematodes exsheath and commence development, releasing their symbiont, which can be subcultured after about 24 h. A second method is to collect under sterile conditions a drop of insect hemolymph from an insect 24 h after infection by *Heterorhabditis*, and to streak it onto nutrient agar. The third method is to crush about 100 surface-disinfected dauer *Heterorhabditis* that have lost their second stage cuticle and to streak the macerate onto nutrient agar (Akhurst, 1980). This latter method is the most rigorous method for assessing the microflora of the intestine of entomopathogenic

nematodes, provided that a suitable control on the effectiveness of the surface disinfection is employed. It reveals the occurrence of *Photorhabdus* in every *Heterorhabditis* sp.

All isolations from nematodes or infected insects should be conducted at or below 28°C.

Human clinical isolates of *Photorhabdus* have been variously obtained from open wounds, tissues aspirated from unerupted lumps, blood, and sputum. These clinical isolates and those from two subspecies of *P. luminescens* can be cultured at 37–38°C and 38–39°C, respectively (Table BXII.γ.245).

MAINTENANCE PROCEDURES

The standard methods of freeze-drying and low temperature storage (liquid nitrogen or at –80°C) used for *Enterobacteriaceae* are also useful for long-term storage of *Photorhabdus* strains; –20°C is unsatisfactory. Cultures do not survive more than a few months in broth or on agar plates at room temperature, and phase variation is likely to occur in this time. Cultures can be routinely maintained for 1 month at 15°C, but storage at 4°C is unsuitable; to prevent phase variation, phase I clones have to be subcultured every week from the neutral red dye-adsorbing clones on MacConkey agar.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

With the exception of the test for annular hemolysis, which should be conducted at 25°C, all biochemical tests used for phenotypic characterization of *Photorhabdus* should be conducted at 28°C, its optimal growth temperature.

Details of the techniques for identifying the phase variants have been summarized (Boemare et al., 1997b). Adsorption of dyes as described by Akhurst (1980) is the most convenient test to characterize the phase variants. MacConkey agar, or better still MacConkey agar without the bile salts, is a good medium for distinguishing phase I variants (red colonies) from phase II variants (off-white or yellow) (Boemare and Akhurst, 1988). On the NBTA medium¹ described by Akhurst (1980), the adsorption of bromothymol blue by *Photorhabdus* may be confused by the pigmentation of strains and the resulting color of the clones can be difficult to distinguish. As most *Photorhabdus* are pigmented, growth on nutrient agar is often sufficient to differentiate clones of the variants, which differ significantly in not only pigmentation but also colony morphology, with phase I being mucoid and convex whereas phase II is non-mucoid and flattened.

To test antibiotic production by *Photorhabdus*, clones of both variants are spot-inoculated on nutrient agar plates. After growth (generally 48 h), cultures are killed by chloroform vapor (30 min) and covered by fresh nutrient semisolid agar (0.6%) inoculated with a bacterial indicator such as *Micrococcus luteus* (Akhurst, 1982a). The inhibition halos of the indicator culture indicate the phase I variants.

Bioluminescence for most of phase I variants can be checked in a darkroom after 10 min for dark-adaptation of the eyes. To assess absence or weakness of such a light production in the phase II variants, by comparison with the phase I variants' luminescence, a scintillation counter, a fluorimeter, or a photomultiplier have to be used. A loopful of an agar culture of each phase variant culture is suspended in 10 ml of distilled water in a scintillation vial for immediate counting with a fully opened window setting (Grimont et al., 1984b).

1. NBTA medium: nutrient agar containing 0.0025% (w/v) bromothymol blue and 0.004% (w/v) triphenyltetrazolium chloride (Akhurst, 1980).

DIFFERENTIATION OF THE GENUS *PHOTORHABDUS* FROM OTHER GENERA

Photorhabdus strains are easily distinguished phenotypically from all the *Xenorhabdus* spp. Bioluminescence and catalase, both physiologically very significant characters, are positive for *Photorhabdus* and negative for *Xenorhabdus*. In most strains, urease is positive and assimilation of DL-lactate is negative for *Photorhabdus* (Table BXII.γ.245); annular hemolysis on sheep blood agar was only observed with *Photorhabdus* strains (Tables BXII.γ.245 and BXII.γ.247). Bioluminescence does not cause confusion with the light-emitting marine bacteria of different families. *Photorhabdus* is differentiated from *Vibrio*, *Alteromonas*, and *Photobacterium* by having peritrichous, not polar, flagella and in not requiring sodium ions for growth.

TAXONOMIC COMMENTS

By DNA–DNA hybridization *Photorhabdus* is only 4% related to *Escherichia coli*, the type species of the type genus for the family (Table BXII.γ.245). However, *Photorhabdus* possesses the enterobacterial common antigen (Ramia et al., 1982). These data indicate that, while most biochemical tests used for differentiation of the *Enterobacteriaceae* are negative for *Photorhabdus* and although it is only distantly related to the genera that comprise the “core” of the family (Farmer, 1984b), *Photorhabdus* should be retained in the family *Enterobacteriaceae*.

It appears that the genus *Photorhabdus* evolved after the main radiation of *Xenorhabdus* species occurred (Forst et al., 1997) (see Fig. BXII.γ.213 in Genus *Xenorhabdus*). This conclusion, originally derived from the analysis of only seven strains (Rainey et al., 1995), was verified when the 16S rDNA analysis was extended to a balanced number of data sets from *Photorhabdus* and *Xenorhabdus* in comparison to other genera and families (Suzuki et al., 1996; Szállás et al., 1997; Fischer-Le Saux et al., 1999). *Photorhabdus*, *Xenorhabdus*, *Proteus*, and *Arsenophonus* can be considered sister genera because they do not branch to a common ancestor with any other group of the family *Enterobacteriaceae* (Liu et al., 1997a).

Poinar isolated a bacterium from a new nematode, *Heterorhabditis bacteriophora* (Poinar, 1975), which was apparently similar to a *Xenorhabdus* sp. but noticeably bioluminescent (Poinar et al., 1977). Poinar et al. proposed to include this luminescent species in the genus as *X. luminescens* (Thomas and Poinar, 1979). The genus *Photorhabdus* was created in 1993, according to the series of arguments described above, to accommodate this species (Boemare et al., 1993). At the present time, it includes three species, *P. luminescens*, *P. temperata*, and *P. asymbiotica*, one of which contains three subspecies, *P. luminescens* subsp. *luminescens*, *P. luminescens* subsp. *temperata*, and *P. luminescens* subsp. *asymbiotica*.

TABLE BXII.γ.247. Differential characteristics of *Photorhabdus*, *Xenorhabdus*, and *Proteus*^a

Characteristic	<i>Xenorhabdus</i>	<i>Photorhabdus</i>	<i>Proteus</i>
Bioluminescence	–	+	–
Catalase	–	+	+
Annular hemolysis on sheep blood agar (25°C)	–	d	–
Urease	–	d	+
Indole	–	d	d
H ₂ S production	–	–	[+]
Nitrate reductase	–	–	+
Acid from D-mannose	+	+	–

^aSymbols: +, 90–100% of strains are positive; [+], 76–89% are positive; d, 26–75% are positive; –, 0–10% are positive.

nescens subsp. *laumondii*, and *P. luminescens* subsp. *akhurstii* (Fischer-Le Saux et al., 1999). According to the rule 65 (2) of the Code of Nomenclature, generic and subgeneric names that are modern compounds from two or more Latin or Greek words take the gender in the original language of the last component of the compound word. Consequently, *Photorhabdus* genus ending by *rhabdus* (from *rhabdos*, rod in Greek as a feminine word), becomes in modern Latin a feminine word, explaining the feminine names of species and subspecies when they are adjectives (Euzéby and Boemare, 2000). Subspecies names referring to a person take the gender of the person, such as *P. luminescens* subsp. *laumondii* and *P. luminescens* subsp. *akhurstii*.

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List of species of the genus *Photorhabdus*

1. ***Photorhabdus luminescens*** (Thomas and Poinar 1979) Boemare, Akhurst and Mourant 1993, 254^{VP} (*Xenorhabdus luminescens* Thomas and Poinar 1979, 354.)

lu.mi.nes' cens. M.L. pres. part. *luminescens* luminescing, for its luminescence.

Large rods (2×0.5 to $6 \times 1.4 \mu\text{m}$). Occurs as two phase variants. Luminous, with luminescence more than 100-fold greater in phase I. Maximum growth temperature in nutrient broth occurs at 35–39°C. Indole positive. Weak acid produced from fructose, N-acetyl-glucosamine, glucose, glycerol, maltose, D-mannose, ribose, and trehalose by most strains. Some strains acidify D-mannitol. Proteinaceous inclusions in the protoplasm of phase I cells; poorly produced in phase II. The natural habitat is in the intestinal lumen of entomopathogenic nematodes of *Heterorhabditis bacteriophora* (Brecon and HP88 groups), and of *Heterorhabditis indica*. Three subspecies have been described.

The mol% G + C of the DNA is: 43–45 (Bd).

Type strain: Hb, ATCC 29999, DSM 3368.

GenBank accession number (16S rRNA): X82248, D78005.

- a. ***Photorhabdus luminescens* subsp. *luminescens*** (Thomas and Poinar 1979) Boemare, Akhurst and Mourant 1993, 254^{VP} (*Xenorhabdus luminescens* Thomas and Poinar 1979, 354.)

Maximum growth in nutrient broth occurs at 38–39°C. Esculin hydrolysis positive and indole weakly positive. DNase, tryptophan deaminase, and urease negative. Annular hemolysis of sheep and horse blood agars. Does not use DL-lactate as sole source of carbon. D-mannitol used as sole source of carbon and energy. Symbiotically associated with nematodes from the group Brecon of *H. bacteriophora*, the type species of the genus *Heterorhabditis* (Poinar, 1975).

The mol% G + C of the DNA is: 43–45 (Bd).

Type strain: Hb, ATCC 29999, DSM 3368.

GenBank accession number (16S rRNA): X82248, D78005.

- b. ***Photorhabdus luminescens* subsp. *akhurstii*** Fischer-Le Saux, Viallard, Brunel, Normand and Boemare 1999, 1654^{VP}
ak.hurs' ti.i. M.L. gen. n. *akhurstii* of Akhurst, referring

to Dr. R. Akhurst, a major contributor to the bacteriological symbionts of entomopathogenic nematodes.

Maximum growth in nutrient broth occurs at 38–39°C. Esculin hydrolysis positive. Tryptophan deaminase and DNase negative. Urease and indole variable. Annular hemolysis observed on sheep blood agar and, in some strains, on horse blood agar. Utilization of DL-lactate as sole source of carbon variable; weak when positive. D-mannitol used and acidified. DL-glycerate utilization negative. Symbiotically associated with the nematode *H. indica* isolated in warm regions; the first strain (strain D1) was isolated from Australia (Darwin, Northern Territory) by Dr. R. Akhurst.

The mol% G + C of the DNA is: 43–45 (Bd).

Type strain: FRG04, CIP 105564.

GenBank accession number (16S rRNA): AJ007359.

- c. ***Photorhabdus luminescens* subsp. *laumondii*** Fischer-Le Saux, Viallard, Brunel, Normand and Boemare 1999, 1654^{VP}
lau.mon' di.i. M.L. gen. n. *laumondii* of Laumond, referring to Dr. C. Laumond, a major contributor to the use of entomopathogenic nematode/bacterial complexes for insect pest control.

Maximum growth in nutrient broth occurs at 35–36°C. Esculin hydrolysis, indole, and DNase positive. Tryptophan deaminase variable and urease mostly positive. Total hemolysis on sheep and horse blood agars; the *Photorhabdus* annular reaction is rare. Does not use L-fucose, DL-glycerate, DL-lactate, or D-mannitol. Symbiotically associated with nematodes of the group HP88 of *H. bacteriophora* isolated in South and North America, southern Europe, and Australia, responding to the satellite DNA probe of the nematode strain HP88 provided by the team of Dr. C. Laumond (Grenier et al., 1996).

The mol% G + C of the DNA is: 43–45 (Bd).

Type strain: TT01, CIP 105565.

GenBank accession number (16S rRNA): AJ007404.

2. ***Photorhabdus asymbiotica*** Fischer-Le Saux, Viallard, Brunel, Normand and Boemare 1999, 1655^{VP}
a.sym.bio' ti.ca. Gr. pref. *a* not; M.L. adj. *symbioticus*, *-a.*, *-um* living together; M.L. fem. adj. *asymbiotica* not symbiotic.

Rods (2×0.5 to 3×1.0 μm). Maximum growth in nutrient broth occurs at 37–38°C. Yellow or brown pigment. No phase I isolates have been detected, and isolates do not absorb dyes, sometimes weakly produce antibiotics, and are negative for lecithinase on egg yolk agar. Positive for urease, esculin hydrolysis, and for Christensen's citrate, but weakly positive on Simmons' citrate. Tryptophan deaminase negative. Indole and DNase negative. Acid produced from fructose, *N*-acetyl-glucosamine, glucose, maltose, *D*-mannose, and ribose; weakly produced from glycerol. Proteinaceous inclusions poorly produced. Tween 40 esterase variable. Annular hemolysis on sheep and horse blood agars. Does not use *L*-fucose, *DL*-lactate, or *D*-mannitol. Natural habitat uncertain. All isolates obtained from human clinical specimens.

The mol% G + C of the DNA is: 43–45 (Bd).

Type strain: 3265-86, ATCC 439500.

GenBank accession number (16S rRNA): Z76755.

3. *Photorhabdus temperata* Fischer-Le Saux, Viillard, Brunel, Normand and Boemare 1999, 1655^{VP}

tem.pe.ra.ta. L. fem. part. adj. *temperata* moderate, because this species grows at moderate temperature.

Large rods (2×0.5 to 6×1.4 μm). Occurs as two phase variants. Highly luminous. Maximum growth temperature in nutrient broth occurs at 33–35°C. DNase positive. Most of the strains are indole negative. Esculin hydrolysis and phenylalanine deaminase mostly positive. Urease variable. Acid produced from fructose, *N*-acetyl-glucosamine, glucose, *D*-mannose, and ribose; weak acid production from glycerol and maltose. Proteinaceous inclusions in protoplasm of phase I cells; poorly produced in phase II. Annular hemolysis often occurs on sheep and horse blood agars. Uses *DL*-glycerate, and does not use *DL*-lactate as sole source of carbon. *D*-mannitol is not used by most strains. The natural habitat is in the intestinal lumen of entomopathogenic nematodes of *Heterorhabditis megidis*, of group NC of *H. bacteriophora*, and of *H. zealandica*.

The mol% G + C of the DNA is: 43–45 (Bd).

Type strain: XINach, CIP 105563.

GenBank accession number (16S rRNA): AJ007405.

Genus XXVII. *Plesiomonas* Habs and Schubert 1962, 324^{AL}

J. MICHAEL JANDA

Pl.si.o.mo' nas. Gr. masc. *n. plesios* neighbor; fem. *n. monas* unit, monad; M.L. fem. *n. Plesiomonas* neighbor monad (to *Aeromonas*).

Rod-shaped, straight cells, $0.8\text{--}1.0 \times 3.0$ μm (Janda, 1998). Endospores not produced. Gram negative. **Motile.** Growth on common basal agars with most strains growing on mineral media containing ammonium salts and glucose as sole nitrogen and carbon sources (Schubert, 1984). Chemoorganotrophic with both a respiratory- and fermentative-type metabolism. **Facultatively anaerobic.** Acid from the catabolism of *D*-glucose, without gas. Anaerogenic fermentation of a limited number of carbohydrates including *m*-inositol. **Oxidase and catalase positive.** **Lysine- and ornithine-decarboxylase and arginine dihydrolase positive.** **NaCl supplementation not required for growth.** **Most strains susceptible to 10 μg and 150 μg of the vibriostatic agent O/129 (2,4-diamino-6,7-diisopropylpteridine).** Few extracellular enzymes elaborated (Schubert, 1984). **Starch not hydrolyzed.** 16S rDNA and 16S rRNA analysis indicate plesiomonads belong to the *Gammaproteobacteria*, with maximum homology to members of the *Enterobacteriaceae* (Martinez-Murcia et al., 1992a; Ruimy et al., 1994). Unique genus-specific signature oligonucleotides have not been identified.

The mol% G + C of the DNA is: 51 (Sebald and Véron, 1963).

Type species: *Plesiomonas shigelloides* (Bader 1954) Habs and Schubert 1962, 324 (*Pseudomonas shigelloides* Bader 1954, 455.)

FURTHER DESCRIPTIVE INFORMATION

Contain 16S rRNA signature oligonucleotides CUAUACCG, YCACAYG (Y = pyrimidine), CUAACUYG, and UCACACAUG at positions 170, 315, 510, and 1410 of the *Escherichia coli* numbering system indicative of *Gammaproteobacteria* (Stackebrandt et al., 1988; Martinez-Murcia et al., 1992a). Although previously classified as *Vibrionaceae*, recent phylogenetic data indicates significant evolutionary divergence between *Plesiomonas* and both the *Vibrio/Photobacterium* and *Aeromonas* groups. 5S rRNA, 16S rRNA/rDNA, and 23S rRNA sequence data demonstrate a close phylogenetic relationship between plesiomonads

and the *Enterobacteriaceae*, especially the genus *Proteus* (MacDonell et al., 1986; East et al., 1992; Martinez-Murcia et al., 1992a; Ruimy et al., 1994).

P. shigelloides typically appears as short, straight rods without observable curvature. Filamentous bacilli, two to three times the normal cell length, can occur in response to various conditions including exposure to β -lactams (Brenden et al., 1988). Capsules not detected. A heat-labile substance, termed a capsular K or masked antigen by Sakazaki (1984), is produced that inhibits agglutination of cells with homologous O antisera (Farmer et al., 1992). Two types of unsheathed flagella produced. Older cultures express lophotrichous flagellation consisting of a tuft of one to five polar flagella with a wavelength of $3.5\text{--}4.0$ μm (Ewing et al., 1961; Schubert, 1984). Young cultures (18 h, 25°C) grown on peptone or nutrient based solid media produce peritrichous, curly, flagella with a shorter periodicity of $1.3\text{--}1.7$ μm (Ewing et al., 1961; Inoue et al., 1991). Up to 68% of young *P. shigelloides* cultures produce lateral flagella when stained with the Flagellar Staining Solution–Shionogi (Inoue and Shimada, 1990; Inoue et al., 1991). Inclusion bodies present in log phase cells consist of polyphosphate granules containing phosphate, potassium, and magnesium (Pastian and Bromel, 1984; Ogawa and Amano, 1987). Laybourn or methylene blue stained inclusions can have bipolar (“safety pin”) or central intracellular locations depending on the growth phase (Ogawa and Amano, 1987; Brenden et al., 1988). Intracellular poly- β -hydroxybutyrate granules not formed.

The enterobacterial common antigen (ECA) found exclusively in members of the *Enterobacteriaceae* is present in *Plesiomonas* (Whang et al., 1972; Ramia et al., 1982). ECA appears to be a heteropolymer consisting of *N*-acetylated amino sugars linked to the lipopolysaccharide moiety (Mäkela and Mayer, 1976). Lipid A analysis reveals a common 1,4'-bis-phosphorylated- β -1,6-linked glucosamine backbone with six fatty acid residues either ester-

or amide-linked (Basu et al., 1985). Cellular fatty acid analysis indicates an overall composition similar to other enteric bacteria and *Vibrionaceae* with hexadecanoate ($C_{16:0}$), hexadecenoate ($C_{16:1}$), and octadecanoate ($C_{18:0}$) as major peaks (Lambert et al., 1983; Yamamoto et al., 1991a). Putrescine, cadaverine, and spermidine are the predominant polyamines synthesized (Yamamoto et al., 1991a).

Ultrastructural analysis of *P. shigelloides* reveals a typical cell wall, cytoplasm, and nucleoplasm characteristic of Gram-negative *Bacteria* (Brenden et al., 1988). Electron-dense granules, 50–150 nm in diameter, can be detected as early as 4 h of incubation with maximum size (>500 nm) reached within 12–24 h (Ogawa and Amano, 1987). Electron-transparent vacuoles have also been described (Brenden et al., 1988). Thin section electron micrographs of ruthenium-stained cells demonstrate a string-like acidic mucopolysaccharide material 35 nm thick emanating from the cell surface (Fig. BXII.γ.207). Fimbriae have not been detected.

Plesiomonads produce uniform turbidity in liquid culture without pellicle formation. Autoagglutination rarely observed. Brain heart infusion broth cultures of *P. shigelloides* produce cells with a high surface charge but low hydrophobicity (Abbott et al., 1991). Colonies on nutrient and blood agars are 1.0–2.0 mm in diameter and are smooth, gray, convex, and opaque with an entire edge. Nonhemolytic. Multiple morphovars arising from individual colonies have been reported (Farmer et al., 1992). Cell-associated or diffusible pigments not elaborated; luminescence not observed. Most *Plesiomonas* produce 1.0–2.0 mm flat, colorless colonies on MacConkey, xylose–lysine–desoxycholate, Hektoen enteric, and desoxycholate agars after 24 h growth (Brenden et al., 1988). Delayed lactose fermentation on MacConkey agar by some *P. shigelloides* leads to lactose-positive variants

arising upon prolonged incubation, which suggests mixed populations. Salmonella–Shigella, eosin methylene blue, and brilliant green agars are inhibitory to some strains (Brenden et al., 1988).

Growth reported over a wide temperature range (8–45°C); optimal growth occurs at 37–38°C (Schubert, 1984; Miller and Koburger, 1986b). Most plesiomonads multiply in liquid media containing 1–4% NaCl; growth in the presence of 5% NaCl reported to be medium-dependent (Miller and Koburger, 1986b). The pH range for growth has been reported to occur between 4.0–9.0 (Schubert, 1984; Miller and Koburger, 1986b). No growth under extremely acidic (pH 3.0) or alkaline (pH 9.5) conditions.

The genus is biochemically homogeneous. Indole formed. Nitrates reduced. The methyl red test is positive. The Voges–Proskauer test (25°C, 37°C) is negative and acetoin is not produced. No growth in KCN. Citrate, gluconate, malonate, and mucate not utilized; sodium acetate is variable. Phenylalanine deaminase and urease activities absent. The string test (0.5% desoxycholate) is negative. No H_2S produced on triple sugar iron agar slants.

Anaerogenic fermentation of D-glucose, *m*-inositol, maltose, and trehalose. Acid from D-galactose, glycerol, lactose, D-mannose, melibiose, and salicin is variable. No fermentation of adonitol, L-arabinose, cellobiose, dextrin, dulcitol, esculin, α-methyl-D-glucoside, inulin, D-mannitol, melezitose, D-sorbitol, sucrose, L-rhamnose, or D-xylose.

Phosphatase and β-galactosidase (ONPG, o-nitrophenyl-β-D-galactopyranoside) positive. β-D-glucuronidase negative. A number of *P. shigelloides* arylamidases and esterases are detected using semiquantitative micromethods with chromogenic substrates (Manafi and Rotter, 1992). Chitinase elaborated (O'Brien and

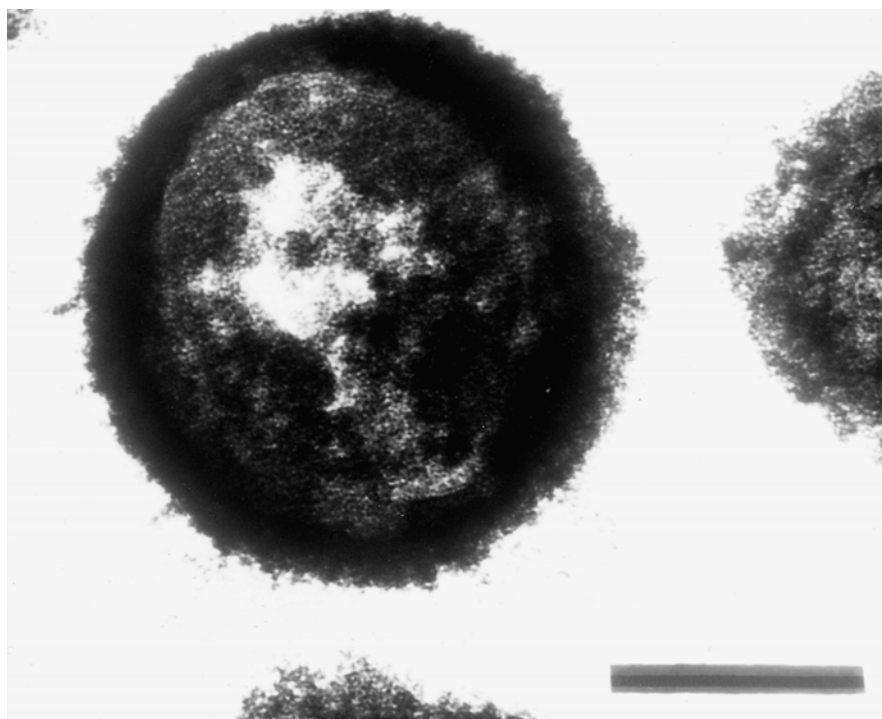


FIGURE BXII.γ.207. Ruthenium-red stained cells of *P. shigelloides* displaying external acidic mucopolysaccharide layer ($\times 34,000$). Bar = 0.5μm. (Reproduced with permission from R.A.Brenden et al., *Reviews of Infectious Disease* 10: 303–316, 1988, ©University of Chicago Press.)

Colwell, 1987). Casein, DNA, elastin, fibrinogen, gelatin, pectate, starch, tyrosine, and Tween 80 are neither degraded nor hydrolyzed.

Formal DNA–DNA hybridization studies on *Plesiomonas* have not been published. Anecdotal information from one epidemiological investigation indicates that all 27 *P. shigelloides* strains constituted a single relatedness group by DNA–DNA hybridization (Holmberg et al., 1986b; Don J. Brenner, personal communication). Between 63 and 100% of *P. shigelloides* strains harbor extrachromosomal elements. Plasmids range in size from 2–8 MDa to very large extrachromosomal elements of 120–312 MDa (Herrington et al., 1987; Ølsvik et al., 1990; Abbott et al., 1991). Plasmid-cured derivatives show loss of expression of β -galactosidase activity (Herrington et al., 1987) and resistance to streptomycin (Marshall et al., 1996). *P. shigelloides* also produces a restriction endonuclease (*Psh* AI) with a novel recognition site of 5'-GACNN/NNGTC (Miyahara et al., 1990). Bacteriocin and phage typing systems have not been described.

An international antigenic scheme recognizes 96 somatic (O) and 48 flagellar (H) antigens of *P. shigelloides* (Aldová and Schubert, 1996). Some serovars show partial or complete identity to *Shigella* somatic antigens. Serovar O17, a common group found in clinical material, reacts with *Shigella* group D antisera (Abbott et al., 1991) and is identical to the O antigen of *S. sonnei* (Sakazaki, 1987). An a, b-a, c type of antigenic relationship also exists between serovar O11 and *S. dysenteriae* 8, serovar O22 and *S. dysenteriae* 7, serovar O23 and *S. boydii* 13, serovar O54 and *S. boydii* 2, and serovar O57 and *S. boydii* 9 (Sakazaki, 1987; Shimada et al., 1994a). Certain plesiomonads share type-specific antigens with *S. flexneri* 6 and common group 1 antigen of *S. dysenteriae* 1 (Albert et al., 1993a). A second unrelated scheme based upon "Schubert" antigens was developed with reference strains from pond water and water insects (Aldová and Schubert, 1996). This scheme is primarily useful in typing environmental isolates; 23 "O" and 5 "H" antigens are recognized.

P. shigelloides is uniformly susceptible *in vitro* to first-, second-, and third-generation cephalosporins, fluoroquinolones, carbapenems, monobactams, and trimethoprim-sulfamethoxazole (Reinhardt and George, 1985; Kain and Kelly, 1989; Clark et al., 1990; Clark, 1992). Widespread resistance to ampicillin, amoxicillin, carbenicillin, mezlocillin, piperacillin, and ticarcillin. Variable results described for tetracycline and aminoglycosides. Most strains produce a β -lactamase using the nitrocefin test (Reinhardt and George, 1985; Clark et al., 1990) but are susceptible to antibiotic- β -lactamase-inhibitor combinations.

The gastrointestinal tract is the primary site from which *P. shigelloides* is recovered. While definitive proof establishing *P. shigelloides* as an enteropathogen is lacking, several lines of evidence support a role in gastroenteritis. These include a low carrier rate, case-controlled investigations, well-circumscribed case reports, and outbreaks of diarrheal disease (Brenden et al., 1988). Carriage rates in asymptomatic individuals range from 0–5.5% with most reported values <0.1% (Arai et al., 1980; Holmberg and Farmer, 1984; Bozsó et al., 1986). Case-controlled investigations have generally found 3- to 20-fold higher isolation rates of plesiomonads from symptomatic patients versus controls (Pitarangsi et al., 1982; Bozsó et al., 1986; Lim et al., 1987). Isolation rates of *P. shigelloides* as high as 22% have been reported from children with bloody diarrhea (Taylor et al., 1986b). Sporadic cases of gastroenteritis have documented high numbers of plesiomonads (3×10^8 CFU/ml) in duodenal biopsies (Penn et al., 1982) and seroconversion to the lipopolysaccharide of infecting strains (van

Loon et al., 1989). Two Japanese outbreaks of *P. shigelloides* diarrheal disease caused by serovars O17:H2 and O24:H5 have been associated with contaminated drinking water (Tsukamoto et al., 1978). A third less well-defined outbreak was linked to oyster consumption (Rutala et al., 1982).

P. shigelloides gastroenteritis has been reported around the world. Diarrheal disease can range from a mild enteritis to frank dysentery (Holmberg et al., 1986b; Kain and Kelly, 1989; Rautelin et al., 1995b). Cholera-like illnesses are rare (Sawle et al., 1986). All age groups are affected. Common symptoms include diarrhea with abdominal pain/cramps. Less frequent symptoms include fever, nausea, and vomiting. Risk factors associated with infection include consumption of raw shellfish, foreign travel, and exposure to reptiles and tropical fish (Davis et al., 1978; Van Damme and Vandepitte, 1980; Holmberg et al., 1986b; Tippen et al., 1989).

Extraintestinal disease principally involves septicemia. At least 20 cases have been reported in the literature to date. Most cases of bacteremia involve persons with underlying conditions including neonatal, hematologic dyscrasias, and liver disease (Clark and Janda, 1991; Delforge et al., 1995; Riley et al., 1996). Meningitis accompanying septicemia is primarily a disease of neonates (Billiet et al., 1989). Prolonged membrane rupture prior to delivery predisposes infants to infection (Clark and Janda, 1991). Infrequent extraintestinal complications of *P. shigelloides* infection include biliary disease (Claesson et al., 1984; Körner et al., 1992), pancreatic abscess (Kennedy et al., 1990), pseudoappendicitis (Fischer et al., 1988), arthritis (Gordon et al., 1983), osteomyelitis (Ingram et al., 1987), and ocular infections (Butt et al., 1997a).

Virulence factors are poorly defined (Janda, 1998). Extracellular factors elaborated by select *P. shigelloides* strains include heat-labile and heat-stable enterotoxins (Manorama et al., 1983; Gardner et al., 1987; Matthews et al., 1988) and cytotoxins (Abbott et al., 1991; Albert et al., 1993a). Most strains produce a cell-associated hemolysin that may be responsible for cytotoxic activity (Janda and Abbott, 1993b). Invasion of HeLa and HEp-2 cells has been reported (Binns et al., 1984; Albert et al., 1993a). Sereny (keratoconjunctivitis) test negative. Siderophores not produced (Daskaleros et al., 1991). Human volunteers fed 10^3 – 10^9 CFU of *P. shigelloides* P012 failed to develop diarrhea (Herrington et al., 1987).

Freshwater environments, including mud and sediment, are the major reservoirs for *P. shigelloides*. A Dutch study (Medema and Schets, 1993) found levels of *P. shigelloides* in surface water samples correlated with fecal pollution (*Escherichia coli*) and the trophic state (chlorophyll *a*, Secchi depth). A subsequent Brazilian investigation did not draw a similar conclusion (de Mondino et al., 1995). A survey of the Suwannee River estuary found most water and sediment samples positive for *P. shigelloides* during three sampling periods (August–February). Flora associated with the ecosystem such as eels, crabs, bream, catfish, crappie, pinfish, clams, and oysters also contained plesiomonads (Miller and Koburger, 1986a). Four of 170 (2.4%) seawater samples were found to contain *P. shigelloides* in one survey (Zakhariev, 1971). Other reported animals found to be colonized with *P. shigelloides* include freshwater fish, tropical fish, dogs, cats, sheep, cows, goats, pigs, monkeys, polecats, turtles, newts, toads, and turkey vultures (Miller and Koburger, 1985; Tippen et al., 1989).

ENRICHMENT AND ISOLATION PROCEDURES

A number of selective media have been designed to recover *Plesiomonas* from environmental samples and contaminated clinical

specimens (e.g., feces). Most of these rely on brilliant green and/or bile salts as inhibitors of competing flora and *m*-inositol, a cyclic polyhydroxyl alcohol, as the differential substrate (von Graevenitz and Bucher, 1983; Farmer et al., 1992). While virtually all *P. shigelloides* are inositol positive, only a limited number of enteric groups (*Enterobacter aerogenes*, *Klebsiella* spp., *Serratia marcescens*) are capable of fermenting this alcoholic sugar.

Inositol-brilliant green-bile salts agar (IBB)¹ has been found to be the best selective media for the recovery of plesiomonads from both clinical (von Graevenitz and Bucher, 1983) and environmental specimens (Miller and Koburger, 1986a). On IBB, *Plesiomonas* colonies appear whitish to pinkish. IBB was 90% sensitive and 100% specific for the recovery of *Plesiomonas* from artificially seeded stool specimens. Optimum *Plesiomonas*/coliform ratios were obtained using IBB (von Graevenitz and Bucher, 1983). However, thermally injured cells may not be satisfactorily recovered using IBB (Miller and Koburger, 1986a). *Plesiomonas* differential agar (PDA), a variation of IBB, was found superior to IBB for the recovery of *P. shigelloides* in a Bangladesh study (Huq et al., 1991). PDA incubated at 42°C produced optimal results with large colonies (3 mm). Alkaline peptone water (pH 8.6) has been found to be the best enrichment medium for plesiomonads by most investigators (von Graevenitz and Bucher, 1983; Janda, unpublished results), although some studies have found this broth to be unsatisfactory (Millership and Chattopadhyay, 1984).

MAINTENANCE PROCEDURES

Plesiomonas strains are sturdier than *Vibrio* or *Aeromonas* isolates but more fragile than common members of the *Enterobacteriaceae*. Working cultures can be maintained on motility deeps for several months. Permanent stock cultures can be prepared by lyophilization or by freezing cultures at -70°C using cryoprotective agents such as bovine serum albumin (fraction V).

PROCEDURES FOR TESTING SPECIAL CHARACTERS

For the demonstration of the properties of *Plesiomonas* strains, the same methods are used as those employed for the characterization of *Aeromonas*, *Vibrio*, and members of the *Enterobacteriaceae* (Schubert, 1984).

DIFFERENTIATION OF THE GENUS *PLESIOMONAS* FROM OTHER GENERA

Although *Plesiomonas* has been transferred to the family *Enterobacteriaceae* based upon phylogenetic relatedness, misidentifica-

tions as a *Vibrio* or *Aeromonas* can occur by virtue of the production of cytochrome oxidase. Key differential tests that separate these taxa are listed in Table BXII.γ.248.

TAXONOMIC COMMENTS

Ferguson and Henderson (1947) initially described *P. shigelloides* as a motile *Enterobacteriaceae* culture ("C27") that possessed the major antigen of *Shigella sonnei* phase I. C2 (now ATCC 14030), however, possessed a number of properties distinct from *S. sonnei* including motility, production of indole, delayed lactose fermentation, and the inability to produce acid from D-mannitol. It was therefore placed in the anaerogenic "Paracolon" group by these authors. Because of polar flagellation, Bader (1954) subsequently placed this C27 in the genus *Pseudomonas*, with the epithet *shigelloides* referring to its *Shigella*-like characteristics (Schubert, 1984). Ewing et al. (1961) proposed that this bacterium be transferred to the genus *Aeromonas* based upon a fermentative metabolism and similarities in morphology, flagella arrangement, and biochemical properties. This recommendation was soon replaced by a proposal to create a new genus for *Aeromonas shigelloides*, *Plesiomonas* (Habs and Schubert, 1962). At least two other proposals have subsequently been made to reassign *P. shigelloides* to either the genus *Fergusonia* or *Vibrio* (Vandepitte et al., 1974).

Current phylogenetic and chemotaxonomic data indicate that plesiomonads are more closely related to the *Enterobacteriaceae* than to any other group (Table BXII.γ.249). These data also include proposed 16S rRNA signature nucleotide differences between the *Enterobacteriaceae* and vibrios (Table BXII.γ.250).

Based on phylogenetic evidence a proposal has been made to reassign *P. shigelloides* to the tribe *Proteeae*, considering the 5S rRNA sequence relatedness to *Proteus mirabilis* and *Proteus vulgaris* (MacDonell and Colwell, 1985). However, this proposal seems untenable as classic defining reactions of the tribe *Proteeae* and the genus *Proteus* are entirely absent in *P. shigelloides* (Table BXII.γ.251).

TABLE BXII.γ.248. Key features separating *Plesiomonas*, *Aeromonas*, and *Vibrio*

Characteristic	<i>Plesiomonas</i>	<i>Aeromonas</i>	<i>Vibrio</i>
Growth in nutrient broth/agar (Difco) containing:			
0% NaCl	+	+	D
6% NaCl	-	-	+
O/129 susceptibility:			
10 µg	d	-	D
150 µg	d	-	D
String test	-	-	+
Gas from glucose	-	D	-

1. Inositol-brilliant green-bile salts agar has the following composition (g/l): proteose peptone (Difco), 10.0; meat extract (Lab Lemco [Oxoid]), 5.0; NaCl, 5.0; bile salts no. 3 (Difco), 8.5; brilliant green (Merck), 0.00033; neutral red (Merck), 0.025; meso-inositol (Merck), 10.0; and agar (Difco), 15.0; pH adjusted to 7.2.

TABLE BXII.γ.249. Characteristics resulting in the reassignment of *P. shigelloides*^{a,b}

Characteristic	<i>Plesiomonas</i>	<i>Enterobacteriaceae</i>	<i>Vibrionaceae</i>	<i>Aeromonadaceae</i>
Possession of ECA	+	+	-	-
Sequence relatedness to <i>Plesiomonas</i> :				
5S rRNA (% homology)	100	96-97	<86	87
16S rRNA (% homology)	100	93-95	91	91
23S rRNA (% homology)	100	91	nd	88

^and, not determined.

^bData from MacDonell et al., 1986; Martinez-Murcia et al., 1992a; East et al., 1992.

TABLE BXII.γ.250. Proposed 16S rRNA signature sequence differences^a

Position (s)	Composition in:		
	<i>Vibrio</i>	<i>E. coli</i>	<i>Plesiomonas</i>
154–167	C–G	U–A	U–A
446–488	U–G	G–C	G–C
614–626	G–C	C–G	C–G
896–903	U–A	C–G	C–G
1123–1149	G–C	U–A	U–A

^aAdapted from Dorsch et al., 1992; Martinez-Murcia et al., 1992a.

FURTHER READING

- Brenden, R.A., M.A. Miller and J.M. Janda. 1988. Clinical disease spectrum and pathogenic factors associated with *Plesiomonas shigelloides* infections in humans. *Rev. Infect. Dis.* 10: 303–316.
- Holmberg, S.D. and J.J. Farmer III. 1984. *Aeromonas hydrophila* and *Plesiomonas shigelloides* as causes of intestinal infections. *Rev. Infect. Dis.* 6: 633–639.
- Miller, M.L. and J.A. Koburger. 1985. *Plesiomonas shigelloides*: an opportunistic food and waterborne pathogen. *J. Food Prot.* 48: 449–457.

TABLE BXII.γ.251. Defining characteristics of the *Proteeae* and *Plesiomonas*^a

Characteristic	<i>Plesiomonas</i>	<i>Proteus</i>	<i>Providencia</i>	<i>Morganella</i>
Phenylalanine deaminase	–	+	+	+
Swarming motility	–	+	–	–
Urea hydrolysis	–	+	D	+
Degradation of L-tyrosine crystals	–	+	+	+
Pigmentation on DL-tryptophan agar	–	+	+	+

^aData from Janda and Abbott (1998a).

List of species of the genus Plesiomonas

1. ***Plesiomonas shigelloides*** (Bader 1954) Habs and Schubert 1962, 324^{AL} (*Pseudomonas shigelloides* Bader 1954, 455.) *shi.gel.loi' des.* M.L. fem. n. *Shigella* a generic name; Gr. suffix *eides* similar; M.L. adj. *shigelloides* *Shigella*-like.

The description of the genus and species are given in Table BXII.γ.252.

Occurs in freshwater and marine environments and inhabitants of those ecosystems including fish, amphibia, and shellfish. Isolated from a variety of animals. Associated with gastroenteritis in healthy adults and children. Occasionally causes systemic infections primarily in persons who are immunocompromised or have underlying illnesses.

The mol% G + C of the DNA is: 51 (Ch).

Type strain: M51, ATCC 14029, CDC 3085-55, DSM 8224, NCIB 9242.

GenBank accession number (16S rRNA): M59159, X74688.

TABLE BXII.γ.252. Characteristics of *Plesiomonas shigelloides*

Test	Reaction or Result
Oxidase	+
Catalase	+
Motility	+
<i>Amino acid decarboxylases (Møller):</i>	
Ornithine decarboxylase	+
Lysine decarboxylase	+
Arginine dihydrolase	+
Methyl red test	+
Voges–Proskauer (25°C, 37°C)	–
Urea hydrolysis	–
ONPG hydrolysis ^a	+
NO ₃ [–] reduced to NO ₂ [–]	+
Utilization of citrate (Simmons)	–
Utilization of malonate, mucate	–
Utilization of acetate	d
Acid production from D-glucose	+
Gas production from D-glucose	–
<i>Acid production from:</i>	
m-Inositol, maltose, trehalose	+
D-Galactose, glycerol, lactose, D-mannose, melibiose, salicin	d
Adonitol, L-arabinose, cellobiose, dextrin, dulcitol, D-mannitol, melezitose, D-sorbitol, sucrose, L-rhamnose, D-xylose	–
Gelatin liquefaction	–
Tyrosine clearing	–
Deoxyribonuclease	–

^aONPG, o-nitrophenyl-β-galactopyranoside.

Genus XXVIII. *Pragia* Aldová, Hausner, Brenner Kocmoud, Schindler, Potužníková, and Petráš 1988a, 187^{VP}

JIRI SCHINDLER, SR.

Pra'gi.a. L. fem. n. *Pragia* of Prague, the city in which strains of the genus were identified.

Gram-negative, oxidase-negative, catalase-positive, peritrichously flagellated rods conforming to the definition of the family *Enterobacteriaceae*. Facultatively anaerobic. Colonies approximately 0.5 mm in diameter are formed on nutrient, Endo, and MacConkey agars. Colonies formed on desoxycholate-citrate agar have a black center. Nonhemolytic on sheep blood agar. A *Shigella*-like odor is produced on nutrient agar. A pigsty-like odor is

produced on Endo agar overlaid with a sloping surface of agar. Growth at 4°C, **but not at 42°C**. Optimal growth is obtained at temperatures between 22 and 37°C. Fermentative, forms acid without gas from D-glucose and D-galactose. Motile, reduces nitrates to nitrite, **utilizes citrate, produces hydrogen sulfide, and oxidizes gluconate**. The majority of strains is positive for the methyl red test and delayed positive for acid production from

glycerol and *myo*-inositol. Negative reactions in tests for indole production, Voges–Proskauer test, phenylalanine deaminase, lysine and ornithine decarboxylases, arginine dihydrolase, urease, deoxyribonuclease, malonate, ONPG, growth in KCN, and acid production from L-arabinose, cellobiose, dulcitol, lactose, maltose, D-mannitol, D-mannose, melibiose, raffinose, L-rhamnose, D-sorbitol, D-sorbose, starch, sucrose, and trehalose. Additional biochemical reactions are given in Table BXII.γ.193 of the chapter on *Enterobacteriaceae*.

The mol% G + C of the DNA is: 46–47.

Type species: **Pragia fontium** Aldová, Hausner, Brenner Kocmoud, Schindler, Potužníková, and Petráš 1988a, 187.

FURTHER DESCRIPTIVE INFORMATION

Pragia is a member of the *Gammaproteobacteria*. *Pragia* DNA is 17% or less related to members of all other genera in *Enterobacteriaceae*. Bacteriocin-like agents called fonticins were observed in five *P. fontium* strains. They were active against other *P. fontium* strains as well as against *Budvicia aquatica*, but were not active

against *Escherichia coli* or *Shigella sonnei*. Similar bacteriocin-like agents, budvicins, with the same specificity and properties have been isolated from *Budvicia aquatica*. Fonticins have very narrow inhibition zones on indicator strains, are heat sensitive (inactivated at temperatures between 42 and 55°C) and are resistant to trypsin (Šmarda, 1987). They are of corpuscular nature and resemble contracted tails of T4 bacteriophage (Šmarda, 1987). Isolated from fresh water springs or fountains in the South Bohemia region of the Czech Republic (Aldová et al., 1983, 1988b). One human isolate not associated with disease. There is no evidence of pathogenicity for humans or animals.

DIFFERENTIATION OF THE GENUS *PRAGIA* FROM OTHER GENERA

Pragia is easily distinguishable from leminorellae, *B. aquatica*, *C. freundii*, and salmonellae based on whole cell protein analysis (Aldová et al., 1988a; Schindler et al., 1992). Biochemical tests useful in differentiating *Budvicia* from other H₂S-positive genera in *Enterobacteriaceae* are shown in Table BXII.γ.253.

List of species of the genus *Pragia*

1. **Pragia fontium** Aldová, Hausner, Brenner Kocmoud, Schindler, Potužníková, and Petráš 1988a, 187.^{VP}
fon' ti.um. L. gen. pl. n. *fontium* from springs or fountains, the source of isolation of all but one strain.

The characteristics are as described for the genus and as listed in Table BXII.γ.193 in the chapter on *Enterobacteriaceae*. Occurs in drinking water. One isolate from the stool

of a healthy human. No indication of pathogenicity for humans or animals.

The mol% G + C of the DNA is: 46–47 (*T_m*).

Type strain: DRL 20125, HG 16, ATCC 49100, CNTCTC Eb11/82, CCUG 18073, CDC 963-83, DSM 5563.

GenBank accession number (16S rRNA): AJ233424.

TABLE BXII.γ.253. Differentiation of *Pragia* from other H₂S-positive genera in *Enterobacteriaceae*^{a,b}

Characteristic	<i>Pragia</i>	<i>Budvicia</i>	<i>Edwardsiella</i>	<i>Salmonella</i>	<i>Citrobacter</i>	<i>Proteus</i>
Lysine decarboxylase	—	—	+	v	—	—
Ornithine decarboxylase	—	—	+	+	v	v
Urease	—	+	—	— ^c	v	+
ONPG	—	+	—	—	+	—
Indole production	—	—	v	+	v	v
Citrate	+	—	—	+	+	v
Motility	+	v	+ ^d	+	+	+
KCN, growth	—	—	—	— ^e	v	+
L-Arabinose, acid	—	v	v	+ ^f	+	—
Gluconate oxidation	+	—	—	—	—	—
Growth, 42°C	—	—	+	—	—	—

^aFor symbols, see standard definitions.

^bReactions carried out at 37°C, unless otherwise noted.

^cSubspecies *Salmonella choleraesuis* subsp. *arizonae* and *Salmonella choleraesuis* subsp. *diarizonae* are positive.

^dExcept for *E. ictaluri*.

^eSubspecies *Salmonella choleraesuis* subsp. *houtenae* is positive.

^fExcept serovars *choleraesuis* and *typhi*.

Genus XXIX. *Proteus* Hauser 1885, 12^{AL}

JOHN L. PENNER

Pro'te.us. Gr. n. *Proteus* an ocean god able to change himself into different shapes.

Straight rods, 0.4–0.8 × 1.0–3.0 μm. Gram negative. **Motile** by peritrichous flagella. **Most strains swarm with periodic cycles of migration producing concentric zones, or spread in a uniform film, over moist surfaces solidified with agar or gelatin.** The organisms in this genus conform to the definition of the family *Enterobacteriaceae*. **They are facultatively anaerobic, chemoorganotrophic, having both a respiratory and a fermentative type of**

metabolism. Optimal growth temperature is 37°C. **Oxidase negative; catalase positive. Methyl red positive; species vary in indole production, Voges–Proskauer, and Simmons citrate tests.** They oxidatively deaminate phenylalanine and tryptophan. Urea is hydrolyzed. **Lysine decarboxylase negative and arginine dihydrolase negative; only *Proteus mirabilis* decarboxylates ornithine.** All but *Proteus myxofaciens* decompose tyrosine to produce a clearing on

agar media in which the insoluble amino acid is incorporated. Grow on KCN. H₂S is usually produced. Malonate is not utilized. D-glucose and a few other carbohydrates are catabolized with production of acid and usually gas. Does not produce acid from inositol or from straight chain tetra-, penta-, or hexahydroxy—alcohols, but generally produces acid from glycerol. One or more species ferment maltose, sucrose, trehalose, and D-xylose. Human pathogens, causing urinary tract infections; also are secondary invaders, causing septic lesions at other sites of the body. Occurs in the intestines of humans and a wide variety of animals; also occurs in manure, soil, and polluted waters. *P. myxofaciens* has been isolated only from gypsy moth larvae. Based on 16S rDNA sequence analysis, *Proteus* belongs to the family *Enterobacteriaceae* within the *Proteobacteria* (Woese et al., 1985; Niebel et al., 1987; Stackebrandt et al., 1988).

The mol% G + C of the DNA is: 38–41 (Falkow et al., 1962).

Type species: ***Proteus vulgaris*** Hauser 1885, 12 emend. Brenner, Hickman-Brenner, Holmes, Hawkey, Penner, Grimont and O'Hara 1995, 870.

FURTHER DESCRIPTIVE INFORMATION

Comparative analysis of 16S ribosomal RNA sequences (the oligonucleotide cataloging method) showed that *P. mirabilis* belongs to the *Gammaproteobacteria*, which includes the family *Enterobacteriaceae* (Woese et al., 1985; Stackebrandt et al., 1988). DNA relatedness studies have shown that members of the genus *Proteus* are only distantly related to the other species in the family *Enterobacteriaceae* and from an evolutionary standpoint they are therefore at the periphery of this family (Brenner et al., 1978). *P. mirabilis* strains are a highly related group. They are 55% related to *P. vulgaris* and to *P. penneri* strains. *P. penneri* consists of a highly related group of indole-negative strains that ferment salicin and hydrolyze esculin. They were classified as *P. vulgaris* biogroup 1 (Brenner et al., 1978) before they were recognized as a new species (Hickman et al., 1982c). At present, *P. vulgaris* consists of two biogroups. *P. vulgaris* biogroup 2 consists of strains that are indole positive, that ferment salicin and hydrolyze esculin. Strains of this biogroup conform to the description of the genus and include the vast majority of the *P. vulgaris* strains isolated from clinical sources. Computerized analysis of electrophoretic protein patterns and DNA relatedness tests showed that *P. vulgaris* biogroup 3 strains (indole positive, salicin negative, and esculin negative) are a heterogeneous group that includes four genomospecies, one of which includes the type strain (Costas et al., 1993; Brenner et al., 1995). Future studies will likely lead to the identification of new species within this biogroup.

On media solidified with gelatin or agar, the bacteria migrate from the point of inoculation to spread over the surface of the medium. The phenomenon, known as swarming, is attributed to the dimorphic nature of the bacteria. The short mononucleoid, peritrichously flagellated rods that occur singly in liquid media, referred to as swimmers, differentiate when transferred to solid media to produce long, nonseptate, multinucleoid swimmers with large increases in the number of flagella (Belas, 1992). Glutamine has been identified as the chemoattractant that signals the surface-induced differentiation (Allison et al., 1993). Differentiation into swarmer cells is associated with marked increases in synthesis of lipopolysaccharide with long O-polymers and a higher fluidity of the outer membrane (Sidorczyk and Zych, 1986). The production also of extracellular slime forming a thin cocoon covering swarms of cells (VanderMolen and Williams, 1977) or capsular polysaccharide containing galacturonic acid

and N-acetylgalactosamine facilitates migration of swarmer cells (Stahl et al., 1983; Gygi et al., 1995). Swarmer cells are the virulent forms of *Proteus* and during differentiation a coordinated increase occurs in the synthesis of urease, hemolysins, and proteases (Jin and Murray, 1987; Allison et al., 1992, 1994).

Three phases have been recognized in the process of swarming. In the first phase, differentiation of swimmer cells into swarmer cells takes place. In the second phase, groups of swarmer cells migrate from the point of inoculation and continue to migrate until either a change in direction occurs or the number of swimmers is reduced by a loss of some along the way. In the third phase, migration ceases, which results in consolidation. At this point, the swarmer cells form septa to become short cells that multiply for a period of time before producing another generation of swarmer cells (Williams and Schwarzhoff, 1978; Rózalski et al., 1997). Cyclic repetitions of the phases leads to concentric rings surrounding the site of inoculation. Some strains (or variants) produce a uniform film (referred to as spreading) without periodic cycles (C variant of Belyavin, 1951; Z variant of Coetzee and Sacks, 1960). Some strains neither spread nor swarm and merely form distinct colonies.

The swarming of *Proteus* makes it difficult to isolate bacteria of other species from pathological specimens plated on agar media and to prevent swarming; inhibitors such as bile salts and detergents are incorporated in the media (Kopp et al., 1966). The compound *p*-nitrophenyl glycerol, at concentrations of 0.1–0.3 mM, inhibits swarming without affecting flagellation or motility, and because of its low toxicity to *Proteus* and other bacteria, its use has been advocated (Williams, 1973). With continued incubation, however, the *Proteus* strains eventually swarm in the presence of this compound and thus isolations of *Campylobacter* species and other bacteria that require longer periods of incubation require other measures such as the incorporation of antibiotics in the isolation media (Firehammer, 1987). The incorporation of charcoal in the medium has been reported to suppress swarming without interfering with the growth of other bacterial species (Alwen and Smith, 1967). Other methods of preventing swarming include reduction of the salt concentration of the media and increasing the concentration of agar to 4% (New Zealand agar) or 7% (Japanese agar).

In the Dienes test, swarms of two strains that do not penetrate into each other but form a sharp line between them indicate that the strains are different but strains that produce swarms that merge into each other without a line of demarcation are interpreted to be the same (Dienes, 1946). This test has been used in epidemiological investigations of *Proteus* infections, mostly of *P. mirabilis*, to determine identity and nonidentity of clinical isolates (Story, 1954; Skirrow, 1969). However, strains of different biovars have been found to swarm into each other (Kippax, 1957), and, in some cases, results obtained with the test fail to correlate with bacteriophage typing results (Hickman and Farmer, 1976). The production of the demarcation line is apparently unrelated to the flagellar (H) antigens (Sourek, 1968; Skirrow, 1969) but appears to be dependent on both the bacteriocins produced by the swarming strains and the bacteriocins to which they are susceptible (Senior, 1977a). The Dienes test is useful in combination with other procedures for epidemiological typing and when its limitations are recognized.

A well-known distinguishing feature of the genus is the production of urease. It is an important taxonomic criterion that differentiates the genus from most other species of *Enterobacteriaceae* except for the genera *Morganella* and *Providencia*, some

species of which also produce the enzyme. The enzyme is constitutive in most strains of *P. mirabilis* and inducible in *Proteus vulgaris*, *Proteus penneri*, and some *P. mirabilis* strains (Mobley et al., 1987; 1995; McLean et al., 1988; Mobley and Hausinger, 1989). The enzyme is a 212–280 kDa protein consisting of three subunits (Jones and Mobley, 1989; Mobley and Hausinger, 1989) and is known to have an important role in the development of human urinary tract infections (Mobley, 1996).

An important distinguishing feature of *Proteus*, *Providencia*, and *Morganella* is their ability to oxidatively deaminate a variety of amino acids, producing keto acids and ammonia (Bernheim et al., 1935; Stumpf and Green, 1944; Singer and Volcani, 1955). The carboxylic acids bind iron and have siderophore activity (Evanylo et al., 1984; Drechsel et al., 1993). Addition of ferric chloride solution to keto acids in aqueous solution produces different colors dependent upon the amino acid from which the keto acid was produced (Singer and Volcani, 1955), and the same colors are produced when ferric chloride solution is added to bacteria grown on nutrient media supplemented with the amino acids. This is the basis of the diagnostic test used in the clinical laboratory to distinguish these three genera from other *Enterobacteriaceae*. Tests for phenylalanine deaminase and tryptophan deaminase are widely used (Henriksen, 1950; Thibault and Le Minor, 1957). The gene encoding an amino acid deaminase (*aad*) in *P. mirabilis* has been identified (Massad et al., 1995). The probe for the *aad* gene does not hybridize with DNA from *Providencia* and *Morganella* strains, suggesting that the deaminases of these three genera are unrelated. As with ureases, an end product of the deaminases is ammonia, which is now known to have an important role in pathogenesis by elevating pH, but the contributions of the deaminases in the pathogenic process have yet to be determined (Rózalski et al., 1997).

The production of amines by *Proteus* species and *Morganella morganii* when grown in nutrient broth is a distinguishing biochemical property of the two genera (Proom and Woiwod, 1951). The inability of *Providencia rettgeri*, classified during that period as *Proteus rettgeri*, to produce amines under these conditions was cited as an important criterion in their proposal for removing this species from the genus *Proteus*. Simplified tests to detect amines for differentiation of these genera from other *Enterobacteriaceae* in the clinical laboratory have not been developed. However, the amines continue to be of interest, particularly with the finding that a carcinogenic nitrosamine (*N*-nitrosodimethylamine) is produced in urine by *P. mirabilis* during the course of a urinary tract infection (Brooks et al., 1972).

The antigens of the genus *Proteus* that are used for differentiating the serovars are the lipopolysaccharide (O) antigens, the flagellar (H) antigens, and the capsular (K or C) antigens. The O antigenic scheme of Kauffmann and Perch for *P. vulgaris* and *P. mirabilis* (Kauffmann, 1966) lists 49 serovars. Tube agglutination tests and cell suspensions boiled for 1 h were used to determine the O specificities. Additional serovars have been identified that are not included in the scheme of Kauffmann and Perch (Larsson and Olling, 1977; Penner and Hennessy, 1980).

Seventeen serovars of the Kauffmann and Perch scheme were found to include only *P. vulgaris* strains, 27 included only *P. mirabilis* strains, and five included strains of both species. Since isolates generally agglutinate in antisera against the same species, separation of the serovars to provide individual schemes for each species facilitated routine serotyping (Penner and Hennessy, 1980). In the system developed by Penner and Hennessy for serotyping *P. vulgaris* and *P. mirabilis*, the passive (indirect) hem-

agglutination technique was employed to determine the specificities of soluble heat extracted antigens. Their system included 22 *P. vulgaris* and 32 *P. mirabilis* strains from the Kauffmann and Perch scheme, five new serovars of *P. vulgaris*, and six new serovars of *P. mirabilis*. However, from the large numbers of untypeable smooth strains isolated in epidemiological studies, it is clear that a substantial number of serovars remain undefined (Larsson and Olling, 1977; Larsson, 1980, 1984; Penner and Hennessy, 1980). Reports differ on the distributions of serotypes of *P. mirabilis*, but among those listed the most frequently occurring are isolates of O serotypes 3, 6, 10, 27, and 28 (Lányi, 1956; de Louvois, 1969; Larsson and Olling, 1977; Penner and Hennessy, 1980; Rózalski et al., 1997). In these earlier studies the strains of *P. vulgaris* were not differentiated into biogroups and thus the collections included a few biogroup 1 strains that would now be classified as *P. penneri* (Hickman et al., 1982c). A system for serotyping *P. penneri* has not been developed. However, one serotypically homogeneous group of strains collected from three countries has been identified and designated group O61 (Sidorczyk et al., 1996). The molecular structures of the lipopolysaccharides of at least 10 different strains of *P. penneri* have been determined (Zych et al., 1997). Examination of the variety of structures suggests the possibility that more serotypes will be defined in the future.

Much research has recently focused on the structures of the lipopolysaccharide (LPS) of *P. vulgaris*, *P. mirabilis*, and *P. penneri*, and progress in this area has been the subject of a major review (Rózalski et al., 1997). The structure of lipid A is comparable to the lipid A of *Escherichia coli* and salmonellae but differs in its fatty acid composition and in the presence of a 4-deoxy-arabinosyl residue that substitutes the phosphate residue of glucosamine (Sidorczyk et al., 1983). The biological activity of *Proteus* lipid A is comparable to that of *E. coli* and salmonellae (Sidorczyk et al., 1978). The core oligosaccharides of *P. vulgaris* and *P. mirabilis* consist of Kdo (3-deoxy-D-manno-octulosonic acid), heptoses, and hexoses, as in *E. coli* and salmonellae, but differ in that they have terminal or chain-linked galacturonic acids that contribute significantly to the immunospecificity of the oligosaccharide (Kotelko, 1986; Sidorczyk et al., 1987). The O-polysaccharide polymers consist of repeating units composed of hexoses, hexosamines, 6-deoxy sugars, uronic acids with amino acids attached to their carboxyl groups, and other noncarbohydrate acidic components (Sidorczyk et al., 1975; Kotelko, 1986; Sidorczyk et al., 1995).

P. vulgaris strains OX19 and OX2 and *P. mirabilis* strain OXK are the test strains for O antigens 1, 2, and 3, respectively, in the antigenic scheme of Kauffmann and Perch (Kauffmann, 1966). Antibodies in sera from patients with rickettsial infections agglutinate either one or two of these strains. This reaction is the basis of the Weil-Felix test, which is a presumptive test for rickettsial infections. Sera from patients with endemic or epidemic typhus agglutinate the OX19 strain, patients infected with the spotted fever group of rickettsiae agglutinate both OX19 and OX2 strains, and patients with scrub typhus agglutinate the OXK strain. The reactions have been suspected to be due to LPS and the finding that sera from patients with spotted fever react with LPS from both the rickettsial strain and the *P. vulgaris* OX2 strain indicates this to be the case (Amano et al., 1993). Since *Proteus* infections may also stimulate antibody production and *P. mirabilis* strains with the O3 antigen are among the most frequently isolated, positive Weil-Felix tests should be interpreted with caution. Furthermore, the heterophile antigens to which the cross-reactions in the Weil-Felix test are attributed also occur in *Legionella*

bozemanii and *Legionella micdadei*, indicating that these antigenic specificities are more widely distributed than originally believed (Sompolsky et al., 1986; Westfall et al., 1986).

Analysis of the flagellar (H) antigens of *P. vulgaris* and *P. mirabilis* was carried out by Perch (1948). A total of 19 were identified and included in the Kauffmann and Perch antigenic scheme (Kauffmann, 1966). Cross-reactions among the H antigens are numerous and complex. Five antigenic factors were identified in the H 1 complex and other H antigens showed even greater complexity. The use of H antigens for routine serotyping of *P. vulgaris* and *P. mirabilis* strains has not been adopted and has been limited essentially to the initial studies of Kauffmann and Perch (Perch, 1948). The flagellar antigens of *P. penneri* have not been investigated.

Capsular antigens, unlike the K antigens of other *Enterobacteriaceae*, were reported to be present in the genus *Proteus* (Namioka and Sakazaki, 1959). These antigens, designated C antigens to distinguish them from the *E. coli* L and B antigens, inhibit O agglutination, retain antibody binding capacity after 1 h at 100°C, and have antigenic specificities restricted to certain O groups. The authors suggested that numerous serological types of C antigens occur in the genus *Proteus*, but a system for identifying them has not been developed. Research has focused on the role of capsules in facilitating swarming (Stahl et al., 1983), and a polysaccharide that evidently has this effect has been identified and chemically characterized (Gygi et al., 1995). Investigations are also under way to examine the role of polysaccharide capsules or glycocalyxes in the formation of biofilms during infections of the urinary tract. It has been hypothesized that the biofilm is a major virulence factor because it enhances bacterial adhesion, surrounds the bacteria to protect them from the immune response and from exposure to antibiotics, and provides a microenvironment in the kidney and bladder that is conducive to stone formation (Nickel et al., 1985; McLean et al., 1988; Clapham et al., 1990).

Since the last edition of the *Manual*, numerous investigations of the surface structures of *P. vulgaris*, *P. mirabilis*, and *P. penneri* have resulted in clarification of the nature of the well-known hemagglutinins and fimbriae and in the discovery of several new types of fimbriae. A greater understanding of the mechanisms of pathogenesis has been gained. The MR/P (mannose resistant *Proteus*-like) fimbriae, also known as type IV fimbriae, have been the most extensively studied. These are highly immunogenic proteins 7 nm in diameter. The MR/P fimbriae have been isolated and purified and the genes involved in their determination have been identified (Sareneva et al., 1990; Bahrani et al., 1991; Bahrani and Mobley, 1993, 1994). Evidence obtained in experiments using the mouse model indicates that these fimbriae are associated with colonization of the upper regions of the urinary tract (Bahrani et al., 1994).

The MR/K (mannose resistant, *Klebsiella*-like) type 3 hemagglutinins are thin (diameter of 4–5 nm) and hemagglutinate tanned erythrocytes. Two antigenic types have been found but neither type is antigenically related to *Klebsiella* fimbriae (Old and Adegbola, 1982). MR/K hemagglutinins are more characteristic of *P. penneri* than of *P. mirabilis* (Yakubu et al., 1989). *Proteus* strains with MR/K hemagglutinins have a high affinity for the materials of which catheters are made, and this feature has been cited as a factor in prolonged chronic urinary tract infections of catheterized patients (Mobley et al., 1988; Yakubu et al., 1989; Roberts et al., 1990).

Of the newly discovered fimbriae, one type designated PMF

for *P. mirabilis* fimbriae, has been suggested to be important in the colonization of the bladder, but not the kidney (Bahrani et al., 1993; Massad et al., 1994b). Another type, referred to as ambient temperature fimbriae (ATF) of *P. mirabilis*, is optimally produced at 23°C and apparently does not have a role in pathogenesis (Massad et al., 1994a). A third type, also expressed by *P. mirabilis*, was referred to as nonagglutinating fimbriae (NAF) (Tolson et al., 1995). It does not have the hemagglutinating properties of MR/P or MR/K fimbriae, but adheres to uroepithelial cells, indicating possible importance as a virulence factor in establishing urinary tract infections.

The ability of *Proteus* species to produce cytotoxic hemolysins has been known since 1919 (Wenner and Rettger, 1919), but definitive studies had to await the development of molecular biological technologies. With the aid of new techniques, it has been established that *Proteus* species produce two distinct hemolysins (Welch, 1987). The HlyA hemolysin is genetically related to the α -hemolysin of *E. coli*, has a molecular size of 110 kDa, and is a Ca^{2+} -dependent, pore-forming cytotoxin (Koronakis et al., 1987; Welch, 1991). The HpmA hemolysin (166 kDa) is related to the ShlA hemolysin of *Serratia marcescens* and is calcium independent (Uphoff and Welch, 1990). A protein, HpmB (63 kDa), which is closely related to the ShlB protein of *S. marcescens* is responsible for the transport and activation of the HpmA protein (Swihart and Welch, 1990b; Uphoff and Welch, 1990). Nearly all the *P. vulgaris* and *P. mirabilis* strains produce the HpmA hemolysin, but only a few *P. vulgaris* produce the HlyA hemolysin (Swihart and Welch, 1990a). Most strains of *P. penneri* produce the HpmA hemolysin and most also produce the HlyA hemolysin (Rózalski and Kotelko, 1987; Lukomski et al., 1990; Senior, 1993). Expression of hemolysins is associated with increased cytotoxicity, cell invasiveness, and a lower LD₅₀ in mice, but is not associated with increased kidney colonization or histopathological changes in murine urinary tract infections (Peerbooms et al., 1983, 1984, 1985; Mobley and Chippendale, 1990).

Proteus spp. are among the few Gram-negative species capable of producing proteolytic enzymes that degrade immunoglobulins. Strains of *P. vulgaris*, *P. mirabilis*, and *P. penneri* have been shown to produce immunoglobulin A (IgA) proteases (Senior et al., 1988; Wassif et al., 1995). Analysis of the proteolytic activity of one *P. mirabilis* strain isolated from a patient with urinary tract infection showed that it has a much wider range of activity than the usual IgA proteases. Both serum and secretory IgA1 and IgA2, both the free and IgA-bound secretory component, and IgG are susceptible targets of this enzyme but its role as a virulence factor in urinary tract infections remains uncertain (Loomes et al., 1990). A polypeptide with protease activity that is secreted by uropathogenic *P. mirabilis* has been implicated as a virulence factor in enhancing the ability of the organisms to infect the urinary tract (Zhao et al., 1999).

The antibiotic susceptibility of *P. myxofaciens* is not known but a feature common to the other three *Proteus* species is an intrinsic resistance to bacitracin and the polymyxins (Potee et al., 1954; Von Graevenitz and Nourbakhsh, 1972; Shimizu et al., 1977). Susceptibility to colistin (colymycin) is rare among *Proteus* species, and if a strain is susceptible to this antibiotic, it is most likely not to be a *Proteus* (Li and Miller, 1970). These cyclic polycationic antibiotics are believed to bind to the charged Kdo-lipid A region of the LPS to cause membrane disruption. Substitution of the phosphate ester bonds of the lipid A with L-arabinoside-4-amine in *Salmonella typhimurium* leads to resistance of the bacteria to polymyxin (Vaara et al., 1981). *Proteus* strains have lipid A that

contains L-arabinoso-4-amine naturally, and this has been suggested as a contributing factor for their natural resistance to these antibiotics (Sidorczyk et al., 1987; Rózsalski et al., 1997).

P. mirabilis is the species most frequently isolated from human infections and is generally also the most susceptible to antibiotics, but some very resistant isolates have been reported (Penner et al., 1982). Increasing resistance, particularly to tetracyclines, has been noted since the 1950s (Potee et al., 1954; Chiu and Hoepflich, 1961). The majority of the strains are susceptible to chloramphenicol, but many carry an inducible chloramphenicol acetylase gene (Charles et al., 1985) that could lead to failure in the treatment of brain abscesses (Hawkey, 1997). The occurrence of both nitrofurantoin-sensitive and nitrofurantoin-resistant strains in clinical specimens has been reported.

P. mirabilis is the most susceptible of the three species to the β -lactam group of antibiotics, although resistant strains occur and the necessity of performing antibiotic susceptibility tests is stressed (Barry and Hoepflich, 1973; Shafi and Datta, 1975). *P. mirabilis*, unlike the other two species, does not appear to carry a chromosomal cephalosporinase or an AmpC type penicillinase, and thus many infections can be treated with either cephalosporins or ampicillin (Livermore, 1995). Only rarely is imipenem-resistant *P. mirabilis* isolated, and it has been suggested to be due to altered penicillin-binding proteins (Neuwirth et al., 1995).

Most strains of *P. mirabilis* are susceptible to aminoglycosides, but some strains may acquire resistance factors (R factors) that cause resistance to gentamicin and kanamycin. Amikacin is effective against *P. mirabilis* strains but some have been reported to be resistant to this antibiotic as well as to gentamicin (Briedis and Robson, 1976; Drasar et al., 1976). Resistant *P. mirabilis* can be anticipated to be more frequently encountered in the future, especially in hospital outbreaks, as increased usage has been shown in the past to lead to increased resistance.

Antibiotics generally used to treat infections caused by *P. mirabilis* are trimethoprim or one of the fluoroquinolones such as ciprofloxacin. However, susceptibility tests should be performed for each isolate (Hawkey, 1997).

P. vulgaris is separable into two biogroups, but most of the data on antibiotic susceptibility do not distinguish strains on this basis, and results of antibiotic susceptibility studies in earlier periods did not distinguish between *P. vulgaris* and *P. penneri*. Assessments of the susceptibilities of *P. vulgaris* have been made based on mixtures of strains. However, it is now clear that *P. vulgaris* strains are more resistant than *P. mirabilis* strains. The high levels of *P. vulgaris* resistance to penicillins and cephalosporins is a feature distinguishing it from *P. mirabilis*. Carbenicillin is the only penicillin that has been shown to have significant activity against *P. vulgaris* strains. Most strains, however, are susceptible to aminoglycosides. A small percentage are resistant to gentamicin and kanamycin, but, in most cases, are susceptible to amikacin. Both chloramphenicol-resistant and chloramphenicol-susceptible *P. vulgaris* strains have been reported to be present in clinical specimens, but it is not known what proportion of these would now be classified as *P. penneri*, which are characteristically resistant to this antibiotic (Hickman et al., 1982c).

The high levels of resistance of *P. penneri* strains to chloramphenicol is a definitive criterion in the description of the species (Hickman et al., 1982c). Although increasing numbers of laboratories are identifying *P. penneri* strains, the numbers reported in susceptibility studies are relatively small for a general assessment of the susceptibility of the species. Reports from several laboratories, however, confirm the resistance to chloram-

phenicol as a species characteristic (Hickman et al., 1982c; Hawkey et al., 1983; Fuksa et al., 1984; Piccolomini et al., 1987). It also appears that the species is more resistant to penicillins than is *P. vulgaris*. In a study of 45 isolates, it was found that ceftizoxime, ceftazidime, moxalactam, and cefoxitin would be effective for treatment of *P. penneri* infections (Fuksa et al., 1984). In another study of 39 isolates, imipenem was found to be the most effective antibiotic against isolates of this species (Piccolomini et al., 1987). Isolates were also found to be susceptible to four aminoglycosides (gentamicin, tobramycin, netilmicin, and amikacin), and their use was recommended for treatment of systemic infections (Fuksa et al., 1984). Isolates of the three species are susceptible to the quinolones, particularly to ciprofloxacin and norfloxacin, but a small number of *P. penneri* isolates show higher levels of resistance (Hawkey and Hawkey, 1984).

P. vulgaris, *P. mirabilis*, and *P. penneri* are important human pathogens causing primary and secondary infections. Of the three species, *P. mirabilis* is most frequently isolated from clinical specimens and is one of the most important pathogens of the urinary tract. Underlying conditions such as diabetes or structural abnormalities of the urinary tract are associated with infections acquired outside the hospital (Grossberg et al., 1962; Wallace and Petersdorf, 1971). Hospital-acquired infections commonly occur in patients with predisposing conditions such as catheterization, urological instrumentation, or surgery of the urinary tract. Infectious strains are transmitted from the intestinal flora of the patient or contracted through transmission from other patients or from a common reservoir (Dutton and Ralston, 1957; Kippax, 1957; Chow et al., 1979). *Proteus* urinary tract infections may give rise to bacteremias that are difficult to treat and often fatal. *Proteus* bacteria may cause infections at other sites of the body if favorable conditions exist. They may cause infections of wounds, burns, and respiratory tracts and have been isolated from the eyes, ears, and the throat. *Proteus* strains may also infect the neonatal umbilical stump, and fatal bacteremias and meningitis may result (Becker, 1962; Librach, 1968; Shortland-Webb, 1968; Burke et al., 1971). *P. mirabilis* has also been isolated from both children and adults with osteomyelitis (Levy and Ingall, 1967; Meyers et al., 1973).

Proteus species have been implicated as causative agents of enteritis and were isolated from 46.2% of infants with diarrhea (Lányi, 1956). However, in light of the fact that approximately one-quarter of the population are intestinal carriers of *Proteus*, it is difficult to assess this finding as convincing evidence of a role for strains of this genus in the etiology of human enteritis. Furthermore, species of the genus *Campylobacter* that are now known to be major causes of human diarrhea were not recognized at the time of the earlier studies.

Recent research on the modification of peptidoglycan (murein) by bacterial enzymes of Gram-positive and Gram-negative bacteria suggest possible involvement of *P. vulgaris* and *P. mirabilis* in the development of arthritic syndromes. It has been demonstrated that one strain of *P. vulgaris* and 17 strains of *P. mirabilis* possess acetyltransferase activity for converting the muramyl residue of peptidoglycan to the 2,6-diacetylmuramyl derivative (Fleck et al., 1971; Gmeiner and Kroll, 1981; Dupont and Clarke, 1991; Clarke and Dupont, 1992). This has the effect of creating resistance of the peptidoglycan to the hydrolytic enzymes in phagocytic cells and serum, resulting in the persistence and circulation in the host of high molecular weight peptidoglycan fragments, which have been implicated in animal models as the inciting agents of rheumatoid arthritis (Chedid et al., 1978; Cro-

martie, 1981; Clarke and Dupont, 1992). It is also noteworthy that antibodies against *Proteus* species have been detected in patients with rheumatoid arthritis (Ebringer et al., 1985). The possibility that organisms of the genus *Proteus* could be involved in the development of rheumatoid arthritis is an interesting concept, but more research is required (Clarke and Dupont, 1992).

Epidemiological typing is performed at present only for *P. vulgaris* and *P. mirabilis*. For reference work, O-serotyping is the method widely accepted as the standard but it is carried out only in a few laboratories that have the large set of typing antisera (Perch, 1948; Kauffmann, 1966; Penner and Hennessy, 1980; Larsson, 1984; Hawkey et al., 1986b). The antisera are not commercially available, and other systems for typing have been introduced and tested with the objective of enabling clinical laboratories to conduct epidemiological studies without the high costs associated with O-serotyping. Among these alternative methods is the Dienes test (described above), which can be readily performed but lacks specificity (Kippax, 1957; Hickman and Farmer, 1976). The incompatibility of two swarms that produce a line of demarcation between them is believed to be due to both the production of and susceptibility to bacteriocins (Senior, 1977a). Typing based on sensitivity to bacteriocins (proticins) has been advocated since 1965 (Cradock-Watson, 1965), and several systems have been developed (Al-Jumaili, 1975; Senior, 1977b; Kusek and Herman, 1980). In a study using the proticine production and proticine sensitivity method, O serotyping, and the Dienes test, a highly discriminating system for identifying *P. mirabilis* and *P. vulgaris* strains, was achieved (Senior and Larsson, 1983). The results also showed that the Dienes test was useful in determining the presence of a common type among a small number of isolates, but not when applied to a large number of isolates for which testing of many combinations is required. In such cases, proticine typing or O serotyping are more appropriate. The greatest discrimination is provided by using both proticine and O serotyping.

Lytic bacteriophage may also be used to differentiate strains of *P. vulgaris* and *P. mirabilis*; several groups of investigators have described typing schemes (Vieu and Capponi, 1965; France and Markham, 1968; Izdebska-Szymona et al., 1971; Schmidt and Jeffries, 1974; Hickman and Farmer, 1976; Bergan, 1978).

A high level of discrimination among *P. mirabilis* strains may also be achieved by testing the strains for growth inhibition in a selected group of chemicals that does not include antibiotics (Kashbur et al., 1974). This method of typing, referred to as resistotyping, was found to compare favorably with serotyping, phage typing, proticine typing, and the Dienes test.

Traditional methods of typing are now being supplanted by new techniques as advances are made in molecular biology. Protein profiles of outer membranes and multilocus enzyme electrophoresis analysis have been used to differentiate strains of *P. vulgaris* and *P. mirabilis* (Kappos et al., 1992). Arbitrarily primed polymerase chain reaction (AP-PCR) and ribotyping have been exploited in a study of *P. mirabilis* isolated in a pediatric hospital (Bingen et al., 1993). With increasing interest in epidemiology, along with expected future simplifications of DNA fingerprinting methods, it is anticipated that more studies will be undertaken to advance our understanding of the epidemiology of these pathogens.

Early studies have shown that *Proteus* strains are widely distributed in nature, occurring in manure, soil, and polluted water, where they are believed to have important roles in the decomposition of organic matter (Wilson and Miles, 1975). They are

also present in the intestinal contents of both wild and domesticated animals as well as in the intestines of patients and healthy humans. Incidences of 17–24% for *P. mirabilis* and 3–5% for *P. vulgaris* in healthy humans have been reported (Rustigian and Stuart, 1945; Krikler, 1953). These results were obtained without knowledge that *P. penneri* existed in the genus. Special notice of indole-negative strains was usually not taken, as these would have been considered as atypical indole-negative *P. vulgaris* or atypical ornithine decarboxylase-negative *P. mirabilis* (Müller, 1986a). Thus, their strains identified as *P. vulgaris* may or may not have included *P. penneri* strains. It is noteworthy that in a recent study of bacteria in fecal and urine contaminated bedding in calf-rearing units, *P. vulgaris* and *P. mirabilis*, but no *P. penneri*, strains were isolated (Hawkey et al., 1986b). It seems premature, therefore, to assume that the distribution of *P. penneri* in nature is similar to that of *P. vulgaris*. However, it has been found that *P. penneri* does occur in human feces. Müller (1986a) isolated 13 (0.9%) *P. penneri* from 1422 healthy humans and 20 (1.6%) from 1271 patients. The ecological importance of *Proteus* in the intestine is not known. The hydrolysis of urea is probably minor in comparison to that produced by the large population of urease-producing anaerobes (Sabbaj et al., 1970; Brown et al., 1971), but the oxidative deamination of amino acids may be significant (Drasar and Hill, 1974).

Other human sites of *Proteus* carriage include the urinary tract, the groin area, the vagina, and the prepuce of male infants. The high carriage rate in the intestine has been implicated in the colonization of the groin area (Ehrenkranz et al., 1989) and subsequent development of autoinfections and cross-infections of the urinary tract (Burke et al., 1971). Urine may be considered a habitat of *Proteus* and the source of hospital acquired infections (Hickman and Farmer, 1976; Warren, 1996). The carriage of *Proteus* in the maternal vagina is reported to be the source of neonatal infections (Bingen et al., 1993). The carriage of *P. mirabilis* in the prepuce of the infant is believed to be the source of infections in male babies (Glennon et al., 1988).

P. myxofaciens has been isolated only from living and dead gypsy moth larvae (*Porthetria dispar*), but a role in the production of disease in these insects has not been established (Cosenza and Podgwaite, 1966).

ENRICHMENT AND ISOLATION PROCEDURES

Isolation media in the clinical laboratory are formulated to inhibit the swarming of *Proteus* spp. and to select the well-known pathogens such as *Salmonella* and *Shigella*. Such media are suitable for the isolation of *Proteus* spp., but special media for the primary isolation of *Proteus* spp. have been designed (Zarett and Doetsch, 1949; Malinowski, 1966; Xilinas et al., 1975; Hawkey et al., 1986a).

For enrichment of *Proteus* species in fecal samples, tetrathionate or selenite broth have been found to increase the rate of isolation of *P. mirabilis* from 8.2% to 23.6% and *P. vulgaris* from 0% to 2.7% when enrichment preceded plating (Hynes, 1942; Rustigian and Stuart, 1945).

MAINTENANCE PROCEDURES

Proteus strains may be maintained on trypticase soy agar at 4°C with monthly transfers or may be preserved indefinitely by lyophilization. At the Centers for Disease Control and Prevention, cultures in tubes of blood agar base or trypticase soy agar sealed with a cork or rubber stopper remain viable at room temperature for many years without transfer (J.J. Farmer and F.W. Hickman, personal communication).

DIFFERENTIATION OF THE GENUS *PROTEUS* FROM OTHER GENERA

Key characteristics for differentiating the genera *Proteus*, *Providencia*, and *Morganella* are shown in Table BXII.γ.254.

FURTHER READING

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- Larsson, P. 1984. Serology of *Proteus mirabilis* and *Proteus vulgaris*. Methods Microbiol. 14: 187–214.
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TABLE BXII.γ.254. Characteristics differentiating *Proteus*, *Morganella*, and *Providencia*^{a,b}

Characteristic	<i>Proteus</i>	<i>Morganella</i>	<i>Providencia</i>
Swarming	+	–	–
H ₂ S production	+	–	–
Gelatin hydrolysis	+	–	–
Lipase (corn oil)	+	–	–
Utilization of citrate (Simmons)	D	–	+
Ornithine decarboxylase	D	+	–
Acid production from:			
Maltose	D	–	–
D-Mannose	–	+	+
Acid from one or more of the following polyhydric alcohols:			
D-Inositol, D-mannitol, adonitol, D-arabitol, erythritol	–	–	+ ^c

^aFor symbols, see standard definitions.

^bTemperature of reactions, 36° ± 1°C. All reactions for 48 h.

^cOne *Providencia* species (*P. rustigianii*) does not produce acid from polyhydric alcohols.

- Mobley, H.L., M.D. Island and R.P. Hausinger. 1995. Molecular biology of microbial ureases. Microbiol. Rev. 59: 451–480.
- Różalski, A., Z. Sidorczyk and K. Kotelko. 1997. Potential virulence factors of *Proteus bacilli*. Microbiol. Mol. Biol. Rev. 61: 65–89.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *PROTEUS*

The key differentiating characteristics for the four species of the genus *Proteus* are shown in Table BXII.γ.255. Additional characteristics of the species are provided in Table BXII.γ.256.

List of species of the genus *Proteus*

- 1. *Proteus vulgaris*** Hauser 1885, 12^{AL} emend. Brenner, Hickman-Brenner, Holmes, Hawkey, Penner, Grimont and O'Hara 1995, 870.
vul.ga'ris. L. adj. *vulgaris* common.

Morphological characteristics as described for the genus. Other characteristics are listed in Tables BXII.γ.254, BXII.γ.255, and BXII.γ.256. Since the last edition of the *Manual*, indole-negative strains formerly classified as *P. vulgaris* biogroup 1 have been assigned to a new species, *Proteus penneri*. *P. vulgaris* at present includes two biogroups. Biogroup 2 strains are positive for indole, salicin, and esculin and have other reactions characteristic of the vast majority of strains in the species. The present type strain and a very small heterogeneous group belong to biogroup 3, which is indole positive but negative in both the salicin and esculin reactions. Additional species will probably be identified in these biogroups.

Some strains are hemolytic on blood agar. Generally resistant to penicillins and cephalosporins. Generally more resistant to antibiotics than *P. mirabilis*. Causes infections of the human urinary tract and other sites but not as frequently as *P. mirabilis*. Widely distributed in nature, occurring in human and animal intestines and in contaminated soil and water.

The mol% G + C of the DNA is: 39.3 ± 1.2 (*T_m*) (Falkow et al., 1962).

Type strain: ATCC 29905, DSM 13387.

- 2. *Proteus mirabilis*** Hauser, 1885, 34^{AL}.
mi.ra'bi.lis. L. adj. *mirabilis* wonderful.

Morphological characteristics are as described for the genus. Other characteristics are listed in Tables BXII.γ.254, BXII.γ.255, and BXII.γ.256. Characteristically indole and maltose negative and ornithine decarboxylase positive. Some strains are hemolytic on blood agar. More frequently isolated from clinical specimens than *P. vulgaris*. Most common site of infection is the human urinary tract. Generally susceptible to penicillins and cephalosporins and more susceptible than *P. vulgaris*.

The mol% G + C of the DNA is: 39.3 ± 1.45 (*T_m*) (Falkow et al., 1962).

Type strain: ATCC 29906.

GenBank accession number (16S rRNA): AF008582.

- 3. *Proteus myxofaciens*** Cosenza and Podgwaite 1966, 188^{AL}.
myx.o.fac'i.ens. Gr. fem. n. *myxa* slime; M.L. masc. n. *faciens* producing; *myxofaciens* slime-producing (bacteria).

Morphological characteristics as described for the genus. Other characters are listed in Tables BXII.γ.254, BXII.γ.255, and BXII.γ.256. Only one isolate studied in detail. Thin film of growth on solid media. Produces highly viscous slime when cultured in trypticase soy broth. Hemolytic on blood agar. Isolated from living and dead gypsy moth larvae (*Porthetria dispar* L.).

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 19692.

TABLE BXII.γ.255. Differentiation of the species and biogroups of the genus *Proteus*^{a,b}

Characteristic	<i>P. vulgaris</i> BG2	<i>P. vulgaris</i> BG3	<i>P. mirabilis</i>	<i>P. myxofaciens</i> ^c	<i>P. penneri</i>
Indole production	+	+	—	—	—
Ornithine decarboxylase	—	—	+	—	—
Acid from:					
Maltose	+	+	—	+	+
D-Xylose	+	+	+	—	+
Salicin	+	—	—	—	—
Esculin hydrolysis	+	—	—	—	—
Tyrosine clearing	+	+	+	—	+
Slime producing ^d	—	—	—	+	—

^aFor symbols, see standard definitions.^bTemperature of all reactions, except slime production, at 36° ± 1°C.^cReactions based on study at one strain (ATCC 19692).^dSlime production at 25°C in trypticase soy broth.**TABLE BXII.γ.256.** Other characteristics of species of genus *Proteus*^{a,b}

Characteristic	<i>P. vulgaris</i> ^c	<i>P. mirabilis</i>	<i>P. myxofaciens</i> ^d	<i>P. penneri</i> ^e
Phenylalanine deaminase	+	+	+	+
Urease	+	+	+	+
NO ₃ [−] reduced to NO ₂ [−]	+	+	+	+
Motility	+	+	+	d
Swarming	+	+	+	d
Gelatin liquefaction (22°C)	+	+	+	d
H ₂ S production (TSI)	+	+	+ (3–4) ^f	—
Growth in KCN	+	+	+	+
Acid from glucose	+	+	+	+
Methyl red test	+	+	+	+
Voges–Proskauer test	—	d	+	—
Citrate utilization (Simmons)	d	d	+	d
Tartrate utilization	+	d	+	+
Acetate utilization	d	d	+	—
Lipase activity (corn oil)	d	d	—	d
Deoxyribonuclease (25°C)	d	d	—	d
Oxidase test	—	—	—	—
ONPG hydrolysis	—	—	—	—
Pectate liquefaction	—	—	—	—
Malonate utilization	—	—	—	—
L-Lysine	—	—	—	—
L-Arginine	—	—	—	—
L-Ornithine	—	+	—	—
Acid production from:				
Adonitol	—	—	—	—
L-Arabinose	—	—	—	—
D-Arabitol	—	—	—	—
Cellobiose	—	—	—	—
Dulcitol	—	—	—	—
Erythritol	—	—	—	—
Glycerol	d	+	+	+ (7)
<i>l</i> -Inositol	—	—	—	—
Lactose	—	—	—	—
D-Mannitol	—	—	—	—
D-Mannose	—	—	—	—
α-CH ₃ -glucoside	d	—	+	+ (7)
Raffinose	—	—	—	—
L-Rhamnose	—	—	—	—
D-Sorbitol	—	—	—	—
Sucrose	+	d	—	+
Esculin hydrolysis	d	—	—	—
Mucate	—	—	—	—
ONPG	—	—	—	—

^aFor symbols, see standard definitions.^bBiochemical reaction at 36° ± 1°C unless otherwise indicated.^cIncludes strains of biogroup 2 and biogroup 3 (Brenner et al., 1978).^dReactions based on study of one strain (Brenner et al., 1978).^eReactions as reported by Hickman et al. (1982c).^fNumber in parentheses indicates number of days for test.

4. **Proteus penneri** Hickman, Steigerwalt, Farmer III and Brenner 1983, 438^{VP} (Effective publication: Hickman, Steigerwalt, Farmer III and Brenner 1982c, 1100.)
penn.er'i. M.L. gen. n. *penneri* of Penner; named after the Canadian microbiologist J.L. Penner.

Morphological characteristics are as described for the genus. Other characteristics are listed in Tables BXII.γ.254,

BXII.γ.255, and BXII.γ.256. Formerly classified as *P. vulgaris* biogroup 1. Strains are characteristically negative in indole, maltose, salicin, and esculin reactions. Some strains are hemolytic on blood agar. A pathogen of the human urinary tract. Very resistant to chloramphenicol.

The mol% G + C of the DNA is: 38 (T_m).

Type strain: ATCC 33519, CDC 1808-73.

Genus XXX. *Providencia* Ewing 1962, 96^{AL}

JOHN L. PENNER

Pro.vi.den'ci.a. M.L. fem. n. *Providencia* *Providencia*, named after the city of Providence, Rhode Island, U.S.A.

Straight rods, 0.6–0.8 × 1.5–2.5 μm, conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. Motile by peritrichous flagella. They are **facultatively anaerobic**, chemorganotrophic, having **both a respiratory and a fermentative type metabolism**. Optimal growth temperature is 37°C. **Oxidatively deaminate phenylalanine and tryptophan. Acid is produced from D-mannose. All except *Providencia rustigianii* produce acid from one or more of the following polyhydric alcohols: inositol, D-mannitol, adonitol, D-arabitol, and erythritol.** Except for *Providencia heimbachae* all are indole positive. Tartrate (Jordan) is utilized by all but a few strains of *P. heimbachae*. Simmons citrate is utilized by all species except *P. heimbachae* and some strains of *P. rustigianii*. Isolated from diarrhetic stools, urinary tract infections, wounds, burns, bacteremias, and contaminated environmental sources. DNA relatedness tests have shown that *Providencia* are more closely related to *Proteus vulgaris* and *Proteus mirabilis* than to other members of the family *Enterobacteriaceae*. Although 16S RNA sequence data are not available for the *Providencia* species, the close relationship of *Providencia* to *Proteus* indicates that *Providencia*, like *Proteus*, belongs to the *Gamma*proteobacteria within the family *Enterobacteriaceae*.

The mol% G + C of the DNA is: 39–42 (Falkow et al., 1962).

Type species: Providencia alcalifaciens (de Salles Gomes 1944) Ewing 1962, 96 (*Eberthella alcalifaciens* de Salles Gomes 1944, 183; *Proteus inconstans* (Ornstein 1921) Shaw and Clarke 1955, 155.)

FURTHER DESCRIPTIVE INFORMATION

Since the last edition of the *Manual*, the genus has been extended to include two new species, *P. rustigianii* and *P. heimbachae*. All *Providencia* species, like those of *Proteus* and *Morganella*, deaminate phenylalanine and other amino acids to produce α-keto and α-hydroxycarboxylic acids (Bernheim et al., 1935; Stumpf and Green, 1944; Singer and Volcani, 1955). The carboxylic acids that actively bind iron have siderophore activity (Drechsel et al., 1993). A probe for the amino acid deaminase (*aad*) of *Proteus mirabilis* does not hybridize with DNA from *Providencia* and *Morganella* strains, indicating unrelatedness of the deaminases of these three genera (Massad et al., 1995). In the clinical laboratory, tests for phenylalanine deaminase are commonly used to identify strains belonging to these three genera (Henriksen, 1950; Thibault and Le Minor, 1957).

Providencia, like *Proteus* and *Morganella*, decomposes tyrosine to produce a clearing on the agar medium in which the insoluble

amino acid is incorporated (Sheth and Kurup, 1975) and produces a reddish-brown pigment on nutrient agar containing 5% tryptophan (Polster and Svobodová, 1964) or other aromatic amino acids (Müller et al., 1986). *Providencia* species, with the exception of *P. rustigianii*, differ from *Proteus* and *Morganella* in their ability to produce acid from inositol and straight-chain tetra-, penta-, or hexahydroxy- alcohols, and the species are differentiated based on their reactions on these substrates. *Providencia rettgeri* is the most noted for this property, as most strains of the species are capable of producing acid from several or all of these polyhydric alcohols (Penner et al., 1975). In contrast, *P. rustigianii* strains do not produce acid from any of the polyols. Yellowish orange-centered colonies are known to be produced on deoxycholate citrate agar by *P. alcalifaciens*, *P. rettgeri*, and *P. stuartii* (Cook, 1948; Buttiaux et al., 1954). *P. heimbachae* and *P. rustigianii* have not been examined for this property. The color is apparently caused by the precipitation of ferric hydroxide because of the alkalinity produced by the growth of the bacteria on the medium (Catsaras et al., 1965).

Urease is produced characteristically by strains of *P. rettgeri* and variably by strains of *P. stuartii* (Penner et al., 1976c; Farmer et al., 1977; Mobley et al., 1985). In a hospital outbreak of urinary tract infections, some patients were found to be infected with both urease-positive and urease-negative strains of *P. stuartii* (Penner et al., 1976c). The proportion of the urease-positive strains in *P. stuartii* has been estimated to be as low as 15–16% (Brenner et al., 1978; Penner et al., 1979b) and as high as 24% and 30% (Mobley et al., 1985; Mobley et al., 1986a). The urease enzyme of at least some *P. stuartii* strains was found to be transmissible (Grant et al., 1981) on an 82-kb plasmid (Mobley et al., 1986b). Its molecular weight has been estimated to be 375,000 ± 35,000 (Mobley et al., 1986b). The presence of the plasmid in endemic strains of some hospitals could be the cause of variations among hospitals in the frequencies of isolations of urease-positive *P. stuartii* (Penner et al., 1979b) and the failure to identify the species correctly with commercial identification systems (Cornaglia et al., 1988). A key test for separating *P. stuartii* from *P. rettgeri*, particularly from those aberrant *P. rettgeri* in fermenting D-arabitol or erythritol or both, is the fermentation of trehalose (Fischer et al., 1989). Of the *Providencia* species, only *P. stuartii* strains ferment this compound, and thus the test serves to separate this species from others of the genus.

The production of indole is an important test in the identi-

fication of *Providencia*. All species give positive reactions in this test except *P. heimbachae* (Müller et al., 1986). However, it should be noted that strains of *P. heimbachae* may produce positive reactions after 1 d of incubation when tryptone or peptone media with 0.1% tryptophan are used (Müller, 1986b). To avoid false-positive reactions tryptone media without added tryptophan has been recommended (Müller, 1986b).

Although the strains of *Providencia* are cited as unable to produce acid from lactose, some variants have been reported. An unusual strain of *P. rettgeri* that fermented lactose was reported to be associated with an outbreak of nosocomial urinary tract infection (Traub et al., 1971) and a group of 31 lactose positive isolates of *P. rettgeri* were obtained from hospitalized patients with urinary tract infections (Richard et al., 1974). In the latter study, it was found that the lactose-positive trait was not transferable by conjugation. Five lactose-positive strains were also identified among a collection of 699 *P. stuartii* obtained from urinary tracts of catheterized patients (Mobley et al., 1985). The lactose fermentation was transferable by a 150-kb plasmid (Mobley et al., 1985). In addition to urease and lactose fermentation, acid production from sucrose has also been shown to be plasmid-mediated in *P. stuartii*, indicating that variants in this species are more likely to be the results of plasmid-encoded traits than is generally recognized (Mobley et al., 1985).

Antigenic structure Like other Gram-negative species, *Providencia* possesses the surface arrays of antigenic molecules that may serve as markers for differentiation of the strains. The lipopolysaccharide (O) antigens have been the most widely used for serotyping. The original serotyping scheme for *Providencia* included both *P. alcalifaciens* and *P. stuartii* and recognized 56 O antigens, 28 flagellar (H) antigens, and two capsular (K) antigens (Ewing et al., 1954). The O serotyping scheme has since been separated according to species and extended, so that currently 46 O antigens for *P. alcalifaciens* and 17 O antigens for *P. stuartii* may be identified (Penner et al., 1976b, 1979a, b). The original scheme proposed for serotyping *P. rettgeri* listed 34 O antigens and 26 H antigens (Namioka and Sakazaki, 1958). The O serotype reference strains (sero strains) of this scheme were included in a revised and extended scheme that currently recognizes 93 O specificities (Penner et al., 1974, 1976a). In the schemes for O serotyping the three *Providencia* species by Penner and colleagues, the passive (indirect) hemagglutination technique was used instead of the slide agglutination technique to identify the O antigens because it provided greater specificity, sensitivity, and reproducibility of results. This is due to the 228-kDa lipoglycoprotein receptor on the erythrocyte membrane that binds lipopolysaccharide specifically and does not bind proteins or polysaccharides (Springer et al., 1973). Thus, the possible participation in passive hemagglutination of flagellar, fimbrial, or capsular antigens is largely excluded from the antigen-antibody reactions. The O serotyping schemes have been applied in a number of epidemiological investigations and in the study that advocated the reassignment of the *P. rettgeri* biogroup 5 strains to *P. stuartii* (Penner et al., 1975; Penner and Hennessy, 1977). At present, serotyping schemes for *P. heimbachae* and *P. rustigianii* have not been developed. However, three strains of *P. rustigianii*, NCTC 9259, NCTC 9195, and NCTC 9235, originally classified as *P. alcalifaciens* biogroup 3 (Ewing et al., 1972), were included in the present serotyping scheme for *P. alcalifaciens* as the reference strains for serotypes O:10, O:14, and O:54, respectively

(Penner et al., 1976b). Strains of these serotypes have not been identified in any epidemiological study on *Providencia* (Penner et al., 1979a).

Although it is highly probable that strains of the same O serotype from the same epidemiological setting are the same and thus important for investigating outbreaks, strains from different sources that have the same O antigens may not be the same. In such cases, a second method of discriminating between isolates is necessary. Using plasmid content to differentiate strains of *P. rettgeri*, it was shown that environmental strains of serotype O:58 were not identical (Hawkey et al., 1986b).

Although flagellar (H) antigens have been defined for some of the serotype reference strains of *P. alcalifaciens* and *P. stuartii* (Ewing et al., 1954) and for some *P. rettgeri* (Namioka and Sakazaki, 1958; Penner and Hinton, 1973), practical use of H typing has been limited to a few isolates examined in the earlier studies. Capsular antigens have also not been used as biomarkers for differentiating strains of this genus. One capsular antigen of *P. rettgeri* has been examined and found to be unlike the typical K antigens of *Enterobacteriaceae* (Namioka and Sakazaki, 1959). It is different in that it retains antibody-binding capacity after heat treatment at 100°C for 1 h and the antigenic specificity is restricted to particular O groups. The authors proposed the designation of C to distinguish this capsular antigen from the *Escherichia coli* L and B antigens and suggested that there are numerous serological types of C antigens.

Numerical analysis of electrophoretic protein patterns Serotyping based on the O antigens has its drawbacks for routine work in the clinical laboratory, because large numbers of commercially unavailable antisera are required. A promising alternative to serotyping is the application of high resolution polyacrylamide gel electrophoresis (PAGE) of proteins combined with computerized analysis of patterns. This approach has been used successfully to determine the PAGE types among strains of *P. alcalifaciens*, *P. rettgeri*, *P. stuartii*, and *P. rustigianii* (Costas et al., 1987, 1989, 1990; Holmes et al., 1988). No strains tested by this procedure were serotyped, and although the two methods have not been correlated, the results obtained by determination of the PAGE types suggest the method is as discriminating as serotyping (Costas et al., 1990). Moreover, computerized analysis of electrophoretic protein patterns correlate well with DNA relatedness studies.

Other methods of differentiating strains of *Providencia* species Results have been shown that rDNA fingerprinting also permits discrimination among strains of *P. stuartii* and their differentiation from other species of *Providencia* and *Proteus* (Owen et al., 1988).

Another alternative to serotyping is to discriminate among strains based on their sensitivity and resistance to bacteriocins. An extensive scheme employing 12 bacteriocins (provincines) to discriminate among isolates of *P. alcalifaciens* and *P. stuartii* demonstrated that 37 sensitivity patterns occurred among 300 isolates of these two species, most of which were *P. stuartii* (Al-Jumaili and Fenwick, 1978). One provicine sensitivity pattern accounted for 32 (~10%) of the isolates, seven patterns accounted for ~40%, and the remaining 29 patterns accounted for 40%. The number of patterns and the distribution of the isolates among the different patterns indicated that this system of typing would be applicable in epidemiological investigations of *Providencia* infections caused by these two species.

Fimbriae and hemagglutination In *Providencia*, as in other *Enterobacteriaceae*, fimbriae of different types are present but their distribution is more complex than in most other genera of the family (Old and Adegbola, 1982). The presence of six different types is demonstrable by electron microscopy. The *Providencia* fimbriae that have been examined in most detail are those of *P. stuartii* (Old and Scott, 1981; Old and Adegbola, 1982; Mobley et al., 1988). Three fimbrial types have been examined for hemagglutinating and adhesive properties. Cells expressing mannose-sensitive (MS) type 2 fimbriae agglutinate guinea pig erythrocytes in the absence but not in the presence of mannose. Mannose-resistant, *Klebsiella*-like (MR/K) type 3 fimbriae agglutinate only tannic acid-treated cells, and mannose-resistant, *Proteus*-like (MR/P) type 4 fimbriae agglutinate both tannic acid-treated and untreated erythrocytes. Evidence suggests that the MR/K fimbriae of this species enable binding to catheter material, an important factor in contributing to its long-term persistence in the urinary tract of infected patients with indwelling catheters (Mobley et al., 1988).

Deoxyribonucleic acid relatedness of *Providencia* species *Providencia* are most closely related to *Proteus*, but both genera are only distantly related to *E. coli* and to other species in the family *Enterobacteriaceae* (Coetzee, 1972; Brenner et al., 1978; Brenner, 1984a). In the biochemical characterization of *P. alcalifaciens* four biogroups were recognized (Ewing et al., 1972). DNA relatedness tests showed that biogroups 1 and 2 are a homogeneous group, but biogroup 3 strains were found to be only 44–49% related to biogroups 1 and 2 and 26–33% related to *P. stuartii* (Brenner et al., 1978; Hickman-Brenner et al., 1983b). The biogroup 3 strains were transferred from *P. alcalifaciens* and assigned to a new species, *P. rustigianii* (Hickman-Brenner et al., 1983b). Another group of strains isolated mostly from feces of several species of penguins resembled most closely the biogroup 3 strains and were placed in another newly described species, *P. friedericianae* (Müller, 1983). In a subsequent collaborative study between Müller and the group at the Centers for Disease Control and Prevention, DNA hybridization testing showed that strains of *P. friedericianae* are very closely related to *P. rustigianii* and it was concluded that the two groups should be included in the same species (Hickman-Brenner et al., 1986). *P. rustigianii* was described first and the species name was validated before *P. friedericianae* and therefore *P. rustigianii* has priority and is the senior of the two subjective synonyms (Hickman-Brenner et al., 1986).

A biogroup of strains from penguins that were initially included in *P. friedericianae* and one strain listed under the CDC Enteric Group 78 (strain CDC 1519-73) were found to constitute a separate hybridization group and have been assigned to *P. heimbachae*, a newly described species (Müller et al., 1986).

The *P. alcalifaciens* biogroup 4 strains, originally reported to be inositol-negative were found on reexamination to be inositol-positive, a characteristic reaction of *P. stuartii*. They are also highly related to *P. stuartii* and have low levels of relatedness to *P. alcalifaciens*, *P. rettgeri*, and to *Proteus*, and have therefore been transferred to *P. stuartii* (Brenner et al., 1978).

In the earlier classification of the *Proteus-Providencia* group, *P. rettgeri* was included in the genus *Proteus* (Ewing and Davis, 1972b). In a biochemical characterization of this species, five biogroups were recognized (Penner et al., 1975). The biogroup 5 strains are urease-positive, a characteristic reaction of strains in the genus *Proteus*, but in other respects they resemble urease-negative *P. stuartii* (Penner et al., 1975, 1976c; Penner and Hen-

nessy, 1977). On the basis of DNA relatedness tests they were found to be inseparable from *P. stuartii* and to be only 30–35% related to typical *Proteus rettgeri* biogroups 1, 2, and 3 and were therefore transferred to *P. stuartii* (Brenner et al., 1978). DNA of the three strains of *P. rettgeri* biogroup 4 (salicin and rhamnose negative) and other strains that have reactions atypical of *P. rettgeri* biogroups 1, 2, and 3 have not been examined for relatedness.

Antimicrobial susceptibilities New insights into species differences in antimicrobial susceptibilities have been noted since the revisions in the classification of genus *Providencia* in 1978 (Brenner et al., 1978). Studies using earlier classifications included urease-positive *P. stuartii* in collections identified as *Proteus rettgeri* in examining strains for antimicrobial susceptibility. Since the general adoption of the revised classification, it has become evident that, with the exception of a few unusual strains, *P. stuartii* is the most resistant species in the genus *Providencia* and one of the most resistant species in the family *Enterobacteriaceae*. *P. alcalifaciens* is the least resistant, being susceptible to cephalosporins, aminoglycosides, and quinolones, whereas *P. rettgeri* is intermediate to *P. alcalifaciens* and *P. stuartii*. Most strains of *P. rettgeri* are susceptible to gentamicin, tobramycin, and amikacin but are markedly resistant to quinolones. Most strains of *P. stuartii* are resistant to gentamicin, tobramycin, and the quinolones (including ciprofloxacin) but are susceptible to amikacin (Penner and Preston, 1980; Penner et al., 1982; Hawkey and Hawkey, 1984). Strains of *P. stuartii* that are also resistant to carbenicillin (due to a 47-kb plasmid) have also been identified (Hawkey et al., 1985). Some of the newer β -lactams, however, may be useful alternatives to amikacin in the treatment of serious *P. stuartii* infections (Hawkey, 1984). Only 11 strains of *P. rustigianii* and 13 of *P. heimbachae* have been examined for antimicrobial susceptibilities. They generally resemble *P. alcalifaciens* in their susceptibility to cephalosporins and aminoglycosides, but a small number differ in resistance to particular antibiotics (Hickman-Brenner et al., 1983b; Müller et al., 1986). 73% of the *P. rustigianii* are resistant to colistin and 100% and 77% of the *P. heimbachae* are resistant to tetracycline and cephalothin, respectively. However, it should not be concluded that these are characteristic species differences because of the small numbers of isolates that were available for study.

Pathogenicity *P. alcalifaciens* has been isolated from human feces, particularly from those of the pediatric population ever since the organism was first described, but whether this species is an etiological agent of human gastroenteritis is still a matter of debate and speculation. Evidence in support of a pathogenic role is the finding in several studies that the organisms are more frequently isolated from patients with diarrhea than from normal individuals (Prakash et al., 1966; Bhat et al., 1971a, b; Albert et al., 1995a, 1998; Guth and Perrella, 1996). Serotyping has shown that isolates of serotype O:3 are more frequently associated with pediatric diarrheas than are other serotypes, and this serotype is not frequently found in healthy children (Carpenter, 1964; Bhat et al., 1971a, b; Penner et al., 1979a). Other serotypes identified less frequently in human cases of diarrhea are O:12, O:13, and O:22 (Stenzel, 1961; Bhat et al., 1971a). In the earlier studies, examinations were not made for the more recently recognized pathogens such as rotaviruses, campylobacteria, pathogenic *E. coli*, *Yersinia enterocolitica*, and intestinal parasites, and therefore the importance of fecal isolates of *P. alcalifaciens* has remained unclear. If the organisms of this species are not causal agents, it

may be that they merely flourish in the intestinal environment created by known diarrhea-causing agents. However, in an investigation of traveler's diarrhea it was found that 10% of returning travelers had stools positive for *P. alcalifaciens*, whereas only 1.3% of stools from a control population with diarrhea were positive (Haynes and Hawkey, 1989). No well-known pathogens were found in any of the samples that were positive for *P. alcalifaciens*, and the authors cited this observation in support of a role for this species in producing human diarrhea. Moreover, strains of this species isolated from diarrheal stools have been shown to be invasive for HEp-2 cells, to be able to produce diarrhea in adult rabbits with removable ileal ties (RITARD model), and to invade the rabbit intestinal mucosa (Albert et al., 1995a, 1998). In addition, some patients infected only with *P. alcalifaciens* experienced manifestations of invasive diarrhea (Albert et al., 1998). In another study, 50% of the *P. alcalifaciens* strains isolated in pure culture or from stool specimens in which no other enteropathogen was identified, were found to be invasive for HeLa cells (Guth and Perrella, 1996). Most of the invasive strains were isolated from diarrheal stools but the invasive characteristic was not present in all *P. alcalifaciens* strains that were isolated from patients with diarrhea. The production of cytotoxins or enterotoxins could not be detected in the invasive *P. alcalifaciens* isolates and their invasiveness could therefore not be attributed to these well-known virulence factors (Albert et al., 1995a, 1998; Guth and Perrella, 1996). It should be noted that an *E. coli* strain that can colonize the small bowel mucosa is unable to produce any of the well-known toxins, but can cause fluid accumulation and diarrhea in the rabbit model (Wanke and Guerrant, 1987). The mechanisms for the production by the *E. coli* strain of the fluid and diarrhea are unknown but the possibility of similar, yet unknown mechanisms, should be considered before discounting an etiological role in the enteropathogenicity of at least some strains of *P. alcalifaciens*. Although evidence is accumulating in support of an etiological role, more direct studies are necessary to provide convincing evidence for the inclusion of this species as a true member of the human enteropathogens.

The urinary tract of the compromised or catheterized patient is the most common site of *P. rettgeri* and *P. stuartii* infections. The rise in importance of these two species is associated with their tendency to cause nosocomial infections and with their marked resistance to numerous antibiotics (Wenzel et al., 1976; Whiteley et al., 1977; Kopf and Freitag, 1979; Penner et al., 1979b, 1981; Kocka et al., 1980; McHale et al., 1981; Warren et al., 1982; Hollick et al., 1984; Warren, 1986). Of the two species, *P. stuartii* is significantly more pathogenic and more resistant (Hawkey, 1984). Earlier studies on *P. rettgeri* often included urease-positive *P. stuartii* strains, thus occluding the significant differences between the two species (Lindsey et al., 1976; Penner et al., 1979b). Since the revision in classification, *P. stuartii* have been the subject of numerous studies to determine the basis of their pathogenicity. It has been shown that the majority of the *P. stuartii* isolated from urinary tract infections adhere to uroepithelial cells *in vitro*, suggesting that this property may account for the persistence of the bacteria in the urinary tract (Mobley et al., 1986a). Further studies have yielded evidence that strains bearing the MR/K fimbriae (mannose-resistant *Klebsiella*-like hemagglutinins) have the exceptional ability of binding to catheters, a characteristic that has been hypothesized to enhance their resistance to antimicrobials and to increase the duration of infection in patients with urinary tract catheters (Mobley et al., 1986a). *P. stuartii* can cause serious

infections in surgical and burn wounds and often give rise to fatal septicemias (Curreri et al., 1973; Milstoc and Steinberg, 1973; Wenzel et al., 1976).

In elderly catheterized patients who often have high levels of urinary indoxyl sulfate and have colonization of their urinary tract with *P. stuartii*, the urinary catheter bag develops an intense purple color. This purple urine bag syndrome has been determined to be due to the presence in the organisms of indoxyl sulfatase activity, which produces indigo through the decomposition of the urinary indoxyl sulfate (Dealler et al., 1988).

Although *P. stuartii* has been isolated from human feces, there is no evidence that the species causes infections of the human gastrointestinal tract, but in an investigation of neonatal diarrhea in calves (calf scours), evidence was obtained that suggested an etiological role for *P. stuartii* (Waldhalm et al., 1969) or a contributing role by acting synergistically with a neonatal calf diarrhea virus in the production of such diarrheas (Waldhalm et al., 1974b).

P. rustigianii strains have been isolated from human feces, but only infrequently. In a collection of 891 *Providencia* isolates made by Ewing et al. (1972), only 11 (<2%) were identified as *P. rustigianii* and three of these were reported to have been isolated from stool cultures (Hickman-Brenner et al., 1983b). Of the 19 isolates of this species studied by Costas et al. (1987), 10 were also from human feces (Costas et al., 1987). These studies indicate that bacteria of this species can colonize the human intestinal tract, but the clinical significance of this is not known. Strains of this species were also isolated from penguin feces by Müller (1983), but the significance of the presence of these organisms in the intestine to the health of the animals has not been investigated.

All 13 isolates of *P. heimbachae* that have been studied are from healthy penguins except one isolate that was obtained from an aborted bovine fetus (Müller et al., 1986). The medical and veterinary importance of these organisms remains unknown.

Habitats Early studies to determine the habitats of the *Providencia* were carried out without the advantage of the recently revised classifications, and the prevailing view was that these organisms are widespread in nature, occurring in the intestines of various animals and polluted environmental sources (Müller, 1972). Uncertainty remains on the significance of their presence in the human intestine and only recently have systematic studies been reported to address this issue. From examinations of 1108 stool cultures of pediatric patients 56 isolates of *P. alcalifaciens* were found, and in the same period of time 25 isolates were recovered in routine from three adult hospitals (Penner et al., 1979a). Clearly organisms of this species can be found in the human intestine without special enrichment techniques, but their association with the production of diarrhea remains to be demonstrated.

It has been suspected that *P. stuartii* may also be found in the human intestine, but it is rarely isolated in the clinical laboratory from stool samples. As shown in two studies using special techniques such as pre-enrichment or selective differential media, the organisms of this species occur in the human intestine more frequently than previously believed and may be an important and previously underestimated nosocomial reservoir (Hawkey et al., 1982a, b; Stickler et al., 1985). The major source of the organisms in the hospital setting was found to be the colonized skin of the groin area of patients that were fecal carriers of *P. stuartii* (Stickler et al., 1985). *P. stuartii* was not isolated from raw

sewage and sewage contaminated water samples (Stickler et al., 1985). In a survey of the *Proteus*, *Providencia*, and *Morganella* in the environment of calf rearing units, 127 isolates were recovered of which 50 were *Providencia* (Hawkey et al., 1986b). One was classified as *P. alcalifaciens*, 35 as *P. rettgeri*, and 14 as *P. stuartii*, and it was suggested that a potential cause of colonization of the human intestine by these organisms is the consumption of meat (Hawkey et al., 1986b).

P. rettgeri was first isolated by Rettger from poultry (Hadley et al., 1918). Its occurrence in the intestines of reptiles, frogs, and a duck, and its presence in natural waters have been reported (Müller, 1972; Penner and Hennessy, 1979a). Of 112 isolates collected from the clinical laboratory of a large hospital, 106 were from urine specimens and six were from stool samples (Penner and Hennessy, 1979a). In a study of the environment of calf rearing units in England, *P. rettgeri* was found to be second to *Proteus vulgaris* as the most frequently occurring species (Hawkey et al., 1986b). It thus appears that the species is widely distributed in nature, but is not a common resident of the human intestinal tract.

Studies to determine the habitats of *P. rustigianii* and *P. heimbachae* have not been conducted. In the first description of *P. rustigianii* only 11 isolates were available. Three of these were isolated from stools, but the sources of the other eight isolates are unknown (Hickman-Brenner et al., 1983b). Another group of *P. rustigianii* isolates (originally classified as *P. friedericianae*) were all isolated from feces of different species of penguins (Müller, 1983). Strains of *P. heimbachae* have been isolated only from fecal specimens of penguins and from one aborted bovine fetus (Müller et al., 1986).

ENRICHMENT AND ISOLATION PROCEDURES

Media used in the clinical laboratory for isolation of *Enterobacteriaceae* may be used to isolate *Providencia*. Tetrathionate or selenite broths may be used for enrichment.

A selective differential medium for the detection of *P. alcalifaciens* has been developed by Senior (1997). After enrichment of fecal samples in tetrathionate broth, the broth cultures are plated on a solid agar medium that contains phenol red dye and three sugars (xylose, mannitol, and galactose) that *P. alcalifaciens* does not ferment. Since other enteric bacteria that are present in the tetrathionate broth ferment one or more of these sugars, they produce lemon-yellow (acid-forming) colonies. Because strains of *P. alcalifaciens* do not ferment sugars in this medium, the strains produce red (alkaline) colonies.

Enrichment in heart infusion broth (2.5%) followed by plating on MacConkey agar containing 5 mg/l gentamicin sulfate has been found useful for the isolation of *P. stuartii* (Hawkey et al., 1982b). A selective differential medium for *P. stuartii* contains colistin (as the selective agent) and inositol in a nutrient agar base with bromothymol blue (as the acid-alkali indicator). After overnight incubation at 37°C, *P. stuartii* produces yellow inositol-fermenting colonies, *Proteus mirabilis* produces green-centered colonies, and *Morganella morganii* produces blue colonies (Stickler et al., 1985).

MAINTENANCE PROCEDURES

Providencia strains may be maintained on trypticase soy agar at 4°C with monthly transfers or may be preserved indefinitely by lyophilization. The Enteric Section, Centers for Disease Control and Prevention, stores cultures at room temperature in tubes of

blood agar base or trypticase soy agar. These tubes are sealed with a cork or rubber stopper, and the cultures have remained viable for many years without transfer (J.J. Farmer III and F. Hickman-Brenner, personal communication).

DIFFERENTIATION OF THE GENUS *PROVIDENCIA* FROM OTHER GENERA

Characteristics and biochemical reactions for differentiating *Providencia* from the closely related genera, *Proteus* and *Morganella*, are described in Table BXII.γ.254 in the chapter on the genus *Proteus*. An important distinguishing characteristic of the *Providencia* (except *P. rustigianii*) is the ability to produce acid from one or more polyhydric alcohols. Biochemical reactions for differentiating *Providencia* from other species in the family *Enterobacteriaceae* are provided in Tables BXII.γ.193, BXII.γ.194, BXII.γ.195, and BXII.γ.196 in the chapter on *Enterobacteriaceae*.

TAXONOMIC COMMENTS

Since the last edition of the *Manual*, two new species have been included in the genus *Providencia*. Strains formerly classified as *P. alcalifaciens* biogroup 3 (Ewing et al., 1972) were found to constitute a separate species based on DNA relatedness tests and were assigned the epithet *rustigianii* in honor of Robert Rustigian who is noted for his early work on the genus *Proteus* (Hickman-Brenner et al., 1983b). A group of well-defined strains isolated from fecal samples obtained from penguins were found to represent a new species of the genus, and the epithet *friedericianae* was proposed to recognize the technical contributions of Friederike Heimbach (Müller, 1983). Both of these species gained standing in nomenclature but studies of strains exchanged between the two laboratories led to the finding that the species were the same. Since *P. rustigianii* was validated earlier it has priority and *P. friedericianae* is therefore a (junior) subjective synonym of *P. rustigianii* (Hickman-Brenner et al., 1983b). A biogroup of *P. friedericianae* and a strain (CDC 1519-73) of the "Enteric Group 78" were found in DNA relatedness tests to constitute a separate species that has been assigned the epithet *heimbachae* in honor of Friederike Heimbach, who isolated the original 12 strains of this species (Müller et al., 1986).

FURTHER READING

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List of species of the genus *Providencia*

1. ***Providencia alcalifaciens*** (de Salles Gomes 1944) Ewing 1962, 96^{AL} (*Eberthella alcalifaciens* de Salles Gomes 1944, 183; *Proteus inconstans* (Ornstein 1921) Shaw and Clarke 1955, 155.)

al.cal.i.fac'i.ens. Fr. n. *alcali* alkali; L. v. *facere* to do, make; L. part. adj. *faciens* making; M.L. part. adj. *alcalifaciens* alkali-producing.

The characteristics are as described for the genus and as listed in Tables BXII.γ.257 and BXII.γ.258. Urease negative. Most strains are susceptible to penicillins, cephalosporins, and aminoglycosides. Organisms are generally isolated from diarrheic stools but also from some normal stools. The species is suspected to be a causal agent of infant diarrhea and has been implicated as a cause of traveler's diarrhea but firm evidence for an etiological role in the production of diarrhea has not been established.

The mol% G + C of the DNA is: 43 (T_m) (Owen et al., 1987).

Type strain: ATCC 9886, DSM 30120.

2. ***Providencia heimbachae*** Müller, O'Hara, Fanning, Hickman-Brenner, Swenson and Brenner 1986, 255.^{VP} *heim.bach'ae.* M.L. gen. fem. n. *heimbachae* of Heimbach, named to honor Friederike Heimbach who isolated the 12 original strains of the species.

The characteristics are as described for the genus and as listed in Tables BXII.γ.257 and BXII.γ.258. It can be differentiated from other *Providencia* species by its negative reactions for Simmons citrate, growth in the presence of KCN, acid production from trehalose, urease, and indole production, and by positive tests for acid production from adonitol, D-arabitol, D-galactose, and L-rhamnose. Generally susceptible to penicillins, cephalosporins, and aminoglycosides, but most strains are resistant to cephalothin and tetracycline. Only 13 strains have been available for study. All were isolated from healthy penguins except for one that was isolated from an aborted bovine fetus. The medical and veterinary significance of these organisms is unknown.

The mol% G + C of the DNA is: 39.6 (T_m) (Owen et al., 1987).

Type strain: ATCC 35613, DSM 3591.

3. ***Providencia rettgeri*** (Hadley, Elkins and Caldwell 1918)

Brenner, Farmer, Steigerwalt, Klykken, Wathen, Hickman and Ewing 1978, 269^{AL} (*Bacterium rettgeri* Hadley, Elkins and Caldwell 1918, 180; *Proteus rettgeri* (Hadley, Elkins and Caldwell 1918) Rustigian and Stuart 1943, 242.)

rett'ge.ri. M.L. gen. n. *rettgeri* of Rettger, named after L.F. Rettger, the American bacteriologist who first isolated the organism in 1904.

The characteristics are as described for the genus and as listed in Tables BXII.γ.257 and BXII.γ.258. Virtually all strains hydrolyze urea and are distinguished from the other *Providencia* by their ability to produce acid from a variety of polyhydric alcohols. Many strains are resistant to the penicillins and cephalosporins, but the strains are generally more susceptible to aminoglycosides than are *P. stuartii* strains. Generally isolated from urine specimens of hospitalized and catheterized patients and less frequently from other sites. May cause nosocomial infections. Rarely isolated from stool specimens.

The mol% G + C of the DNA is: 40.5 (T_m) (Owen et al., 1987).

Type strain: ATCC 29944, DSM 4542.

4. ***Providencia rustigianii*** Hickman-Brenner, Farmer, Steigerwalt and Brenner 1983a, 673^{VP} (Effective publication: Hickman-Brenner, Farmer, Steigerwalt and Brenner 1983b, 1060.)

rus.tig.i.an'i.i. M.L. gen. n. *rustigianii* of Rustigian, named in honor of Robert Rustigian who did early studies on the *Proteus* group.

The characteristics are as described for the genus and as listed in Tables BXII.γ.257 and BXII.γ.258. Strains are differentiated from other *Providencia* species by their production of acid from D-galactose but not from adonitol, D-inositol, and trehalose. Only 11 strains have been examined for antibiotic susceptibility. Most are susceptible to cephalosporins and synthetic penicillins but resistant to natural penicillin and colistin. They may colonize the human intestinal tract but the medical significance of this is not known.

The mol% G + C of the DNA is: 41.8 (T_m) (Owen et al., 1987).

Type strain: ATCC 33673, DSM 4541.

5. ***Providencia stuartii*** (Buttiaux, Osteux, Fresnoy and Mor-

TABLE BXII.γ.257. Differential characteristics of the genus *Providencia*

Characteristic ^a	1. <i>P. alcalifaciens</i>	2. <i>P. heimbachae</i>	3. <i>P. rettgeri</i>	4. <i>P. rustigianii</i>	5. <i>P. stuartii</i>
Indole production	+	—	+	+	+
Citrate (Simmons)	+	—	+	d	+
Urea	—	—	+	—	d
Motility	+	d	+	d	d
Growth in KCN	+	—	+	+	+
Gas from D-Glucose	d	—	—	d ^b	—
<i>Acid production from:</i>					
Adonitol	+	+	+	—	—
D-Arabitol	—	+	+	—	—
D-Galactose	—	+	+	+	+
D-Inositol	—	d	+	—	+
D-Mannitol	—	—	+	—	—
L-Rhamnose	—	+	d	—	—
Trehalose	—	—	—	—	+

^aAll tests were performed at 36° ± 1°C.

^bMinimal gas production.

TABLE BXII.γ.258. Other characteristics of the species of the genus *Providencia*

Characteristic ^a	1. <i>P. alcalifaciens</i> ^b	2. <i>P. heimbachae</i> ^c	3. <i>P. rettgeri</i> ^b	4. <i>P. rustigianii</i> ^d	5. <i>P. stuartii</i> ^b
Methyl red	+	d	+	d	+
Voges-Proskauer	—	—	—	—	—
H ₂ S on TSI	—	—	—	—	—
Phenylalanine	+	+	+	+	+
L-Lysine (Møller's)	—	—	—	—	—
L-Arginine (Møller's)	—	—	—	—	—
L-Ornithine (Møller's)	—	—	—	—	—
Gelatin (22°C)	—	—	—	—	—
Malonate	—	—	—	—	—
D-Glucose (acid production)	+	+	+	+	+
D-Glucose (gas production)	d	d (7) ^e	d	d	—
<i>Acid from:</i>					
L-Arabinose	—	—	—	—	—
Dulcitol	—	—	—	—	—
Erythritol	—	—	d	—	—
Glycerol	d	d (7)	d	—	d
Lactose	—	—	—	—	—
Maltose	—	+ (7)	—	—	—
D-Mannose	+	+	+	+	+
Melibiose	—	—	—	—	—
α-CH ₃ glucoside	—	—	—	—	—
Raffinose	—	—	—	—	—
Salicin	—	—	d	—	—
D-Sorbitol	—	—	—	—	—
Sucrose	d	—	d	+ (7)	d
D-Xylose	—	d (7)	d	—	—
Esculin hydrolysis	—	—	d	—	—
Acid from mucate	—	—	—	—	—
Tartrate (Jordan's)	+	d	+	+	+
Acetate utilization	d	—	d	—	d
Lipase (corn oil)	—	—	—	—	—
DNase 25°C	—	—	—	—	—
NO ₃ →NO ₂	+	+	+	+	+
Oxidase production	—	—	—	—	—
ONPG production	—	—	—	—	—
Tyrosine clearing	+	+	+	+	+

^aBiochemistry reactions at 36° ± 1°C unless otherwise stated.^bData from Brenner et al. (1978).^cData from Müller et al. (1986).^dData from Hickman-Brenner et al. (1983b).^eNumbers in parentheses indicate number of days for test.

iametz 1954) Ewing 1962, 96^{AL} (*Proteus stuartii* Buttiaux, Osteux, Fresnoy and Moriametz 1954, 385; *Proteus inconstans* (Ornstein 1921) Shaw and Clarke 1955, 155.)

stu.ar'ti.i. M.L. gen. n. *stuartii* of Stuart, named after C.A. Stuart, American bacteriologist who did much of the early work on *Providencia*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.257 and BXII.γ.258. Highly resistant to antibiotics. Some strains are among the most resistant of the *Enterobacteriaceae*. Strains may be urease positive or urease negative. Isolated most often from urine speci-

mens of hospitalized and catheterized patients. Some strains have the ability to adhere to catheters and thus enhance resistance to treatment and increase the duration of the infection. Less frequently isolated from wounds, burns, and bacteremias. They occur in small numbers in the intestines of some individuals and are rarely isolated routinely from stool cultures in the clinical laboratory but serve as the source for the colonization of the skin of the groin area, which may lead to autoinfections and cross-infections.

The mol% G + C of the DNA is: 40.7 (T_m) (Owen et al., 1987).

Type strain: ATCC 29914, DSM 4539.

Genus XXXI. Rahnella Izard, Gavini, Trinel and Leclerc 1981c, 382^{VP} (Effective publication: Izard, Gavini, Trinel and Leclerc 1979, 174)

PETER KÄMPFER

Rah' nel.la. M.L. dim. ending -ella; M.L. fem. n. *Rahnella* named after Otto Rahn, the German-American microbiologist and who proposed the name *Enterobacteriaceae* in 1937.

Straight rods 0.5–0.7 × 2–3 μm, conforming to the general definition of the family *Enterobacteriaceae*. **Gram negative, motile by peritrichous flagella when grown at 25°C.** Facultatively anaerobic.

Chemoorganotrophic. **Psychrotolerant**, growing at 4°C. D-Glucose is fermented with the production of acid and, for the majority of strains, gas. Nitrate is reduced to nitrite. Oxidase neg-

ative, catalase positive. **Negative for lysine and ornithine decarboxylases and for arginine dihydrolase. Most strains are (weakly) positive for phenylalanine deaminase (after 48 h), methyl red, and Voges-Proskauer reaction. Acids are produced from various carbohydrates, including L-arabinose, cellobiose, lactose, maltose, mannose, L-rhamnose, raffinose, D-xylose, and salicin.** Sequence analyses of the 16S rDNA of four *Rahnella* strains clearly placed the genus within the *Gammaproteobacteria* in the family *Enterobacteriaceae*. Highest 16S rDNA sequence similarity to *Yersinia enterocolitica* ATCC 9610^T (97.7%). By omitting the hyper-variable regions V1 and V5 (corresponding to *E. coli* base positions 147–1490; Brosius et al., 1978), *Hafnia alvei* ATCC 13337^T showed the highest sequence similarity (96.63%) to *R. aquatilis* ATCC 33071^T. Most often isolated from fresh water, but also found in the intestine of snails and various other environmental habitats, including soils and the rhizosphere. Can occasionally be isolated from foods or human clinical specimens, including wound infections, bacteremias, feces from patients with acute gastroenteritis, and septicemia, especially from immunocompromised patients. The genus currently includes three genomospecies (*R. aquatilis* [*Rahnella* genomospecies 1], *Rahnella* genomospecies 2, and *Rahnella* genomospecies 3), which cannot be phenotypically differentiated.

The mol% G + C of the DNA is: 51–56.

Type species: *Rahnella aquatilis* Izard, Gavini, Trinel and Leclerc 1981c, 382 (Effective publication: Izard, Gavini, Trinel and Leclerc 1979, 174.)

FURTHER DESCRIPTIVE INFORMATION

A comparison of the total 16S rDNA sequence of the type strains of *R. aquatilis* with other representatives of the family *Enterobacteriaceae* showed highest similarity (97.7%) with the type strain of *Yersinia enterocolitica*. Eight representative strains representing all three *Rahnella* genomospecies were used for 16S rRNA gene amplification and sequencing (Brenner et al., 1998). The hypervariable regions V1 and V5 were omitted for sequence comparison resulting in a sequence that corresponded to *E. coli* base positions 147–1490 (Brosius et al., 1978). The eight strains showed sequence similarities ranging from 97.7–100% and fell in three distinct groups (Brenner et al., 1998). Three additional strains sequences already deposited in GenBank also grouped with *R. aquatilis* type strain, exhibiting 99.7–100% sequence similarity. The second group contained five strains used in the DNA–DNA relatedness experiments (Brenner et al., 1998), four of which were in genomospecies 2 and one in genomospecies 1 (*Rahnella aquatilis sensu stricto*). Two further strains, for which sequences were deposited in GenBank, were also found in this group. This observation supports the advisability of performing DNA–DNA hybridization experiments on strains showing 97% or greater sequence similarity, before considering them to belong to the same species (Brenner et al., 1998). The third group contained only strain DSM 30078, representing *Rahnella* genomospecies 3.

The phenotypic description of the genus *Rahnella* is largely based on the studies of Gavini et al. (1976a), Izard et al. (1979), Farmer (1984b), Farmer et al. (1985a), Brenner (1992a), and Brenner et al. (1998). In the study of Brenner et al. (1998), a total of 51 *Rahnella* strains were included, and the results indicated that the strains comprise three closely related species (genomospecies), which cannot be differentiated phenotypically. Their subdivision is described below (see Taxonomic Comments).

Rahnella strains grow readily on all kinds of media and no growth requirements have been described. A nutrient-rich medium (e.g., tryptone–soy agar, sheep-blood agar, or nutrient agar) gives best results. Incubation at temperatures ranging from 25–35°C results often in better growth than incubation at 37°C. The majority of strains are psychrotolerant, growing (slowly) at temperatures of 4°C. Growth was reported even at temperatures of 1°C ± 1°C (Davis and Eyles, 1992).

Members of the genus *Rahnella* ferment lactose and produce β-galactosidase (ONPG test). Acid production from D-xylose is observed in parallel with strong β-xylosidase activity. β-glucuronidase is negative in all strains. Indole production, hydrogen sulfide formation, tyrosine deaminase, urease, and DNase tests are negative. Nitrogen fixation (Berge et al., 1991; Kim et al., 1998) and mineral phosphate solubilization (Kim et al., 1998) has been reported in some strains of *R. aquatilis*. Various carbon sources, including carbohydrates, organic acids, and amino acids, are utilized as sole sources of carbon (Brenner et al., 1998).

Isoprenoid quinone Q-8 is the predominant quinone and small amounts of the menaquinone MK-8 are found in *R. aquatilis* ATCC 33071^T (P. Kämpfer, unpublished results). The fatty acid composition as determined by gas chromatographic analysis of all 51 strains used in the study of Brenner et al. (1998) differed only slightly (P. Kämpfer, unpublished results). All strains contained the fatty acids C_{12:0}, C_{14:0}, C_{16:0}, C_{17:0} cyclo, C_{19:0} cyclo, summed feature (C_{16:1} iso I and/or C_{14:0} 3OH), summed feature (C_{16:1} ω7c and/or C_{15:0} iso 2OH), and summed feature (C_{18:1} ω7c, C_{18:1} ω9t, and/or C_{18:1} ω12t), a fatty acid type typical of a member of the family *Enterobacteriaceae*. The dominant polyamines, as determined by high-performance liquid chromatography, were putrescine and cadaverine (Hamana, 1996). Minor amounts of diaminopropane were found in both strains studied; one strain also contained agmatine, whereas the other contained spermidine.

Members of the genus *Rahnella* are usually sensitive to aminoglycosides, cefotaxime, doxycycline, trimethoprim, gentamicin, tetracycline, and tobramycin, but resistant to chloramphenicol. Susceptibility to β-lactams differs among strains. Antimicrobial susceptibility data are summarized by Funke and Rosner (1995). They state that aminopenicillins and first-generation cephalosporins have only limited activity against *R. aquatilis*, whereas ureidopenicillins and carbapenems are more active. *R. aquatilis* is usually susceptible to aminoglycosides and quinolones. *Rahnella* produces enterobacterial common antigen (Böttger et al., 1987).

The pathogenicity of *Rahnella* in humans and animals is not clearly established. Strains isolated from humans appear to be opportunistic rather than true pathogens.

Rahnella are widely distributed in nature. They may be isolated from foods and occasionally from a wide variety of human sources. The natural habitat of *Rahnella* seems to be water, and all of the isolates of Gavini et al. (1976a) were from waters in France. Several American water isolates have also been identified as *R. aquatilis*. The study of the intestinal content of snails for the presence of *Enterobacteriaceae* (Brenner et al., 1998) resulted in the isolation of infrequently seen enterobacterial species, among them *Rahnella* spp. Rhodes et al. (1998) found *Rahnella* spp. in the intestinal content associated with 12,000-year-old mastodon remains. The study of mountain soils for psychrotrophic *Enterobacteriaceae* resulted in the isolation of *Rahnella* (Horie et al., 1985), and nitrogen-fixing strains were isolated from the rhizosphere of wheat and maize (Berge et al., 1991; Heulin et al., 1994). Heron et al. (1993) found *Rahnella* isolated among enterobacteria associated with grass and silages. The microbiolog-

ical study of prepared salad vegetables and sprouts resulted in the isolation of *Rahnella* (Geiges et al., 1990). In minced meat, fish, and milk, *Rahnella* were frequently encountered by Lindberg et al. (1998), and one fish isolate was found to contain the gene of *E. coli* heat-labile toxin (lt). Hamze et al. (1991) found *R. aquatilis* as a potential contaminant in lager beer breweries. Two strains of *Rahnella* were isolated from buckwheat seeds (Iimura and Hosono, 1996).

R. aquatilis may also occasionally occur in human clinical specimens. The first case reported was a strain from a burn wound (Farmer et al., 1985a). Maraki et al. (1994) reported a surgical wound infection with *Rahnella*. Two *Rahnella* isolates were recovered from the feces of two patients with acute gastroenteritis, one of whom was an AIDS patient (Reina and Lopez, 1996a). The strains were resistant to ampicillin, cephalothin, and cefoxitin. Funke and Rosner (1995) reported *R. aquatilis* bacteremia in an HIV-infected intravenous drug-abuser. Caraccio et al. (1994) isolated *Rahnella* from a bacteremic patient with chronic renal failure. Matsukura et al. (1996) isolated *Rahnella aquatilis* from a blood culture obtained from a case of endocarditis. *R. aquatilis* was found in the blood of a diabetic patient and a patient with laryngeal carcinoma (Oh and Tay, 1995). Both patients recovered from the infection after treatment with parenteral antibiotics. Additional cases were described by Goubau et al. (1988), Harrell et al. (1989), and Hoppe et al. (1993). Literature on *R. aquatilis* infections in humans are reviewed by Alballa et al. (1992), Maraki et al. (1994), and Funke and Rosner (1995).

ENRICHMENT AND ISOLATION PROCEDURES

Media used for the isolation of *Rahnella* are similar to those used for other members of the family *Enterobacteriaceae*. No specific selective medium is available for *Rahnella* species. As pointed out by Grimont and Grimont (1992), the requirements for a selective medium can be summarized as follows: (1) a precise and stable delineation of the taxon (genus or species); (2) known and stable common nutritional properties uncommon outside the group (genus or species); and (3) a special, in most cases clinical, or public health requirement for such a medium. Because of the low clinical significance and the presence of *Rahnella* in various habitats, the selective isolation of *Rahnella* is rarely required, except for epidemiological studies. In most cases, organisms belonging to the genus *Rahnella* are isolated with differential media not inhibitory for *Enterobacteriaceae*, such as MacConkey agar, bromothymol blue lactose agar, or phenol red lactose agar. *Rahnella* strains grow readily on nutrient-rich media, and no particular growth requirements have been described.

MAINTENANCE PROCEDURES

Rahnella strains can be maintained in tryptone soy agar stab or on nutrient agar when kept at room temperature in the dark. They can be preserved by storage in broth containing 10% glycerol or in calf or bovine serum at -80°C . Lyophilization seems to be the best procedure for preservation.

DIFFERENTIATION OF THE GENUS *RAHNELLA* FROM OTHER GENERA

The genus *Rahnella* has no single distinguishing feature useful for differentiation from the other genera of *Enterobacteriaceae* (Table BXII.γ.193 of the *Enterobacteriaceae*). *Rahnella* does not produce a yellow pigment, is negative for lysine and ornithine decarboxylases, and is weakly positive for phenylalanine deaminase. These characters may be helpful in differentiation of *Rahnella* from the genera *Pantoea* and *Erwinia*.

TAXONOMIC COMMENTS

Based on a numerical taxonomic study of *Enterobacteriaceae*, Gavini et al. (1976a) defined a new group within the family and gave it the vernacular name "group H2". Subsequently, Izard et al. (1979) used DNA–DNA hybridization to compare strains of group H2 to each other and to named species of *Enterobacteriaceae*. Based on the close relatedness within group H2 and the low relatedness to other *Enterobacteriaceae*, they proposed the new genus *Rahnella* with one species *Rahnella aquatilis*, and designated a holotype strain. The name *Rahnella* was chosen to honor Otto Rahn, the German-American microbiologist for his many contributions to systematic bacteriology and for his proposal of the family name *Enterobacteriaceae* in 1937. Because all of the isolates investigated by Izard et al. (1979) were from water, the species name *aquatilis* was chosen. The names *Rahnella* and *R. aquatilis* were effectively published (Izard et al., 1979) but were not validated in the *International Journal of Systematic Bacteriology* before 1 January 1980. They did not appear on the Approved Lists of Bacterial Names (Skerman et al., 1980), but both names have now been validly published (Izard et al., 1981c) and have standing in nomenclature. From 1979 to now, *Rahnella* was isolated from various other environmental habitats, including soils and the rhizosphere, and from food, indicating a wide distribution in nature. Furthermore, several clinical isolates were reported, in most cases from immunocompromised patients. Between 1984 and 1988, Müller et al. (1995a, b, 1996) screened the intestinal contents of snails and slugs for the presence of *Enterobacteriaceae* and found several *Rahnella* isolates in addition to the genera *Buttiauxella*, *Khuyvera*, and others. In a subsequent study 51 *Rahnella* and *Rahnella*-like organisms were subjected to an extensive DNA–DNA hybridization study. The results indicated that the strains comprised three closely related (genomo-)species within the genus, one corresponding to *R. aquatilis*. The relative binding ratios obtained from DNA–DNA hybridization studies between the *Rahnella* genomospecies are given in Table BXII.γ.259. For the two new species the vernacular names *Rahnella* genomospecies 2 and *Rahnella* genomospecies 3 were proposed (Brenner et al., 1998). It was not possible to differentiate these genomospecies on the basis of physiological and biochemical tests.

ACKNOWLEDGMENTS

I thank Don J. Brenner for his helpful comments and critical reading of this chapter, Hans E. Müller for interesting discussions and furnishing

TABLE BXII.γ.259. Relative binding ratios of DNAs at 60°C (given as percentages/ranges of percentages) between the genomospecies of the genus *Rahnella*^a

Test	<i>R. aquatilis</i> (<i>Rahnella</i> genomospecies 1)	<i>Rahnella</i> genomospecies 2	<i>Rahnella</i> genomospecies 3
<i>R. aquatilis</i>	88 (71–100)	64 (42–81)	54 (45–64)
<i>Rahnella</i> genomospecies 2		93 (70–100)	54 (49–61)
<i>Rahnella</i> genomospecies 3			100

^aData adapted from Brenner et al. (1998).

TABLE BXII.γ.260. Characteristics of (genomo)species of the genus *Rahnella*^{a,b}

Biochemical reaction	<i>Rahnella aquatilis</i> (Genomospecies 1)	<i>Rahnella</i> (Genomospecies 2)	<i>Rahnella</i> (Genomospecies 3)
Methyl red	[+]	d	+
Voges-Proskauer	+	d	+
Citrate utilization	[+] (+)	d ([+])	—
Phenylalanine deaminase	+	d	+
Arginine dihydrolase	—	—	—
Lysine decarboxylase	—	—	—
Ornithine decarboxylase	— ([—])	—	—
Motility	d	d	—
Gelatin liquefaction (22°C)	—	— (d)	—
KCN, growth	— ([—])	—	—
Malonate	[+]	d ([+])	+
Gas from D-glucose	[+]	[+]	—
<i>Acid from:</i>			
D-Arabitol	—	—	+
Cellobiose	+	+	+
Dulcitol	[+] (+)	[+] (+)	—
Maltose	+	+	+
D-Mannose	+	+	+
Melibiose	+	+	+
Methyl-α-D-glucoside	—	—	—
Raffinose	+	+	+
L-Rhamnose	+	+	—
D-Sorbitol	+	+	+
Sucrose	+	+	+
D-Xylose	+	+	(+)
Glycerol	— (d)	— (d)	—
Esculin hydrolysis	+	+	+
Mucate	d	—	—
Tartrate	d	[+]	+
Acetate	[—]	—	—
Lipase (corn oil)	—	—	—
Lipase (Tween 80)	[+]	d	—
<i>Utilization of:</i>			
N-Acetyl-D-galactosamine	—	—	—
D-Fructose	[+]	d	+
Sucrose	+	+	+
Acetate	—	d	—
cis-Aconitate	d	[—]	—
trans-Aconitate	—	—	—
4-Aminobutyrate	—	—	—
DL-Lactate	+	+	—
2-Oxoglutarate	d	—	—
L-Alanine	+	d	+
L-Aspartate	+	+	+
L-Histidine	d	—	+
L-Ornithine	d	—	—
L-Phenylalanine	—	—	—
L-Proline	d	—	—
Phenylacetate	—	—	—
<i>Hydrolysis of:</i>			
pNP-β-D-Galactopyranoside	—	—	—
pNP-α-D-Glucopyranoside	d	—	—
pNP-β-D-Glucopyranoside	+	[—]	+
2-Deoxythymidine-5'-pNP-phosphate	—	[—]	—
L-Glutamate-γ-3-carboxy-pNA	+	[+]	—
L-Proline-pNA	[+]	—	+

^aSymbols: +, positive for 90–100% of strains; [+], positive for 75–89% of strains; d, positive for 26–74% of strains; [—], positive for 11–25% of strains; —, positive for 0–10% of strains.

^bData adapted from Brenner et al. (1998). All reactions, unless otherwise stated were done at 36° ± 1°C and read after 48 h. Numbers in parentheses indicate the percentage of strains giving a positive reaction within 7 d. All strains grew at 4°C, 25°C, and 37°C, and gave positive reactions within 48 h in tests for fermentation of L-arabinose, D-galactose, D-glucose, lactose, D-mannitol, salicin, and trehalose, and in the tests for nitrate reduction to nitrite, catalase, motility at 25°C, and gelatin liquefaction at 22°C. All strains were oxidase-negative and gave negative reactions after 7 d in tests for fermentation of adonitol, erythritol, and *i*-inositol, and in tests for DNase, hydrogen sulfide, indole production, oxidase, tyrosine deaminase, urease, and yellow pigment. All strains gave positive reactions in tests for utilization of N-acetyl-D-glucosamine, L-arabinose, p-arbutin, D-cellobiose, D-galactose gluconate, D-glucose, D-maltose, D-mannose, α-D-melibiose, L-rhamnose, D-ribose, salicin, D-trehalose, D-xylose, D-mannitol, D-sorbitol, citrate, fumarate, L-malate, pyruvate, and L-serine, and positive reactions in tests for hydrolysis of pNP-β-D-galactopyranoside, pNP-β-D-xylopyranoside, bis-pNP-phosphate, pNP-phenyl-phosphonate, pNP-phosphoryl choline, and L-alanine-pNA (pNP = para-nitrophenyl; pNA = para-nitroanilide). All strains gave negative reactions in tests for utilization of adonitol, *i*-inositol, putrescine, propionate, adipate, glutarate, DL-3-hydroxybutyrate, mesaconate, suberate, β-alanine, L-leucine, L-tryptophan, 3-hydroxybenzoate, and 4-hydroxybenzoate.

with strains, and Wolfgang Ludwig for critical comments and his experience in 16S rRNA sequence analyses.

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List of species of the genus *Rahnella*

1. *Rahnella aquatilis* Izard, Gavini, Trinel and Leclerc 1981c, 382^{VP} (Effective publication: Izard, Gavini, Trinel and Leclerc 1979, 174.)

a.qua'ti.lis. L. adj. *aquatilis* living in water.

Represents *Rahnella* genomospecies 1 of the study of Brenner et al. (1998). The characteristics are as described for the genus. Detailed characteristics are given in Table BXII.γ.260.

Occurs in freshwater, but also in environmental habitats like soil. May occasionally be isolated from human clinical

specimens; in most cases from immunocompromised hosts. Clinical significance is unknown. Mean DNA–DNA relatedness among the 21 strains studied by Brenner et al. (1998), including 9 of the 10 original water isolates described by Izard et al. (1979) was 88% in 60°C reactions (range 72–100%). Mean relatedness in 75°C reactions was 81% (range 61–100%).

The mol% G + C of the DNA is: 51–56 (T_m).

Type strain: 133, ATCC 33071, CIP 78-65, DSM 4594.

GenBank accession number (16S rRNA): AJ233426.

Other Organisms

1. *Rahnella* genomospecies 2.

This species was proposed on the basis on DNA–DNA hybridization experiments and comprises 30 strains (Brenner et al., 1998). Phenotypically it is indistinguishable from *Rahnella aquatilis*. A combination of the utilization tests with L-histidine, L-ornithine, and L-proline (Table BXII.γ.260; Brenner et al., 1998) may be helpful in differentiation from *Rahnella aquatilis*, but does not allow a clear (genomo)-species identification. All 30 strains studied by Brenner et al. (1998) were 83% or more related in 60°C reactions (range 70–100%). In 75°C reactions the strains were 74% or more related, with one exception showing only 68% relatedness to the labeled reference organism.

Strains were isolated from water, intestinal contents of

snails, and clinical material, including stored donated human blood.

Deposited strain: SM S7/1-576.

2. *Rahnella* genomospecies 3.

This species was also proposed based on DNA–DNA hybridization experiments (Brenner et al., 1998) and is represented by a single strain: DSM 30078 (isolated from minced meat). Phenotypically it can be separated from *Rahnella aquatilis* and *Rahnella* genomospecies 2 (Table BXII.γ.260), but without results from different strains one cannot determine whether the results are strain- or species-specific.

Deposited strain: DSM 30078.

Genus XXXII. *Saccharobacter* Yaping, Xiaoyang and Jiaqu 1990, 412^{VP}

DON J. BRENNER

Sac.cha.ro.bac'ter. L. n. *saccharum* sugar; M.L. n. *bacter* a rod; M.L. masc. n. *Saccharobacter* a sugar rod.

Small straight rods, $0.5\text{--}0.9 \times 1.0\text{--}1.0 \mu\text{m}$. **Gram negative, motile by peritrichous flagella.** Facultatively anaerobic. Chemoorganotrophic. Colonies on glucose-yeast extract agar are opaque and milky white, smooth, and low convex with entire margins. Oxidase negative, catalase positive. Nonpigmented. Nonsporeforming. No lipid granules present. Optimal growth temperature range is from 30–46°C. **D-glucose is fermented with the production of ethanol, CO₂, and small amounts of acids, but no hydrogen. Indole negative, methyl red positive, Voges-Proskauer test positive, citrate positive.** H₂S production, gelatin hydrolysis, urease, nitrogen reduction, lysine decarboxylase, and ornithine decarboxylase are negative; arginine dihydrolase is delayed positive (4 d). Acid is produced from D-glucose, L-arabinose, esculin, D-fructose, D-galactose, maltose, D-mannitol, melibiose, L-rhamnose, L-sorbose, starch, sucrose, trehalose, D-xylose. Lactose fer-

mentation is delayed positive after 9 d. Dulcitol, gluconate, myo-inositol, and raffinose are not fermented. Malonate is used as a sole carbon source; ammonium sulfate, yeast extract, urea, phenylalanine, glutamine, and tryptone serve as sole nitrogen sources, but sodium glutamate does not. Produces β-galactosidase and grows in the presence of KCN. **Grows in the presence of 35% D-glucose and 6% NaCl, but not 40% D-glucose or 8% NaCl.** 16S rDNA sequence has not been determined.

The mol% G + C of the DNA is: 63.5.

Type species: *Saccharobacter fermentatus* Yaping, Xiaoyang, and Jiaqu 1990, 412.

FURTHER DESCRIPTIVE INFORMATION

Carbohydrates are presumably degraded by the Embden-Meyerhof-Parnas pathway, as indicated by the absence of 6-phos-

phogluconate dehydratase and 2-keto-3-deoxygluconate-6-phosphate aldolase and the presence of fructose diphosphate aldolase. Degrades 1 mol of glucose to approximately 2 mol of ethanol and 2 mol of CO₂.

MAINTENANCE PROCEDURES

Cultures are grown at 30°C and maintained on glucose-peptone-yeast extract agar. Lyophilization is used for long-term storage.

DIFFERENTIATION OF THE GENUS *SACCHAROBACTER* FROM OTHER GENERA

Saccharobacter is easily differentiated from *Zymomonas*, another ethanol-producing genus. *Saccharobacter* is peritrichously flagellated, whereas *Zymomonas* has polar flagella. *Zymomonas* ferments only D-glucose, fructose, and sucrose, whereas *Saccharobacter* ferments a number of other carbohydrates (see above). *Zymomonas* degrades sugars by the Entner-Duoderoff pathway, while *Saccharobacter* ferments sugars by the Embden-Meyerhof-Parnas pathway. *Saccharobacter* can easily be differentiated from various species of *Enterobacteriaceae* based on its indole, methyl red, Voges-Proskauer, and citrate (IMViC) reactions, together with its negative nitrate reductase reaction, its positive phenylalanine de-

aminase reaction, and its sugar fermentation pattern (see chapter on *Enterobacteriaceae*).

TAXONOMIC COMMENTS

Since 16S rDNA sequencing has not been done on the type species, it is impossible to accurately place the genus *Saccharobacter* phylogenetically. *Saccharobacter* appears to be phenotypically similar to members of the family *Enterobacteriaceae* based on a number of key characteristics. These include negative oxidase and positive catalase reactions, peritrichous flagella, facultatively anaerobic metabolism, and fermentation of D-glucose and other carbohydrates. It differs from *Enterobacteriaceae* in the end products of fermentation, which in *Enterobacteriaceae* are mixed acids or 2,3-butanediol, with no more than 0.5 mol of ethanol from 1 mol of D-glucose. *Saccharobacter* also differs from most, if not all, members of *Enterobacteriaceae* by the high mol% G + C content.

FURTHER READING

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List of species of the genus *Saccharobacter*

1. ***Saccharobacter fermentatus*** Yaping, Xiaoyang and Jiaqi 1990, 412^{VP}
fer.men.ta'tus. M.L. adj. *fermentatus* fermentative.

The characteristics are those given for the genus. All

extant strains were isolated from squeezed leaf juice of agave. Isolated from squeezed leaf juice of agave in Wuhan, People's Republic of China.

The mol% G + C of the DNA is: 63.5 (*T_m*).

Type strain: WVB8512.

Genus XXXIII. *Salmonella* Lignières 1900, 389^{AL}

MICHEL Y. POPOFF AND LÉON E. LE MINOR

Sal.mon.el'la. M.L. *-ella* dim. ending; M.L. fem. n. *Salmonella* named after D.E. Salmon, an American bacteriologist.

Straight rods, 0.7–1.5 × 2.0–5.0 μm, conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. **Usually motile** (peritrichous flagella). Facultatively anaerobic. Colonies are generally 2–4 mm in diameter. Nitrates are reduced to nitrites. **Gas is usually produced from D-glucose**. Hydrogen sulfide is usually produced on triple-sugar iron agar. Indole negative. **Citrate is usually utilized as a sole carbon source**. Lysine and ornithine decarboxylase (Møller's) reactions are usually positive. Urease negative. Phenylalanine and tryptophan are not oxidatively deaminated. Sucrose, salicin, inositol, and amygdalin are usually not fermented. Lipase and deoxyribonuclease are not produced. Pathogenic for humans, causing enteric fevers, gastroenteritis, and septicemia; may also infect many animal species besides humans. Some serovars are strictly host-adapted.

The mol% G + C of the DNA is: 50–53.

Type species: ***Salmonella choleraesuis*** (Smith 1894) Weldin 1927, 155 (*Bacillus cholerae suis* Smith 1894, 9.)

FURTHER DESCRIPTIVE INFORMATION

Although most salmonellae are motile, the serovar Gallinarum (or serovar Pullorum) is always nonmotile.

Certain *Salmonella* serovars, such as serovar Abortusovis, may form unusually small colonies (~1 mm diameter), whereas most types form larger colonies (2–4 mm).

Most salmonellae are aerogenic. However, serovar Typhi, an important exception, never produces gas. Anaerogenic variants of normally gas-producing *Salmonella* serovars may occur; this is particularly common with serovar Dublin.

Hydrogen sulfide is produced by most salmonellae, but a few serovars do not produce it (e.g., most strains of serovar Paratyphi A and some strains of serovar Choleraesuis).

Citrate is generally utilized by salmonellae, but some serovars do not use it (particularly host-adapted serovars such as serovar Typhi and serovar Paratyphi A).

The lysine decarboxylase reaction (Møller's) is positive for most salmonellae, but an important exception is serovar Paratyphi A. Most salmonellae are also positive for ornithine decarboxylase reaction (Møller's), but serovar Typhi is negative.

Other biochemical characteristics of the genus are indicated in Tables BXII.γ.193, BXII.γ.194, and BXII.γ.196 in the chapter on the family *Enterobacteriaceae*. Subdivision of the genus *Salmonella* into species and subspecies (Le Minor and Popoff, 1987; Reeves et al., 1989a) based on biochemical characteristics is shown in Table BXII.γ.261. Kauffmann (1960, 1963a, b, 1964) subdivided the genus *Salmonella* into "subgenera" (see Taxonomic Comments). These subdivisions correspond more closely to species or subspecies in other groups of bacteria, but whatever rank is assigned to them, the worthiness of these subdivisions was confirmed by Rohde (1965, 1966, 1967).

TABLE BXII.γ.261. Differential characteristics of the *Salmonella* species and subspecies^a

Characteristic	<i>S. enterica</i> subsp. <i>enterica</i>	<i>S. enterica</i> subsp. <i>arizonae</i>	<i>S. enterica</i> subsp. <i>diarizonae</i>	<i>S. enterica</i> subsp. <i>houstenae</i>	<i>S. enterica</i> subsp. <i>indica</i>	<i>S. enterica</i> subsp. <i>salamae</i>	<i>S. bongori</i>
Dulcitol	+	—	—	—	d	+	+
ONPG (2 h)	—	+	+	—	d	—	+
Malonate	—	+	+	—	—	+	—
Gelatinase	—	+	+	+	+	+	—
Sorbitol	+	+	+	+	—	+	+
Culture with KCN	—	—	—	+	—	—	+
L-(+)-Tartrate ^b	+	—	—	—	—	—	—
Galacturonate	—	—	+	+	+	+	+
γ-Glutamyltransferase	+ ^c	—	+	+	+	+	+
β-Glucuronidase	d	—	+	—	d	d	—
Mucate	+	+	— (70%)	—	+	+	+
Salicine	—	—	—	+	—	—	—
Lactose	—	— (75%)	— (75%)	—	d	—	—
Lysis by phage O1	+	—	+	—	+	+	d
<i>Habitat:</i>							
Warm-blooded animals	+						
Cold-blooded animals		+	+	+	+	+	+

^aSymbols: +, positive for 90% or more of strains in 1–2 days; d, positive for 11–89% of strains in 1–2 days; —, positive for 0–10% of strains in 1–2 days, unless otherwise indicated in the table. The temperature for all reactions is 37°C.

^bD-tartrate.

^cTyphimurium d, Dublin —.

Division into serovars The antigenic formulae of *Salmonella* serovars are listed in the Kauffmann–White scheme (Table BXII.γ.262). They are composed of numbers and letters given to the different O (somatic), Vi (capsular), and H (flagellar) antigens. Only those antigens of primary diagnostic importance are indicated in the Kauffmann–White scheme. Antigenic formulae (for example 6,7,[Vi]:c:1,5) represent the O antigenic factors, the Vi capsular antigen when present, the first phase of the H antigen, and the second phase of the H antigen, respectively. Those formulae with major O antigenic factors in common are collected into an O group and arranged alphabetically by the first phase of the H antigen within the group.

Lysogenization by certain converting phages may produce changes in the O antigenic formulae of salmonellae. In groups O:2, O:4, and O:9, presence of O:1 antigenic factor is associated with lysogenization (Iseki and Kashiwagi, 1955, 1957; Zinder, 1957; Stocker, 1958), but presence or absence of this factor in strains of these groups does not change the name of the serovar (for example, serovar Typhimurium applies to both O:1-positive and O:1-negative strains). Factors associated with phage conversion are underlined in the Kauffmann–White scheme. Converting phages of *Salmonella* are identical in morphology (Vieu et al., 1965), but their action is limited to certain O groups and they are serologically different from one another (Le Minor, 1968).

The specificities of O factors in *Salmonella* are determined by the composition and structure of the polysaccharides. Specificity is modified during smooth to rough mutation and by bacteriophage conversions (see reviews by Lüderitz et al., 1971; Stocker and Mäkelä, 1971, 1978). The only difference between the 4,12 and the 9,12 O-specific repeating units is in the di-deoxyhexose branch unit attached to the mannose, which is abequose in O:4,12 and tyvelose in O:9,12. In the conversion of O:3,10 to O:3,15, the terminal acetyl radical of the chain is suppressed and the α-linkage between galactose and mannose is transformed into a β-linkage. Other modifications of the specificity of O antigens may occur after mutation(s), resulting in new specificities called

T1 and T2 by Kauffmann (1956) or in different R types (reviewed by Lüderitz et al., 1971; Stocker and Mäkelä, 1971, 1978).

Surface antigens, commonly observed in other members of the *Enterobacteriaceae* family (e.g., *Escherichia coli* and *Klebsiella*), may be found in some *Salmonella* serovars. Surface antigens in *Salmonella* may mask O antigens, and the culture will not agglutinate with O antisera. The Vi antigen is a surface antigen found mainly in serovar Typhi and serovar Paratyphi C, as well as in a few strains of serovar Dublin. Heating at 100°C generally solubilizes Vi antigen, and heat-treated bacteria agglutinate with proper O antisera. Biogenesis of the Vi polysaccharide is governed by a set of genes located at the *viaB* locus, which is specific to Vi-expressing strains. Expression of Vi antigen is controlled by two two-component regulatory systems, OmpR-EnvZ and RcsB-RcsC (reviewed by Virlogeux-Payant and Popoff, 1996). One signal for this complex regulatory control was shown to be osmolarity (Pickard et al., 1994).

Subdivision of serovars Biovars are different sugar fermentation patterns shown by strains of the same serovar. They are determined by the presence or absence of enzymes and hence are genetically determined. Biovars may serve as markers and be of interest epidemiologically (for example, the xylose-positive and xylose-negative character of serovar Typhi).

Phagovars are determined by the sensitivity of cultures to a series of bacteriophages at appropriate dilutions. Phage typing of serovar Typhi and other salmonellae that possess the Vi antigen (serovar Paratyphi C and a few strains of serovar Dublin) is based on a series of adapted phages from phage Vi II of Craigie and Yen (1938). Phage typing of serovar Paratyphi B (Felix and Callow, 1943) and serovar Typhimurium (Anderson, 1964) uses a different series of phages. Analogous methods have been proposed for other serovars of *Salmonella*, some of them making use of the lysogenicity of the strains.

Other subdivisions of serovars may be based on production of, or on sensitivity to, bacteriocins, and on resistance to antibiotics.

TABLE BXII.γ.262. Antigenic formulae of the serovars of the genus *Salmonella*: the Kauffmann-White scheme^a

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Group O:2 (A)			
Paratyphi A	<u>1,2,12</u>	a	[1,5]
Nitra	<u>2,12</u>	g,m	
Kiel	<u>1,2,12</u>	g,p	
Koessen	<u>2,12</u>	l,v	1,5
Group O:4(B)			
Kisangani	<u>1,4,[5],12</u>	a	1,2
Hessarek	<u>4,12,27</u>	a	1,5
Fulica	<u>4,[5],12</u>	a	
Arechavaleta	<u>4,[5],12</u>	a	1,7
Bispebjerg	<u>1,4,[5],12</u>	a	e,n,x
Tinda	<u>1,4,12, 27</u>	a	e,n,z ₁₅
II	<u>1,4,[5],12, 27</u>	a	e,n,x
Huettwillen	<u>1,4,12</u>	a	l,w
Nakuru	<u>1,4,12, 27</u>	a	z ₆
II	<u>1,4,12, 27</u>	a	z ₃₉
Paratyphi B	<u>1,4,[5],12</u>	b	1,2
Limete	<u>1,4,12, 27</u>	b	1,5
II	<u>4,12</u>	b	1,5
Canada	<u>4,12,27</u>	b	1,6
Uppsala	<u>1,4,12, 27</u>	b	1,7
Abony	<u>1,4,[5],12, 27</u>	b	e,n,x
II	<u>1,4,12, 27</u>	b	[e,n,x]
Wagenia	<u>1,4,12, 27</u>	b	e,n,z ₁₅
Wien	<u>1,4,12, 27</u>	b	l,w
Tripoli	<u>1,4,12, 27</u>	b	z ₆
Schleissheim	<u>4,12,27</u>	b	
Legon	<u>1,4,12, 27</u>	c	1,5
Abortusovis	<u>4,12</u>	c	1,6
Altendorf	<u>4,12,27</u>	c	1,7
Bissau	<u>4,12</u>	c	e,n,x
Jericho	<u>1,4,12, 27</u>	c	e,n,z ₁₅
Hallfold	<u>1,4,12, 27</u>	c	l,w
Bury	<u>4,12,27</u>	c	z ₆
Stanley	<u>1,4,[5],12, 27</u>	d	1,2
Eppendorf	<u>1,4,12, 27</u>	d	1,5
Brezany	<u>1,4,12, 27</u>	d	1,6
Schwarzengrund	<u>1,4,12, 27</u>	d	1,7
II	<u>4,12</u>	d	e,n,x
Sarajane	<u>1,4,[5],12, 27</u>	d	e,n,x
Duisburg	<u>1,4,12, 27</u>	d	e,n,z ₁₅
Mons	<u>1,4,12, 27</u>	d	l,w
Ayinde	<u>1,4,12, 27</u>	d	z ₆
Saintpaul	<u>1,4,[5],12</u>	e,h	1,2
Reading	<u>1,4,[5],12</u>	e,h	1,5
Eko	<u>4,12</u>	e,h	1,6
Kaapstad	<u>4,12</u>	e,h	1,7
Chester	<u>1,4,[5],12</u>	e,h	e,n,x
Sandiego	<u>4,[5],12</u>	e,h	e,n,z ₁₅
II	<u>4,12</u>	e,n,x	1,2,7
II	<u>1,4,12, 27</u>	e,n,x	1,[5],7
Derby	<u>1,4,[5],12</u>	f,g	[1,2]
Agona	<u>1,4,12</u>	f,g,s	[1,2]
II	<u>1,4,[5],12</u>	f,g,t	z ₆ ,z ₄₂
Essen	<u>4,12</u>	g,m	
Hato	<u>1,4,[5],12</u>	g,m,s	
II	<u>1,4,12, 27</u>	g,[m],[s],t	e,n,x
II	<u>1,4,12, 27</u>	g,[m],t	[1,5]
II	<u>4,12</u>	g,m,t	z ₃₉
California	<u>4,12</u>	g,m,t	[z ₆₇]
Kingston	<u>1,4,[5],12, 27</u>	g,s,t	[1,2]
Budapest	<u>1,4,12, 27</u>	g,t	
Travis	<u>4,[5],12</u>	g,z ₅₁	1,7
Tennysen	<u>4,5,12</u>	g,z ₅₁	e,n,z ₁
II	<u>4,12</u>	g,z ₆₂	
Banana	<u>1,4,[5],12</u>	m,t	[1,5]
Madras	<u>4,[5],12</u>	m,t	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Typhimurium	<u>1,4,[5],12</u>	i	1,2
Lagos	<u>1,4,[5],12</u>	i	1,5
Agama	<u>4,12</u>	i	1,6
Farsta	<u>4,12</u>	i	e,n,x
Tsevie	<u>4,12</u>	i	e,n,z ₁₅
Gloucester	<u>1,4,12, 27</u>	i	l,w
Tumodi	<u>1,4,12</u>	i	z ₆
II	<u>4,12,27</u>	i	z ₃₅
Massenya	<u>1,4,12, 27</u>	k	1,5
Neumuenster	<u>1,4,12, 27</u>	k	1,6
II	<u>1,4,12, 27</u>	k	1,6
Ljubljana	<u>4,12,27</u>	k	e,n,x
Texas	<u>4,[5],12</u>	k	e,n,z ₁₅
Fyris	<u>4,[5],12</u>	l,v	1,2
Azteca	<u>4,[5],12,27</u>	l,v	1,5
Clackamas	<u>4,12</u>	l,v	1,6
Bredeney	<u>1,4,12, 27</u>	l,v	1,7
Kimuenza	<u>1,4,12, 27</u>	l,v	e,n,x
II	<u>1,4,12, 27</u>	l,v	e,n,x
Brandenburg	<u>1,4,[5],12, 27</u>	l,v	e,n,z ₁₅
II	<u>1,4,12, 27</u>	l,v	z ₃₉
Mono	<u>4,12</u>	l,w	1,5
Togo	<u>4,12</u>	l,w	1,6
II	<u>4,12</u>	l,w	e,n,x
Ayton	<u>1,4,12, 27</u>	l,w	z ₆
Kunduchi	<u>1,4,[5],12, 27</u>	l,[z ₁₃],[z ₂₈]	1,2
Tyresoe	<u>1,4,12, 27</u>	l,[z ₁₃],z ₂₈	1,5
Haduna	<u>4,12</u>	l,z ₁₃ ,[z ₂₈]	1,6
Kubacha	<u>1,4,12, 27</u>	l,z ₁₃ ,z ₂₈	1,7
Kano	<u>1,4,12, 27</u>	l,z ₁₃ ,z ₂₈	e,n,x
Vom	<u>1,4,12, 27</u>	l,z ₁₃ ,z ₂₈	e,n,z ₁₅
Reinickendorf	<u>4,12</u>	l,z ₂₈	e,n,x
II	<u>4,12</u>	l,z ₂₈	
Heidelberg	<u>1,4,[5],12</u>	r	1,2
Bradford	<u>4,12,27</u>	r	1,5
Winneba	<u>4,12</u>	r	1,6
Remo	<u>1,4,12, 27</u>	r	1,7
Bochum	<u>4,[5],12</u>	r	l,w
Southampton	<u>1,4,12, 27</u>	r	z ₆
Drogana	<u>1,4,12, 27</u>	r,i	e,n,z ₁₅
Africana	<u>4,12</u>	r,i	l,w
Coeln	<u>1,4,[5],12</u>	y	1,2
Trachau	<u>4,12,27</u>	y	1,5
Finaghy	<u>4,12</u>	y	1,6
Teddington	<u>1,4,12, 27</u>	y	1,7
Ball	<u>1,4,12, 27</u>	y	e,n,x
Jos	<u>1,4,12, 27</u>	y	e,n,z ₁₅
Kamorou	<u>4,12,27</u>	y	z ₆
Shubra	<u>4,[5],12</u>	z	1,2
Kiambu	<u>1,4,12</u>	z	1,5
II	<u>1,4,12, 27</u>	z	1,5
Loubomo	<u>4,12</u>	z	1,6
Indiana	<u>1,4,12</u>	z	1,7
II	<u>4,12</u>	z	1,7
Neftenbach	<u>4,12</u>	z	e,n,x
II	<u>1,4,12, 27</u>	z	e,n,x
Koenigstuhl	<u>1,4,[5],12</u>	z	e,n,z ₁₅
Preston	<u>1,4,12</u>	z	l,w
Entebbe	<u>1,4,12, 27</u>	z	z ₆
II	<u>4,12</u>	z	z ₃₉
Stanleyville	<u>1,4,[5],12, 27</u>	z ₄ ,z ₂₃	[1,2]
Vuadens	<u>4,12,27</u>	z ₄ ,z ₂₃	z ₆
Kalamu	<u>4,[5],12</u>	z ₄ ,z ₂₄	[1,5]
Haifa	<u>1,4,[5],12</u>	z ₁₀	1,2
Ituri	<u>1,4,12</u>	z ₁₀	1,5
Tudu	<u>4,12</u>	z ₁₀	1,6
Albert	<u>4,12</u>	z ₁₀	e,n,x
Tokoin	<u>4,12</u>	z ₁₀	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Mura	1,4,12	z ₁₀	l,w
Fortune	1,4,12, 27	z ₁₀	z ₆
Vellore	1,4,12, 27	z ₁₀	z ₃₅
Brancaster	1,4,12, 27	z ₂₉	
II	1,4,12	z ₂₉	e,n,x
Pasing	4,12	z ₃₅	1,5
Tafo	1,4,12, 27	z ₃₅	1,7
Sloterdijk	1,4,12, 27	z ₃₅	z ₆
Yaounde	1,4,12, 27	z ₃₅	e,n,z ₁₅
Tejas	4,12	z ₃₆	
Wilhelmsburg	1,4,[5],12, 27	z ₃₈	[e,n,z ₁₅]
II	1,4,12, 27	z ₃₉	1,[5],7
Thayngen	1,4,12, 27	z ₄₁	1,(2),5
Maska	1,4,12, 27	z ₄₁	e,n,z ₁₅
Abortusequi	4,12		e,n,x
Group O:7 (C1)^d			
Sanjuan	6,7	a	1,5
II	6,7,14	a	1,5
Umlhali	6,7	a	1,6
Austin	6,7	a	1,7
Oslo	6,7,14	a	e,n,x
Denver	6,7	a	e,n,z ₁₅
Coleypark	6,7,14	a	l,w
Damman	6,7	a	z ₆
II	6,7	a	z ₆
II	6,7	a	z ₄₂
Brazzaville	6,7	b	1,2
Edinburg	6,7,14	b	1,5
Adime	6,7	b	1,6
Koumra	6,7	b	1,7
Lockleaze	6,7,14	b	e,n,x
Georgia	6,7	b	e,n,z ₁₅
II	6,7	b	[e,n,x]:z ₄₂
Ohio	6,7,14	b	l,w
Leopoldville	6,7	b	z ₆
Kotte	6,7	b	z ₃₅
II	6,7	b	z ₃₉
Hissar	6,7,14	c	1,2
Paratyphi C	6,7,[Vi]	c	1,5
Choleraesuis	6,7	c	1,5
Typhisuis	6,7	c	1,5
Birkenhead	6,7	c	1,6
Schwabach	6,7	c	1,7
Namibia	6,7	c	e,n,x
Kaduna	6,7,14	c	e,n,z ₁₅
Kisii	6,7	d	1,2
Isangi	6,7,14	d	1,5
Kivu	6,7	d	1,6
Kambole	6,7	d	1,[2],7
Amersfoort	6,7,14	d	e,n,x
Gombe	6,7,14	d	e,n,z ₁₅
Livingstone	6,7,14	d	l,w
Wil	6,7	d	l,z ₁₃ ,z ₂₈
Nieukerk	6,7,14	d	z ₆
II	6,7	d	z ₄₂
Larochelle	6,7	e,h	1,2
Lomita	6,7	e,h	1,5
Norwich	6,7	e,h	1,6
Nola	6,7	e,h	1,7
Braenderup	6,7,14	e,h	e,n,z ₁₅
II	6,7	e,n,x	1,6:z ₄₂
Rissen	6,7,14	f,g	
Eingedi	6,7	f,g,t	1,2,7
Afula	6,7	f,g,t	e,n,x
Montevideo	6,7,14	g,m,[p],s	[1,2,7]
II	6,7	g,m,[s],t	e,n,x
II	6,7	(g),m,[s],t	1,5
II	6,7	g,[m],s,t	[z ₄₂]

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Othmarschen	6,7,14	g,m,[t]	
Menston	6,7	g,s,[t]	[1,6]
II	6,7	g,t	e,n,x:z ₄₂
Riggil	6,7	g,(t)	
Alamo	6,7	g,z ₅₁	1,5
IV	6,7	g,z ₅₁	
Haelsingborg	6,7	m,p,t,[u]	
Winston	6,7	m,t	1,6
Oakey	6,7	m,t	z ₆₄
II	6,7	m,t	
Oranienburg	6,7,14	m,t	[z ₅₇]
Augustenborg	6,7,14	i	1,2
Oritamerin	6,7	i	1,5
Garoli	6,7	i	1,6
Lika	6,7	i	1,7
Athinai	6,7	i	e,n,z ₁₅
Norton	6,7	i	l,w
Stuttgart	6,7,14	i	z ₆
Galiema	6,7,14	k	1,2
Thompson	6,7,14	k	1,5
Daytona	6,7	k	1,6
Baiboukoum	6,7	k	1,7
Singapore	6,7	k	e,n,x
Escanaba	6,7	k	e,n,z ₁₅
IIIb	6,7	(k)	z:[z ₅₄]
II	6,7	k	[z ₆]
Concord	6,7	l,v	1,2
Irumu	6,7	l,v	1,5
Mkamba	6,7	l,v	1,6
Kortrijk	6,7	l,v	1,7
Bonn	6,7	l,v	e,n,x
Potsdam	6,7,14	l,v	e,n,z ₁₅
Gdansk	6,7,14	l,v	z ₆
Coromandel	6,7	l,v	z ₃₅
IIIb	6,7	l,v	z ₅₃
Gabon	6,7	l,w	1,2
Colorado	6,7	l,w	1,5
II	6,7	l,w	1,5,7
II	6,7	l,w	z ₄₂
Nessziona	6,7	l,z ₁₃	1,5
Kenya	6,7	l,z ₁₃	e,n,x
Neukoelln	6,7	l,z ₁₃ ,[z ₂₈]	e,n,z ₁₅
Makiso	6,7	l,z ₁₃ ,z ₂₈	z ₆
Strathcona	6,7	l,z ₁₃ ,z ₂₈	1,7
II	6,7	l,z ₂₈	1,5:[z ₄₂]
II	6,7	l,z ₂₈	e,n,x
II	6,7	l,z ₂₈	z ₆
Virchow	6,7	r	1,2
Infantis	6,7,14	r	1,5
Nigeria	6,7	r	1,6
Colindale	6,7	r	1,7
Papuana	6,7	r	e,n,z ₁₅
Grampian	6,7	r	l,w
Richmond	6,7	y	1,2
Bareilly	6,7,14	y	1,5
Oyonnax	6,7	y	1,6
Gatow	6,7	y	1,7
Hartford	6,7	y	e,n,x:[z ₆₇]
Mikawasima	6,7,14	y	e,n,z ₁₅
Chile	6,7	z	1,2
Poitiers	6,7	z	1,5
II	6,7	z	1,5
Oakland	6,7	z	1,6,[7]
Cayar	6,7	z	e,n,x
II	6,7	z	e,n,x
Businga	6,7	z	e,n,z ₁₅
Bruck	6,7	z	l,w
II	6,7	z	z ₆

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	6,7	z	z ₃₉
II	6,7	z	z ₄₂
Obogu	6,7	z ₄ ,z ₂₃	1,5
Planckendael	6,7	z ₄ ,z ₂₃	1,6
Aequatoria	6,7	z ₄ ,z ₂₃	e,n,z ₁₅
Goma	6,7	z ₄ ,z ₂₃	z ₆
IV	6,7	z ₄ ,z ₂₃	
II	6,7	z ₄ ,z ₂₄	z ₄₂
Somone	6,7	z ₄ ,z ₂₄	
IV	6,7	z ₄ ,z ₂₄	
II	6,7	z ₆	1,7
Menden	6,7	z ₁₀	1,2
Inganda	6,7	z ₁₀	1,5
Eschweiler	6,7	z ₁₀	1,6
Ngili	6,7	z ₁₀	1,7
Djugu	6,7	z ₁₀	e,n,x
Mbandaka	6,7,14	z ₁₀	e,n,z ₁₅
Jerusalem	6,7,14	z ₁₀	1,w
Redba	6,7	z ₁₀	z ₆
Omuna	6,7	z ₁₀	z ₃₅
Tennessee	6,7,14	z ₂₉	[1,2,7]
II	6,7	z ₂₉	[z ₄₂]
Tienba	6,7	z ₃₅	1,6
Palime	6,7	z ₃₅	e,n,z ₁₅
Tampico	6,7	z ₃₆	e,n,z ₁₅
II	6,7	z ₃₆	z ₄₂
IV	6,7	z ₃₆	
Rumford	6,7	z ₃₈	1,2
Lille	6,7,14	z ₃₈	
IIIb	6,7,14	z ₃₉	1,2
II	6,7	z ₃₉	1,5,7
VI	6,7	z ₄₁	1,7
Hillsborough	6,7	z ₄₁	1,w
Tamilnadu	6,7	z ₄₁	z ₃₅
II	6,7	z ₄₂	1,7
Bulovka	6,7	z ₄₄	
II	6,7		1,6
Group O:8 (C2-C3)^c			
Be	8,20	a	[z ₆]
Valdosta	6,8	a	1,2
Doncaster	6,8	a	1,5
Curacao	6,8	a	1,6
Nordufer	6,8	a	1,7
Narashino	6,8	a	e,n,x
II	6,8	a	e,n,x
Leith	6,8	a	e,n,z ₁₅
II	6,8	a	z ₃₉
II	6,8	a	z ₅₂
Djelfa	8	b	1,2
Skansen	6,8	b	1,2
Korbol	8,20	b	1,5
Nagoya	6,8	b	1,5
II	6,8	b	1,5
Stourbridge	6,8	b	1,6
Sanga	8	b	1,7
Eboko	6,8	b	1,7
Konstanz	8	b	e,n,x
Gatuni	6,8	b	e,n,x
Shipley	8,20	b	e,n,z ₁₅
Presov	6,8	b	e,n,z ₁₅
Bukuru	6,8	b	1,w
Tounouma	8,20	b	z ₆
Banalia	6,8	b	z ₆
Wingrove	6,8	c	1,2
Utah	6,8	c	1,5
Bronx	6,8	c	1,6
Belfast	6,8	c	1,7
Alexanderpolder	8	c	1,w

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Santiago	8,20	c	e,n,x
Belem	6,8	c	e,n,x
Quiniela	6,8	c	e,n,z ₁₅
Tado	8,20	c	z ₆
Virginia	8	d	1,2
Muenchen	6,8	d	1,2:[z ₆₇]
Yovokome	8,20	d	1,5
Manhattan	6,8	d	1,5
Portanigra	8,20	d	1,7
Dunkwa	6,8	d	1,7
Sterrenbos	6,8	d	e,n,x
Herston	6,8	d	e,n,z ₁₅
Labadi	8,20	d	z ₆
II	6,8	d	z ₆ ,z ₄₂
Bardo	8	e,h	1,2
Newport	6,8,20	e,h	1,2:[z ₆₇]
Ferruch	8	e,h	1,5
Kottbus	6,8	e,h	1,5
Cremieu	6,8	e,h	1,6
Atakpame	8,20	e,h	1,7
Tshiongwe	6,8	e,h	e,n,z ₁₅
Rechovot	8,20	e,h	z ₆
Sandow	6,8	f,g	e,n,z ₁₅
II	6,8	f,g,m,t	[e,n,x]
Emek	8,20	g,m,s	
Chincol	6,8	g,m,[s]	[e,n,x]
II	6,8	g,m,t	1,7
Reubeuss	8,20	g,m,t	
Alminko	8,20	g,s,t	
Nanergou	6,8	g,s,t	
Yokoe	8,20	m,t	
II	6,8	m,t	1,5
II	6,8	m,t	e,n,x
Bassa	6,8	m,t	
Lindenbourg	6,8	i	1,2
Bargny	8,20	i	1,5
Takoradi	6,8	i	1,5
Warnow	6,8	i	1,6
Malmoe	6,8	i	1,7
Bonariensis	6,8	i	e,n,x
Aba	6,8	i	e,n,z ₁₅
Magherafelt	8,20	i	1,w
Cyprus	6,8	i	1,w
Kentucky	8,20	i	z ₆
Kallo	6,8	k	1,2
Haardt	8	k	1,5
Blockley	6,8	k	1,5
Schwerin	6,8	k	e,n,x
Charlottenburg	6,8	k	e,n,z ₁₅
Pakistan	8	l,v	1,2
Litchfield	6,8	l,v	1,2
Loanda	6,8	l,v	1,5
Amherstiana	8	l,v	1,6
Manchester	6,8	l,v	1,7
Holcomb	6,8	l,v	e,n,x
II	6,8	l,v	e,n,x
Edmonton	6,8	l,v	e,n,z ₁₅
Fayed	6,8	l,w	1,2
II	6,8	l,w	z ₆ ,z ₄₂
Hiduddify	6,8	l,z ₁₃ ,z ₂₈	1,5
Breukelen	6,8	l,z ₁₃ ,[z ₂₈]	e,n,z ₁₅
II	6,8	l,z ₂₈	e,n,x
Bsilla	6,8	r	1,2
Hindmarsh	8,20	r	1,5
Bovismorbificans	6,8,20	r,[i]	1,5
Noya	8	r	1,7
Akanji	6,8	r	1,7
Cocody	8,20	r,i	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Hidalgo	6,8	r,[i]	e,n,z ₁₅
Brikama	8,20	r,[i]	l,w
Goldcoast	6,8	r	l,w
Altona	8,20	r,[i]	z ₆
Giza	8,20	y	1,2
Brunei	8,20	y	1,5
Tananarive	6,8	y	1,5
Bulgaria	6,8	y	1,6
II	6,8	y	1,6;z ₄₂
Alagbon	8	y	1,7
Inchpark	6,8	y	1,7
Sunnycove	8	y	e,n,x
Daarle	6,8	y	e,n,x
Praha	6,8	y	e,n,z ₁₅
Kraligen	8,20	y	z ₆
Benue	6,8	y	l,w
Sindelfingen	8,20	y	l,w
Mowanjum	6,8	z	1,5
II	6,8	z	1,5
Phaliron	8	z	e,n,z ₁₅
Kalumburu	6,8	z	e,n,z ₁₅
Kuru	6,8	z	l,w
Daula	8,20	z	z ₆
Bellevue	8	z ₄ ,z ₂₃	1,7
Lezennes	6,8	z ₄ ,z ₂₃	1,7
Breda	6,8	z ₄ ,z ₂₃	e,n,x
Chailey	6,8	z ₄ ,z ₂₃	e,n,z ₁₅
Dabou	8,20	z ₄ ,z ₂₃	l,w
Corvallis	8,20	z ₄ ,z ₂₃	[z ₆]
Albany	8,20	z ₄ ,z ₂₄	
Duesseldorf	6,8	z ₄ ,z ₂₄	
Tallahassee	6,8	z ₄ ,z ₃₂	
Bazenheid	8,20	z ₁₀	1,2
Zerifin	6,8	z ₁₀	1,2
Paris	8,20	z ₁₀	1,5
Mapo	6,8	z ₁₀	1,5
Cleveland	6,8	z ₁₀	1,7
Istanbul	8	z ₁₀	e,n,x
Hadar	6,8	z ₁₀	e,n,x
Chomedey	8,20	z ₁₀	e,n,z ₁₅
Glostrup	6,8	z ₁₀	e,n,z ₁₅
Remiremont	8,20	z ₁₀	l,w
Molade	8,20	z ₁₀	z ₆
Wippra	6,8	z ₁₀	z ₆
II	6,8	z ₂₉	1,5
II	8	z ₂₉	e,n,x;z ₄₂
Tamale	8,20	z ₂₉	[e,n,z ₁₅]
Uno	6,8	z ₂₉	[e,n,z ₁₅]
II	6,8	z ₂₉	e,n,x
Kolda	8,20	z ₃₅	1,2
Yarm	6,8	z ₃₅	1,2
Angers	8,20	z ₃₅	z ₆
Apeyeme	8,20	z ₃₈	
Diogoye	8,20	z ₄₁	z ₆
Aesch	6,8	z ₆₀	1,2
Group O:9 (D1)			
Sendai	1,9,12	a	1,5
Miami	1,9,12	a	1,5
II	9,12	a	1,5
Os	9,12	a	1,6
Saarbruecken	1,9,12	a	1,7
Lomalinda	1,9,12	a	e,n,x
II	1,9,12	a	e,n,x
Durban	9,12	a	e,n,z ₁₅
II	9,12	a	z ₃₉
II	1,9,12	a	z ₄₂
Onarimon	1,9,12	b	1,2
Frintrop	1,9,12	b	1,5

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	1,9,12	b	e,n,x
II	1,9,12	b	z ₆
II	1,9,12	b	z ₃₉
Goeteborg	9,12	c	1,5
Ipeko	9,12	c	1,6
Elokate	9,12	c	1,7
Alabama	9,12	c	e,n,z ₁₅
Ridge	9,12	c	z ₆
Ndolo	1,9,12	d	1,5
Tarshyne	9,12	d	1,6
Eschberg	9,12	d	1,7
II	9,12	d	e,n,x
Bangui	9,12	d	e,n,z ₁₅
Zega	9,12	d	z ₆
Jaffna	1,9,12	d	z ₃₅
II	9,12	d	z ₃₉
Typhi	9,12[Vi]	d	
Bournemouth	9,12	e,h	1,2
Eastbourne	1,9,12	e,h	1,5
Westafrika	9,12	e,h	1,7
Israel	9,12	e,h	e,n,z ₁₅
II	9,12	e,n,x	1,[5],7
II	9,12	e,n,x	1,6
Berta	1,9,12	[f],g,[t]	
Enteritidis	1,9,12	g,m	
Blegdam	9,12	g,m,q	
II	1,9,12	g,m,[s],t	[1,5,7]:[z ₄₂]
II	1,9,12	g,m,s,t	e,n,x
Dublin	1,9,12[Vi]	g,p	
Naestved	1,9,12	g,p,s	
Rostock	1,9,12	g,p,u	
Moscow	9,12	g,q	
II	9,12	g,s,t	e,n,x
Newmexico	9,12	g,z ₅₁	1,5
II	1,9,12	g,z ₆₂	[e,n,x]
Antarctica	9,12	g,z ₆₃	
II	9,12	m,t	e,n,x
Pensacola	1,9,12	m,t	[1,2]
II	1,9,12	m,t	1,5
II	1,9,12	m,t	z ₃₉
Seremban	9,12	i	1,5
Claibornei	1,9,12	k	1,5
Goverdhan	9,12	k	1,6
Mendoza	9,12	l,v	1,2
Panama	1,9,12	l,v	1,5
Kapemba	9,12	l,v	1,7
Zaiman	9,12	l,v	e,n,x
II	9,12	l,v	e,n,x
Goettingen	9,12	l,v	e,n,z ₁₅
II	9,12	l,v	z ₃₉
Victoria	1,9,12	l,w	1,5
II	1,9,12	l,w	e,n,x
Itami	9,12	l,z ₁₃	1,5
Miyazaki	9,12	l,z ₁₃	1,7
Napoli	1,9,12	l,z ₁₃	e,n,x
Javiana	1,9,12	l,z ₂₈	1,5
Kotu	9,12	l,z ₂₈	1,6
II	9,12	l,z ₂₈	1,5:[z ₄₂]
II	9,12	l,z ₂₈	e,n,x
Jamaica	9,12	r	1,5
Camberwell	9,12	r	1,7
Campinense	9,12	r	e,n,z ₁₅
Lome	9,12	r	z ₆
Powell	9,12	y	1,7
II	1,9,12	y	z ₃₉
Mulhouse	1,9,12	z	1,2
Lawndale	1,9,12	z	1,5
Kimpese	9,12	z	1,6

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	1,9,12	z	1,7
II	1,9,12	z	z ₆
II	9,12	z	z ₃₉
Wangata	1,9,12	z ₄ ,z ₂₃	[1,7]
Natal	9,12	z ₄ ,z ₂₄	
Franken	9,12	z ₆	z ₆₇
Portland	9,12	z ₁₀	1,5
Treguier	9,12	z ₁₀	z ₆
Ruanda	9,12	z ₁₀	e,n,z ₁₅
II	9,12	z ₂₉	1,5
II	1,9,12	z ₂₉	e,n,x
Penarth	9,12	z ₃₅	z ₆
Elomrane	1,9,12	z ₃₈	
II	1,9,12	z ₃₉	1,7
Ottawa	1,9,12	z ₄₁	1,5
II	1,9,12	z ₄₂	1,[5],7
Gallinarum	1,9,12		
Group O:9,46 (D2)			
Baildon	9,46	a	e,n,x
Doba	9,46	a	e,n,z ₁₅
Cheltenham	9,46	b	1,5
Zadar	9,46	b	1,6
Worb	9,46	b	e,n,x
II	9,46	b	e,n,x
Bamboye	9,46	b	1,w
Linguere	9,46	b	z ₆
Kolar	9,46	b	z ₃₅
Itutaba	9,46	c	z ₆
Ontario	9,46	d	1,5
Quentin	9,46	d	1,6
Strasbourg	9,46	d	1,7
Olten	9,46	d	e,n,z ₁₅
Plymouth	9,46	d	z ₆
Bergedorf	9,46	e,h	1,2
Waedenswil	9,46	e,h	1,5
Guerin	9,46	e,h	z ₆
II	9,46	e,n,x	1,5,7
Wernigerode	9,46	f,g	
Hillingdon	9,46	g,m	
Macclesfield	9,46	g,m,s	1,2,7
II	9,46	g,[m],[s],t	[e,n,x]
Gateshead	9,46	g,s,t	
II	9,46	g,z ₆₂	
II	9,46	m,t	e,n,x
Sangalkam	9,46	m,t	
Mathura	9,46	i	e,n,z ₁₅
Potto	9,46	i	z ₆
Marylebone	9,46	k	1,2
Cochin	9,46	k	1,5
Ceyco	9,46	k	z ₃₅
India	9,46	l,v	1,5
Geraldton	9,46	l,v	1,6
Toronto	9,46	l,v	e,n,x
Ackwepe	9,46	l,w	
Nordrhein	9,46	l,z ₁₃ ,z ₂₈	e,n,z ₁₅
Deckstein	9,46	r	1,7
Shoreditch	9,46	r	e,n,z ₁₅
Sokode	9,46	r	z ₆
Benin	9,46	y	1,7
Irchel	9,46	y	e,n,x
Nantes	9,46	y	1,w
Mayday	9,46	y	z ₆
II	9,46	z	1,5
II	9,46	z	e,n,x
Bambylor	9,46	z	e,n,z ₁₅
Ekotedo	9,46	z ₄ ,z ₂₃	
II	9,46	z ₄ ,z ₂₄	z ₃₉ ,z ₄₂
Ngaparou	9,46	z ₄ ,z ₂₄	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Lishabi	9,46	z ₁₀	1,7
Inglis	9,46	z ₁₀	e,n,x
Mahina	9,46	z ₁₀	e,n,z ₁₅
Louisiana	9,46	z ₁₀	z ₆
II	9,46	z ₁₀	z ₆
II	9,46	z ₁₀	z ₃₉
Ouakam	9,46	z ₂₉	
Hillegersberg	9,46	z ₃₅	1,5
Basingstoke	9,46	z ₃₅	e,n,z ₁₅
Trimdon	9,46	z ₃₅	z ₆
Fresno	9,46	z ₃₈	
II	9,46	z ₃₉	1,7
Wuppertal	9,46	z ₄₁	
Group O:9,46,27 (D3)			
II	1,9,12,46,27	a	z ₆
II	1,9,12,46,27	c	z ₃₉
II	9,12,46,27	g,t	e,n,x
II	1,9,12,46,27	l,z ₁₃ ,z ₂₈	z ₃₉
II	1,9,12,46,27	y	z ₃₉
II	1,9,12,46,27	z ₄ ,z ₂₄	1,5
II	1,9,12,46,27	z ₁₀	1,5
II	1,9,12,46,27	z ₁₀	e,n,x
II	1,9,12,46,27	z ₁₀	z ₃₉
Group O:3,10 (E1)^f			
Aminatu	3,10	a	1,2
Goelzau	3,10[15]	a	1,5
Oxford	3,10[15] [15,34]	a	1,7
Masembe	3,10	a	e,n,x
II	3,10	a	e,n,x
Galil	3,10	a	e,n,z ₁₅
II	3,10	a	1,v
II	3,10	a	z ₃₉
Kalina	3,10	b	1,2
Butantan	3,10[15] [15,34]	b	1,5
Allerton	3,10	b	1,6
Huvudsta	3,10	b	1,7
Benfica	3,10	b	e,n,x
II	3,10	b	e,n,x
Yaba	3,10[15]	b	e,n,z ₁₅
Epicrates	3,10	b	1,w
Wilmington	3,10	b	z ₆
Westminster	3,10[15]	b	z ₃₅
II	3,10	b	z ₃₉
Asylanta	3,10	c	1,2
Gbadago	3,10[15]	c	1,5
Ikayi	3,10[15]	c	1,6
Pramiso	3,10	c	1,7
Agege	3,10	c	e,n,z ₁₅
Anderlecht	3,10	c	1,w
Okefoko	3,10	c	z ₆
Stormont	3,10	d	1,2
Shangani	3,10[15]	d	1,5
Lekke	3,10	d	1,6
Onireke	3,10	d	1,7
Souza	3,10[15]	d	e,n,x
II	3,10	d	e,n,x
Madjorio	3,10	d	e,n,z ₁₅
Birmingham	3,10[15]	d	1,w
Weybridge	3,10	d	z ₆
Maron	3,10	d	z ₃₅
Vejle	3,10[15]	e,h	1,2
Muenster	3,10[15] [15,34]	e,h	1,5
Anatum	3,10[15] [15,34]	e,h	1,6
Nyborg	3,10[15]	e,h	1,7
Newlands	3,10[15,34]	e,h	e,n,x
Lamberhurst	3,10	e,h	e,n,z ₁₅
Meleagridis	3,10[15] [15,34]	e,h	1,w
Sekondi	3,10	e,h	z ₆

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	3,10	e,n,x	1,7
Regent	3,10	f,g,[s]	[1,6]
Alfort	3,10	f,g	e,n,x
Suberu	3,10	g,m	
Amsterdam	3,10[15][15,34]	g,m,s	
II	3,10[15]	g,m,s,t	[1,5]
Westhampton	3,10[15][15,34]	g,s,t	
Bloomsbury	3,10	g,t	1,5
II	3,10	g,t	
II	3,10	m,t	1,5
Southbank	3,10[15][15,34]	m,t	[1,6]
II	3,10	m,t	e,n,x
Cuckmere	3,10	i	1,2
Amounderness	3,10	i	1,5
Tibati	3,10	i	1,6
Truro	3,10	i	1,7
Bessi	3,10	i	e,n,x
Falkensee	3,10[15]	i	e,n,z ₁₅
Hoboken	3,10	i	1,w
Yeerongpilly	3,10	i	z ₆
Wimborne	3,10	k	1,2
Zanzibar	3,10[15]	k	1,5
Serrekunda	3,10	k	1,7
Yundum	3,10	k	e,n,x
Marienthal	3,10	k	e,n,z ₁₅
Newrochelle	3,10	k	1,w
Nchanga	3,10[15]	l,v	1,2
Sinstorf	3,10	l,v	1,5
London	3,10[15]	l,v	1,6
Give	3,10[15][15,34]	[d],l,v	1,7
II	3,10	l,v	e,n,x
Ruzizi	3,10	l,v	e,n,z-15
II	3,10	l,v	z ₆
Sinchew	3,10	l,v	z ₃₅
Assinie	3,10	l,w	z ₆
Freiburg	3,10	l,z ₁₃	1,2
Uganda	3,10[15]	l,z ₁₃	1,5
Fallowfield	3,10	l,z ₁₃ ,z ₂₈	e,n,z ₁₅
Hoghton	3,10	l,z ₁₃ ,z ₂₈	z ₆
II	3,10	l,z ₂₈	1,5
Joal	3,10	l,z ₂₈	1,7
Lamin	3,10	l,z ₂₈	e,n,x
II	3,10	l,z ₂₈	e,n,x
II	3,10	l,z ₂₈	z ₃₉
Ughelli	3,10	r	1,5
Elisabethville	3,10[15]	r	1,7
Simi	3,10	r	e,n,z ₁₅
Wettevreden	3,10[15]	r	z ₆
Seegefeld	3,10	r,i	1,2
Dumfries	3,10	r,i	1,6
Amager	3,10[15]	y	1,2
Orion	3,10[15][15,34]	y	1,5
Mokola	3,10	y	1,7
Ohlstedt	3,10[15]	y	e,n,x
Bolton	3,10	y	e,n,z ₁₅
Langensalza	3,10	y	1,w
Stockholm	3,10[15]	y	z ₆
Fufu	3,10	z	1,5
II	3,10	z	1,5
Harleystreet	3,10	z	1,6
Huddinge	3,10	z	1,7
II	3,10	z	e,n,x
Clerkenwell	3,10	z	1,w
Landwasser	3,10	z	z ₆
II	3,10	z	z ₃₉
Adabraka	3,10	z ₄ ,z ₂₃	[1,7]
Wagadugu	3,10	z ₄ ,z ₂₃	z ₆
Florian	3,10[15]	z ₄ ,z ₂₄	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	3,10	z ₄ ,z ₂₄	
Okerara	3,10	z ₁₀	1,2
Lexington	3,10[15][15,34]	z ₁₀	1,5
Harrisonburg	3,10[15][15,34]	z ₁₀	1,6
Coquilhatville	3,10	z ₁₀	1,7
Kristianstad	3,10	z ₁₀	e,n,z ₁₅
Biafra	3,10	z ₁₀	z ₆
Everleigh	3,10	z ₂₉	e,n,x
II	3,10	z ₂₉	[e,n,x]
Jedburgh	3,10[15]	z ₂₉	
Zongo	3,10	z ₃₅	1,7
Shannon	3,10	z ₃₅	1,w
Cairina	3,10	z ₃₅	z ₆
Macallen	3,10	z ₃₆	
Bolombo	3,10	z ₃₈	[z ₆]
II	3,10	z ₃₈	z ₄₂
II	3,10	z ₃₉	1,[5],7
Pietersburg	3,10[15,34]	z ₆₉	1,7
Group O:1,3,19 (E4)			
Niumi	1,3,19	a	1,5
Juba	1,3,19	a	1,7
Gwoza	1,3,19	a	e,n,z ₁₅
Alkmaar	1,3,19	a	1,w
Gnesta	1,3,19	b	1,5
Visby	1,3,19	b	1,6
Tambacounda	1,3,19	b	e,n,x
Kande	1,3,19	b	e,n,z ₁₅
Broughton	1,3,19	b	1,w
Accra	1,3,19	b	z ₆
Eastglam	1,3,19	c	1,5
Bida	1,3,19	c	1,6
Madiago	1,3,19	c	1,7
Ahmadi	1,3,19	d	1,5
Liverpool	1,3,19	d	e,n,z ₁₅
Tilburg	1,3,19	d	1,w
Niloese	1,3,19	d	z ₆
Vilvoorde	1,3,19	e,h	1,5
Hayindogo	1,3,19	e,h	1,6
Sanktmarx	1,3,19	e,h	1,7
Sao	1,3,19	e,h	e,n,z ₁₅
Calabar	1,3,19	e,h	1,w
Rideau	1,3,19	f,g	
Petahtikve	1,3,19	f,g,t	1,7
Maiduguri	1,3,19	f,g,t	e,n,z ₁₅
Kouka	1,3,19	g,m,[t]	
Senftenberg	1,3,19	g,[s],t	
Cannstatt	1,3,19	m,t	
Stratford	1,3,19	i	1,2
Chichester	1,3,19	i	1,6
Machaga	1,3,19	i	e,n,x
Avonmouth	1,3,19	i	e,n,z ₁₅
Zuilen	1,3,19	i	1,w
Taksony	1,3,19	i	z ₆
Oersterbro	1,3,19	k	1,5
Bethune	1,3,19	k	1,7
Ngor	1,3,19	l,v	1,5
Parkroyal	1,3,19	l,v	1,7
Svedvi	1,3,19	l,v	e,n,z ₁₅
Fulda	1,3,19	l,w	1,5
Westerstede	1,3,19	l,z ₁₃	1,2
Winterthur	1,3,19	l,z ₁₃	1,6
Lokstedt	1,3,19	l,z ₁₃ ,z ₂₈	1,2
Stuivenberg	1,3,19	l,[z ₁₃]z ₂₈	1,5
Bedford	1,3,19	l,z ₁₃ ,z ₂₈	e,n,z ₁₅
Tomelilla	1,3,19	l,z ₂₈	1,7
Kindia	1,3,19	l,z ₂₈	e,n,x
Yalding	1,3,19	r	e,n,z ₁₅
Fareham	1,3,19	r,i	1,w

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Gatineau	1,3,19	y	1,5
Thies	1,3,19	y	1,7
Slade	1,3,19	y	e,n,z ₁₅
Kinson	1,3,19	y	e,n,x
Krefeld	1,3,19	y	1,w
Korlebu	1,3,19	z	1,5
Kainji	1,3,19	z	1,6
Lerum	1,3,19	z	1,7
Schoeneberg	1,3,19	z	e,n,z ₁₅
Carno	1,3,19	z	1,w
Hongkong	1,3,19	z	z ₆
Sambre	1,3,19	z ₄ ,z ₂₄	
Dallgow	1,3,19	z ₁₀	e,n,z ₁₅
Llandoff	1,3,19	z ₂₉	[z ₆]
Ochiogu	1,3,19	z ₃₈	[e,n,z ₁₅]
Chittagong	1,3,10,19	b	z ₃₅
Bilu	1,3,10,19	f,g,t	1,(2),7
Ilugun	1,3,10,19	z ₄ ,z ₂₃	z ₆
Dessau	1,3,15,19	g,s,t	
Cannonhill	1,3,15,19	y	e,n,x
Group O:11 (F)			
II	11	a	d:e,n,z ₁₅
Gallen	11	a	1,2
Marseille	11	a	1,5
VI	11	a	1,5
Toowong	11	a	1,7
Luciana	11	a	e,n,z ₁₅
Epina	11	a	1,z ₁₃ ,z ₂₈
II	11	a	z ₆ ,z ₄₂
Atento	11	b	1,2
Leeuwarden	11	b	1,5
Wohlen	11	b	1,6
VI	11	b	1,7
VI	11	b	e,n,x
Pharr	11	b	e,n,z ₁₅
Chiredzi	11	c	1,5
Brindisi	11	c	1,6
II	11	c	e,n,z ₁₅
Woodinville	11	c	e,n,x
Ati	11	d	1,2
Gustavia	11	d	1,5
Chandans	11	d	[e,n,x]:[r]
Findorff	11	d	z ₆
Chingola	11	e,h	1,2
Adamstua	11	e,h	1,6
Redhill	11	e,h	1,z ₁₃ ,z ₂₈
Abuja	11	g,m	1,5
Missouri	11	g,s,t	
II	11	g,[m],s,t	z ₃₉
IV	11	g,z ₅₁	
Moers	11	m,t	
II	11	m,t	e,n,x
Aberdeen	11	i	1,2
Brijbhumi	11	i	1,5
Heerlen	11	i	1,6
Veneziana	11	i	e,n,x
Pretoria	11	k	1,2
Abacetuba	11	k	1,5
Sharon	11	k	1,6
Colobane	11	k	1,7
Kisarawe	11	k	e,n,x,[z ₁₅]
Mannheim	11	k	1,w
Amba	11	k	1,z ₁₃ ,z ₂₈
IIIb	11	k	z ₅₃
Stendal	11	l,v	1,2
Maracaibo	11	l,v	1,5
Fann	11	l,v	e,n,x
Bullbay	11	l,v	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIb	11	l,v	z
IIIb	11	l,v	z ₅₃
Glidji	11	l,w	1,5
Tours	11	1,z ₁₃	1,2
Connecticut	11	1,z ₁₃ ,z ₂₈	1,5
Osnabrueck	11	1,z ₁₃ ,z ₂₈	e,n,x
II	11	1,z ₂₈	e,n,x
Senegal	11	r	1,5
Rubislaw	11	r	e,n,x
Clanvillian	11	r	e,n,z ₁₅
Euston	11	r,i	e,n,x,z ₁₅
Volta	11	r	1,z ₁₃ ,z ₂₈
Solt	11	y	1,5
Jalisco	11	y	1,7
Herzliya	11	y	e,n,x
Crewe	11	z	1,5
Maroua	11	z	1,7
II	11	z	e,n,x
Nyanza	11	z	z ₆ : [z ₈₃]
II	11	z	z ₃₉
Remete	11	z ₄ ,z ₂₃	1,6
Etterbeek	11	z ₄ ,z ₂₃	e,n,z ₁₅
IIIa	11	z ₄ ,z ₂₃	
IV	11	z ₄ ,z ₂₃	
Yehuda	11	z ₄ ,z ₂₄	
IV	11	z ₄ ,z ₃₂	
Wentworth	11	z ₁₀	1,2
Straengnaes	11	z ₁₀	1,5
Telhashomer	11	z ₁₀	e,n,x
Lene	11	z ₃₈	
Maastricht	11	z ₄₁	1,2
II	11		1,5
Group O:13 (G)^g			
Chagoua	1,13,23	a	1,5
II	1,13,23	a	1,5
Mim	13,22	a	1,6
II	13,22	a	e,n,x
Wyldegreen	1,13,23	a	1,w
Marshall	13,22	a	1,z ₁₃ ,z ₂₈
II	1,13,23	a	z ₄₂
Ibadan	13,22	b	1,5
Mississippi	1,13,23	b	1,5
Oudwijk	13,22	b	1,6
II	1,13,23	b	[1,5]:z ₄₂
Bracknell	13,23	b	1,6
Rottmest	1,13,22	b	1,7
Vaertan	13,22	b	e,n,x
Ullevi	1,13,23	b	e,n,x
Bahati	13,22	b	e,n,z ₁₅
Durham	13,23	b	e,n,z ₁₅
Sanktjohann	13,23	b	1,w
II	1,13,22	b	z ₄₂
Haouaria	13,22	c	e,n,x,z ₁₅
Handen	1,13,23	d	1,2
Mishmarhaemek	1,13,23	d	1,5
Friedenau	13,22	d	1,6
Wichita	1,13,23	d	1,6
Grumpensis	1,13,23	d	1,7
II	13,23	d	e,n,x
Diguel	1,13,22	d	e,n,z ₁₅
Telelkebir	13,23	d	e,n,z ₁₅
Putten	13,23	d	1,w
Isuge	13,23	d	z ₆
Tschangu	1,13,23	e,h	1,5
Willemstad	1,13,22	e,h	1,6
Vridi	1,13,23	e,h	1,w
II	1,13,23	e,n,x	1,[5],7
Raus	13,22	f,g	e,n,x

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Havana	1,13,23	f,g,[s]	
Bron	13,22	g,m	[e,n,z ₁₅]
Agbeni	1,13,23	g,m,[s],[t]	
II	1,13,22	g,m,t	[1,5]
II	1,13,23	g,m,s,t	1,5
II	1,13,23	g,m,[s],t	[e,n,x]
II	1,13,23	g,m,s,t	z ₄₂
Congo	13,23	g,m,s,t	
Newyork	13,22	g,s,t	
Okatie	13,23	g,[s],t	
II	1,13,22	g,t	1,5
II	1,13,23	g,t	1,5
II	1,13,23	g,[s],t	z ₄₂
IIIa	1,13,23	g,z ₅₁	
Washington	13,22	m,t	
II	1,13,23	m,t	1,5
II	1,13,23	m,t	e,n,x
II	13,22	m,t	z ₄₂ :z ₃₉
II	1,13,23	m,t	z ₄₂
Kintambo	13,23	m,t	
V	1,13,22	i	
Idikan	1,13,23	i	1,5
Jukestown	13,23	i	e,n,z ₁₅
Kedougou	1,13,23	i	1,w
II	13,22	k	1,5:z ₄₂
Marburg	13,23	k	
II	13,23	k	z ₄₁
Lovelace	13,22	l,v	1,5
IIIb	13,22	l,v	1,5,7
Borbeck	13,22	l,v	1,6
Nanga	1,13,23	l,v	e,n,z ₁₅
II	13,23	l,w	e,n,x
Taiping	13,22	l,z ₁₃	e,n,z ₁₅
II	13,22	l,z ₂₈	1,5
II	13,23	l,z ₂₈	1,5
II	13,23	l,z ₂₈	z ₆
II	1,13,23	l,z ₂₈	z ₄₂
V	13,22	r	
Adjame	13,23	r	1,6
Linton	13,23	r	e,n,z ₁₅
Tanger	1,13,22	y	1,6
Yarrabah	13,23	y	1,7
Ordenez	1,13,23	y	1,w
Tunis	1,13,23	y	z ₆
II	1,13,23	z	1,5
Poona	1,13,22	z	1,6
Farmsen	13,23	z	1,6
Bristol	13,22	z	1,7
Tanzania	1,13,22	z	e,n,z ₁₅
Worthington	1,13,23	z	1,w
II	1,13,23	z	z ₄₂
II	13,22	z	
Ried	1,13,22	z ₄ :z ₂₃	[e,n,z ₁₅]
IIIa	13,22	z ₄ :z ₂₃	
Ajiobo	13,23	z ₄ :z ₂₃	
IIIa	13,23	z ₄ :z ₂₃ ,[z ₃₂]	
Romanby	1,13,23	z ₄ :z ₂₄	
IIIa	1,13,23	z ₄ :z ₂₄	
Roodepoort	1,13,22	z ₁₀	1,5
II	1,13,22	z ₁₀	z ₆
Sapele	13,23	z ₁₀	e,n,z ₁₅
Demerara	13,23	z ₁₀	1,w
II	13,22	z ₂₉	1,5
II	13,22	z ₂₉	e,n,x
II	1,13,23	z ₂₉	e,n,x
Agoueve	13,22	z ₂₉	
Cubana	1,13,23	z ₂₉	
Mampong	13,22	z ₃₅	1,6

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Nimes	13,22	z ₃₅	e,n,z ₁₅
Anna	13,23	z ₃₅	e,n,z ₁₅
Leiden	13,22	z ₃₈	
Fanti	13,23	z ₃₈	
II	13,22	z ₃₉	1,7
II	1,13,23	z ₃₉	1,5,7
II	1,13,23	[z ₄₂]	1,[5],7
II	13,23		1,6
Group O:6,14 (H)			
Garba	1,6,14,25	a	1,5
VI	[1],6,14	a	1,5
VI	1,6,14,25	a	e,n,x
Banjul	1,6,14,25	a	e,n,z ₁₅
Ndjamena	1,6,14,25	b	1,2
Kuntair	1,6,14,25	b	1,5
Tucson	[1],6,14,[25]	b	1,7
IIIb	(6),14	b	e,n,x
Blijdorp	1,6,14,25	c	1,5
Kassberg	1,6,14,25	c	1,6
Runby	1,6,14,25	c	e,n,x
Minna	1,6,14,25	c	1,w
Finkenwerder	[1],6,14,[25]	d	1,5
Woodhull	1,6,14,25	d	1,6
Midway	6,14,24	d	1,7
Florida	[1],6,14,[25]	d	1,7
Lindern	6,14,[24]	d	e,n,x
Charity	[1],6,14,[25]	d	e,n,x
Teko	1,6,14,25	d	e,n,z ₁₅
Encino	1,6,14,25	d	1,z ₁₃ :z ₂₈
Albuquerque	1,6,14,24	d	z ₆
Bahrenfeld	6,14,24	e,h	1,5
Onderstepoort	1,6,14,[25]	e,h	1,5
Magumeri	1,6,14,25	e,h	1,6
Beaudesert	[1],6,14,[25]	e,h	1,7
Warragul	[1],6,14,[25]	g,m	
Caracas	[1],6,14,[25]	g,m,s	
Sylvania	[1],6,14,[25]	g,p	
Catanzaro	6,14	g,s,t	
II	1,6,14	m,t	1,5
II	6,14	m,t	e,n,x
Kaitaan	1,6,14,25	m,t	
Mampeza	1,6,14,25	i	1,5
Buzu	[1],6,14,[25]	i	1,7
Schalkwijk	6,14	i	e,n,z ₁₅
Moussoro	1,6,14,25	i	e,n,z ₁₅
Harburg	[1],6,14,[25]	k	1,5
II	6,14,[24]	k	1,6
II	6,14	k	e,n,x
IIIb	(6),14	k	z
II	1,6,14	k	z ₆ :z ₄₂
IIIb	(6),14	k	z ₅₃
Boecker	[1],6,14,[25]	l,v	1,7
Horsham	1,6,14,[25]	l,v	e,n,x
IIIb	(6),14	l,v	z
IIIb	(6),14	l,v	z ₃₅
IIIb	(6),14	l,v	z ₅₃
Aflao	1,6,14,25	l,z ₂₈	e,n,x
Istoria	1,6,14,25	r,i	1,5
IIIb	(6),14	r	z
Surat	[1],6,14,[25]	r,[i]	e,n,z ₁₅
Carrau	6,14,[24]	y	1,7
Madelia	1,6,14,25	y	1,7
Fischerkietz	1,6,14,25	y	e,n,x
Mornington	1,6,14,25	y	e,n,z ₁₅
Homosassa	1,6,14,25	z	1,5
Kanifing	1,6,14,25	z	1,6
Soahanina	6,14,24	z	e,n,x
Sundsvall	[1],6,14,[25]	z	e,n,x

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Royan	1,6,14,25	z	e,n,z ₁₅
Poano	1,6,14,25	z	l,z ₁₃ ,z ₂₈
Arapahoe	6,14	z ₄ ,z ₂₃	1,5
Bouso	1,6,14,25	z ₄ ,z ₂₃	e,n,z ₁₅
IV	6,14	z ₄ ,z ₂₃	
Chichiri	6,14,24	z ₄ ,z ₂₄	
Uzaramo	1,6,14,25	z ₄ ,z ₂₄	
Nessa	1,6,14,25	z ₁₀	1,2
VI	1,6,14,25	z ₁₀	1,(2),7
II	1,6,14	z ₁₀	1,5
Laredo	1,6,14,25	z ₁₀	1,6
IIIb	(6),14	z ₁₀	e,n,x,z ₁₅
IIIb	(6),14	z ₁₀	z
II	1,6,14	z ₁₀	z ₆ ,z ₄₂
IIIb	6,14	z ₁₀	z ₅₃
Potosi	6,14	z ₃₆	1,5
Sara	1,6,14,25	z ₃₈	e,n,x
II	1,6,14	z ₄₂	1,6
IIIb	6,14	z ₅₂	e,n,x,z ₁₅
IIIb	1,6,14,25	z ₅₂	z ₃₅
Group 0:16 (I)			
Hannover	16	a	1,2
Brazil	16	a	1,5
Amunigun	16	a	1,6
Nyeko	16	a	1,7
Togba	16	a	e,n,x
Fischerhuetten	16	a	e,n,z ₁₅
Heron	16	a	z ₆
Hull	16	b	1,2
Wa	16	b	1,5
Glasgow	16	b	1,6
Hvittingfoss	16	b	e,n,x
II	16	b	e,n,x
Sangera	16	b	e,n,z ₁₅
Vegesack	16	b	1,w
Malstatt	16	b	z ₆
II	16	b	z ₃₉
II	16	b	z ₄₂
Vancouver	16	c	1,5
Gafsa	16	c	1,6
Shamba	16	c	e,n,x
Hithergreen	16	c	e,n,z ₁₅
Yoruba	16	c	1,w
Oldenburg	16	d	1,2
Sculcoates	16	d	1,5
II	16	d	1,5
Sherbrooke	16	d	1,6
Gaminara	16	d	1,7
Barranquilla	16	d	e,n,x
Nottingham	16	d	e,n,z ₁₅
Caen	16	d	1,w
Barmbek	16	d	z ₆
Malakal	16	e,h	1,2
Saboya	16	e,h	1,5
Rhydyfelin	16	e,h	e,n,x
Weston	16	e,h	z ₆
II	16	e,n,x	1,(5),7
II	16	e,n,x	1,6;z ₄₂
Tees	16	f,g	
Adeoyo	16	g,m,[t]	
Nikolaifleet	16	g,m,s	
II	16	g,[m],[s],t	[1,5]:[z ₄₂]
II	16	g,[m],[s],t	[e,n,x]
Cardoner	16	g,s,t	
II	16	m,t	e,n,x
II	16	m,t	[z ₄₂]
Mpouto	16	m,t	
Amina	16	i	1,5

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Agbara	16	i	1,6
Wisbech	16	i	1,7
Frankfurt	16	i	e,n,z ₁₅
Pisa	16	i	1,w
Abobo	16	i	z ₆
IIIb	16	i	z ₃₅
Szentes	16	k	1,2
Nuatja	16	k	e,n,x
Orientalis	16	k	e,n,z ₁₅
IIIb	16	k	z
IIIb	16	(k)	z ₃₅
IIIb	16	k	z ₅₃
IIIb	16	l,v	1,5,7
Shanghai	16	l,v	1,6
Welikade	16	l,v	1,7
Salford	16	l,v	e,n,x
Burgas	16	l,v	e,n,z ₁₅
IIIb	16	l,v	z:[z ₆₁]
Losangeles	16	l,v	z ₆
IIIb	16	l,v	z ₃₅
IIIb	16	l,v	z ₅₃
Zigong	16	l,w	1,5
Westeinde	16	l,w	1,6
Brooklyn	16	l,w	e,n,x
Lomnava	16	l,w	e,n,z ₁₅
II	16	l,w	z ₆
Mandera	16	l,z ₁₃	e,n,z ₁₅
Enugu	16	l,[z ₁₃],z ₂₈	[1,5]
Battle	16	l,z ₁₃ ,z ₂₈	1,6
Ablogame	16	l,z ₁₃ ,z ₂₈	z ₆
II	16	l,z ₂₈	z ₄₂
Rovaniemi	16	r,i	1,5
Ivory	16	r	1,6
Brunflo	16	r	1,7
Annedal	16	r,i	e,n,x
Zwickau	16	r,i	e,n,z ₁₅
Saphra	16	y	1,5
Akuafo	16	y	1,6
Kikoma	16	y	e,n,x
Avignon	16	y	e,n,z ₁₅
Gerland	16	z	1,5
Fortlamy	16	z	1,6
Lingwala	16	z	1,7
II	16	z	e,n,x
Brevik	16	z	e,n,[x],z ₁₅
Bouake	16	z	z ₆
II	16	z	z ₄₂
Kibi	16	z ₄ ,z ₂₃	[1,6]
II	16	z ₄ ,z ₂₃	
IV	16	z ₄ ,z ₂₃	
II	16	z ₄ ,z ₂₄	
IV	16	z ₄ ,z ₃₂	
II	16	z ₆	1,6
Badagry	16	z ₁₀	1,5
IIIb	16	z ₁₀	1,5,7
Lisboa	16	z ₁₀	1,6
IIIb	16	z ₁₀	e,n,x,z ₁₅
Redlands	16	z ₁₀	e,n,z ₁₅
Angouleme	16	z ₁₀	z ₆
Saloniki	16	z ₂₉	
II	16	z ₂₉	1,5
II	16	z ₂₉	e,n,x
Trier	16	z ₃₅	1,6
Dakota	16	z ₃₅	e,n,z ₁₅
II	16	z ₃₅	e,n,x
IV	16	z ₃₆	
II	16	z ₃₆	e,n,z ₁₅
Naware	16	z ₃₈	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Grancanaria	16	z ₃₉	[1,6]
II	16	z ₄₂	1,(5),7
IIIb	16	z ₅₂	z ₃₅
Group O:17 (J)			
Bonames	17	a	1,2
Jangwani	17	a	1,5
Kinondoni	17	a	e,n,x
Kirkee	17	b	1,2
Dahra	17	b	1,5
II	17	b	e,n,x,z ₁₅
Bignona	17	b	e,n,z ₁₅
II	17	b	z ₆
Luedinghausen	17	c	1,5
Victoriaborg	17	c	1,6
II	17	c	z ₃₉
Berlin	17	d	1,5
Karlshamn	17	d	e,n,z ₁₅
Niamey	17	d	1,w
Jubilee	17	e,h	1,2
II	17	e,n,x,z ₁₅	1,6
II	17	e,n,x,z ₁₅	1,[5],7
II	17	g,m,s,t	
Lowestoft	17	g,s,t	
II	17	g,t	[e,n,x,z ₁₅]
II	17	g,t	z ₃₉
Bama	17	m,t	
II	17	m,t	
Ahanou	17	i	1,7
IIIb	17	i	z ₃₅
Irenea	17	k	1,5
Warri	17	k	1,7
Matadi	17	k	e,n,x
Zaria	17	k	e,n,z ₁₅
IIIb	17	k	z
II	17	k	
Morotai	17	l,v	1,2
Michigan	17	l,v	1,5
Lancaster	17	l,v	1,7
Carmel	17	l,v	e,n,x
IIIb	17	l,v	e,n,x,z ₁₅
IIIb	17	l,v	z ₃₅
Granlo	17	l,z ₂₈	e,n,x
Lode	17	r	1,2
IIIb	17	r	z
II	17	y	
Tendeba	17	y	e,n,x
Hadejia	17	y	e,n,z ₁₅
Gori	17	z	1,2
Warengo	17	z	1,5
II	17	z	1,7
Tchamba	17	z	e,n,z ₁₅
II	17	z	l,w,z ₄₂
IIIa	17	z ₄ ,z ₂₃	
IIIa	17	z ₄ ,z ₂₃ ,z ₃₂	
IIIa	17	z ₄ ,z ₂₄	
IIIa	17	z ₄ ,z ₃₂	
Djibouti	17	z ₁₀	e,n,x
IIIb	17	z ₁₀	e,n,x,z ₁₅
IIIb	17	z ₁₀	z
II	17	z ₁₀	
Kandla	17	z ₂₉	
IIIa	17	z ₂₉	
IV	17	z ₂₉	
Aachen	17	z ₃₅	1,6
IIIa	17	z ₃₆	
IV	17	z ₃₆	
Group O:18 (K)			
Brazos	6, 14,18	a	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Fluntern	6, 14,18	b	1,5
Rawash	6, 14,18	c	e,n,x
Groenekan	18	d	1,5
Usumbura	18	d	1,7
Pontypridd	18	g,m	
IIIa	18	g,z ₅₁	
II	18	m,t	1,5
Langenhorn	18	m,t	
Memphis	18	k	1,5
IIIb	18	(k)	z ₅₃
IIIb	18	(k)	z ₅₄
IIIb	18	l,v	e,n,x,z ₁₅
Orlando	18	l,v	e,n,z ₁₅
IIIb	18	l,v	z
IIIb	18	l,v	z ₅₃
Toulon	18	l,w	e,n,z ₁₅
Tennenlohe	18	r	1,5
IIIb	18	r	z
II	18	y	e,n,x,z ₁₅
Potengi	18	z	
Cerro	6, 14,18	z ₄ ,z ₂₃	[1,5]
Aarhus	18	z ₄ ,z ₂₃	z ₆₄
II	18	z ₄ ,z ₂₃	
IIIa	18	z ₄ ,z ₂₃	
Blukwa	6, 14,18	z ₄ ,z ₂₄	
II	18	z ₄ ,z ₂₄	
IIIa	18	z ₄ ,z ₃₂	
IIIb	18	z ₁₀	e,n,x,z ₁₅
Leer	18	z ₁₀	1,5
Carnac	18	z ₁₀	z ₆
II	18	z ₁₀	z ₆
II	18	z ₃₆	
IV	18	z ₃₆ ,z ₃₈	
Sinthia	18	z ₃₈	
Delmenhorst	18	z ₇₁	
Cotia	18		1,6
Group O:21 (L)			
Assen	21	a	[1,5]
II	21	b	1,5
Ghana	21	b	1,6
Minnesota	21	b	e,n,x
Hydra	21	c	1,6
Rhone	21	c	e,n,x
II	21	c	e,n,x
IIIb	21	c	e,n,x,z ₁₅
Spartel	21	d	1,5
Magwa	21	d	e,n,x
Madison	21	d	z ₆
Good	21	f,g	e,n,x
II	21	g,[m],[s],t	
IIIa	21	g,z ₅₁	
IV	21	g,z ₅₁	
II	21	m,t	
Diourbel	21	i	1,2
IIIb	21	i	1,5,7
IIIb	21	i	e,n,x,z ₁₅
IIIb	21	k	e,n,x,z ₁₅
IIIb	21	k	z
Surrey	21	k	1,2,5
IIIb	21	l,v	z
IIIb	21	l,v	z ₅₇
Keve	21	l,w	
Jambur	21	l,z ₂₈	e,n,z ₁₅
Mountmagnet	21	r	
IIIb	21	r	z
Ibaragi	21	y	1,2
Ruiru	21	y	e,n,x
II	21	z	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Baguida	21	Z ₄ ,Z ₂₃	
IIIa	21	Z ₄ ,Z ₂₃	
IV	21	Z ₄ ,Z ₂₃	
II	21	Z ₄ ,Z ₂₄	
IIIa	21	Z ₄ ,Z ₂₄	
IV	21	Z ₄ ,Z ₃₂	
IIIb	21	Z ₁₀	e,n,x,z ₁₅
IIIb	21	Z ₁₀	z
II	21	Z ₁₀	[Z ₆]
IIIb	21	Z ₁₀	Z ₅₃
IIIa	21	Z ₂₉	
Gambaga	21	Z ₃₅	e,n,z ₁₅
IV	21	Z ₃₆	
IIIb	21	Z ₆₅	e,n,x,z ₁₅
Group O:28 (M)			
Solna	28	a	1,5
Dakar	28	a	1,6
Bakau	28	a	1,7
Seattle	28	a	e,n,x
II	28	a	e,n,x
Honelis	28	a	e,n,z ₁₅
Dibra	28	a	Z ₆
Moero	28	b	1,5
Ashanti	28	b	1,6
Bokanjac	28	b	1,7
Soumbedioune	28	b	e,n,x
II	28	b	e,n,x
Langford	28	b	e,n,z ₁₅
Freefalls	28	b	1,w
II	28	b	Z ₆
Hermannswerder	28	c	1,5
Eberswalde	28	c	1,6
Halle	28	c	1,7
Dresden	28	c	e,n,x
Wedding	28	c	e,n,z ₁₅
Techimani	28	c	Z ₆
Amoutive	28	d	1,5
Hatfield	28	d	1,6
Mundonobo	28	d	1,7
Mocamedes	28	d	e,n,x
Patience	28	d	e,n,z ₁₅
Cullingworth	28	d	1,w
Kpeme	28	e,h	1,7
Gozo	28	e,h	e,n,z ₁₅
II	28	e,n,x	1,7
Friedrichsfelde	28	f,g	
Yardley	28	g,m	1,6
Abadina	28	g,m	[e,n,z ₁₅]
II	28	g,(m),[s],t	1,5
Croft	28	g,m,s	[e,n,z ₁₅]
II	28	g,m,t	e,n,x
II	28	g,m,t	Z ₃₉
II	28	g,s,t	e,n,x
Ona	28	g,s,t	
II	28	m,t	[e,n,x]
Vinohrady	28	m,t	[e,n,z ₁₅]
Morillons	28	m,t	1,6
Doorn	28	i	1,2
Cotham	28	i	1,5
Volkmarisdorf	28	i	1,6
Dieuppeul	28	i	1,7
Warnemuende	28	i	e,n,x
Kuessel	28	i	e,n,z ₁₅
Douala	28	i	1,w
Guildford	28	k	1,2
Ilala	28	k	1,5
Adamstown	28	k	1,6
Ikeja	28	k	1,7

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Taunton	28	k	e,n,x
Ank	28	k	e,n,z ₁₅
Leoben	28	1,v	1,5
Vitkin	28	1,v	e,n,x
Nashua	28	1,v	e,n,z ₁₅
Ramsey	28	1,w	1,6
Catalunia	28	1,z ₁₃ ,Z ₂₈	1,5
Penilla	28	1,z ₁₃ ,Z ₂₈	e,n,z ₁₅
II	28	1,z ₂₈	1,5
Fajara	28	1,z ₂₈	e,n,x
Bassadji	28	r	1,6
Kibusi	28	r	e,n,x
II	28	r	e,n,z ₁₅
Fairfield	28	r	1,w
Chicago	28	r,[i]	1,5
Banco	28	r,i	1,7
Sanktgeorg	28	r,[i]	e,n,z ₁₅
Oskarshamn	28	y	1,2
Nima	28	y	1,5
Pomona	28	y	1,7:[z ₆₀]
Kitenge	28	y	e,n,x
Telaviv	28	y	e,n,z ₁₅
Shomolu	28	y	1,w
Selby	28	y	Z ₆
Vanier	28	z	1,5
II	28	z	1,5
Doel	28	z	1,6
Ezra	28	z	1,7
Brisbane	28	z	e,n,z ₁₅
II	28	z	Z ₃₉
Cannobio	28	Z ₄ ,Z ₂₃	1,5
Teltow	28	Z ₄ ,Z ₂₃	1,6
Babelsberg	28	Z ₄ ,Z ₂₃	[e,n,z ₁₅]
Rogy	28	Z ₁₀	1,2
Farakan	28	Z ₁₀	1,5
Libreville	28	Z ₁₀	1,6
Malaysia	28	Z ₁₀	1,7
Umbilo	28	Z ₁₀	e,n,x
Luckenwalde	28	Z ₁₀	e,n,z ₁₅
Moroto	28	Z ₁₀	1,w
IIIb	28	Z ₁₀	z
Djermaia	28	Z ₂₉	
II	28	Z ₂₉	1,5
II	28	Z ₂₉	e,n,x
Konolfingen	28	Z ₃₅	1,6
Balili	28	Z ₃₅	1,7
Santander	28	Z ₃₅	e,n,z ₁₅
Aderike	28	Z ₃₈	e,n,z ₁₅
Group O:30 (N)			
Overvecht	30	a	1,2
Zehlendorf	30	a	1,5
Guarapiranga	30	a	e,n,x
Doulassame	30	a	e,n,z ₁₅
II	30	a	Z ₃₉
Louga	30	b	1,2
Aschersleben	30	b	1,5
Urbana	30	b	e,n,x
Neudorf	30	b	e,n,z ₁₅
II	30	b	Z ₆
Zaire	30	c	1,7
Morningside	30	c	e,n,z ₁₅
II	30	c	Z ₃₉
Messina	30	d	1,5
Livulu	30	e,h	1,2
Torhout	30	e,h	1,5
Godesberg	30	g,m,[t]	
II	30	g,m,s	e,n,x
Giessen	30	g,m,s	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Sternschanze	30	g,s,t	
II	30	g,t	
Wayne	30	g,z ₅₁	
II	30	m,t	
Landau	30	i	1,2
Morehead	30	i	1,5
Mjordan	30	i	e,n,z ₁₅
Soerenga	30	i	l,w
Hilversum	30	k	1,2
Ramatgan	30	k	1,5
Aqua	30	k	1,6
Angoda	30	k	e,n,x
Odozi	30	k	e,n,[x],z ₁₅
II	30	k	e,n,x,z ₁₅
Ligeo	30	l,v	1,2
Donna	30	l,v	1,5
Ockenheim	30	l,z ₁₃ ,z ₂₈	1,6
Morocco	30	l,z ₁₃ ,z ₂₈	e,n,z ₁₅
II	30	l,z ₂₈	z ₆
Grandhaven	30	r	1,2
Gege	30	r	1,5
Matopeni	30	y	1,2
Bietri	30	y	1,5
Steinplatz	30	y	1,6
Baguirmi	30	y	e,n,x
Nijmegen	30	y	e,n,z ₁₅
Stoneferry	30	z ₄ ,z ₂₃	
Bodjonegoro	30	z ₄ ,z ₂₄	
II	30	z ₆	1,6
Sada	30	z ₁₀	1,2
Senneville	30	z ₁₀	1,5
Kumasi	30	z ₁₀	e,n,z ₁₅
II	30	z ₁₀	e,n,x,z ₁₅
Aragua	30	z ₂₉	
Kokoli	30	z ₃₅	1,6
Wuiti	30	z ₃₅	e,n,z ₁₅
Ago	30	z ₃₈	
II	30	z ₃₉	1,7
Group O:35 (O)			
Umlhatazana	35	a	e,n,z ₁₅
Tchad	35	b	
Gouloumbo	35	c	1,5
Yolo	35	c	[e,n,z ₁₅]
II	35	d	1,5
Dembe	35	d	l,w
Gassi	35	e,h	z ₆
Adelaide	35	f,g	
Ealing	35	g,m,s	
II	35	g,m,s,t	
Ebrie	35	g,m,t	
Anecho	35	g,s,t	
II	35	g,t	1,5
II	35	g,t	z ₄₂
Agodi	35	g,t	
IIIa	35	g,z ₅₁	
Monschau	35	m,t	
II	35	m,t	
IIIb	35	i	e,n,x,z ₁₅
Gambia	35	i	e,n,z ₁₅
Bandia	35	i	l,w
IIIb	35	i	z
IIIb	35	i	z ₃₅
IIIb	35	i	z ₅₃
IIIb	35	k	e,n,x,z ₁₅
IIIb	35	k	z
IIIb	35	(k)	z
IIIb	35	(k)	z ₃₅
IIIb	35	(k)	z ₅₄
IIIb	35	(k)	z ₅₄
IIIb	35	(k)	1,2
IIIb	35	(k)	1,5
IIIb	35	(k)	1,5,7
IIIb	35	(k)	1,6
IIIb	35	(k)	e,n,x,z ₁₅
IIIb	35	(k)	z:[z ₅₇]
IIIb	35	k	z ₃₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIb	35	l,v	1,5,7
IIIb	35	l,v	z ₃₅ :[z ₆₇]
II	35	l,z ₂₈	
IIIb	35	r	e,n,x,z ₁₅
Massakory	35	r	l,w
IIIb	35	r	z
IIIb	35	r	z ₃₅
IIIb	35	r	z ₆₁
Alachua	35	z ₄ ,z ₂₃	
IIIa	35	z ₄ ,z ₂₃	
Westphalia	35	z ₄ ,z ₂₄	
IIIa	35	z ₄ ,z ₃₂	
Camberene	35	z ₁₀	1,5
Enschede	35	z ₁₀	l,w
Ligna	35	z ₁₀	z ₆
IIIb	35	z ₁₀	z ₃₅
II	35	z ₂₉	e,n,x
Widemarsh	35	z ₂₉	
IIIa	35	z ₂₉	
IIIa	35	z ₃₆	
Haga	35	z ₃₈	
IIIb	35	z ₅₂	1,5,7
IIIb	35	z ₅₂	e,n,x,z ₁₅
IIIb	35	z ₅₂	z
IIIb	35	z ₅₂	z ₃₅
Group O:38 (P)			
Oran	38	a	e,n,z ₁₅
II	38	b	1,2
Rittersbach	38	b	e,n,z ₁₅
Sheffield	38	c	1,5
Kidderminster	38	c	1,6
II	38	d	[1,5]
Thiaroye	38	e,h	1,2
Kasenyi	38	e,h	1,5
Korovi	38	g,m,[s]	
II	38	g,t	
IIIa	38	g,z ₅₁	
IV	38	g,z ₅₁	
Rothenburgsort	38	m,t	
Mgulani	38	i	1,2
Lansing	38	i	1,5
IIIb	38	i	z
IIIb	38	i	z ₅₃
Echa	38	k	1,2
Mango	38	k	1,5
Inverness	38	k	1,6
Njala	38	k	e,n,x
IIIb	38	k	e,n,x,z ₁₅
IIIb	38	k	z
IIIb	38	k	z ₅₃
IIIb	38	(k)	1,5,7
IIIb	38	(k)	z
IIIb	38	(k)	z ₃₅
IIIb	38	(k)	z ₅₄
IIIb	38	(k)	z ₅₄
Alger	38	l,v	1,2
Kimberley	38	l,v	1,5
Taylor	38	l,v	e,n,z ₁₅
Roan	38	l,v	e,n,x
IIIb	38	l,v	z
IIIb	38	l,v	z ₃₅
IIIb	38	l,v	z ₅₃ :[z ₅₄]
Lindi	38	r	1,5
IIIb	38	r	1,5,7
Emmastad	38	r	1,6
IIIb	38	r	e,n,x,z ₁₅
IIIb	38	r	z:[z ₅₇]
IIIb	38	r	z ₃₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Freetown	38	y	1,5
Colombo	38	y	1,6
Perth	38	y	e,n,x
Stachus	38	z	
Yoff	38	z ₄ ,z ₂₃	1,2
IIIa	38	z ₄ ,z ₂₃	
IV	38	z ₄ ,z ₂₃	
Bangkok	38	z ₄ ,z ₂₄	
Neunkirchen	38	z ₁₀	
IIIb	38	z ₁₀	z
IIIb	38	z ₁₀	z ₅₃
Klouto	38	z ₃₈	
IIIb	38	z ₅₂	z ₃₅
IIIb	38	z ₅₂	z ₅₃
IIIb	38	z ₅₃	
IIIb	38	z ₆₁	[z ₅₃]
Group O:39 (Q)			
II	39	a	z ₃₉
Wandsworth	39	b	1,2
Abidjan	39	b	1,w
II	39	c	e,n,x
Logone	39	d	1,5
Mara	39	e,h	1,5
II	39	e,n,x	1,7
II	39	[g],m,t	[e,n,x]
Hofit	39	i	1,5
Cumberland	39	i	e,n,x
Alma	39	i	e,n,z ₁₅
Champaign	39	k	1,5
II	39	1,v	1,5
Kokomlemle	39	1,v	e,n,x
Oerlikon	39	1,v	e,n,z ₁₅
II	39	1,z ₂₈	e,n,x
II	39	1,z ₂₈	z ₃₉
Anfo	39	y	1,2
Windermere	39	y	1,5
Hegau	39	z ₁₀	
II	39		1,7
Group O:40 (R)			
Shikmonah	40	a	1,5
Greiz	40	a	z ₆
II	1,40	a	z ₆
II	40	a	z ₃₉
Rio grande	40	b	1,5
Saugus	40	b	1,7
Johannesburg	1,40	b	e,n,x
Duval	1,40	b	e,n,z ₁₅
Benguella	40	b	z ₆
II	40	b	
II	1,40	c	e,n,x,z ₁₅
II	1,40	c	z ₃₉
Driffield	1,40	d	1,5
II	40	d	
Tilene	1,40	e,h	1,2
II	1,40	e,n,x	1,[5],7
II	1,40	e,n,x,z ₁₅	1,6
Bijlmer	1,40	g,m	
II	1,40	g,[m],[s],[t]	e,n,x
II	1,40	g,[m],[s],t	1,5
II	1,40	g,t	e,n,x,z ₁₅
II	40	g,t	z ₃₉
IV	1,40	g,t	
II	1,40	g,[m],[s],t	z ₄₂
IIIa	40	g,z ₅₁	
IIIb	40	g,z ₅₁	e,n,x,z ₁₅
IV	1,40	g,z ₅₁	
II	40	m,t	z ₃₉
II	1,40	m,t	z ₄₂

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IV	40	m,t	
IIIb	40	i	1,5,7
Goulfey	1,40	k	1,5
Allandale	1,40	k	1,6
Hann	40	k	e,n,x
II	1,40	k	e,n,x,z ₁₅
IIIb	40	k	z:z ₅₇
II	40	k	z ₆
IIIb	40	k	z ₅₃
Millesi	1,40	1,v	1,2
Canary	40	1,v	1,6
II	40	1,v	e,n,x
IIIb	40	1,v	z
IIIb	40	1,v	z ₅₃
Overchurch	1,40	1,w	[1,2]
Tiko	1,40	1,z ₁₃ ,z ₂₈	1,2
Bukavu	1,40	1,z ₂₈	1,5
II	1,40	1,z ₂₈	1,5:z ₄₂
Santhiaba	40	1,z ₂₈	1,6
II	1,40	1,z ₂₈	z ₃₉
Odienné	40	y	1,5
II	1,40	z	1,5
Casamance	40	z	e,n,x
Nowawes	40	z	z ₆
II	1,40	z	z ₆
II	1,40	z	z ₃₉
II	40	z	z ₄₂
IIIa	40	z ₄ ,z ₂₃	
IV	1,40	z ₄ ,z ₂₃	
II	40	z ₄ ,z ₂₄	z ₃₉
IIIa	40	z ₄ ,z ₂₄	
IV	40	z ₄ ,z ₂₄	
IIIa	40	z ₄ ,z ₃₂	
IV	40	z ₄ ,z ₃₂	
II	1,40	z ₆	1,5
Trotha	40	z ₁₀	z ₆
IIIb	40	z ₁₀	z ₃₅
Omifisan	1,40	z ₂₉	
IIIa	40	z ₂₉	
II	1,40	z ₃₅	e,n,x,z ₁₅
Yekepa	1,40	z ₃₅	e,n,z ₁₅
V	1,40	z ₃₅	
IIIa	40	z ₃₆	
II	1,40	z ₃₉	1,5:z ₄₂
II	1,40	z ₃₉	1,6
IIIb	40	z ₃₉	1,6
II	40	z ₃₉	1,7
Karamoja	1,40	z ₄₁	1,2
II	1,40	z ₄₂	1,6
II	1,40	[z ₄₂]	1,(5),7
V	1,40	z ₈₁	
Group O:41 (S)			
Burundi	41	a	
II	41	b	1,5
Vaugirard	41	b	1,6
VI	41	b	1,7
Vietnam	41	b	z ₆
Sica	41	b	e,n,z ₁₅
IIIb	41	c	e,n,x,z ₁₅
II	41	c	z ₆
Egusi	41	d	1,5
II	41	d	z ₆
II	41	g,m,s,t	z ₆
II	41	g,t	
IIIa	41	g,z ₅₁	
Leatherhead	41	m,t	1,6
Samaru	41	i	1,5
Verona	41	i	1,6

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Ferlo	41	k	1,6
II	41	k	1,6
II	41	k	z ₆
IIIb	41	(k)	z ₃₅
II	41	1,z ₁₃ ,z ₂₈	e,n,x,z ₁₅
Lubumbashi	41	r	1,5
II	41	z	1,5
Bofflens	41	z ₄ ,z ₂₃	1,7
Waycross	41	z ₄ ,z ₂₃	[e,n,z ₁₅]
IIIa	41	z ₄ ,z ₂₃	
IV	41	z ₄ ,z ₂₃	
IIIa	41	z ₄ ,z ₂₃ ,z ₃₂	
Ipswich	41	z ₄ ,z ₂₄	1,5
IIIa	41	z ₄ ,z ₂₄	
IIIa	41	z ₄ ,z ₃₂	
II	41	z ₁₀	1,2
Leipzig	41	z ₁₀	1,5
Landala	41	z ₁₀	1,6
Inpraw	41	z ₁₀	e,n,x
II	41	z ₁₀	e,n,x,z ₁₅
II	41	z ₁₀	z ₆
Lodz	41	z ₂₉	
IIIa	41	z ₂₉	
IV	41	z ₂₉	
Ahoutoue	41	z ₃₅	1,6
IIIa	41	z ₃₆	
Offa	41	z ₃₈	
IV	41	z ₅₂	
II	41		1,6
Group O:42 (T)			
Faji	1,42	a	e,n,z ₁₅
II	42	b	1,5
Orbe	42	b	1,6
II	42	b	e,n,x,z ₁₅
Tomegbe	1,42	b	e,n,z ₁₅
Egusitoo	1,42	b	z ₆
II	42	b	z ₆
Antwerpen	1,42	c	e,n,z ₁₅
Kampala	1,42	c	z ₆
II	42	d	z ₆
II	42	e,n,x	1,6
II	42	g,t	
Maricopa	1,42	g,z ₅₁	1,5
IIIa	42	g,z ₅₁	
IV	1,42	g,z ₅₁	
II	42	m,t	[e,n,x,z ₁₅]
Waral	1,42	m,t	
Kaneshie	1,42	i	1,w
Borromea	42	i	1,6
Middlesbrough	1,42	i	z ₆
Haferbreite	42	k	1,6
IIIb	42	k	e,n,x,z ₁₅
IIIb	42	k	z
Gwale	1,42	k	z ₆
IIIb	42	(k)	z ₃₅
IIIb	42	1,v	1,5,7
II	42	1,v	e,n,x,z ₁₅
IIIb	42	1,v	e,n,x,z ₁₅
Coogee	42	1,v	e,n,z ₁₅
IIIb	42	1,v	z
IIIb	42	1,v	z ₅₃
II	1,42	1,w	e,n,x
II	1,42	1,[z ₁₃],z ₂₈	z ₆
Sipane	1,42	r	e,n,z ₁₅
Brive	1,42	r	1,w
IIIb	42	r	z
IIIb	42	r	z ₅₃
II	42	r	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIa	42	r	
Spalentor	1,42	y	e,n,z ₁₅
Harvestehude	1,42	y	z ₆
II	42	z	1,5
Ursenbach	1,42	z	1,6
II	42	z	e,n,x,z ₁₅
Melbourne	42	z	e,n,z ₁₅
II	42	z	z ₆
Gera	1,42	z ₄ ,z ₂₃	1,6
Broc	42	z ₄ ,z ₂₃	e,n,z ₁₅
IIIa	42	z ₄ ,z ₂₃	
Toricada	1,42	z ₄ ,z ₂₄	
IIIa	42	z ₄ ,z ₂₄	
IV	1,42	z ₄ ,z ₂₄	
II	42	z ₆	1,6
II	42	z ₁₀	1,2
II	42	z ₁₀	e,n,x,z ₁₅
IIIb	42	z ₁₀	e,n,x,z ₁₅
IIIb	42	z ₁₀	z
Loenga	1,42	z ₁₀	z ₆
II	42	z ₁₀	z ₆
IIIb	42	z ₁₀	z ₃₅
IIIb	42	z ₁₀	z ₆₇
Djama	1,42	z ₂₉	[1,5]
Kahla	1,42	z ₃₅	1,6
Hennekamp	42	z ₃₅	e,n,z ₁₅
Tema	1,42	z ₃₅	z ₆
Weslaco	42	z ₃₆	
IV	42	z ₃₆	
Vogan	1,42	z ₃₈	z ₆
Taset	1,42	z ₄₁	
IIIb	42	z ₅₂	z
Group O:43 (U)			
Graz	43	a	1,2
Berkeley	43	a	1,5
II	43	a	1,5
II	43	a	z ₆
Niederoderwitz	43	b	
II	43	b	z ₄₂
Montreal	43	c	1,5
Orleans	43	d	1,5
II	43	d	e,n,x,z ₁₅
II	43	d	z ₃₉
II	43	d	z ₄₂
II	43	e,n,x,z ₁₅	1,(5),7
II	43	e,n,x,z ₁₅	1,6
Milwaukee	43	f,g,[t]	
II	43	g,m,[s],t	[z ₄₂]
II	43	g,t	[1,5]
IIIa	43	g,z ₅₁	
IV	43	g,z ₅₁	
II	43	g,z ₆₂	e,n,x
Mbao	43	i	1,2
Voulte	43	i	e,n,x
Thetford	43	k	1,2
Ahuza	43	k	1,5
IIIb	43	k	z
IIIb	43	1,v	z ₅₃
Sudan	43	1,z ₁₃	
II	43	1,z ₁₃ ,z ₂₈	1,5
IIIb	43	r	e,n,x,z ₁₅
IIIb	43	r	z
IIIb	43	r	z ₅₃
Farcha	43	y	1,2
Kingabwa	43	y	1,5
Ogbete	43	z	1,5
II	43	z	1,5
Arusha	43	z	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	43	Z ₄ ,Z ₂₃	
IIIa	43	Z ₄ ,Z ₂₃	
IV	43	Z ₄ ,Z ₂₃	
IIIa	43	Z ₄ ,Z ₂₄	
IV	43	Z ₄ ,Z ₂₄	
IV	43	Z ₄ ,Z ₃₂	
Adana	43	Z ₁₀	1,5
II	43	Z ₂₉	e,n,x
II	43	Z ₂₉	Z ₄₂
Makiling	43	Z ₂₉	
IV	43	Z ₂₉	
Ahepe	43	Z ₃₅	1,6
IIIa	43	Z ₃₆	
IV	43	Z ₃₆ ,Z ₃₈	
Irigny	43	Z ₃₈	
II	43	Z ₄₂	1,5,7
IIIb	43	Z ₅₂	Z ₅₃
Group O:44 (V)			
IV	44	a	
Niakhar	44	a	1,5
Tiergarten	44	a	e,n,x
Niarembe	44	a	1,w
Sedgwick	44	b	e,n,z ₁₅
Madigan	44	c	1,5
Quebec	44	c	e,n,z ₁₅
Bobo	44	d	1,5
Kermel	44	d	e,n,x
Fischerstrasse	44	d	e,n,z ₁₅
Palamaner	1,44	d	Z ₃₅
II	1,44	e,n,x	1,6
Vleuten	44	f,g	
Gamaba	1,44	g,m,[s]	
Splott	44	g,s,t	
II	44	g,t	Z ₄₂
IIIb	44	g,t	1,5;Z ₄₂
Carswell	44	g,Z ₅₁	
IV	44	g,Z ₅₁	-
Muguga	44	m,t	
Maritzburg	1,44	i	e,n,z ₁₅
Lawra	44	k	e,n,z ₁₅
Malika	44	l,Z ₂₈	1,5
Brefet	44	r	e,n,z ₁₅
V	44	r	
Uhlenhorst	44	z	1,w
Bolama	44	z	e,n,x
Kua	44	Z ₄ ,Z ₂₃	
Ploufragan	1,44	Z ₄ ,Z ₂₃	e,n,z ₁₅
II	44	Z ₄ ,Z ₂₃	
IIIa	44	Z ₄ ,Z ₂₃	
IV	44	Z ₄ ,Z ₂₃	
IIIa	44	Z ₄ ,Z ₂₃ ,Z ₃₂	
Christiansborg	44	Z ₄ ,Z ₂₄	
IIIa	44	Z ₄ ,Z ₂₄	
IV	44	Z ₄ ,Z ₂₄	
IIIa	44	Z ₄ ,Z ₃₂	
IV	1,44	Z ₄ ,Z ₃₂	
Guinea	1,44	Z ₁₀	1,7
Llobregat	44	Z ₁₀	e,n,x
II	44	Z ₂₉	e,n,x;Z ₄₂
Zinder	44	Z ₂₉	
IV	44	Z ₂₉	
IV	44	Z ₂₉	
Koketime	44	Z ₃₆ ,[Z ₃₈]	
II	1,44	Z ₃₈	
V	44	Z ₃₉	e,n,x,z ₁₅
V	44	Z ₃₉	
Group O:45 (W)			
VI	45	a	e,n,x
Meekatharra	45	a	e,n,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	45	a	Z ₁₀
Riverside	45	b	1,5
Fomeco	45	b	e,n,z ₁₅
Deversoir	45	c	e,n,x
Dugbe	45	d	1,6
Karachi	45	d	e,n,x
Warmen	45	d	e,n,z ₁₅
Sueldorf	45	f,g	
Tornow	45	g,m,[s],[t]	
II	45	g,m,s,t	1,5
II	45	g,m,s,t	e,n,x
II	45	g,m,t	e,n,x,z ₁₅
Binningen	45	g,s,t	
IIIa	45	g,Z ₅₁	
IV	45	g,Z ₅₁	
II	45	m,t	1,5
Apapa	45	m,t	
Verviers	45	k	1,5
Casablanca	45	k	1,7
Cairns	45	k	e,n,z ₁₅
Imo	45	l,v	[e,n,z ₁₅]
Kofandoka	45	r	e,n,z ₁₅
II	45	z	1,5
Yopougon	45	z	e,n,z ₁₅
II	45	z	Z ₃₉
IIIa	45	Z ₄ ,Z ₂₃	
IV	45	Z ₄ ,Z ₂₃	
Transvaal	45	Z ₄ ,Z ₂₄	
IIIa	45	Z ₄ ,Z ₂₄	
IIIa	45	Z ₄ ,Z ₃₂	
Aprad	45	Z ₁₀	
Jodhpur	45	Z ₂₉	
II	45	Z ₂₉	1,5
II	45	Z ₂₉	e,n,x
II	45	Z ₂₉	Z ₄₂
IIIa	45	Z ₂₉	
Lattenkamp	45	Z ₃₅	1,5
Balcones	45	Z ₃₆	
IV	45	Z ₃₆ ,Z ₃	
Group O:47 (X)			
II	47	a	1,5
II	47	a	e,n,x,z ₁₅
Wenatchee	47	b	1,2
II	47	b	1,5
II	47	b	e,n,x,z ₁₅
Sya	47	b	Z ₆
II	47	b	Z ₆
IIIb	47	c	1,5,7
Kodjovi	47	c	1,6
IIIb	47	c	e,n,x,z ₁₅ ;[Z ₅₇]
IIIb	47	c	z
IIIb	47	c	Z ₃₅
Stellingen	47	d	e,n,x
II	47	d	Z ₃₉
II	47	e,n,x,z ₁₅	1,6
Sljeme	1,47	f,g	
Luke	1,47	g,m	
II	47	g,t	e,n,x
IIIa	47	g,Z ₅₁	
Mesbit	47	m,t	e,n,z ₁₅
IIIb	47	i	e,n,x,z ₁₅
Bergen	47	i	e,n,z ₁₅
IIIb	47	i	z
IIIb	47	i	Z ₃₅
IIIb	47	i	Z ₅₃ ;[Z ₅₇]
Staoueli	47	k	1,2
Bootle	47	k	1,5
IIIb	47	k	1,5,7

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Dahomey	47	k	1,6
IIIb	47	k	e,n,x,z ₁₅
Lyon	47	k	e,n,z ₁₅
IIIb	47	k	z
IIIb	47	k	z ₃₅
IIIb	47	k	z ₅₃
IIIb	47	l,v	1,5,(7)
IIIb	47	l,v	e,n,x,z ₁₅
IIIb	47	l,v	z
IIIb	47	l,v	z ₃₅
IIIb	47	l,v	z ₅₃
IIIb	47	l,v	z ₅₇
IV	47	l,v	
Teshie	1,47	l,z ₁₃ ,z ₂₈	e,n,z ₁₅
IIIb	47	r	e,n,x,z ₁₅
Dapango	47	r	1,2
IIIb	47	r	1,5,7
IIIb	47	r	z
IIIb	47	r	z ₃₅
IIIb	47	r	z ₅₃ :[z ₆₀]
IIIa	47	r	
Moualine	47	y	1,6
Blitta	47	y	e,n,x
Mountpleasant	47	z	1,5
Kaolack	47	z	1,6
II	47	z	e,n,x,z ₁₅
II	47	z	z ₆
Tabligbo	47	z ₄ ,z ₂₃	e,n,z ₁₅
Binche	47	z ₄ ,z ₂₃	l,w
Bere	47	z ₄ ,z ₂₃	z ₆
IIIa	47	z ₄ ,z ₂₃	
Tamberma	47	z ₄ ,z ₂₄	
II	47	z ₆	1,6
IIIb	47	z ₁₀	1,5,7
Namoda	47	z ₁₀	e,n,z ₁₅
IIIb	47	z ₁₀	z
IIIb	47	z ₁₀	z ₃₅
II	47	z ₂₉	e,n,x,z ₁₅
Ekpoui	47	z ₂₉	
IIIa	47	z ₂₉	
Bingerville	47	z ₃₅	e,n,z ₁₅
IV	47	z ₃₆	
Alexanderplatz	47	z ₃₈	
Quinhon	47	z ₄₄	
IIIb	47	z ₅₂	1,5,7
IIIb	47	z ₅₂	e,n,x,z ₁₅
IIIb	47	z ₅₂	z
IIIb	47	z ₅₂	z ₃₅
Group O:48 (Y)			
Hisingen	48	a	1,5,7
II	48	a	z ₆
II	48	a	z ₃₉
II	48	b	z ₆
V	48	b	
IIIb	48	c	z
II	48	d	1,2
II	48	d	z ₆
Buckeye	48	d	
Fitzroy	48	e,h	1,5
II	48	e,n,x,z ₁₅	z ₆
II	48	g,m,t	
IIIa	48	g,z ₅₁	
IV	48	g,z ₅₁	
IIIb	48	i	z
IIIb	48	i	z ₃₅ :[z ₅₇]
IIIb	48	i	z ₅₃
IIIb	48	i	z ₆₁
V	48	i	

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIb	48	k	1,5,(7)
II	48	k	e,n,x,z ₁₅
IIIb	48	k	e,n,x,z ₁₅
Dahlem	48	k	e,n,z ₁₅
IIIb	48	k	z
IIIb	48	k	z ₃₅
II	48	k	z ₃₉
IIIb	48	k	z ₅₃
IIIb	48	(k)	z ₅₃
Australia	48	l,v	1,5
IIIb	48	l,v	1,5,(7)
IIIb	48	l,v	z
IIIb	48	r	e,n,x,z ₁₅
IIIb	48	r	z
Toucra	48	z	1,5
II	48	z	1,5
IIIb	48	z	1,5,7
IIIa	48	z ₄ ,z ₂₃	
IV	48	z ₄ ,z ₂₃	
IIIa	48	z ₄ ,z ₂₃ ,z ₃₂	
Djakarta	48	z ₄ ,z ₂₄	
IIIa	48	z ₄ ,z ₂₄	
IIIa	48	z ₄ ,z ₃₂	
IV	48	z ₄ ,z ₃₂	
II	48	z ₁₀	[1,5]
VI	48	z ₁₀	1,5
Isaszeg	48	z ₁₀	e,n,x
IIIb	48	z ₁₀	e,n,x,z ₁₅
IIIb	48	z ₁₀	z
II	48	z ₂₉	
IV	48	z ₂₉	
IIIb	48	z ₃₅	z ₅₂
V	48	z ₃₅	
IIIa	48	z ₃₆	
IV	48	z ₃₆ ,[z ₃₈]	
V	48	z ₃₉	
V	48	z ₄₁	
IIIb	48	z ₅₂	e,n,x,z ₁₅
IIIb	48	z ₅₂	z
V	48	z ₆₅	
V	48	z ₈₁	
Group O:50 (Z)			
IV	50	a	
Rochdale	50	b	e,n,x
II	50	b	z ₆
IV	50	b	
Hemingford	50	d	1,5
IV	50	d	
II	50	e,n,x	1,7
II	50	g,[m],s,t	[1,5]
IV	50	g,z ₅₁	
II	50	g,z ₆₂	e,n,x
II	50	m,t	z ₆ ,z ₄₂
IIIb	50	i	1,5,7
IIIb	50	i	e,n,x,z ₁₅
IIIb	50	i	z
IIIb	50	k	1,5,7
II	50	k	e,n,x,z ₄₂
IIIb	50	k	e,n,x,z ₁₅
IIIb	50	k	z
IIIb	50	(k)	z
II	50	k	z ₆
IIIb	50	k	z ₃₅
IIIb	50	(k)	z ₃₅
IIIb	50	k	z ₅₃
Fass	50	l,v	1,2
IIIb	50	l,v	e,n,x,z ₁₅
IIIb	50	l,v	z

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIb	50	l,v	z ₃₅
VI	50	l,v	z ₆₇
II	50	l,w	e,n,x,z ₁₅ ;z ₄₂
II	50	l,z ₂₈	z ₄₂
IIIb	50	r	1,5,(7)
IIIb	50	r	e,n,x,z ₁₅
IIIb	50	r	z
IIIb	50	r	z ₃₅
IIIb	50	r	z ₅₃
Dougi	50	y	1,6
II	50	z	e,n,x
IIIb	50	z	z ₅₂
IIIa	50	z ₄ ;z ₂₃	
IV	50	z ₄ ;z ₂₃	
IIIa	50	z ₄ ;z ₂₃ ;z ₃₂	
IIIa	50	z ₄ ;z ₂₄	
IV	50	z ₄ ;z ₂₄	
IIIa	50	z ₄ ;z ₃₂	
IV	50	z ₄ ;z ₃₂	
IIIb	50	z ₁₀	z
II	50	z ₁₀	z ₆ ;z ₄₂
IIIb	50	z ₁₀	z ₅₃
Ivorycoast	50	z ₂₉	
IIIa	50	z ₂₉	
IIIa	50	z ₃₆	
II	50	z ₄₂	1,7
IIIb	50	z ₅₂	1,5,7
IIIb	50	z ₅₂	z ₃₅
IIIb	50	z ₅₂	z ₅₃
Group O:51			
IV	51	a	
Tione	51	a	e,n,x
Karaya	51	b	1,5
IV	51	b	
II	51	c	
Gokul	1,51	d	1,5
Meskin	51	e,h	1,2
II	51	g,s,t	e,n,x
IIIa	51	g,z ₅₁	
Kabete	51	i	1,5
Dan	51	k	e,n,z ₁₅
IIIb	51	k	z ₃₅
Harcourt	51	l,v	1,2
Overschie	51	l,v	1,5
Dadzie	51	l,v	e,n,x
IIIb	51	l,v	z
Moundou	51	l,z ₂₈	1,5
II	51	l,z ₂₈	z ₆
II	51	l,z ₂₈	z ₃₉
Lutetia	51	r,i	l,z ₁₃ ;z ₂₈
Antsalova	51	z	1,5
Treforest	1,51	z	1,6
Lechler	51	z	e,n,z ₁₅
IIIa	51	z ₄ ;z ₂₃	
IV	51	z ₄ ;z ₂₃	
IIIa	51	z ₄ ;z ₂₄	
IIIa	51	z ₄ ;z ₃₂	
Bergues	51	z ₁₀	1,5
II	51	z ₂₉	e,n,x,z ₁₅
II	51		1,7
Group O:52			
Uithof	52	a	1,5
Ord	52	a	e,n,z ₁₅
Molesey	52	b	1,5
Flottbek	52	b	e,n,x
II	52	c	k
Utrecht	52	d	1,5
II	52	d	e,n,x,z ₁₅

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
II	52	d	z ₃₉
Butare	52	e,h	1,6
Derkle	52	e,h	1,7
Saintemarie	52	g,t	
II	52	g,t	
Bordeaux	52	k	1,5
IIIb	52	k	z ₃₅
IIIb	52	(k)	z ₃₅
IIIb	52	k	z ₅₃
IIIb	52	l,v	z ₅₃
II	52	z	z ₃₉
IIIb	52	z	z ₅₂
II	52	z ₃₉	1,5,7
II	52	z ₄₄	1,5,7
Group O:53			
II	53	c	1,5
II	53	d	1,5
II	1,53	d	z ₃₉
II	53	d	z ₄₂
IIIa	53	g,z ₅₁	
IV	1,53	g,z ₅₁	
IIIb	53	i	z
IIIb	53	k	e,n,x,z ₁₅
IIIb	53	k	z
IIIb	53	(k)	z
IIIb	53	(k)	z ₃₅
IIIb	53	k	z ₅₃
IIIb	53	l,v	e,n,x,z ₁₅
IIIb	53	l,v	z
IIIb	53	l,v	z ₃₅
II	53	l,z ₂₈	e,n,x
II	53	l,z ₂₈	z ₆
II	53	l,z ₂₈	z ₃₉
IIIb	53	r	z
IIIb	53	r	z ₃₅
IIIb	53	r	z ₆₈
II	53	z	1,5
IIIb	53	z	1,5,(7)
II	53	z	z ₆
IIIa	53	z ₄ ;z ₂₃	
IV	53	z ₄ ;z ₂₃	
IIIa	53	z ₄ ;z ₂₃ ;z ₃₂	
II	53	z ₄ ;z ₂₄	
IIIa	53	z ₄ ;z ₂₄	
IIIb	53	z ₁₀	z
IIIb	53	z ₁₀	z ₃₅
IIIa	53	z ₂₉	
IV	1,53	z ₃₆ ;z ₃₈	
IIIb	53	z ₅₂	z ₃₅
IIIb	53	z ₅₂	z ₅₃
Leda	53		1,6
Group O:54			
Tonev	21,54	b	e,n,x
Winnipeg	54	e,h	1,5
Rosleben	3,54	e,h	1,6
Borreze	54	f,g,s	
Uccle	3,54	g,s,t	
Newholland	4,12,54	m,t	
Poeseldorf	8,20,54	i	z ₆
Ochsenwerder	6,7,54	k	1,5
Czernyng	54	r	1,5
Steinwerder	3,15,54	y	1,5
Yerba	54	z ₄ ;z ₂₃	
Canton	54	z ₁₀	e,n,x
Barry	54	z ₁₀	e,n,z ₁₅
Group O:55			
II	55	k	z ₃₉

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
Group O:56			
II	56	b	
II	56	d	
II	56	e,n,x	1,7
II	56	l,v	z ₃₉
II	56	l,z ₂₈	
II	56	z	z ₆
IIIa	56	z ₄ ,z ₂₃	
IIIa	56	z ₄ ,z ₂₃ ,z ₃₂	
II	56	z ₁₀	e,n,x
IIIa	56	z ₂₉	
Group O:57			
Antonio	57	a	z ₆
II	57	a	z ₄₂
Maryland	57	b	1,7
Batonrouge	57	b	e,n,x,z ₁₅
IIIb	57	c	e,n,x,z ₁₅
IIIb	57	c	z:[z ₆₀]
II	57	d	1,5
II	57	g,[m],s,t	z ₄₂
II	57	g,t	
IIIb	57	i	e,n,x,z ₁₅
IIIb	57	i	z
IIIb	57	k	e,n,x,z ₁₅
IV	57	z ₄ ,z ₂₃	
IIIb	57	z ₁₀	z
II	57	z ₂₉	z ₄₂
II	57	z ₃₉	e,n,x,z ₁₅
II	57	z ₄₂	1,6:z ₅₃
Group O:58			
II	58	a	z ₆
II	58	b	1,5
II	58	c	z ₆
II	58	d	z ₆
IIIb	58	i	e,n,x,z ₁₅
IIIb	58	k	z
IIIb	58	l,v	e,n,x,z ₁₅
IIIb	58	l,v	z ₃₅
II	58	l,z ₁₃ ,z ₂₈	1,5
II	58	l,z ₁₃ ,z ₂₈	z ₆
IIIb	58	r	e,n,x,z ₁₅
IIIb	58	r	z
IIIb	58	r	z ₅₃ :[z ₅₇]
II	58	z ₆	1,6
II	58	z ₁₀	1,6
IIIb	58	z ₁₀	e,n,x,z ₁₅
II	58	z ₁₀	z ₆
IIIb	58	z ₁₀	z ₅₃
II	58	z ₃₉	e,n,x,z ₁₅
IIIb	58	z ₅₂	z
IIIb	58	z ₅₂	z ₃₅
Group O:59			
IIIb	59	c	e,n,x,z ₁₅
IIIb	59	i	e,n,x,z ₁₅
IIIb	59	i	z
IIIb	59	i	z ₃₅
IIIb	59	(k)	e,n,x,z ₁₅
II	59	k	(z)
IIIb	59	(k)	z
IIIb	59	(k)	z ₃₅
IIIb	59	k	z ₅₃
IIIb	59	l,v	z
IIIb	59	l,v	z ₅₃
IIIb	59	r	z ₃₅
II	1,59	z	z ₆
IIIa	59	z ₄ ,z ₂₃	
IIIb	59	z ₁₀	z ₅₃
IIIb	59	z ₁₀	z ₅₇

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIa	59	z ₂₉	
IIIa	59	z ₃₆	
IIIb	59	z ₅₂	z ₅₃
Group O:60			
II	60	b	
II	60	g,m,t	z ₆
IIIb	60	i	e,n,x,z ₁₅
IIIb	60	i	z
IIIb	60	i	z ₃₅
IIIb	60	k	z
IIIb	60	k	z ₃₅
IIIb	60	(k)	z ₅₃
IIIb	60	l,v	z
IIIb	60	r	e,n,x,z ₁₅
IIIb	60	r	z
IIIb	60	r	z ₃₅
IIIb	60	r	z ₅₃
II	60	z	e,n,x
IIIb	60	z ₁₀	z
IIIb	60	z ₁₀	z ₃₅
IIIb	60	z ₁₀	z ₅₃
II	60	z ₂₉	e,n,x
V	60	z ₄₁	
IIIb	60	z ₅₂	1,5,[7]
IIIb	60	z ₅₂	z
IIIb	60	z ₅₂	z ₃₅
IIIb	60	z ₅₂	z ₅₃
Group O:61			
IIIb	61	c	1,5,(7)
IIIb	61	c	z ₃₅
IIIb	61	i	e,n,x,z ₁₅
IIIb	61	i	z
IIIb	61	i	z ₃₅
IIIb	61	i	z ₅₃
IIIb	61	k	1,5,(7)
IIIb	61	k	z ₃₅
IIIb	61	(k)	z ₅₃
IIIb	61	l,v	1,5,7:[z ₅₇]
IIIb	61	l,v	z
IIIb	61	l,v	z ₃₅
IIIb	61	r	1,5,7
IIIb	61	r	z
IIIb	61	r	z ₃₅
IIIb	61	r	z ₅₃
IIIb	61	z ₁₀	z ₃₅
V	61	z ₃₅	
IIIb	61	z ₅₂	1,5,7
IIIb	61	z ₅₂	z
IIIb	61	z ₅₂	z ₃₅
IIIb	61	z ₅₂	z ₅₃
Group O:62			
IIIa	62	g,z ₅₁	
IIIa	62	z ₄ ,z ₂₃	
IIIa	62	z ₄ ,z ₃₂	
IIIa	62	z ₂₉	
IIIa	62	z ₃₆	
Group O:63			
IIIa	63	g,z ₅₁	
IIIa	63	z ₄ ,z ₂₃	
IIIa	63	z ₄ ,z ₃₂	
IIIa	63	z ₃₆	
Group O:65			
IIIb	65	c	1,5,7
IIIb	65	c	z
IIIb	65	c	z ₅₃
II	65	g,t	
IIIb	65	i	e,n,x,z ₁₅
IIIb	65	(k)	z

(continued)

TABLE BXII.γ.262. (cont.)

Type ^c	Antigenic formulas ^b		
	Somatic (O) antigen	Flagellar (H) antigen	
		Phase 1	Phase 2
IIIb	65	(k)	Z ₃₅
IIIb	65	(k)	Z ₅₃
IIIb	65	l,v	c,n,x,Z ₁₅
IIIb	65	l,v	z
IIIb	65	l,v	Z ₃₅
IIIb	65	l,v	Z ₅₃
IIIb	65	r	Z ₃₅
IIIb	65	Z ₁₀	c,n,x,Z ₁₅
IIIb	65	Z ₁₀	z
IIIb	65	Z ₅₂	c,n,x,Z ₁₅
IIIb	65	Z ₅₂	z
IIIb	65	Z ₅₂	Z ₃₅
IIIb	65	Z ₅₂	Z ₅₃
II	65		1,6
Group O:66			
V	66	Z ₃₅	
V	66	Z ₃₉	
V	66	Z ₄₁	
V	66	Z ₆₅	
V	66	Z ₈₁	
Group O:67			
Crossness	67	r	1,2

^aSymbols: [], O (not underlined) or H factor that may be present or absent without relation to phage conversion, i.e., factor [5] of O:4 (B) group. When H factors are in square brackets, this means that they are exceptionally found in wild strains. For example, most strains of Paratyphi A possess the monophasic antigen phase 1 H:a. In rare cases, diphasic strains with H:1,5 as H antigen phase 2 may be isolated. For this reason, [1,5] is mentioned in square brackets in the formula of this serovar. (), O or H factor weakly agglutinable. The factor (k) corresponding to Arizona factor H:22 is weakly agglutinable by k standard serum, but is normally agglutinable by k polyvalent serum. Symbols for somatic factors determined by phage conversion are underlined (example 6, 14, 18). They are present only if the culture is lysogenized by the corresponding converting phage. These factors are usually added to the factors present in nonconverted strain (for example 6, 7 → 6, 7, 14). In O:3,10 group, factors 15 or 15, 34 take the place of factor 10. For this reason, these factors are underlined and quoted into square brackets in this group. These underlined factors are mentioned in the table for serovars in which they were found. It is likely that this situation may be encountered for all serovars in the same group O.

^bThe subfactors of O factors—40, 47, 48, and 50—are no longer mentioned, as their identification is unnecessary in current practice. O and H factors having the same symbol in the Kauffmann–White scheme are always related, but not always identical in different serovars. Table BXII.γ.262 of the antigenic formulae of the *Salmonella* is a scheme established for a diagnostic purpose. Details unnecessary for the diagnosis of serovars are not given in this scheme (e.g., R-phase of the H antigen are not indicated).

^cNames of the serovars of *S. enterica* subsp. *enterica*. For other subspecies of *S. enterica*, the subspecies to which the serovars belong are indicated by the following symbols: II, serovars of *S. enterica* subsp. *salamae*; IIIa, serovars of *S. enterica* subsp. *arizonae*; IIIb, serovars of *S. enterica* subsp. *diarizonae*; IV, serovars of *S. enterica* subsp. *houstenae*; VI, serovars of *S. enterica* subsp. *indica*; V, serovars of *S. bongori* (the symbol "V" was retained to avoid confusion with serovar name of *S. enterica* subsp. *enterica*).

^dStrains of this group maybe lysogenized by phage 14 (O:6,7 → O:6,7,14). The strains possessing O:6,7,14 had been classified in a special group, C4. They are now classified into group C1. Names formerly given to serovars of group C4 are deleted.

^eGroups O:6,8(C2) and O:8(C3) differentiated only by presence or absence of O:6 factor, were lumped together in a single group O:8.

^fStrains of this group maybe lysogenized by phage e15 (O:3,10 → O:3,15), then by phage e34 (O:3,15 → O:3,15,34). In these cases, factors O:15 or O:15,34 take the place of factor O:10 which is no longer agglutinable. The strains of O structure O:3,15 were formerly classified in a special group, E2, and the strains of O structure O:3,15,34 in another group, E3. They are now classified with the strains O:3,10 into group E1. Factors O:15 and O:15,34 are given in square brackets when they have been found in wild strains.

^gGroups formerly called O:13,22(G1) and O:13,23 (G2) were lumped together in a single group O:13.

Genetics Analysis of 16S rRNA gene sequences indicates that salmonellae belong to the *Gammaproteobacteria* (Chang et al., 1997). *S. enterica* and *S. bongori* (Table BXII.γ.261) were further separated by 16S rDNA analysis and found to be closely related

to the *E. coli* and *Shigella* complex by both 16S and 23S rDNA analyses (Christensen et al., 1998).

A total of 1160 genes are located on the genetic map of Typhimurium strain LT2: 1081 are on the circular chromosome, 29 on the 90-kb virulence-associated plasmid, and 50 are not yet mapped (see review by Sanderson et al., 1996). The linkage map of serovar Typhimurium was first determined by F-mediated conjugation. Hfr strains may be selected after F plasmid transfer. Conjugative chromosomal transfer may occur from *Salmonella* to *E. coli*, from *E. coli* to *Salmonella*, and from one serovar of *Salmonella* to another. Chromosomal genes responsible for O, Vi, and H antigens can be transferred from one genus to the other (Iino and Lederberg, 1964). Crosses may be used to localize the regions of the bacterial chromosome that specify avirulence for mice (Krishnapillai and Baron, 1964) or to study the role of O antigen factors in the virulence of *Salmonella* (Mäkelä et al., 1973).

The physical map of the chromosome of several *Salmonella* serovars is now available. Comparisons of these maps show that serovar Typhimurium (Liu and Sanderson, 1992; Liu et al., 1993b), serovar Enteritidis (Liu et al., 1993a), serovar Paratyphi B (Liu et al., 1994a), serovar Paratyphi A (Liu and Sanderson, 1995a), and serovar Typhi (Liu and Sanderson, 1995b) share a common basic genomic structure: genome sizes are all between 4600 and 4800 kb, the order of genes on chromosome segments is usually the same, and all have seven *rrn* operons. Relative to other *Salmonella* serovars, the genomic cleavage map of serovar Paratyphi A shows an insertion of about 100 kb between *rrnH-G* and *proB*, and an inversion of half the genome between *rrnH* and *rrnG*. The chromosome of serovar Typhi has undergone major genetic rearrangements (Liu and Sanderson, 1995b), including (1) homologous recombination between the seven *rrn* operons; (2) inversion that covers the replication terminus region; and (3) at least three insertions, one of which is up to 118 kb long and contains the *viaB* locus for Vi antigen biosynthesis.

As with other *Enterobacteriaceae*, salmonellae may harbor "foreign" replicons (such as temperate phages and plasmids) that may code for virulence determinants, for antibiotic resistance, for antigenic changes of O antigen, and for metabolic characteristics commonly used in diagnostic identification, e.g., lactose or sucrose fermentation (Le Minor et al., 1973, 1974). Thus it is unwise to exclude *Salmonella* solely based on a positive lactose or sucrose reaction. It is also more difficult to identify salmonellae when a pleiotrophic mutation occurs, such as one that simultaneously affects nitrate, tetrathionate, and thiosulfate reductase as well as hydrogenlyase (Le Minor, et al., 1969).

About 5% of *Salmonella* strains produce bacteriocins active against *E. coli*, *Shigella*, and/or *Salmonella* (Fredericq, 1948). Most of these bacteriocins adsorb to the same receptor as that for colicins B, E1, E2, or I. *Salmonella* bacteriocins differ from colicins *sensu stricto* by their activity spectra on colicin indicator strains. Some of these *Salmonella* bacteriocins are not even active against colicin indicator strains but are active against *Salmonella* strains only (Hamon and Péron, 1966).

Susceptibility to the O1 phage Most strains of the genus *Salmonella* are susceptible to the O1 phage of Felix and Callow (1943). This phage is highly specific for *Salmonella*, lysing more than 98% of the strains studied in routine *Salmonella* diagnosis (Kallings, 1967). Susceptibility of *Salmonella* species and subspecies to phage O1 is reported in Table BXII.γ.261. Mutations conferring resistance to O1 phage have been studied by Lindberg and Holme (1969), MacPhee et al. (1975), and Hudson et al. (1978).

A *Salmonella* phage that infects only flagellated bacteria was isolated by Sertic and Boulgakov (1936). Sensitivity to this phage depends on the antigenic specificity of the H antigen. For example, bacteria with antigens of the "G" complex are resistant (Meynell, 1961).

Pathogenicity *Salmonella* serovars may be strictly adapted to one particular host (these serovars are auxotrophic), may be ubiquitous (found in a large number of animal species), or may be of still unknown pathogenicity.

Serovars adapted to humans (e.g., serovar Typhi, serovar Paratyphi A, and serovar Sendai) usually cause severe diseases with septicemia-typhoidic syndrome. They are not naturally pathogenic for other animal species. Salmonellosis is transmitted from person to person, without an intermediate host, through fecal contamination of water and food. The incidence is higher in developing countries with poor hygiene. Typhoid fever, a systemic infection caused by serovar Typhi, remains a major public concern. The World Health Organization has estimated that there are more than 16.6 million typhoid cases per year worldwide, causing 600,000 deaths yearly (Ivanoff and Levine, 1997).

Other serovars are adapted to one animal species: serovar Abortusovis is adapted to sheep and is a major cause of abortion in ewes; serovar Typhisuis and serovar Gallinarum are adapted to swine and poultry, respectively.

Ubiquitous *Salmonella* serovars (e.g., serovar Typhimurium) are mainly responsible for food-borne infections. Low infective doses ($<10^3$) are sufficient to cause clinical symptoms (Blaser and Newman, 1982). Salmonellosis of newborns and infants (who are more susceptible to infections than adults) presents diverse clinical symptoms, from a grave typhoid-like illness with septicemia to a mild or asymptomatic infection. In pediatric wards, the infection is usually transmitted by the hands of personnel.

The entrance of a serovar into a food chain may be the origin of its importation into a country. For example, many countries have become infected with serovar Hadar introduced by imported turkeys, or by serovar Enteritidis introduced by imported poultry and eggs.

After recovery from a clinical case of salmonellosis, some patients, although asymptomatic, remain carriers for weeks, months, or years (i.e., continue to eliminate salmonellae in feces). Carriage contributes to the dissemination of salmonellosis, especially if the diagnosis of the carrier state is not monitored by periodic stool cultures. Antibiotics that are active in curing the disease are usually ineffective in the treatment of the carrier state.

Strains of *Salmonella* from urine are often of the R form. Bilharziosis, a parasitic infection caused by *Schistosoma*, has to be controlled in *Salmonella* carriers (Lo Verde et al., 1980). Sickle-cell anemia must be suspected in cases of osteomyelitis due to *Salmonella* in black children (Vandepitte et al., 1953).

The first step in *Salmonella* pathogenicity is the invasion of the small bowel mucosa. Electron microscopic studies showed that *Salmonella* rapidly adhered to and entered M cells of the follicle-associated epithelium and subsequently invaded absorptive enterocytes. This essential feature of *Salmonella* pathogenesis may be conceptualized as a two-step process. Bacteria may adhere to the target cell first, thereupon inducing an endocytic event that results in internalization of *Salmonella*. The intimate interaction between the bacterium and the host cell appears to trigger cytoskeletal rearrangements and membrane ruffling of the epithelial cell, which ultimately leads to bacterial uptake by macropinocytosis. Complexity of the mechanisms governing *Salmonella* entry into epithelial cells is reflected by the large number of loci

involved in this process. Most of them are clustered at centisome 63 of the *Salmonella* chromosome (reviewed by Galan, 1996a, 1996b). In the second step of *Salmonella* pathogenesis, bacteria gain access to the mesenteric lymph nodes, drain through the lymphatics to the thoracic duct into the blood, and ultimately infect the liver and spleen. It is now well established that pathogenic serovars of *Salmonella* other than serovar Typhi contain a plasmid that is essential for systemic dissemination in the appropriate host. Although the role of this virulence-associated plasmid is not fully understood, it is currently thought that plasmid products enhance bacterial growth within the reticuloendothelial system of the host (reviewed by Gulig et al., 1993). In human beings, systemic dissemination of serovar Typhi in the reticuloendothelial system requires expression of the Vi antigen. It was shown that the presence of Vi antigen was associated with resistance to the bactericidal effect of serum, resistance to activation of complement by the alternative pathway, resistance to opsonization by inhibition of C3b binding to bacteria, and resistance to post-phagocytic oxidative burst (reviewed by Virlogeux-Payant and Popoff, 1996).

Antibiotic and drug sensitivity Similarly to *E. coli*, *Salmonella* strains can readily acquire plasmids that contain genes conferring resistance to antibiotics. Multiple resistance is selected for when antibiotics are used extensively in hospitals or added to animal feed. The same plasmids may be found in strains of human or animal origin (Anderson et al., 1975). Since around 1990, strains of serovar Typhi have emerged that are resistant to most previously useful oral antibiotics. The antimicrobials that remain effective are relatively expensive (e.g., fluoroquinolones and ceftriaxone) and some must be administered parentally (e.g., ceftriaxone), thereby posing a quandary for developing countries (reviewed by Ivanoff and Levine, 1997). Similarly, multiple antibiotic-resistant strains of serovar Typhimurium definitive type (DT) 104 have emerged recently (Wall et al., 1994; Glynn et al., 1998; Poppe et al., 1998). They are pathogenic for humans and animals, particularly cattle. The DT104 strains were isolated in the United States, Canada, and Europe. More prudent use of antimicrobial agents in farm animals and more effective disease prevention on farms are necessary to reduce the dissemination of this multi-drug-resistant pathogen (Glynn et al., 1998).

Ecology Although some *Salmonella* serovars are strictly host adapted, the majority have a wide host range (e.g., serovar Typhimurium). Some are localized in a particular region of the globe (e.g., serovar Sendai in the Far East, serovar Berta in North America), but others are ubiquitous (e.g., serovar Typhimurium). Strains belonging to *S. enterica* subsp. *salamae*, subsp. *arizonae*, and subsp. *diarizonae* are frequently isolated from the intestinal contents of cold-blooded animals and only rarely from human beings and warm-blooded animals. Strains of subsp. *houtenae* and *S. bongori* are isolated chiefly from the environment and are rarely pathogenic for humans.

ENRICHMENT AND ISOLATION PROCEDURES

Isolation from blood is done according to the classical method for hemoculture. A biphasic culture bottle containing a vertical agar layer along one side and a broth medium at the bottom (Castaneda, 1947; Hall et al., 1979; Krieg and Gerhardt, 1981) prepared with tryptic soy agar/broth containing 2% sodium citrate is convenient. Isolated colonies grow on the agar layer. Identification is usually done by (a) diagnosis of the family *Enterobacteriaceae*, (b) diagnosis of the genus *Salmonella* (diagnosis of the subspecies for strains isolated from blood cultures is not

routinely necessary, because almost all blood isolates belong to subsp. *enterica*), (c) diagnosis of the serovar, (d) determination of the antibiotic susceptibility pattern, and (e) further study of the biovar and phagovar if indicated.

Selective procedures are needed for the isolation of *Salmonella* from specimens containing mixed bacterial flora (fecal samples, autopsy samples, food, environmental samples, etc.). Enrichment (i.e., an increased ratio of *Salmonella* cells to other bacterial cells during incubation) is obtained using liquid nutrient media containing selective agents that inhibit or retard growth of bacteria other than *Salmonella*. Use of enrichment media is essential when the number of salmonellae in a sample is very low, i.e., when the probability of finding colonies by direct isolation is low. Three media may be recommended for general use: (a) the tetrathionate medium of Muller (1923); (b) Muller's medium modified by Kauffmann (1935) by addition of bile and brilliant green; and (c) selenite F broth devised by Leifson (1936). Tetrathionate and selenite broth are suitable for all *Salmonella* serovars. Tetrathionate bile brilliant-green medium is suitable for all serovars, except host-adapted serovars such as serovar Typhi. Enrichment media should be heavily inoculated, e.g., 0.5 ml of fecal suspension per 10 ml of medium. After incubation for 18 h at 37°C, a loopful of enrichment culture is streaked onto agar plating medium.

The same enrichment media may be used for detection of salmonellae in water. The simplest method is to add one volume of the water sample to an equal volume of double-strength medium. For detecting salmonellae in food, a generally suitable procedure is to inoculate 25 g of the suspected food into 225 ml of selenite F broth, incubate for 24 h, and isolate on selective agar media. In the case of a dehydrated food, nutrient broth containing the sample is incubated overnight before inoculation of enrichment media.

Agar media are used for isolation of salmonellae. Streaking a loopful of enrichment culture or a suspension of the sample (e.g., stool) should be done carefully to obtain the greatest number of perfectly isolated colonies. Because the most discriminating character is lactose fermentation, the majority of media for isolation contain lactose and a pH indicator. In addition, the media contain selective agents to inhibit the growth of non-*Salmonella* organisms and the swarming of *Proteus mirabilis* and *P. vulgaris*. Some media also contain ferrous citrate detection of H₂S-producing bacteria.

Examples of media of moderate selectivity are (a) MacConkey agar, which contains lactose, neutral red, and the selective inhibitors crystal violet and bile salts; lactose-positive colonies are red, lactose-negative colonies are colorless; and (b) desoxycholate citrate agar, which contains lactose, neutral red, and the selective agent desoxycholate; ferric ammonium citrate is included as an indicator of H₂S production; lactose-positive colonies are red; lactose-negative colonies are colorless; if H₂S is produced, the inner part of the colony is black.

Examples of media of higher selectivity are the following: (a) Salmonella-Shigella (SS) agar, which contains lactose, neutral red, and the selective agents brilliant green and bile salts. Ferric citrate is an indicator of H₂S production. The appearance of colonies is the same as on desoxycholate citrate agar. (b) Brilliant green agar, which contains lactose, phenol red, and the selective agent brilliant green. This medium is easy to prepare and is suitable for all salmonellae except host-adapted serovars. It is not suitable for shigellae. Lactose-positive colonies are green, lactose-negative colonies are pink. All of the above-mentioned media were reviewed by Kauffmann (1966) and Edwards and Ewing (1972). (c) Hektoen enteric medium (King and Metzger, 1968), which contains lactose, sucrose and salicin, a mixture of

bromothymol blue and Andrade's pH indicator, ferric citrate to detect H₂S production, and sodium desoxycholate as a selective inhibitor. Colonies that do not ferment any of the three sugars (e.g., *Salmonella*) are blue-green, with a black center if H₂S is produced. Colonies fermenting one or more of the sugars (e.g., *E. coli*, *Enterobacter cloacae*) are salmon-colored. This medium is suitable for all *Salmonella* serovars and for shigellae.

A general procedure for the detection of salmonellae in feces or food is as follows. A suspension of the sample in saline is streaked onto the chosen isolation medium and also inoculated into an enrichment broth. After overnight incubation, the plating medium is examined for suspect colonies (lactose negative, H₂S positive or negative); also, a loopful of the enrichment culture is streaked onto another plate of selective agar medium. After overnight incubation, this plate is also examined for suspect colonies. A quick screening of several suspect colonies is done by inoculating each into a few drops of urea medium and incubating at 37°C for 2 h. Biochemical characterization is continued only for urease-negative colonies (urease-positive colonies growing at 18 h are likely to be *Proteus*) by inoculating, e.g., triple sugar-iron (TSI) or Kligler-Hajna medium. *Salmonella* must be differentiated mainly from *Citrobacter freundii*, *P. mirabilis*, and, in food bacteriology, *Alteromonas putrefaciens*. To detect *Salmonella* belonging to subsp. *arizonae* and serovar Diarizonae, attention should be given to lactose-positive, H₂S-positive colonies on plating media.

MAINTENANCE PROCEDURES

Salmonella cultures remain viable for many years when stored on peptone agar (meat extract, 5.0 g; peptone, 10.0 g; NaCl, 3.0 g; Na₂HPO₄·12H₂O, 2.0 g; agar, 10.0 g; distilled water, 1000 ml; pH 7.4) and distributed into small, tightly stoppered, screw-capped tubes. This medium is stab-inoculated and kept in the dark at room temperature. Lyophilization and freezing at or below -70°C give very good results. For lyophilization, it is necessary to isolate each subculture and to select a colony with the desired serologic characteristics.

DIFFERENTIATION OF THE GENUS *SALMONELLA* FROM OTHER GENERA

Characteristics useful for differentiating the genus *Salmonella* from other *Enterobacteriaceae* are given in Tables BXII.γ.193, BXII.γ.194, and BXII.γ.196 of the chapter on the family *Enterobacteriaceae*.

TAXONOMIC COMMENTS

The species concept in the genus *Salmonella* has evolved in several overlapping phases. The typhoid bacillus was first observed by Eberth (1880) in spleen sections and mesenteric lymph nodes from a patient who died from typhoid fever. During the following decade, other bacteria were isolated from clinical cases of typhoid fever, but they were distinct from the typhoid bacillus both culturally and serologically. The genus *Salmonella* was created by Lignières (1900). Hence, *Salmonella* strains were considered different species when isolated from different conditions or different hosts. However, it was soon realized that a number of these so-called species were ubiquitous, and emphasis shifted to antigenic properties.

Serological analysis of *Salmonella* O and H antigens was initiated by White (1926). This work was extended by Kauffmann (1961), who defined species as "a group of related sero-fermentative phage types". Consequently, each *Salmonella* serovar was considered as a species. But, as most serovars could not be distinguished by biochemical tests, Kauffmann (1966) divided the

genus *Salmonella* into four subgenera (I–IV) and continued to apply his concept of “one serovar–one species”. Nevertheless, phenotypic studies and numerical taxonomy showed that, apart from host-adapted serovars, *Salmonella* serovars within each subgenus were biochemically indistinguishable (reviewed by Le Minor and Popoff, 1987). For this reason, Kauffmann’s subgenera were considered to be species (Le Minor et al., 1970b).

Later, DNA relatedness studies demonstrated that all *Salmonella* serovars formed a single DNA hybridization group with seven subgroups delineated by studies of the thermal stability of hybrids (Crosa et al., 1973a; Stoleru et al., 1976; Le Minor et al., 1986). The seven DNA subgroups could be differentiated by using biochemical characteristics. Correlation between DNA subgroups and Kauffmann’s subgenera was close except that subgenus III was split into two DNA subgroups containing monophasic and diphasic serovars, respectively.

Based on these numerical taxonomy and DNA relatedness studies, it was proposed that the genus *Salmonella* should consist of a single species that could be divided into seven subspecies (I–VII), although subspecies V might possibly represent a second species (Le Minor et al., 1982, 1986). The strict application of the Bacteriological Code (Rules Revision Committee, Judicial Commission, International Committee on Systematic Bacteriology, 1985) led Le Minor et al. (1982, 1986) to propose *S. choleraesuis* as the species name, having seven subspecies designated subsp. *choleraesuis*, subsp. *salamae*, subsp. *arizonae*, subsp. *diarizonae*, subsp. *houtenae*, subsp. *bongori*, and subsp. *indica*. Subspecies names were validated by announcement in the *International Journal of Systematic Bacteriology* (1985, 1987). However, the name *S.*

choleraesuis could be considered as an ambiguous name since this name was used for the serovar possessing the antigenic formula 6,7:c:1,5 (Table BXII.γ.262). To avoid further confusion, it was proposed that the type species of the genus *Salmonella* be *S. enterica* (Le Minor and Popoff, 1987). This name, first proposed by Kauffmann and Edwards (1952), received unanimous support from the members of the Subcommittee of *Enterobacteriaceae* of the International Committee on Systematic Bacteriology at the XIV International Congress of Microbiology (Manchester, U.K., 1986). However, the request to change the name of the type species to *S. enterica* was not approved by the Judicial Commission. A second request has been made and, in the interim, the WHO collaborating center for reference and research on *Salmonella* and national *Salmonella* reference laboratories are using *S. enterica*.

Finally, by using multilocus enzyme analysis, it was shown that *S. enterica* subsp. *bongori* had evolved considerably from other subspecies (Reeves et al., 1989a). Based on this divergence and on DNA relatedness data (Le Minor et al., 1982; 1986), Reeves et al. (1989a) proposed that *S. enterica* subsp. *bongori* be elevated to the level of species in the new combination *S. bongori*.

The practice of giving names to the serovars of *S. enterica* subsp. *enterica* should continue since the diagnostic use of the Kauffmann–White scheme (Table BXII.γ.262) is overridingly important and since these names are very familiar to microbiologists and physicians. However, these names must no longer be italicized and the first letter must be a capital. Serovars of the other subspecies of *S. enterica* and those of *S. bongori* should be designated only by their antigenic formulae.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *SALMONELLA*

Table BXII.γ.261 presents the biochemical characteristics differentiating the species and subspecies of the genus *Salmonella*. The antigenic formulae of *Salmonella* serovars (i.e., the Kauffmann–

White scheme) is given in Table BXII.γ.262. An alphabetical listing of serovars indicating the O groups is presented in Table BXII.γ.263.

List of species of the genus *Salmonella*

1. *Salmonella enterica* (ex Kauffmann and Edwards 1952) Le Minor and Popoff 1987, 466.

en.te.ri' ca. Gr. n. *enteron* gut; L. adj. *enterica* of the gut.

The description of *S. enterica* is that of *S. choleraesuis sensu* (Le Minor et al., 1982). The characteristics are as described for the genus (see Tables BXII.γ.193, BXII.γ.194, and BXII.γ.196 of the chapter on the family *Enterobacteriaceae* and as listed in Table BXII.γ.261. *S. enterica* can be divided into six subspecies (Le Minor et al., 1982, 1986; Reeves et al. 1989a).

The type strain carries a virulence-associated plasmid of about 90 kb. Antigenic formula 4,5,12:i:1,2 (serovar Typhimurium).

The mol% G + C of the DNA is: 50–53 (hydrolysis and chromatography, Bd, T_m).

Type strain: LT2, ATCC 43971, CIP 60.62, NCIB 11450.

a. *Salmonella enterica* subsp. *enterica* (ex Kauffmann and Edwards 1952) Le Minor and Popoff 1987, 467.

Distinguished from other *S. enterica* subspecies by the characteristics reported in Table BXII.γ.261. Contains at least 1443 serovars (Popoff et al., 1998). Isolated mainly from humans and warm-blooded animals.

A list of selected serovars belonging to *S. enterica* subsp. *enterica* is given below.

The mol% G + C of the DNA is: see species description.

Type strain: LT2, ATCC 43971, CIP 60.62, NCIB 11450.

i. *Salmonella enterica* subsp. *enterica* serovar *Choleraesuis* (Smith 1894) Weldin 1927, 155 (*Bacillus cholerae suis* Smith 1894, 9.)

chol.er.ae.su' is. Gr. n. *cholera* cholera; L. n. *sus* hog. M.L. gen. n. *suis* of a hog; M.L. gen. n. *choleraesuis* of hog cholera.

Antigenic formula: 6,7:c:1,5. The detailed O antigen formula is normally 6₂,7, but this may be transformed by lysogenization into 6₁,7 or 6₂,7,14. Arabinose and trehalose are not fermented; dulcitol is slowly and irregularly fermented. Those strains that produce H₂S are designated as serovar *Choleraesuis* biovar Kunzendorf. Pathogenic for humans and animals.

The mol% G + C of the DNA is: see species description.

Deposited strain: ATCC 13312, NCTC 5735.

ii. *Salmonella enterica* subsp. *enterica* serovar *Enteritidis* (Gaertner 1888) Castellani and Chalmers 1919, 939 (*Bacterium enteritidis* Gaertner 1888, 573.)

en.te.ri'ti.dis. Gr. n. *enteron* gut, intestine; M.L. n. *enteritis* enteritis, inflammation of the intestine; M.L. gen. n. *enteritidis* of enteritis.

Antigenic formula: 1,9,12:g,m:–. The presence of factor O1 is connected with lysogenization. Ubiquitous and frequently the cause of infections in humans and

TABLE BXII.γ.263. Alphabetical list of names of *Salmonella* serovars, indicating the O groups

Serovar	O Group
Aachen	J
Aarhus	K
Aba	C2–C3
Abadina	M
Abacetuba	F
Aberdeen	F
Abidjan	Q
Ablogame	I
Abobo	I
Abony	B
Abortusequi	B
Abortusovis	B
Abuja	F
Accra	E4
Ackwepe	D2
Adabraka	E1
Adamstown	M
Adamstua	F
Adana	U
Adelaide	O
Adeoyo	I
Aderike	M
Adime	C1
Adjame	G
Aequatoria	C1
Aesch	C2–C3
Aflao	H
Africana	B
Afula	C1
Agama	B
Agbara	I
Agbeni	G
Agege	E1
Ago	N
Agodi	O
Agona	B
Agoueve	G
Ahanou	J
Ahepe	U
Ahmadi	E4
Ahoutoue	S
Ahuza	U
Ajiobo	G
Akanji	C2–C3
Akuafo	I
Alabama	D1
Alachua	O
Alagbon	C2–C3
Alamo	C1
Albany	C2–C3
Albert	B
Albuquerque	H
Alexanderplatz	X
Alexanderpolder	C2–C3
Alfort	E1
Alger	P
Alkmaar	E4
Allandale	R
Allerton	E1
Alma	Q
Alminko	C2–C3
Altendorf	B
Altona	C2–C3
Amager	E1
Amba	F
Amersfoort	C1
Amherstiana	C2–C3
Amina	I
Aminatu	E1
Amounderness	E1

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Amoutive	M
Amsterdam	E1
Amunigun	I
Anatum	E1
Anderlecht	E1
Anecho	O
Anfo	Q
Angers	C2–C3
Angoda	N
Angouleme	I
Ank	M
Anna	G
Annedal	I
Antarctica	D1
Antonio	57
Antsalova	51
Antwerpen	T
Apapa	W
Apeyeme	C2–C3
Aprad	W
Aqua	N
Aragua	N
Arapahoe	H
Arechavaleta	B
Arusha	U
Aschersleben	N
Ashanti	M
Assen	L
Assinie	E1
Asylanta	E1
Atakpame	C2–C3
Atento	F
Athinai	C1
Ati	F
Augustenborg	C1
Austin	C1
Australia	Y
Avignon	I
Avonmouth	E4
Ayinde	B
Ayton	B
Azteca	B
Babelsberg	M
Badagry	I
Baguida	L
Baguirmi	N
Bahati	G
Bahrenfeld	H
Baiboukoum	C1
Baildon	D2
Bakau	M
Balcones	W
Balili	M
Ball	B
Bama	J
Bamboye	D2
Bambylor	D2
Banalia	C2–C3
Banana	B
Banco	M
Bandia	O
Bangkok	P
Bangui	D1
Banjul	H
Bardo	C2–C3
Bareilly	C1
Bargny	C2–C3
Barmbek	I
Barranquilla	I
Barry	54
Basingstoke	D2

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Bassa	C2-C3
Bassadji	M
Batonrouge	57
Battle	I
Bazenheid	C2-C3
Be	C2-C3
Beaudesert	H
Bedford	E4
Belem	C2-C3
Belfast	C2-C3
Bellevue	C2-C3
Benfica	E1
Benguella	R
Benin	D2
Benue	C2-C3
Bere	X
Bergedorf	D2
Bergen	X
Bergues	51
Berkeley	U
Berlin	J
Berta	D1
Bessi	E1
Bethune	E4
Biafra	E1
Bida	E4
Bietri	N
Bignona	J
Bijlmer	R
Bilu	E4
Binche	X
Bingerville	X
Binningen	W
Birkenhead	C1
Birmingham	E1
Bispebjerg	B
Bissau	B
Blegdam	D1
Blijdorp	H
Blitta	X
Blockley	C2-C3
Bloomsbury	E1
Blukwa	K
Bobo	V
Bochum	B
Bodjonegoro	N
Boecker	H
Bofflens	S
Bokanjac	M
Bolama	V
Bolombo	E1
Bolton	E1
Bonames	J
Bonariensis	C2-C3
Bonn	C1
Bootle	X
Borbeck	G
Bordeaux	52
Borreze	54
Borromea	T
Bouake	I
Bournemouth	D1
Bousso	H
Bovismorbificans	C2-C3
Bracknell	G
Bradford	B
Braenderup	C1
Brancaster	B
Brandenburg	B
Brazil	I
Brazos	K

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Brazzaville	C1
Breda	C2-C3
Bredeney	B
Brefet	V
Breukelen	C2-C3
Brevik	I
Brezany	B
Brijbhumi	F
Brikama	C2-C3
Brindisi	F
Brisbane	M
Bristol	G
Brive	T
Broc	T
Bron	G
Bronx	C2-C3
Brooklyn	I
Broughton	E4
Bruck	C1
Brunei	C2-C3
Brunflo	I
Bsilla	C2-C3
Buckeye	Y
Budapest	B
Bukavu	R
Bukuru	C2-C3
Bulgaria	C2-C3
Bullbay	F
Bulovka	C1
Burgas	I
Burundi	S
Bury	B
Businga	C1
Butantan	E1
Butare	52
Buzu	H
Caen	I
Cairina	E1
Cairns	W
Calabar	E4
California	B
Camberene	O
Camberwell	D1
Campinense	D1
Canada	B
Canary	R
Cannobio	M
Cannonhill	E4
Cannstatt	E4
Canton	54
Caracas	H
Cardoner	I
Carmel	J
Carnac	K
Carno	E4
Carrau	H
Carswell	V
Casablanca	W
Casamance	R
Catalunia	M
Catanzaro	H
Cayar	C1
Cerro	K
Ceyco	D2
Chagoua	G
Chailey	C2-C3
Champaign	Q
Chandans	F
Charity	H
Charlottenburg	C2-C3
Cheltenham	D2

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Chester	B
Chicago	M
Chichester	E4
Chichiri	H
Chile	C1
Chincol	C2–C3
Chingola	F
Chiredzi	F
Chittagong	E4
Choleraesuis	C1
Chomedey	C2–C3
Christiansborg	V
Clackamas	B
Claibornei	D1
Clanvillian	F
Clerkenwell	E1
Cleveland	C2–C3
Cochin	D2
Cocody	C2–C3
Coeln	B
Coleypark	C1
Colindale	C1
Colobane	F
Colombo	P
Colorado	C1
Concord	C1
Congo	G
Connecticut	F
Coogee	T
Coquilhatville	E1
Coromandel	C1
Corvallis	C2–C3
Cotham	M
Cotia	K
Cremieu	C2–C3
Crewe	F
Croft	M
Crossness	67
Cubana	G
Cuckmere	E1
Cullingworth	M
Cumberland	Q
Curacao	C2–C3
Cyprus	C2–C3
Czernyring	54
Daarle	C2–C3
Dabou	C2–C3
Dadzie	51
Dahlem	Y
Dahomey	X
Dahra	J
Dakar	M
Dakota	I
Dallgow	E4
Damman	C1
Dan	51
Dapango	X
Daula	C2–C3
Daytona	C1
Deckstein	D2
Delmenhorst	K
Dembe	O
Demerara	G
Denver	C1
Derby	B
Derkle	52
Dessau	E4
Deversoir	W
Dibra	M
Dieuppeul	M
Diguel	G

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Diogoye	C2–C3
Diourbel	L
Djakarta	Y
Djama	T
Djelfa	C2–C3
Djermaia	M
Djibouti	J
Djugu	C1
Doba	D2
Doel	M
Doncaster	C2–C3
Donna	N
Doorn	M
Douala	M
Dougi	Z
Doulassame	N
Dresden	M
Driffield	R
Drogana	B
Dublin	D1
Duesseldorf	C2–C3
Dugbe	W
Duisburg	B
Dumfries	E1
Dunkwa	C2–C3
Durban	D1
Durham	G
Duval	R
Ealing	O
Eastbourne	D1
Eastglam	E4
Eberswalde	M
Eboko	C2–C3
Ebrie	O
Echa	P
Edinburg	C1
Edmonton	C2–C3
Egusi	S
Egusitoo	T
Eingedi	C1
Eko	B
Ekotedo	D2
Ekpoui	X
Elisabethville	E1
Elokate	D1
Elomrane	D1
Emek	C2–C3
Emmastad	P
Encino	H
Enschede	O
Entebbe	B
Enteritidis	D1
Enugu	I
Epicrates	E1
Epinay	F
Eppendorf	B
Escanaba	C1
Eschberg	D1
Eschweiler	C1
Essen	B
Etterbeek	F
Euston	F
Everleigh	E1
Ezra	M
Fairfield	M
Fajara	M
Faji	T
Falkensee	E1
Fallowfield	E1
Fann	F
Fanti	G

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Farakan	M
Farcha	U
Fareham	E4
Farmsen	G
Farsta	B
Fass	Z
Fayed	C2-C3
Ferlo	S
Ferruch	C2-C3
Finaghy	B
Findorff	F
Finkenwerder	H
Fischerhuetten	I
Fischerkietz	H
Fischerstrasse	V
Fitzroy	Y
Florian	E1
Florida	H
Flottbek	52
Fluntern	K
Fomeco	W
Fortlamy	I
Fortune	B
Franken	D1
Frankfurt	I
Freefalls	M
Freetown	P
Freiburg	E1
Fresno	D2
Friedenau	G
Friedrichsfelde	M
Frintrop	D1
Fufu	E1
Fulda	E4
Fulica	B
Fyris	B
Gabon	C1
Gafsa	I
Galiema	C1
Galil	E1
Gallen	F
Gallinarum	D1
Gamaba	V
Gambaga	L
Gambia	O
Gaminara	I
Garba	H
Garoli	C1
Gassi	O
Gateshead	D2
Gatineau	E4
Gatow	C1
Gatuni	C2-C3
Gbadago	E1
Gdansk	C1
Gege	N
Georgia	C1
Gera	T
Geraldton	D2
Gerland	I
Ghana	L
Giessen	N
Give	E1
Giza	C2-C3
Glasgow	I
Glidji	F
Glostrup	C2-C3
Gloucester	B
Gnesta	E4
Godesberg	N
Goelzau	E1

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Goeteborg	D1
Goettingen	D1
Gokul	51
Goldcoast	C2-C3
Goma	C1
Gombe	C1
Good	L
Gori	J
Goulfey	R
Gouloumbo	O
Goverdhan	D1
Gozo	M
Grampian	C1
Grancanaria	I
Grandhaven	N
Granlo	J
Graz	U
Greiz	R
Groenekan	K
Grumpensis	G
Guarapiranga	N
Guerin	D2
Guildford	M
Guinea	V
Gustavia	F
Gwale	T
Gwoza	E4
Haardt	C2-C3
Hadar	C2-C3
Hadejia	J
Haduna	B
Haelsingborg	C1
Haferbreite	T
Haga	O
Haifa	B
Halle	M
Hallfold	B
Handen	G
Hann	R
Hannover	I
Haouaria	G
Harburg	H
Harcourt	51
Harleystreet	E1
Harrisonburg	E1
Hartford	C1
Harvesthude	T
Hatfield	M
Hato	B
Havana	G
Hayindogo	E4
Heerlen	F
Hegau	Q
Heidelberg	B
Hemingford	Z
Hennekamp	T
Hermannswerder	M
Heron	I
Herston	C2-C3
Herzliya	F
Hessarek	B
Hidalgo	C2-C3
Hiduddify	C2-C3
Hillegersberg	D2
Hillingdon	D2
Hillsborough	C1
Hilversum	N
Hindmarsh	C2-C3
Hisingen	Y
Hissar	C1
Hithergreen	I

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Hoboken	E1
Hofit	Q
Hoghton	E1
Holcomb	C2–C3
Homosassa	H
Honelis	M
Hongkong	E4
Horsham	H
Huddinge	E1
Huettwillen	B
Hull	I
Huvudsta	E1
Hvittingfoss	I
Hydra	L
Ibadan	G
Ibaragi	L
Idikan	G
Ikayi	E1
Ikeja	M
Ilala	M
Ilugun	E4
Imo	W
Inchpark	C2–C3
India	D2
Indiana	B
Infantis	C1
Inganda	C1
Inglis	D2
Inpraw	S
Inverness	P
Ipeko	D1
Ipswich	S
Irchel	D2
Irenea	J
Irigny	U
Irumu	C1
Isangi	C1
Isaszeg	Y
Israel	D1
Istanbul	C2–C3
Istoria	H
Isuge	G
Itami	D1
Ituri	B
Itutaba	D2
Ivory	I
Ivorycoast	Z
Jaffna	D1
Jalisco	F
Jamaica	D1
Jambur	L
Jangwani	J
Javiana	D1
Jedburgh	E1
Jericho	B
Jerusalem	C1
Joal	E1
Jodhpur	W
Johannesburg	R
Jos	B
Juba	E4
Jubilee	J
Jukestown	G
Kaapstad	B
Kabete	51
Kaduna	C1
Kahla	T
Kainji	E4
Kaitaan	H
Kalamu	B
Kalina	E1

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Kallo	C2–C3
Kalumburu	C2–C3
Kambole	C1
Kamoru	B
Kampala	T
Kande	E4
Kandla	J
Kaneshie	T
Kanifing	H
Kano	B
Kaolack	X
Kapemba	D1
Karachi	W
Karamoja	R
Karaya	51
Karlshamn	J
Kasenyi	P
Kassberg	H
Kedougou	G
Kentucky	C2–C3
Kenya	C1
Kermel	V
Keve	L
Kiambu	B
Kibi	I
Kibusi	M
Kidderminster	P
Kiel	A
Kikoma	I
Kimberley	P
Kimpese	D1
Kimuenza	B
Kindia	E4
Kingabwa	U
Kingston	B
Kinondoni	J
Kinson	E4
Kintambo	G
Kirkee	J
Kisangani	B
Kisarawe	F
Kisii	C1
Kitenge	M
Kivu	C1
Klouto	P
Kodjovi	X
Koenigstuhl	B
Koessen	A
Kofandoka	W
Koketime	V
Kokoli	N
Kokomlemle	Q
Kolar	D2
Kolda	C2–C3
Konolfingen	M
Konstanz	C2–C3
Korbol	C2–C3
Korlebu	E4
Korovi	P
Kortrijk	C1
Kottbus	C2–C3
Kotte	C1
Kotu	D1
Kouka	E4
Koumra	C1
Kpeme	M
Kraligen	C2–C3
Krefeld	E4
Kristianstad	E1
Kua	V
Kubacha	B

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Kuessel	M
Kumasi	N
Kunduchi	B
Kuntair	H
Kuru	C2-C3
Labadi	C2-C3
Lagos	B
Lamberhurst	E1
Lamin	E1
Lancaster	J
Landala	S
Landau	N
Landwasser	E1
Langenhorn	K
Langensalza	E1
Langford	M
Lansing	P
Laredo	H
Larochelle	C1
Lattenkamp	W
Lawndale	D1
Lawra	V
Leatherhead	S
Lechler	51
Leda	53
Leer	K
Leeuwarden	F
Ligon	B
Leiden	G
Leipzig	S
Leith	C2-C3
Lekke	E1
Lene	F
Leoben	M
Leopoldville	C1
Lerum	E4
Lexington	E1
Lezennes	C2-C3
Libreville	M
Ligeo	N
Ligna	O
Lika	C1
Lille	C1
Limete	B
Lindenburg	C2-C3
Lindern	H
Lindi	P
Linguere	D2
Lingwala	I
Linton	G
Lisboa	I
Lishabi	D2
Litchfield	C2-C3
Liverpool	E4
Livingstone	C1
Livulu	N
Ljubljana	B
Llandoff	E4
Llobregat	V
Loanda	C2-C3
Lockleaze	C1
Lode	J
Lodz	S
Loenga	T
Logone	Q
Lokstedt	E4
Lomalinda	D1
Lome	D1
Lomita	C1
Lomnava	I
London	E1

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Losangeles	I
Loubomo	B
Louga	N
Louisiana	D2
Lovelace	G
Lowestoft	J
Lubumbashi	S
Luciana	F
Luckenwalde	M
Luedinghausen	J
Luke	X
Lutetia	51
Lyon	X
Maastricht	F
Macallen	E1
Macclesfield	D2
Machaga	E4
Madelia	H
Madiago	E4
Madigan	V
Madison	L
Madjorio	E1
Madras	B
Magherafelt	C2-C3
Magumeri	H
Magwa	L
Mahina	D2
Maiduguri	E4
Makiling	U
Makiso	C1
Malakal	I
Malaysia	M
Malika	V
Malmoe	C2-C3
Malstatt	I
Mampeza	H
Mampong	G
Manchester	C2-C3
Mandera	I
Mango	P
Manhattan	C2-C3
Mannheim	F
Mapo	C2-C3
Mara	Q
Maracaibo	F
Marburg	G
Maricopa	T
Marienthal	E1
Maritzburg	V
Maron	E1
Maroua	F
Marseille	F
Marshall	G
Maryland	57
Marylebone	D2
Masembe	E1
Maska	B
Massakory	O
Massenya	B
Matadi	J
Mathura	D2
Matopeni	N
Mayday	D2
Mbandaka	C1
Mbao	U
Meekatharra	W
Melbourne	T
Meleagridis	E1
Memphis	K
Menden	C1
Mendoza	D1

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Menston	C1
Mesbit	X
Meskin	51
Messina	N
Mgulani	P
Miami	D1
Michigan	J
Middlesbrough	T
Midway	H
Mikawasima	C1
Millesi	R
Milwaukee	U
Mim	G
Minna	H
Minnesota	L
Mishmarhaemek	G
Mississippi	G
Missouri	F
Miyazaki	D1
Mjordan	N
Mkamba	C1
Mocamedes	M
Moero	M
Moers	F
Mokola	E1
Molade	C2-C3
Molesey	52
Mono	B
Mons	B
Monschaui	O
Montevideo	C1
Montreal	U
Morehead	N
Morillons	M
Morningside	N
Mornington	H
Morocco	N
Morotai	J
Moroto	M
Moscow	D1
Moualine	X
Moundou	51
Mountmagnet	L
Mountpleasant	X
Moussoro	H
Mowanjum	C2-C3
Mpouto	I
Muenchen	C2-C3
Muenster	E1
Muguga	V
Mulhouse	D1
Mundonobo	M
Mura	B
Naestved	D1
Nagoya	C2-C3
Nakuru	B
Namibia	C1
Namoda	X
Nanergou	C2-C3
Nanga	G
Nantes	D2
Napoli	D1
Narashino	C2-C3
Nashua	M
Natal	D1
Naware	I
Nchanga	E1
Ndjamena	H
Ndolo	D1
Neftenbach	B
Nessa	H

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Nessziona	C1
Neudorf	N
Neukoelln	C1
Neumuenster	B
Neunkirchen	P
Newholland	54
Newlands	E1
Newmexico	D1
Newport	C2-C3
Newrochelle	E1
Newyork	G
Ngaparou	D2
Ngili	C1
Ngor	E4
Niakhar	V
Niamey	J
Niarembe	V
Niederoderwitz	U
Nieukerk	C1
Nigeria	C1
Nijmegen	N
Nikolaifleet	I
Niloese	E4
Nima	M
Nimes	G
Nitra	A
Niumi	E4
Njala	P
Nola	C1
Nordrhein	D2
Nordufer	C2-C3
Norton	C1
Norwich	C1
Nottingham	I
Nowawes	R
Noya	C2-C3
Nuatja	I
Nyanza	F
Nyborg	E1
Nyeko	I
Oakey	C1
Oakland	C1
Obogu	C1
Ochiogu	E4
Ochsenwerder	54
Ockenheim	N
Odiene	R
Odozi	N
Oerlikon	Q
Oersterbro	E4
Offa	S
Ogbete	U
Ohio	C1
Ohlstedt	E1
Okatie	G
Okefoko	E1
Okerara	E1
Oldenburg	I
Olten	D2
Omifisan	R
Omuna	C1
Ona	M
Onarimon	D1
Onderstepoort	H
Onireke	E1
Ontario	D2
Oran	P
Oranienburg	C1
Orbe	T
Ord	52
Ordenez	G

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Orientalis	I
Orion	E1
Oritamerin	C1
Orlando	K
Orleans	U
Os	D1
Oskarshamn	M
Oslo	C1
Osnabrueck	F
Othmarschen	C1
Ottawa	D1
Ouakam	D2
Oudwijk	G
Overchurch	R
Overschie	51
Overvecht	N
Oxford	E1
Oyonnax	C1
Pakistan	C2-C3
Palamaner	V
Palime	C1
Panama	D1
Papuana	C1
Paratyphi A	A
Paratyphi B	B
Paratyphi C	C1
Paris	C2-C3
Parkroyal	E4
Pasing	B
Patience	M
Penarth	D1
Penilla	M
Pensacola	D1
Perth	P
Petahtikve	E4
Phaliron	C2-C3
Pharr	F
Pietersburg	E1
Pisa	I
Planckendael	C1
Ploufragan	V
Plymouth	D2
Poano	H
Poeseldorf	54
Poitiers	C1
Pomona	M
Pontypridd	K
Poona	G
Portanigra	C2-C3
Portland	D1
Potengi	K
Potosi	H
Potsdam	C1
Potto	D2
Powell	D1
Praha	C2-C3
Pramiso	E1
Presov	C2-C3
Preston	B
Pretoria	F
Putten	G
Quebec	V
Quentin	D2
Quinhon	X
Quiniela	C2-C3
Ramatgan	N
Ramsey	M
Raus	G
Rawash	K
Reading	B
Rehovot	C2-C3

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Redba	C1
Redhill	F
Redlands	I
Regent	E1
Reinickendorf	B
Remete	F
Remiremont	C2-C3
Remo	B
Reubeuss	C2-C3
Rhone	L
Rhydyfelin	I
Richmond	C1
Rideau	E4
Ridge	D1
Ried	G
Riggil	C1
Riogrande	R
Rissen	C1
Rittersbach	P
Riverside	W
Roan	P
Rochdale	Z
Rogy	M
Romanby	G
Roodepoort	G
Rosleben	54
Rostock	D1
Rothenburgsort	P
Rottneest	G
Rovaniemi	I
Royan	H
Ruanda	D1
Rubislaw	F
Ruiru	L
Rumford	C1
Runby	H
Ruzizi	E1
Saarbruecken	D1
Saboya	I
Sada	N
Saintemarie	52
Saintpaul	B
Salford	I
Saloniki	I
Samaru	S
Sambre	E4
Sandiego	B
Sandow	C2-C3
Sanga	C2-C3
Sangalkam	D2
Sangera	I
Sanjuan	C1
Sanktgeorg	M
Sanktjohann	G
Sanktmarx	E4
Santander	M
Santhiaba	R
Santiago	C2-C3
Sao	E4
Sapele	G
Saphra	I
Sara	H
Sarajane	B
Saugus	R
Schalkwijk	H
Schleissheim	B
Schoeneberg	E4
Schwabach	C1
Schwarzengrund	B
Schwerin	C2-C3
Sculcoates	I

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Seattle	M
Sedgwick	V
Seegefeld	E1
Sekondi	E1
Selby	M
Sendai	D1
Senegal	F
Senftenberg	E4
Senneville	N
Seremban	D1
Serrekunda	E1
Shamba	I
Shangani	E1
Shanghai	I
Shannon	E1
Sharon	F
Sheffield	P
Sherbrooke	I
Shikmonah	R
Shipley	C2–C3
Shomolu	M
Shoreditch	D2
Shubra	B
Sica	S
Simi	E1
Sinchew	E1
Sindelfingen	C2–C3
Singapore	C1
Sinstorf	E1
Sinthia	K
Sipane	T
Skansen	C2–C3
Slade	E4
Sljeme	X
Sloterdijk	B
Soahanina	H
Soerenga	N
Sokode	D2
Solna	M
Solt	F
Somone	C1
Soumbédioune	M
Southampton	B
Southbank	E1
Souza	E1
Spalantor	T
Spartel	L
Splott	V
Stachus	P
Stanley	B
Stanleyville	B
Staoueli	X
Steinplatz	N
Steinwerder	54
Stellingen	X
Stendal	F
Sternschanze	N
Sterrenbos	C2–C3
Stockholm	E1
Stoneferry	N
Stormont	E1
Stourbridge	C2–C3
Straengnaes	F
Strasbourg	D2
Stratford	E4
Strathcona	C1
Stuivenberg	E4
Stuttgart	C1
Suberu	E1
Sudan	U
Suëldorf	W

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Sundsvall	H
Sunnycove	C2–C3
Surat	H
Surrey	L
Svedvi	E4
Sya	X
Sylvania	H
Szentes	I
Tabligbo	X
Tado	C2–C3
Tafo	B
Taiping	G
Takoradi	C2–C3
Taksony	E4
Tallahassee	C2–C3
Tamale	C2–C3
Tambacounda	E4
Tamberma	X
Tamilnadu	C1
Tampico	C1
Tananarive	C2–C3
Tanger	G
Tanzania	G
Tarshyne	D1
Taset	T
Taunton	M
Taylor	P
Tchad	O
Tchamba	J
Techimani	M
Teddington	B
Tees	I
Tejas	B
Teko	H
Telaviv	M
Telelkebir	G
Telhashomer	F
Teltow	M
Tema	T
Tendeba	J
Tennenlohe	K
Tennessee	C1
Tennyson	B
Teshie	X
Texas	B
Thayngen	B
Thetford	U
Thiaroye	P
Thies	E4
Thompson	C1
Tibati	E1
Tienba	C1
Tiergarten	V
Tiko	R
Tilburg	E4
Tilene	R
Tinda	B
Tione	51
Togba	I
Togo	B
Tokoin	B
Tomegbe	T
Tomelilla	E4
Tonev	54
Toowong	F
Torhout	N
Toricada	T
Tornow	W
Toronto	D2
Toucra	Y
Toulon	K

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Tounouma	C2-C3
Tours	F
Trachau	B
Transvaal	W
Travis	B
Treforest	51
Treguier	D1
Trier	I
Trimdon	D2
Tripoli	B
Trotha	R
Truro	E1
Tschangu	G
Tsevie	B
Tshiongwe	C2-C3
Tucson	H
Tudu	B
Tumodi	B
Tunis	G
Typeb	O
Typhi	D1
Typhimurium	B
Typhisuis	C1
Tyresoe	B
Uccle	54
Uganda	E1
Ughelli	E1
Uhlenhorst	V
Uithof	52
Ullevi	G
Umbilo	M
Umhlali	C1
Umhlatazana	O
Uno	C2-C3
Uppsala	B
Urbana	N
Ursenbach	T
Usumbura	K
Utah	C2-C3
Utrecht	52
Uzaramo	H
Vaertan	G
Valdosta	C2-C3
Vancouver	I
Vanier	M
Vaugirard	S
Vege sack	I
Vejle	E1
Vellore	B
Veneziana	F
Verona	S
Verviers	W
Victoria	D1
Victoriaborg	J
Vietnam	S
Vilvoorde	E4
Vinohrady	M
Virchow	C1
Virginia	C2-C3
Visby	E4
Vitkin	M
Vleuten	V
Vogan	T
Volkmar sdorf	M
Volta	F
Vom	B
Voulte	U
Vridi	G
Vuadens	B
Wa	I
Waedenswil	D2

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Wagadugu	E1
Wagenia	B
Wandsworth	Q
Wangata	D1
Waral	T
Warengo	J
Warm sen	W
Warnemuende	M
Warnow	C2-C3
Warragul	H
Warri	J
Washington	G
Waycross	S
Wayne	N
Wedding	M
Welikade	I
Weltevreden	E1
Wenatchee	X
Wentworth	F
Wernigerode	D2
Weslaco	T
Westafrica	D1
Westeinde	I
Westerstede	E4
Westhampton	E1
Westminster	E1
Weston	I
Westphalia	O
Weybridge	E1
Wichita	G
Widemarsh	O
Wien	B
Wil	C1
Wilhelmsburg	B
Willemstad	G
Wilmington	E1
Wimborne	E1
Windermere	Q
Wingrove	C2-C3
Winneba	B
Winnipeg	54
Winston	C1
Winterthur	E4
Wippra	C2-C3
Wisbech	I
Wohlen	F
Woodhull	H
Woodinville	F
Worb	D2
Worthington	G
Wuiti	N
Wuppertal	D2
Wyldegreen	G
Yaba	E1
Yalding	E4
Yaounde	B
Yardley	M
Yarm	C2-C3
Yarrabah	G
Yeerongpilly	E1
Yehuda	F
Yekepa	R
Yerba	54
Yoff	P
Yokoe	C2-C3
Yolo	O
Yopougon	W
Yoruba	I
Yovokome	C2-C3
Yundum	E1
Zadar	D2

(continued)

TABLE BXII.γ.263. (cont.)

Serovar	O Group
Zaiman	D1
Zaire	N
Zanzibar	E1
Zaria	J
Zega	D1
Zehlendorf	N
Zerifin	C2-C3
Zigong	I
Zinder	V
Zongo	E1
Zuilen	E4
Zwickau	I

animals; since the last decade, very frequent agent of *Salmonella* gastroenteritis in humans.

The mol% G + C of the DNA is: see species description.

Deposited strain: ATCC 13076.

- iii. ***Salmonella enterica* subsp. *enterica* serovar *Gallinarum*** (Klein 1889) Bergey, Harrison, Breed, Hammer and Huntoon 1925, 236 (*Bacillus gallinarum* Klein 1889, 689; *Bacterium pullorum* Rettger 1909, 123; *Salmonella gallinarum-pullorum* Taylor, Bensted, Boyd, Carpenter, Dowson, Lovell, Taylor, Thornton, Wilson and Shaw 1952, 140.)
gal.li.na'rum. L. n. *gallina* hen; L. gen. pl. n. *gallinarum* of hens.

Antigenic formula: 1,9,12:—:—. The presence of factor O1 is connected with lysogenization. Always non-motile. May be subdivided into biovars based on fermentation characteristics, production of gas, and production of H₂S. Does not grow on a minimal defined medium. Isolated chiefly from chickens and other birds. Causative agent of fowl typhoid.

The mol% G + C of the DNA is: see species description.

- iv. ***Salmonella enterica* subsp. *enterica* serovar *Paratyphi A*** (Brion and Kayser 1902) Castellani and Chalmers 1919, 939 (*Bacterium paratyphi* Kayser 1902, 426; *Bacterium paratyphi* typhus A Brion and Kayser 1902, 613.)
pa.ra.ty'phi. Gr. prep. *para* alongside of; Gr. n. *typhus* a stupor; M.L. gen. n. *paratyphi* A of type A typhoid-like infection.

Antigenic formula: 1,2,12:a:—:—. The presence of factor O1 is connected with lysogenization. Aerogenic. Ferments arabinose but no xylose. The majority of strains do not produce H₂S, and in this respect serovar Paratyphi A is unlike most other salmonellae. Lysine decarboxylase reaction is weak or negative. Pathogenic only for humans.

The mol% G + C of the DNA is: see species description.

- v. ***Salmonella enterica* subsp. *enterica* serovar *Paratyphi B*** (Brion and Kayser 1902) Bergey, Harrison, Breed, Hammer and Huntoon 1923, 213 (*Bacterium paratyphi* typhus B Brion and Kayser 1902, 613; *Bacillus schottmuelleri* Winslow, Kligler and Rothberg 1919, 479.)
pa.ra.ty'phi. Gr. prep. *para* alongside of; Gr. n. *typhus* a stupor; M.L. gen. n. *paratyphi* B of type B typhoid-like infection.

Antigenic formula: 1,4,[5],12:b:1,2. The presence of

factor O1 is connected with lysogenization. Produces a slime layer when grown on a medium containing 0.5% glucose and 0.2 M sodium phosphate, pH 7. Negative for D-tartrate. Causes enteric fever in humans and very rarely infects animals. A variant known as biovar Java is positive for D-tartrate, fails to produce a slime layer, and usually causes enteritis in humans and not uncommonly in animals as well (Kauffmann, 1941). Some strains are intermediate between these two extremes.

The mol% G + C of the DNA is: see species description.

- vi. ***Salmonella enterica* subsp. *enterica* serovar *Paratyphi C*** Hirschfeld 1919, 296 (Paratyphoid C bacillus Hirschfeld 1919, 296; *Salmonella hirschfeldii* Weldin 1927, 161; *Salmonella paratyphi-C* International Salmonella Subcommittee 1934.)
pa.ra.ty'phi. Gr. prep. *para* alongside of; Gr. n. *typhus* a stupor; M.L. gen. n. *paratyphi* C of type C typhoid-like infection.

Antigenic formula: 6,7,[Vi]:c:1,5. Ferments dulcitol and trehalose; produces H₂S. Arabinose fermentation is variable.

The mol% G + C of the DNA is: see species description.

- vii. ***Salmonella enterica* subsp. *enterica* serovar *Typhi*** (Schroeter 1886) Warren and Scott 1930, 416 (*Bacillus typhi* Schroeter 1886, 165.)
ty'phi. Gr. n. *typhus* a stupor; M.L. gen. n. *typhi* of typhoid.

Antigenic formula: 9,12,[Vi]:d:—:—. Wild strains may possess H antigen z66 instead of H antigen d (Guinée et al., 1981). Does not grow on Simmons' citrate medium or on a minimal defined medium; requires at least tryptophan as a growth factor. Does not produce gas from glucose or other sugars. Fermentation of xylose is variable. Many strains are agglutinated by anti-Vi serum and are inagglutinable by anti-O9 serum; their colonies (V colonies) are opaque and have an iridescent appearance when examined by transmitted light. Colonies of intermediate appearance agglutinable by both Vi and O antisera may occur (VW colonies). Pathogenic only for humans, causing typhoid (enteric) fever; transmitted by water or food contaminated by human excreta.

The mol% G + C of the DNA is: see species description.

Deposited strain: ATCC 19430.

- viii. ***Salmonella enterica* subsp. *enterica* serovar *Typhimurium*** (Loeffler 1892) Castellani and Chalmers 1919, 939 (*Bacterium typhimurium* Loeffler 1892, 134.)
ty.phi.mu'ri.um. Gr. n. *typhus* a stupor; L. n. *mus* mouse; L. gen. pl. n. *murium* of mice; M.L. gen. pl. n. *typhimurium* typhoid of mice.

Antigenic formula: 1,4,[5],12:i:1,2. The presence of factor O1 follows lysogenization by a converting phage named *iota* or PLT22. Ubiquitous and frequently the cause of infections in humans and animals; very frequent agent of *Salmonella* gastroenteritis in man.

The mol% G + C of the DNA is: see species description.

Deposited strain: ATCC 13311.

- b. *Salmonella enterica* subsp. *arizonae* (Borman 1957) Le Minor and Popoff 1987, 467 (*Salmonella choleraesuis* subsp. *arizonae* (Borman 1957) Le Minor, Véron and Popoff 1985, 375; *Salmonella arizonae* (Borman 1957) Kauffmann in van Oye 1964; *Paracolonobacterium arizonae* Borman 1957, 347.)

a.ri.zo'nae. M.L. gen. n. *arizonae* of Arizona, a state in the United States.

Distinguished from other *S. enterica* subspecies by the characteristics reported in Table BXII.γ.261. Contains at least 94 serovars (Popoff et al., 1998). All serovars belonging to this subspecies are monophasic for the H antigen. Isolated mainly from cold-blooded animals and environment.

Antigenic formula: 51:z₄,z₂₃:—.

The mol% G + C of the DNA is: see species description.

Type strain: ATCC 13314, CIP 82.30, NCTC 8297.

- c. *Salmonella enterica* subsp. *diarizonae* (Le Minor, Véron and Popoff 1985) Le Minor and Popoff 1987, 467 (*Salmonella choleraesuis* subsp. *diarizonae* Le Minor, Véron and Popoff 1985, 375.)

di.a.ri.zo'nae. Gr. adj. *dis* twice, two; M.L. gen. n. *arizonae* of Arizona, a state in the United States.

Distinguished from other *S. enterica* subspecies by the characteristics reported in Table BXII.γ.261. Contains at least 323 serovars (Popoff et al., 1998). All serovars belonging to this subspecies are diphasic for the H antigen. Isolated mainly from cold-blooded animals and environment.

Antigenic formula: 6,7:l,v:z₅₃.

The mol% G + C of the DNA is: see species description.

Type strain: ATCC 43973, CIP 82.31, NCTC 10060.

- d. *Salmonella enterica* subsp. *houtenae* (Kauffmann 1962) Le Minor and Popoff 1987, 467 (*Salmonella choleraesuis* subsp. *houtenae* (Kauffmann 1962) Le Minor, Rohde and Taylor 1970b, 209; *Salmonella houtenae* Kauffmann 1962, 353.)

hou'te.nae. M.L. gen. n. *houtenae* of Houten, a town in Holland.

Distinguished from other *S. enterica* subspecies by the characteristics reported in Table BXII.γ.261. Contains at least 70 serovars (Popoff et al., 1998). All serovars belonging to this subspecies are monophasic for the H antigen. Isolated mainly from cold-blooded animals and environment.

Antigenic formula: 41:g,z₅₁:—.

The mol% G + C of the DNA is: see species description.

Type strain: ATCC 43974, CIP 82.32, NCTC 10401.

- e. *Salmonella enterica* subsp. *indica* (Le Minor, Popoff, Laurent and Hermant 1987) Le Minor and Popoff 1987, 467 (*Salmonella choleraesuis* subsp. *indica* Le Minor, Popoff, Laurent and Hermant 1987, 179.)

in.di'ca. L. adj. *indica* of India.

Distinguished from other *S. enterica* subspecies by the characteristics reported in Table BXII.γ.261. Contains at least 11 serovars (Popoff et al., 1998). Isolated mainly from cold-blooded animals and environment.

Antigenic formula: 1,6,14,25:a:e,n,x.

The mol% G + C of the DNA is: see species description.

Type strain: ATCC 43976, CIP 102501.

- f. *Salmonella enterica* subsp. *salamae* (Le Minor, Rohde and Taylor 1970b) Le Minor and Popoff 1987, 467 (*Salmonella choleraesuis* subsp. *salamae* (Le Minor, Rohde and Taylor 1970b) Le Minor, Véron and Popoff 1985, 375) (*Salmonella salamae* Le Minor, Rohde and Taylor 1970b, 209.)

sa.la'mae. M.L. gen. n. *salamae* of (Dare-es) salaam.

Distinguished from other *S. enterica* subspecies by the characteristics reported in Table BXII.γ.261. Contains at least 488 serovars (Popoff et al., 1998). Isolated mainly from cold-blooded animals and environment.

Antigenic formula: 1,9,12:l,w:e,n,x.

The mol% G + C of the DNA is: see species description.

Type strain: ATCC 43972, CIP 82.29, NCTC 5773.

2. *Salmonella bongori* (Le Minor, Véron and Popoff 1985); Reeves, Evins, Heiba, Plikaytis and Farmer 1989b, 371^{VP} (Effective publication: Reeves, Evins, Heiba, Plikaytis and Farmer 1989a, 319) (*Salmonella choleraesuis* subsp. *bongori* Le Minor, Véron and Popoff 1985, 375.)

bon'gori. M.L. gen. n. *bongori* of Bongor, a town in Chad.

The characteristics are as described for the genus (see also Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*) and as listed in Table BXII.γ.261. All serovars belonging to *S. bongori* are monophasic for the H antigen. Contains at least 20 serovars (Popoff et al., 1998). Isolated mainly from cold-blooded animals and environment.

Antigenic formula: 66:z₄₁:—.

The mol% G + C of the DNA is: 51.8 ± 0.6 (*T_m*).

Type strain: ATCC 43975, CIP 82.33.

GenBank accession number (16S rRNA): AF029227.

Genus XXXIV. *Serratia* Bizio 1823, 288^{AL}

FRANCINE GRIMONT AND PATRICK A.D. GRIMONT

Ser.ra'ti.a. M.L. fem. n. *Serratia* named after Serafino Serrati, an Italian physicist.

Straight rods, 0.5–0.8 × 0.9–2.0 µm in length, with rounded ends. Conform to the general definition of the family *Enterobacteriaceae*. Gram negative, generally **motile**, by means of peritrichous flagella. **Facultatively anaerobic**. **Nitrate and chlorate are reduced anaerobically**. **Growth factors are generally not required**. Colonies on nutrient agar are most often opaque, somewhat iridescent, and either **white, pink, or red in color**. Almost all strains can grow at temperatures between 10 and 36°C, at pH 5–9, and in the presence of 0–4% (w/v) NaCl. The catalase reaction is strongly positive. **D-Glucose is fermented** through the Embden–

Meyerhof pathway. The major glucose entry route involves a phosphoenolpyruvate-dependent phosphotransferase system with both enzyme II^{Glc} (glucose permease) and enzyme II^{Man} (mannose permease). Glucose is also oxidized to gluconate in the presence of pyrroloquinoline quinone. **Gluconate is oxidized to 2-ketogluconate**. **Acetoin is produced from pyruvate** by all species except *S. fonticola*. Fructose, D-galactose, maltose, D-mannitol, D-mannose, ribose, and trehalose are fermented and utilized as sole carbon sources. L-fucose is fermented and utilized as sole carbon source by all species except *S. fonticola*. L-sorbose

is not fermented or utilized as sole carbon source. All species but *S. fonticola* fail to ferment or utilize dulcitol and tagatose. *N*-acetylglucosamine, D-alanine, L-alanine, citrate, D-galacturonate, D-glucosamine, D-glucuronate, 2-ketogluconate, L-proline, putrescine, L-serine are utilized as sole carbon sources by most strains. Caprate, caproate, **caprylate**, and tyrosine are **utilized** as sole carbon sources by all species except *S. fonticola*. 5-Aminovalerate, butyrate, *m*-coumarate, ethanolamine and tryptamine are not utilized as sole carbon sources. All species except *S. fonticola* fail to utilize 3-phenylpropionate. All species except *S. entomophila* fail to utilize itaconate. Phenylalanine, histidine, and tryptophan deaminases and thiosulfate reductase (H_2S from thiosulfate) are not produced. ***o*-Nitrophenyl- β -D-galactopyranoside (ONPG) is hydrolyzed by most strains.** Esculin is hydrolyzed by most strains except *S. proteamaculans* subsp. *quinovora*. **Extracellular enzymes** of all species except *S. fonticola* **hydrolyze DNA, lipids** (tributyrin, corn oil) **and proteins** (gelatin, casein), but not starch (in four days), polygalacturonic acid, or pectin. Tween-80 is hydrolyzed by all species except *S. odorifera*. The organisms **occur in the natural environment** (soil, water, plant surfaces) or **as opportunistic human pathogens**.

The mol% G + C of the DNA is: 52–60.

Type species: *Serratia marcescens* Bizio 1823, 288.

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic affinities among *Serratia* species have been studied (Fig. BXII.γ.208) (Dauga et al., 1990; Spröer et al., 1999). Three clusters can be observed in all studies: cluster I includes *S. marcescens* and *S. rubidaea*, cluster II includes psychrotrophic species (*S. proteamaculans*, *S. grimesii*, *S. liquefaciens*, *S. plymuthica*, and *S. fonticola*), and cluster III includes *S. ficaria* and *S. entomophila*. Depending on the analysis, *S. odorifera* may branch with cluster I or after clusters II and III have merged. Sequence comparisons leave no doubt as to the affiliation of *S. fonticola* to the genus *Serratia* and especially to the psychrophilic group of species.

Cells of *Serratia* rarely show a visible capsule in India ink mounts, although mucoid colonies can be observed in *S. plymuthica* and occasionally in other *Serratia* species; however, cells of *S. odorifera* possess a microcapsule which can be evidenced by the quellung reaction (capsular swelling) using *Klebsiella* anti-capsule K4 or K68 sera (Richard, 1979). Polysaccharides excreted by cells of *S. marcescens* can be extracted from the cell surface layer or from the culture medium. These polysaccharides contain chiefly D-glucose and glucuronic acid and lower proportions of

D-mannose, heptose, L-fucose and L-rhamnose (Adams and Martin, 1964; Adams and Young, 1965).

The major fatty acid components found in whole-cell methanolsates are $C_{14:0} 3OH$, $C_{16:0}$, $C_{16:1}$, $C_{18:1}$, and $C_{18:2}$, contributing 50–80% of the components in each strain. $C_{14:0}$ contributes 3.7–9.4% whereas other components contribute less than 3% each (Bergan et al., 1983).

Colony diameters are ~1.5–2.0 mm after overnight growth on nutrient agar. Swarming does not occur.

Two different pigments can be produced by various *Serratia* strains: prodigiosin and pyrimine (Williams and Qadri, 1980). Prodigiosin, a nondiffusible, water-insoluble, red pigment bound to the cell envelope, is produced by two biogroups (A1 and A2) of *S. marcescens* and by most strains of *S. plymuthica* and *S. rubidaea*. Prodigiosin-producing colonies are totally red or show either a red center, a red margin, or red sectors. The exact color given by the pigment depends upon cultural conditions (e.g., amino acids, carbohydrates, pH, inorganic ions, temperature) and may include orange, pink, red, or magenta. Prodigiosin is best produced on peptone-glycerol agar at 20–35°C. The temperature range for pigment production is 12–36°C. Prodigiosin is not produced anaerobically. Chemically, prodigiosin is a tripyrrole derivative, 2-methyl-3-amy-6-methoxyprodigiosene (prodigiosene is 5-(2-pyrrolyl)-2,2'-dipyrrolylmethene). In the cell, prodigiosin is formed by condensation of a volatile 2-methyl-3-amylopyrrol (MAP) and a nonvolatile 4-methoxy-2,2'-bipyrrole-5-carboxaldehyde (MBC). Several classes of nonpigmented mutants that are blocked on either the MAP pathway or the MBC pathway have been isolated. Syntrophic pigmentation may occur when two different class mutants are grown side by side (Williams and Qadri, 1980).

Pyrimine, a water-soluble, diffusible pink pigment (Williams and Qadri, 1980), is produced by some strains of *S. marcescens* biogroup A4. Ferrous iron is required for the production of pyrimine. Pyrimine is L-2-(2-pyridyl)-D'-pyrroline-5-carboxylic acid. When pyrimine is produced, the agar medium turns pink while the colonies are white to pinkish.

A yellow diffusible pigment, 2-hydroxy-5-carboxymethylmuconic acid semialdehyde, is produced from the meta cleavage of 3,4-dihydroxyphenylacetic acid (3,4-DHP) by the enzyme 3,4-DHP 2,3-dioxygenase (Trias et al., 1988), induced by tyrosine in all *S. marcescens* strains. At present, only *S. marcescens* strains of biotype A8a which have lost the ability to grow on aromatic compounds can produce the yellow pigment. A new reddish violet pigment, a peptide-ferropyrroline complex, produced by a *S. marcescens* O5:H1 was described by Suzuki et al. (1993b).

Cultures can produce two kinds of odors, a fishy to urinary odor attributed to trimethylamine (mixed with some NH_3), or a musty, potato-like odor attributed to alkyl-methoxypyrazine. The musty odor produced by *S. odorifera*, *S. ficaria*, and a few strains of *S. rubidaea* is due to 3-isopropyl-2-methoxy-5-methylpyrazine (Gallois and Grimont, 1985). All other strains and species produce the fishy-urinary odor.

All species except *S. marcescens* and *S. rubidaea* can grow readily at 4–5°C and several grow at 40°C (*S. marcescens* and several strains of *S. rubidaea* and *S. odorifera*); however, the temperature of 37°C is not favorable for the isolation of *S. plymuthica*. When *S. liquefaciens* and *S. plymuthica* are studied, many tests that are positive at 28–35°C give negative results at 37°C (e.g., Voges-Proskauer, decarboxylases, tetrathionate reductase tests).

A strong catalase activity, which can be evidenced with 3% (or

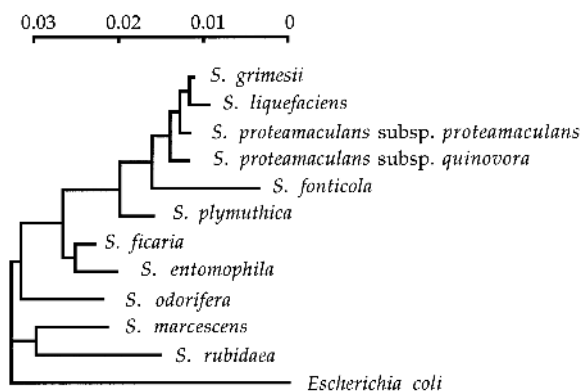


FIGURE BXII.γ.208. Phylogenetic tree of the genus *Serratia*. The tree was built by the neighbor-joining method (Saitou and Nei, 1987). Scale is in K_{nuc} according to Jukes and Cantor (1969).

less) H_2O_2 , is produced by *Serratia* strains (Taylor and Achanzar, 1972).

There is no sodium ion requirement for growth in the genus *Serratia*; however, the optimum concentration of NaCl for growth is ~0.5% (w/v) for *S. marcescens* or 1% (w/v) for *S. rubidaea* (Grimont and Grimont, unpublished results). Tolerance to NaCl ranges from 5 to 6% (w/v) for *S. plymuthica*, to 10% (w/v) for *S. rubidaea*.

In Biotype-100 strips (BioMerieux, Craponne, France) or a minimal medium containing ammonium sulfate as the nitrogen source, the following compounds serve universally as sole carbon sources for all *Serratia* strains: *N*-acetylglucosamine, *cis*-aconitate, citrate, *D*-fructose, *D*-galactose, *D*-galacturonate, *D*-gluconate, *D*-glucosamine, *D*-glucose, *myo*-inositol, 2-ketogluconate, maltose, maltotriose, *D*-mannitol, *D*-mannose, *D*-ribose, and *D*-trehalose. Most strains of all species can utilize *D*-alanine, *L*-alanine, *D*-glucuronate, glycerate, glycerol, lactate, putrescine, *L*-serine, and *L*-tryptophan. All species except some or all strains of *S. fonticola* can utilize *L*-aspartate, *L*-fucose, fumarate, *L*-glutamate, *L*-malate, *L*-proline, and succinate. Utilization of caprate, caproate, and caprylate by all species except *S. fonticola* is better observed in a minimal agar containing ammonium sulfate as the nitrogen source, rather than in Biotype-100 strips. The following compounds are never utilized as sole carbon sources (Biotype-100 strips): *DL*-5-aminovalerate, *m*-coumarate, ethanolamine, glutarate, sorbose, and tryptamine. The following carbon sources are never utilized by *Serratia* species except some or all *S. fonticola* strains: dulcitol, 3-phenylpropionate, and tagatose (Grimont et al., 1977b, 1978b, 1979b; Grimont and Grimont, 1978b).

Characteristic extracellular enzymes are produced by most species. All species except *S. fonticola* can hydrolyze DNA, gelatin, soluble casein, tributyrin, and corn oil. Only rare strains fail to produce one or more of these extracellular enzymes. All species, except *S. odorifera*, can hydrolyze Tween 80. Chitin is hydrolyzed by all species except *S. rubidaea*, *S. odorifera*, and *S. fonticola*. Lecithin is also hydrolyzed by many strains. Spot-inoculated starch agar (Starch agar: nutrient agar containing 0.5% [w/v] soluble starch), incubated for four days and then flooded with Lugol's iodine, shows no zone of clearing (Grimont et al., 1977b); however, longer incubation (6–14 days) may allow detection of some amylase-producing strains (M. Popoff, personal communication).

A red-pigmented *S. marcescens* has been found to produce a carboxymethyl cellulase (Thayer, 1978). Depolymerization of a carboxymethylcellulose gel is faster with *S. marcescens*, *S. rubidaea*, and *S. liquefaciens* than with *S. odorifera*, *S. ficaria*, and *S. plymuthica* (unpublished results).

Up to 11 proteinases have been revealed by agar gel electrophoresis. Each strain produces one to four different proteinases. Different species have different proteinase patterns (Grimont et al., 1977a). These have been used to type strains (Grimont and Grimont, 1978c). Isoelectric points of the 11 proteinases are between pH 3.6 and pH 6.0 (Grimont and Grimont, 1978b).

Fructose, maltose, *D*-mannitol, *D*-mannose, ribose, and trehalose are fermented by all strains. Most strains ferment glycerol and *myo*-inositol. Fermentation of *D*-glucose is not prevented by 0.001 M iodoacetate (Grimont et al., 1977b, 1978b, 1979b; Grimont and Grimont, 1978b), an inhibitor of the Embden–Meyerhof–Parnas glycolytic pathway and other enzymic reactions. *Serratia* species can produce gluconate-6-phosphate dehydrase and 2-keto-3-deoxygluconate-6-phosphate aldolase (Kerstens and De Ley, 1968), which are the characteristic enzymes of the Entner–Doudoroff pathway.

Under aerobic conditions, gluconate is produced from *D*-glucose in the presence of iodoacetate by all *Serratia* species due to a glucose dehydrogenase, when pyrroloquinoline quinone (PQQ) is added. Only *S. marcescens*, *S. odorifera*, *S. rubidaea*, *S. entomophila*, *S. ficaria*, and some strains of *S. plymuthica* can produce gluconate without added PQQ (Bouvet et al., 1989).

A reducing compound, 2-ketogluconate, is also produced from gluconate by all species due to a gluconate dehydrogenase (Grimont et al., 1977b, 1978a, 1979b; Bouvet et al., 1989).

2,5-Diketogluconate is produced from 2-ketogluconate at 20°C (not at 30°C) due to a 2-ketogluconate dehydrogenase by *S. marcescens*, *S. liquefaciens*, and *S. grimesii* (Bouvet et al., 1989).

In anaerobic glycerol dissimilation, a glycerol dehydrogenase induced by glycerol and not by hydroxyacetone is present in *S. marcescens*, *S. proteamaculans*, *S. liquefaciens*, *S. grimesii*, and *S. fonticola*, but not in *S. rubidaea*, *S. ficaria*, *S. odorifera*, *S. entomophila*, and *S. plymuthica* (Bouvet et al., 1995a). The Voges–Proskauer (VP) test, when done on a 3-day-old culture in Clark–Lubs medium, is negative for 40% of strains of *S. plymuthica*, although acetoin can be detected after incubation for 18 h by use of a sensitive method (Richard, 1972). These strains, which are VP-negative after 3 days of incubation, can utilize 2,3-butanediol as a sole carbon source (Grimont et al., 1978b). *Serratia* strains, other than *S. fonticola*, that cannot produce acetoin from pyruvate (under any experimental conditions) are very rare. *S. fonticola* is always negative for acetoin. A tiny gas bubble is commonly produced by *S. marcescens* in a peptone–water–glucose medium with Durham tube. *S. plymuthica* and *S. liquefaciens* produce a large amount of gas. The end products of glucose fermentation by *S. marcescens* are 2,3-butanediol, ethanol, formate, lactate, succinate, and CO_2 with small amounts of acetate, acetoin, and glycerol and very little or no H_2 (Neish et al., 1948; White and Starr, 1971). The end products yielded by *S. plymuthica* are 2,3-butanediol, ethanol, lactate, succinate, CO_2 , H_2 , and small amounts of formate, acetate, acetoin, and glycerol (Neish et al., 1948). The 2,3-butanediol produced by *S. marcescens* is mostly a *meso*-isomer, whereas *S. plymuthica* is unique in producing a *levo*-rotatory 2,3-butanediol (Neish et al., 1948).

Transduction systems in *S. marcescens* have been described (Kaplan and Brendel, 1969; Matsumoto et al., 1973). The earliest genetic transfer described in *S. marcescens* (Belser and Bunting, 1956) is also suggestive of a transduction mechanism. *S. marcescens* was transformed with plasmid pBR322 by Reid et al. (1982). Transformants were selected based on resistance to high levels of ampicillin.

All species of the genus *Serratia* have been delineated by DNA–DNA hybridization. All species of the genus except *S. fonticola* share a number of phenotypic properties. *S. fonticola* was included in the genus *Serratia* because of significant DNA relatedness with other *Serratia* species (Steigerwalt et al., 1976; Gavini et al., 1979); this inclusion was confirmed by comparison of 16S rDNA gene sequences (Dauga et al., 1990). A summary of DNA relatedness within the genus *Serratia* is given in Table BXII.γ.264.

Lactose plasmids have been demonstrated in *S. liquefaciens* (Le Minor et al., 1974) and in *S. marcescens* (C. Coynault, personal communication). Antibiotic resistance plasmids of incompatibility groups S (= H_2), C, L/M, P, W, and FII have been identified in *S. marcescens*. Plasmids of groups M and N have been found in *S. liquefaciens* (Hedges, 1980). Replicon typing, using cloned DNA probes, identified plasmid groups FIB, FIC, FIIA, H12, L/M, N, B/O, P, W, Y, and Com9 in multiresistant *S. marcescens* strains (Llanes et al., 1994).

TABLE BXII-7.264. DNA relatedness among *Serratia* species^{a,b}

Species	Source of labeled DNA										
	<i>S. marcescens</i> ^c	<i>S. entomophila</i>	<i>S. fonticola</i>	<i>S. ficaria</i> ^d	<i>S. grimesii</i>	<i>S. liquefaciens</i>	<i>S. odorifera</i> ^e	<i>S. phymuthica</i>	<i>S. proteamaculans</i> subsp. <i>proteamaculans</i>	<i>S. proteamaculans</i> subsp. <i>quinovora</i>	<i>S. rubidaea</i>
<i>S. marcescens</i>	92 ± 5.5 d = 0-3.2 ND	42 ± 6.3 d = 7.5-8.0	50 ± 5.1	46 ± 4.7 d = 8.5-9.5	36	29	31 ± 2.8 d = 10.5-11.5	29	34	33	26 ± 2.1
<i>S. entomophila</i>		84 ± 8.2 d = 0-1.0	ND	62 ± 5.4 d = 6.0-9.0	ND	ND	ND	ND	ND	ND	ND
<i>S. fonticola</i>	57 ± 4.9 d = 12.0-14.0	28	88 ± 5.5	40 d = 10.0	22	23	38 ± 2.8 d = 13.0	21	29	22	16
<i>S. ficaria</i>	ND	47 ± 6.4 d = 5.5-8.5	ND	91 ± 4.2 d = 0-2.0	44	38	ND	44	41	47 d = 10.0	27 ± 3.4
<i>S. grimesii</i>	53 ± 1.4 d = 12.0-12.5	34 ± 6.0	ND	29	97 ± 6.4 d = 0-4.0	36 ± 3.8 d = 9.5-14.0	32 ± 1.1 d = 13.5	36 ± 4.6 d = 11.0-15.5	42 ± 5.5 d = 9.5-10.5	44 ± 3.4 d = 8.5-15.0	18 ± 2.6
<i>S. liquefaciens</i>	62 ± 1.1 d = 10.5-12.5	48	41 ± 2.1	33 ± 1.7	47 ± 7.0 d = 10.0-11.5	86 ± 7.5 d = 0-3.0	39 ± 3.5 d = 11.5	44 ± 5.1 d = 8.5-12.5	45 ± 7.1 d = 7.5-9.5	48 ± 5.7 d = 8.0-9.5	19 ± 5.0
<i>S. odorifera</i>	ND	40	ND	33 ± 2.8 d = 16.5	29	26	78 ± 13.9 d = 0-4.0	28	28	25	21 ± 4.2
<i>S. phymuthica</i>	62 d = 10.0	42	44 ± 5.7	43 ± 4.6 d = 13.0	43 ± 4.1 d = 9.5-12.0	39 ± 5.2 d = 8.5-10.0	38 ± 2.0 d = 11.0	85 ± 14.6 d = 0-4.5	52 ± 2.3 d = 8.5-9.5	44 ± 4.7 d = 9.0-12.0	20 ± 2.5
<i>S. proteamaculans</i> subsp. <i>proteamaculans</i>	ND	40	ND	41 ± 5.6	52 ± 5.6 d = 9.0-13.0	42 ± 5.4 d = 9.0-13.0	ND	45 ± 5.7 d = 9.0-12.0	81 ± 11.0 d = 0-6.5	64 ± 5.3 d = 5.5-12.0	21 ± 4.8
<i>S. proteamaculans</i> subsp. <i>quinovora</i>	ND	ND	ND	ND	52 ± 3.5 d = 8.5-10.0	44 ± 2.5 d = 8.5-10.5	ND	48 ± 4.4 d = 9.0-12.0	58 ± 7.4 d = 5.0-6.0	88 ± 6.7 d = 0-4.5	ND
<i>S. rubidaea</i>	49 ± 1.7 d = 10.0-11.0	34	38 ± 18.7	26 ± 3.0 d = 12.0	24	24	32 ± 3.2 d = 10.0	26	26	30	88 ± 20.6 d = 0-6.0

^aAll DNA relatedness values obtained with the S1 nuclease method except otherwise stated. Divergence (d) given rounded to the nearest 0.5.^bData from Steigerwalt et al. (1976), Grimont et al. (1978a, 1979b, 1982b, 1988), and Gavini et al. (1979). Data in the *S. rubidaea* column are unpublished.^cDNA relatedness values obtained with the hydroxyapatite method.^dDNA relatedness values obtained with the filter method.

Bacteriophages active on *Serratia* are easily found in river water or sewage. Phages that are active on one species of *Serratia* are usually active on strains of other species of that genus, but rarely on strains of other genera (Grimont and Grimont, 1978b). Lysozyme is very common in *Serratia* species (Prinsloo, 1966). Several phage typing systems have been studied (Pillich et al., 1964; Hamilton and Brown, 1972; Farmer, 1974; Grimont, 1977a).

Bacteriocins produced by *Serratia* are of two kinds: (a) a trypsin-resistant, acid-sensitive (pH 2) structure (Hamon and Péron, 1961) called "group A bacteriocin" by Prinsloo (1966) and later found by electron microscopy to resemble phage tails (Traub, 1972); and (b) a trypsin-sensitive, acid-resistant protein (Hamon and Péron, 1961) called "group B bacteriocin" by Prinsloo (1966). Bacteriocins produced by one species of *Serratia* frequently cross-react with other species of this same genus. *Serratia* bacteriocins are also frequently active on *Escherichia coli* K12. *S. marcescens* strains produce group A and/or group B bacteriocins. *S. rubidaea* strains produce only group A bacteriocins. *S. liquefaciens* and *S. ficaria* produce only group B bacteriocins. *S. odorifera* produces neither group A nor group B bacteriocins (Hamon and Péron, 1979; Y. Hamon, personal communication). Bacteriocin typing can be used for epidemiological purposes (Traub, 1980).

The antigenic structure of *S. marcescens* has been described. The present scheme consists of 28 somatic antigens (O1 to O28) and 25 flagellar antigens (H1 to H25) (Edwards and Ewing, 1972; Le Minor and Pigache, 1978; Traub and Fukushima, 1979a, b; Le Minor and Sauvageot-Pigache, 1981; Traub, 1991; Aucken et al., 1996). Subdivision of antigens O5 (into O5a, O5b, O5c), O10 (into O10a, O10b), and O16 (into O16a, O16b, O16c, O16d) has been proposed (Le Minor and Sauvageot-Pigache, 1981). Cross-reactions between factors O6 and O14 are very extensive and the distinction between these two factors does not seem worthwhile. Serovar O27 cross-reacts with the O4 serovar strain, and serovar O28 with the O5 serovar strain. The O-typing scheme was improved by separating capsular material from O-antigen (Gaston and Pitt, 1989a, b). H antigens are monophasic in *S. marcescens* (Aucken et al., 1996).

Four serovars (O1:H1, O2:H1, O3:H1, and O4:H1), all sharing a common H antigen, were described among *S. ficaria* strains (Grimont and Deval, 1982).

Resistance to cephalothin, colistin, and polymyxin (with respect to achievable serum levels of antibiotics) is very frequent in the genus and almost constant in *S. marcescens*. With the antibiotic disk method, a zone phenomenon develops around disks impregnated with colistin and polymyxin: the inhibition zone contains colonies close to the disk. However, this zone phenomenon is not restricted only to *Serratia*. Resistance to tetracycline and ampicillin is very frequent in *S. marcescens* and rare in other *Serratia* species. Plasmid-determined resistance to aminoglycoside antibiotics, carbenicillin, chloramphenicol, trimethoprim, sulfonamides, and mercury ions can be found in clinical strains of *S. marcescens*. "Third generation" cephalosporins are still active on *S. marcescens* and in a multicenter survey in the USA, fewer than 8% of *S. marcescens* isolates were resistant to piperacillin, 3–4% were resistant to ceftazidime, ceftriaxone, and/or cefotaxime, and 0.3% were resistant to imipenem (Jones, 1998). Resistance to cetyl trimethylammonium chloride (1.5 mg/ml) and thallus acetate (0.8 mg/ml) is very frequent (Grimont et al., 1977b). Of all the *Serratia* species, *S. marcescens* is the most resistant to antibiotics, antiseptics, and metal ions; *S. plymuthica* is the least resistant to these antimicrobials.

Healthy humans are not often infected by *Serratia*. *S. marcescens*

is a prominent opportunistic pathogen for hospitalized human patients. At present, *S. marcescens* is the only known nosocomial species of *Serratia*. Clinically, *Serratia* infections do not differ from infections by other opportunistic pathogens (von Graevenitz, 1980). Other *Serratia* species can be involved in respiratory tract infection or colonization and septicemia, especially when these bacteria are accidentally injected into the body (e.g., contaminated perfusion or irrigation liquid) (Grimont and Grimont, 1978b).

S. marcescens and *S. liquefaciens* are known to infect and cause mortality in a variety of insects which can be serious pests of crops, ornamentals, and turf throughout the world (Klein and Kaya, 1995). Commercial utilization of *S. entomophila* against the grass grub *Costelytra zealandica* in New Zealand pastures has been achieved (Klein and Jackson, 1992). This bacterium turns the grubs a honey or amber color. Pathogenic strains of *S. entomophila* colonize the larva gut, adhere to the crop, and induce starvation, which causes depletion of the fat bodies. Pathogenicity is correlated with the production of lecithinase, proteinase, and chitinase (Kaska, 1976; Lysenko, 1976).

Mastitis in cows and other animal infections have been associated with *Serratia* species (Grimont and Grimont, 1978b). Pathogenicity in experimental animals is of the type expected of a Gram-negative bacterium. Experimental depression of phagocytic cell number or function in animals enhances susceptibility to *Serratia* infections (Simberloff, 1980).

A typical hypersensitivity reaction is produced by inoculation of plants such as tobacco and king protea with *Serratia* (Lakso and Starr, 1970; Grimont et al. 1978b). *S. proteamaculans* was isolated from a leaf spot disease of *Protea cynaroides* (Paine and Stanfield, 1919) and *S. marcescens* (under the name *Erwinia amylovora* var. *alfalfae*) was isolated from a root disease of alfalfa (Shinde and Lukezic, 1974a).

Serratia species occur on plants, in the digestive tract of rodents (unpublished data), and in soil and water. *S. ficaria* is especially associated with the fig/fig-wasp ecosystem (Grimont et al., 1979b).

ENRICHMENT AND ISOLATION PROCEDURES

Fecal samples (diluted with distilled water) or plant material washings are inoculated onto caprylate–thallus (CT) agar (Starr et al., 1976). After 2–5 d, the growth is removed by scraping and tested for deoxyribonuclease (DNase) activity. DNase-positive cultures are then purified by streaking on a nonselective medium (e.g., tryptic soy agar). Different colonial types are tested for DNase, and DNase-positive isolates are then thoroughly characterized and identified. This procedure allows isolation of all *Serratia* species as defined in this chapter. *Providencia*, *Acinetobacter*, and fluorescent *Pseudomonas* strains can grow on CT agar when samples contain large numbers of these organisms. Other selective media based on DNase production and antibiotic resistance have been proposed (Cate, 1972; Berkowitz and Lee, 1973; Farmer et al., 1973). These antibiotic-containing media are efficient for the isolation of *S. marcescens* but may not be as reliable for more sensitive species (e.g., *S. plymuthica*).

MAINTENANCE PROCEDURES

For short-term preservation (several months), heavy suspensions of bacteria in sterile distilled water are made from bacterial growth scraped with a platinum loop from a nutrient agar slant. The suspensions are stored at room temperature. For longer preservation (several years), screw-capped tubes containing semi-

solid nutrient agar are stab-inoculated. After overnight growth at 30°C, the tubes are tightly closed and kept at room temperature in the dark. Maintenance failure may occur if the tube is not protected from desiccation by a rubber seal in the screw cap. Rubber corks dipped in melted paraffin wax may be preferred in place of screw caps.

Bacterial suspensions in brain–heart infusion supplemented with 50% glycerol and glass beads can be frozen at –80°C (cryoconservation). When needed, a glass bead can be taken without thawing the cryoconservation and transferred to a sterile broth for subculture. For long-term preservation (over 5 years), freeze-drying is preferred.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Carbon source utilization test This test is done by using Biotype-100 strips (bioMérieux, Craponne, France) that contains 99 pure carbon sources. Bottles of Biotype medium 1 are inoculated with a calibrated bacterial suspension (Grimont et al., 1996). The strips are incubated at 30°C for 4 d. Growth is scored visually after 2 and 4 days by comparison with the control cupule (without carbon source). The incubation day when growth is observed (1–4) is recorded for each cupule. Program Recognizer (Taxotron package, Institut Pasteur, Paris, France) can be used to enter the Biotype-100 data in an Apple Macintosh for automatic identification (Grimont et al., 1996).

Glucose oxidation test A 1-liter portion of glucose oxidation medium is composed of basal medium containing 8 g of nutrient broth (Difco Laboratories, Detroit, Mich.), 0.02 g of bromocresol purple, and 0.62 g of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (2.5 mM). To 9.2 ml of autoclaved (121°C, 15 min) basal medium is added 0.4 ml of a filter-sterilized 1 M D-glucose solution (final concentration, 40 mM). This medium is supplemented with 0.4 ml of a fresh, sterile 25 mM iodoacetate solution (final concentration, 1 mM). The glucose oxidation medium is distributed (0.5-ml portions) into glass tubes (11 by 75 mm), which are plugged with sterile cotton wool. Bacteria grown overnight at 20 or 30°C on tryptocasein soy agar (Biorad, Marnes-la Coquette, France) supplemented with 0.2% (w/v) D-glucose are collected with a platinum loop, suspended in a sterile 2.5 mM MgSO_4 solution and adjusted to an absorbance at 600 nm of about 4. Glucose oxidation medium is then inoculated with 50 µl of bacterial suspension and vigorously shaken at 270 strokes per minute overnight at 20 or 30°C. The test is positive when a yellow color develops (acid production). In a negative test, the medium remains purple. When negative, the glucose oxidation test is repeated in the presence of 10 µM pyrroloquinoline quinone (PQQ). The control strains used are the type strains of *S. marcescens* (positive without requirement for PQQ), *S. liquefaciens* (positive only when PQQ is supplied), and *Hafnia alvei* (negative) (Bouvet et al., 1989).

Gluconate- and 2-ketogluconate dehydrogenase tests The gluconate dehydrogenase test is done as follows (Bouvet et al., 1989). Bacteria are grown overnight at 20°C on tryptocasein soy agar supplemented with 0.2% D-gluconate, then collected with a platinum loop and suspended in sterile distilled water to an absorbance (at 600 nm) of about 4. The reaction medium contains 0.2 M acetate buffer (pH 5), 1% (w/v) Triton X-100, 2.5 mM MgSO_4 , 75 mM gluconate and (added immediately before use) 1 mM iodoacetate. A control medium contains the same ingredients except gluconate. The reaction medium and the control medium are dispensed (100-µl portions) into 96-well microtiter plates (Dynatech AG, Denkendorf, FRG). Bacterial suspen-

sions (10 µl) are added to the reaction and control media and the microtiter plates are incubated at 20°C for 20 min. Then, 10-µl portions of a 0.1 M potassium ferricyanide solution (kept in the dark at room temperature for no longer than 1 week) are added to the wells and the plates are gently shaken and incubated at 20°C for 40 min. Then 50-µl portions of a reagent ($\text{Fe}_2(\text{SO}_4)_3$, 0.6 g; SDS, 0.36 g; 85% phosphoric acid, 11.4 ml; distilled water to 100 ml) are added to the wells. The plates are examined for the development of a green to blue color (due to Prussian blue) within 15 min at room temperature. The color in the uninoculated control medium remains yellow. The suggested control strains used are *Escherichia coli* K12 (negative) and the type strain of *Serratia marcescens* (positive).

The 2-ketogluconate dehydrogenase test is the same as above, except that the control and reaction media are adjusted to pH 4.0 and 2-ketogluconate is used in place of gluconate in the reaction medium. Suggested control strains are *Escherichia coli* K12 (negative) and the type strain of *Serratia marcescens* (positive at 20°C, not at 30°C) (Bouvet et al., 1989).

Glycerol dehydrogenase test (Bouvet et al., 1995a) For the detection of glycerol dehydrogenase, bacteria are grown overnight at 30°C on tryptocasein soy agar plates (TCS, BioRad, Marnes-la-Coquette, France) supplemented with 1% glycerol or 70 mM hydroxyacetone. Bacterial growth is collected, suspended in reaction buffer [0.1 M K_2CO_3 and 30 mM $(\text{NH}_4)_2\text{SO}_4$, adjusted to pH 9.0] to an absorbance of 0.6, and bacterial suspensions are dispensed into 96-well microtiter plates. Then, 30-µl aliquots of a reagent (NAD^+ , 210 mg; glycerol, 600 ml; nitro-blue tetrazolium, 42 mg; phenazine methosulfate, 2 mg; distilled water to 10 ml) are added to the wells and incubated in the dark with shaking. The plate is examined for a purple color developed within 15 to 30 min (presence of a glycerol dehydrogenase). The color in the uninoculated control medium remains yellow. Suggested control strains are the type strain of *Serratia liquefaciens* (glycerol dehydrogenase induced by glycerol, not by hydroxyacetone) and the type strain of *Enterobacter asburiae* (dehydrogenase induced by hydroxyacetone, not by glycerol) (Bouvet et al., 1995a).

Voges–Proskauer test (Richard’s modification) Clark and Lubs medium (BBL) is dispensed in large 22 × 215 mm tubes (0.5 ml per tube) and inoculated with 0.05 ml of a heavy bacterial suspension in distilled water. After incubation at 30°C for 18 h, 0.5 ml of α-naphthol solution (6% w/v alcoholic solution) and 0.5 ml of 4 M NaOH are added. The tubes are shaken, heated a few seconds in a Bunsen flame, and examined for a red color (Richard, 1972).

Tetrathionate reduction The medium of Le Minor et al. (1970a) contains: peptone (Difco), 10.0 g, NaCl, 5.0 g; $\text{K}_2\text{S}_4\text{O}_6$, 5.0 g; bromothymol blue (0.2% aqueous solution), 25 ml; and distilled water to 1 liter. Adjust the pH to 7.4, sterilize by filtration, and dispense into 12 × 120 mm tubes (4 ml per tube). The size of the tubes (for a rather limited aeration) is critical. Inoculated tubes are incubated at 30°C for 24 h and examined for a yellow color (tetrathionate reduction).

β-Xylosidase Paper disks (0.5 cm) are loaded with 0.1 ml of a 2% (w/v) aqueous solution of p-nitrophenyl-β-D-xylopyranoside and kept dry in a tightly-capped flask at 4°C. The test is performed exactly like the β-galactosidase test, but with p-nitrophenyl-β-D-xylopyranoside disks in place of ONPG disks (Brisou et al., 1972).

H-Immobilization test The motility of each isolate to be typed must be enhanced by passage through a 0.3% semisolid agar U-tube.

The following autoclaved semisolid medium is dispensed in 2.0-ml volumes into small (92 × 13 mm) screw-capped tubes: tryptic peptone, 20.0 g; D-mannitol, 2.0 g; KNO₃, 1.5 g; phenol red solution (1%), 4 ml; agar, 4.5 g; distilled water, 1000 ml; pH 7.4. The tubes of semisolid medium are melted (boiling water bath), cooled to 50°C in a water bath, supplemented with 0.05 ml of each serum dilution under sterile conditions, and allowed to gel.

Tubes with serum dilutions (and control tubes without serum) are stab-inoculated with a highly motile culture. After overnight incubation, tubes are examined for immobilization. This H-immobilization test is very specific and much easier to perform than the classical H-agglutination (Le Minor and Pigache, 1977).

DIFFERENTIATION OF THE GENUS *SERRATIA* FROM OTHER GENERA

Table BXII.γ.265 provides the primary characteristics that can be used to differentiate the genus *Serratia* from biochemically similar taxa.

TAXONOMIC COMMENTS

A number of changes have been made since the eighth edition of the *Manual of Determinative Bacteriology*, in which it was indicated that the genus *Serratia* was composed of only one species, *S. marcescens* (the type species). Objective approaches such as numerical taxonomy and DNA relatedness applied to strains recovered from diverse habitats delineated an increasing number of species in the genus. Seven species were mentioned in the first edition of the *Bergey's Manual of Systematic Bacteriology* (Grimont and Grimont, 1984) and 10 species were mentioned in the second edition of *The Prokaryotes* (Grimont and Grimont, 1995). Ten species are currently known to belong in the genus *Serratia*.

Transfer of *Enterobacter liquefaciens* to the genus *Serratia* was first proposed by Barbe (1969) and supported by studies on bacteriocin cross-reactions between *S. marcescens* and *E. liquefaciens*

(Hamon et al., 1970). Valid publication of the new combination *S. liquefaciens* followed a numerical taxonomy study (Bascomb et al., 1971).

A phenon named "biovar 2" (Bascomb et al., 1971) and "phenon B" (Grimont and Dulong de Rosnay, 1972) was thought identical to "*Bacterium rubidaea*" Stapp 1940 and named *S. rubidaea* (Ewing et al., 1973). The same phenon was also identified as *S. marinorubra* Zobell and Upham 1944 (Grimont et al., 1977b). *S. rubidaea* and *S. marinorubra* were based on different type strains (ATCC 27593 and ATCC 27614, respectively). The *Approved Lists of Bacterial Names*, however, gives both names (*S. rubidaea* and *S. marinorubra*) with the same type strain (viz. ATCC 27614, the type strain of *S. rubidaea*). Hence, both names, which were subjective synonyms, are now objective synonyms and redundant. To avoid further confusion, the name *S. rubidaea* (Stapp) Ewing (1986a) should now be used exclusively to designate the same (*S. rubidaea*-*S. marinorubra*) taxon.

The ancient species *S. plymuthica* (Lehmann and Neumann 1896) Breed, Murray, and Hitchens 1948 was shown to be a valid species by numerical taxonomy (Grimont et al., 1977b) and by DNA-DNA hybridization (Grimont et al., 1978a).

Three species, *S. odorifera* (Grimont et al., 1978a), *S. ficaria* (Grimont et al., 1979b), and *S. entomophila* (Grimont et al., 1988), were defined by DNA relatedness, carbon source utilization tests, and by standard biochemical tests. DNA relatedness studies have shown that *S. marcescens*, *S. plymuthica*, *S. rubidaea*, *S. odorifera*, and *S. ficaria* are homogeneous and discrete genomospecies (Steigerwalt et al., 1976; Grimont et al., 1978a, 1979b, 1988).

S. liquefaciens sensu lato was shown to be heterogeneous (Steigerwalt et al., 1976) and later found to be composed of several genomospecies. One biovar (Clc) of *S. liquefaciens* was identified as *Erwinia proteamaculans* (Paine and Stanfield 1919) Dye 1966a and renamed *S. proteamaculans* (Grimont et al., 1978b). When DNA binding ratios were examined without studies on the thermal stability of hybridized molecules, *S. proteamaculans* was thought to be a subjective synonym of *S. liquefaciens* (Grimont et al., 1978b). Reexamination of DNA relatedness (including thermal stability studies) in *S. liquefaciens sensu lato* disclosed at least

TABLE BXII.γ.265. Differential characteristics of the genus *Serratia* and other biochemically similar taxa^a

Characteristics	<i>Serratia</i> spp. ^b	<i>Serratia fonticola</i>	<i>Pantoea</i>	<i>Enterobacter cloacae</i>	<i>Pectobacterium</i> ^c	<i>Klebsiella</i> ^d
<i>Carbon source utilization:</i> ^e						
4-Aminobutyrate (M)	+	—	+	d	—	d
4-Aminovalerate (B,M)	—	—	—	—	—	d
Arginine (M)	—	—	—	—	—	+
Caprate (M)	+	—	—	—	—	—
Caproate (M)	+	—	—	—	—	—
Caprylate (M)	+	—	—	—	—	—
D-Dulcitol (B,M)	—	+	—	—	—	—
L-Fucose (B,M)	+	—	—	—	—	+
3-Phenylpropionate (B,M)	—	+	—	—	—	—
Tagatose (B,M)	—	+	—	—	—	d
Tyrosine (M)	+	—	—	—	—	d
Voges-Proskauer test	+	—	+	+	+	+
Gelatin hydrolyzed	+	—	d	d	d	d
Tributyrin hydrolyzed	+	—	—	—	—	—
Deoxyribonuclease	+	—	—	—	d	—
Gluconate dehydrogenase	+	+	+	—	—	d
Mol% G+C of DNA	52–60	49–52	53–56	53	51–54	54–57

^aSymbols: see standard definitions.

^bExcept *S. fonticola*.

^cAccording to Bouvet et al. (1989).

^dIncluding *Klebsiella pneumoniae* and *K. mobilis* (*Enterobacter aerogenes*).

^eDetermined in Biotype-100 (B) or minimal agar (M).

three genomospecies: *S. liquefaciens sensu stricto*, *S. proteamaculans* (Grimont et al., 1982b), and a third group containing strain ATCC 14460 and named *S. grimesii* (Grimont et al., 1982a, b).

A group of strains called "*Citrobacter lysine* + " or "*Citrobacter*-like" was found to be significantly related to the genus *Serratia* in DNA-DNA hybridization studies (Crosa et al., 1974). This genomospecies has been named *Serratia fonticola* Gavini et al. 1979; however, a difficulty is that *S. fonticola* does not have the key characteristics of the genus *Serratia*. Furthermore, *Serratia* phages that are active on strains of any *Serratia* species (as defined herein) have been found to be inactive on all *S. fonticola* strains tested (unpublished data). Bacteriocins from *Serratia* are also inactive on *S. fonticola* (Hamon, personal communication). However, the 16S rRNA gene sequence of *S. fonticola* branches within the psychrotolerant *Serratia* cluster (*S. liquefaciens*, *S. proteama-*

culans, *S. grimesii*, and *S. plymuthica*) and this justifies the inclusion of *S. fonticola* in the genus *Serratia* (Dauga et al., 1990).

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *SERRATIA*

The differential characteristics of the species of *Serratia* are indicated in Table BXII.γ.266. Other characteristics of the species are listed in Table BXII.γ.267.

List of species of the genus *Serratia*

1. *Serratia marcescens* Bizio 1823, 288^{AL}

mar.ces' *cens.* M.L. v. *marcesco* to fade; L. part. adj. *marcescens* fading away.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin or pyrimine can be produced.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

A biotyping system based on pigment production, tetrathionate reduction and utilization of *meso*-erythritol, trigonelline, quinate, benzoate, 3-hydroxybenzoate, 4-hydroxybenzoate, and DL-carnitine as sole carbon sources, has been described (Grimont and Grimont, 1978a). DL-carnitine is not in Biotype-100 strips and can be replaced by D-malate and *meso*-tartrate. Groups of biovars (called biogroups) (Table BXII.γ.268) correspond to definite, nonoverlapping sets of serovars (Table BXII.γ.269) (Grimont et al., 1979a).

Nonpigmented biogroups A3 and A4 are ubiquitous. Nonpigmented biogroups A5/8 and TCT are almost confined to hospitalized patients. Pigmented biogroups A1 and A2/6 are found in the natural environment and occasionally in human patients.

The mol% G + C of the DNA is: 57.5-60 (T_m , Bd).

Type strain: ATCC 13880, CIP 103235, DSM 30121, DSM 47, JCM 1239, NCDC 813-60, NCIB 9155, NCTC 10211.

GenBank accession number (16S rRNA): AJ233431, M59160.

2. *Serratia entomophila* Grimont, Jackson, Ageron and Noonan 1988, 5^{VP}

en.to.mó'phi.la. Gr. n. *entomon* insect; Gr. v. *phylein* love; L. fem. adj. *entomophila* insect loving.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is not produced.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

Two biotypes can be delineated (Table BXII.γ.270).

Isolated from larvae of *Costelytra zealandica* (grass grub) with amber disease, and from the environment. No strain

has been identified as being involved in a human, animal (other than insect), or plant disease.

The mol% G + C of the DNA is: 58 (T_m).

Type strain: A1, ATCC 43705, CIP 102919, DSM 12358.

GenBank accession number (16S rRNA): AJ233427.

3. *Serratia ficaria* Grimont, Grimont and Starr 1981d, 216^{VP} (Effective publication: Grimont, Grimont and Starr 1979b, 282.)

fi.ca'ri.a. M.L. fem. adj. *ficaria* of figs.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is not produced. Cultures give off a musty, potato-like odor.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

Associated with the fig/fig-wasp biological cycle. Occasionally found on plants other than fig trees.

The mol% G + C of the DNA is: 59.6 (T_m).

Type strain: ATCC 33105, CIP 79-23, DSM 4569, ICPB 4050, JCM 1241.

GenBank accession number (16S rRNA): AJ233428, AB004745.

4. *Serratia fonticola* Gavini, Ferragut, Izard, Trinel, Leclerc, Lefebvre and Mossel 1979, 98^{AL}

fon.ti'co.la. M.L. n. *fons*, *fontis* spring, fountain; L. suff. *-cola* dweller; M.L. n. *fonticola* spring-dweller.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is not produced.

Does not share the key characteristics of the genus *Serratia*.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

Occurs in freshwater.

The mol% G + C of the DNA is: 48.8-52.5 (T_m).

Type strain: ATCC 29844, CIP 78.64, CCUG 37824, DSM 4576, JCM 1242, LMG 7882.

GenBank accession number (16S rRNA): AJ233429.

TABLE BXII.γ.266. Characteristics differentiating the species of the genus *Serratia*^a

Characteristics	1. <i>S. marcescens</i>	2. <i>S. entomophila</i>	3. <i>S. ficaria</i>	4. <i>S. fonticola</i>	5. <i>S. grimesii</i>	6. <i>S. liquefaciens</i>	7. <i>S. odorifera</i>	8. <i>S. plymuthica</i>	9. <i>S. proteamaculans</i>	10. <i>S. rubidaea</i>
Prodigiosin production	d	—	—	—	—	—	—	d	—	+
Potato-like odor	—	—	+	—	—	—	+	—	—	d
Indole production	—	—	—	—	—	—	+	—	—	—
Lysine decarboxylase	+	—	—	+	+	+	+	—	+	d
Ornithine decarboxylase (Møller)	+	—	—	+	+	+	d	—	+	—
Arginine decarboxylase (Møller)	—	—	—	—	+	—	—	—	—	—
Tween 80 hydrolysis	+	+	+	+	+	+	—	+	+	+
Malonate test	—	—	—	+	—	—	—	—	—	d
<i>Carbon source utilization:</i>										
Adonitol	+	+	+	+	—	—	+	—	d	+
L-Arabinose	—	—	+	+	+	+	+	+	+	+
D-Arabitol	—	d	+	+	—	—	—	—	—	+
L-Arabitol	+	d	+	+	—	—	+	—	—	—
Betaine	—	—	—	—	—	—	—	d	—	+
D-Cellobiose	—	+	+	d	d	+	+	+	d	+
Dulcitol	—	—	—	+	—	—	—	—	—	—
±Erythritol	d	—	+	+	—	—	d	—	d	+
L-Fucose	+	+	+	—	+	+	+	+	+	+
Gentiobiose	—	+	+	+	d	+	d	+	+	+
Itaconate	—	+	—	—	—	—	—	—	—	—
5-Ketogluconate	+	+	+	d	+	+	d	d	+	—
Malitol	—	—	+	d	+	d	—	+	+	+
D-Melezitose	—	—	+	—	d	d	—	+	+	d
D-Melibiose	—	—	+	+	+	+	+	+	+	+
Mucate	—	—	+	d	—	—	+	d	—	+
Palatinose	—	—	+	+	+	d	—	+	+	+
3-Phenylpropionate	—	—	—	d	—	—	—	—	—	—
Quinate	d	d	+	—	—	—	—	+	—	+
D-Raffinose	—	—	+	+	+	+	d	+	+	+
L-Rhamnose	—	—	+	+	—	—	+	—	d	—
D-Saccharate	—	—	+	+	—	—	+	d	—	+
D-Sorbitol	+	—	+	+	+	+	+	d	d	—
D-Tagatose	—	—	—	+	—	—	—	—	—	—
meso-Tartrate	d	—	—	—	—	+	+	—	—	—
Trigonelline	d	—	d	—	—	—	+	—	—	+
D-Turanose	—	—	+	d	+	d	—	+	+	+
Xylitol	+	—	+	d	—	—	d	—	—	d
D-Xylose	—	d	+	d	+	+	+	+	+	+

^aData from Grimont and Grimont (1995) and Gavini et al. (1979). For symbols see standard definitions.

5. ***Serratia grimesii*** Grimont, Grimont and Irino 1983b, 438^{VP} (Effective publication: Grimont, Grimont and Irino 1982a, 73.)

gri.me' sii. M.L. gen. masculine form of Grimes.

S. grimesii has been isolated from the natural environment and from human clinical specimens.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

Biogroup ADC might represent a subspecies of *S. grimesii*.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 14460, CIP 103361, DSM 30063, IFO 13537, JCM 5910.

GenBank accession number (16S rRNA): AJ233430.

6. ***Serratia liquefaciens*** (Grimes and Hennerty 1931) Bascomb, Lapage, Willcox and Curtis 1971, 293^{AL} ("Aerobacter liquefaciens" Grimes and Hennerty 1931, 93.)

li.que.fa' ciens. M.L. part. adj. *liquefaciens* dissolving.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is not produced.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

S. liquefaciens sensu lato was composed of several biovars and some of these were found to constitute genomospecies which were subsequently given species status: biovar C1ab (including the type strain of *S. liquefaciens*) corresponded to *S. liquefaciens sensu stricto*; biovars C1c (including the type strain of *S. proteamaculans*), EB, RB, and RQ corresponded to *S. proteamaculans*; and biovars C1d and ADC corresponded to *S. grimesii* (Table BXII.γ.271) (Grimont et al., 1977b, 1982a, b).

S. liquefaciens is the most prevalent *Serratia* species in the natural environment (plants, digestive tract of rodents). Occasionally encountered as an opportunistic pathogen.

The mol% G + C of the DNA is: 53–54 (*T_m*, Bd).

Type strain: ATCC 27592, CIP 103238, DSM 4487, JCM 1245, LMG 7884, NCTC 12962.

GenBank accession number (16S rRNA): AJ306725, AB004752.

It should be mentioned that the strain formerly considered to be the type strain of "*Aerobacter liquefaciens*" or "*Aerobacter lipolyticus*" by Grimes (1961) was ATCC 14460 (now the type strain of *S. grimesii*). Since this strain was considered atypical compared to other strains labeled as *S. liquefaciens*, another strain (ATCC 27592) was given as type strain of *S. liquefaciens* in the Approved Lists.

TABLE BXII.γ.267. Other characteristics of the species of the genus *Serratia*^a

Characteristics	1. <i>S. marcescens</i>	2. <i>S. entomophila</i>	3. <i>S. ficaria</i>	4. <i>S. fonticola</i>	5. <i>S. grimesii</i>	6. <i>S. liquefaciens</i>	7. <i>S. odorifera</i>	8. <i>S. plymuthica</i>	9. <i>S. proteamaculans</i>	10. <i>S. rubidaea</i>
<i>Carbon source utilization:</i>										
N-Acetyl-D-glucosamine	+	+	+	+	+	+	+	+	+	+
cis-Aconitate	+	+	+	+	+	+	+	+	+	+
trans-Aconitate	+	+	+	d	—	—	+	d	d	+
D-Alanine	+	+	+	d	+	+	+	+	+	+
L-Alanine	+	+	+	d	d	+	+	+	+	+
4-Aminobutyrate	d	d	+	—	+	+	+	d	+	+
5-Aminovalerate	—	—	—	—	—	—	—	—	—	—
L-Aspartate	+	+	+	d	+	+	+	+	+	+
Benzoate	d	d	d	—	d	—	—	d	d	d
Caprate	+	d	+	—	d	+	d	—	d	d
Caprylate	d	d	+	—	d	d	d	—	d	d
Citrate	+	+	+	+	+	+	+	+	+	+
m-Coumarate	—	—	—	—	—	—	—	—	—	—
Ethanolamine	—	—	—	—	—	—	—	—	—	—
D-Fructose	+	+	+	+	+	+	+	+	+	+
Fumarate	+	+	+	d	+	+	+	+	+	+
D-Galactose	+	+	+	+	+	+	+	+	+	+
D-Galacturonate	+	+	+	+	+	+	+	+	+	+
Gentisate	d	—	—	d	—	—	—	—	d	—
D-Gluconate	+	+	+	+	+	+	+	+	+	+
D-Glucosamine	+	+	+	+	+	+	+	+	+	+
D-Glucose	+	+	+	+	+	+	+	+	+	+
D-Glucuronate	+	+	+	+	+	+	+	+	d	+
L-Glutamate	+	+	+	d	+	+	+	+	+	+
Glutarate	—	—	—	—	—	—	—	—	—	—
DL-Glycerate	+	+	+	+	+	+	+	d	+	+
Glycerol	+	+	+	+	+	+	+	+	+	+
Histamine	—	—	—	—	—	—	—	—	—	d
3-Hydroxybenzoate	d	—	—	d	—	—	—	—	—	—
4-Hydroxybenzoate	d	—	d	—	—	—	—	—	—	—
3-Hydroxybutyrate	d	—	d	—	—	—	d	—	—	—
myo-Inositol	+	+	+	+	+	+	+	+	+	+
2-Ketogluconate	+	+	+	+	+	+	+	+	+	+
2-Ketoglutarate	d	d	d	d	+	d	+	d	d	d
L-Lactate	+	+	d	d	+	+	+	d	+	+
Lactose	—	—	d	d	+	d	+	+	d	+
Lactulose	—	—	—	+	d	—	d	d	—	d
D-Lyxose	+	d	d	d	+	+	—	—	+	—
D-Malate	d	d	—	d	d	+	+	d	—	d
L-Malate	+	+	+	d	+	+	+	+	+	+
Malonate	—	—	—	d	—	—	—	—	—	d
Maltose	+	+	+	+	+	+	+	+	+	+
Maltotriose	+	+	+	+	+	+	+	+	+	+
D-Mannitol	+	+	+	+	+	+	+	+	+	+
D-Mannose	+	+	+	+	+	+	+	+	+	+
1-O-Methyl-α-galactoside	—	—	+	+	+	+	+	+	+	+
1-O-Methyl-β-galactoside	d	—	d	+	d	d	+	+	—	+
3-O-Methyl-D-glucose	—	—	—	d	d	d	d	d	+	—
1-O-Methyl-α-D-glucoside	—	—	d	d	—	d	—	d	d	+
1-O-Methyl-β-D-glucoside	+	+	+	+	+	+	+	+	+	+
Phenylacetate	d	d	d	d	d	+	d	d	d	+
L-Proline	+	+	+	d	d	+	+	+	+	+
Protocatechuate	d	—	d	—	d	d	—	+	d	d
Propionate	d	d	d	—	—	—	—	d	—	—
Putrescine	+	+	+	+	+	+	+	d	+	+
D-Ribose	+	+	+	+	+	+	+	+	+	+
L-Serine	+	+	+	d	+	+	+	+	+	+
L-Sorbose	—	—	—	—	—	—	—	—	—	—
Succinate	+	+	+	d	+	+	+	+	+	+
Sucrose	+	+	+	d	+	+	d	+	+	+
D-Tartrate	—	—	—	d	—	—	d	—	—	d
L-Tartrate	—	—	—	—	—	—	d	—	—	d
D-Trehalose	+	+	+	+	+	+	+	+	+	+
Tricarballoylate	—	—	—	d	—	—	—	—	—	d
Tryptamine	—	—	—	—	—	—	—	—	—	—
L-Tyrosine	d	d	+	—	d	d	+	d	d	+

(continued)

TABLE BXII.γ.267. (cont.)

Characteristics	1. <i>S. marcescens</i>	2. <i>S. entomophila</i>	3. <i>S. ficaria</i>	4. <i>S. fonticola</i>	5. <i>S. grimesii</i>	6. <i>S. liquefaciens</i>	7. <i>S. odorifera</i>	8. <i>S. plymuthica</i>	9. <i>S. proteamaculans</i>	10. <i>S. rubidaea</i>
<i>Oxidation of:</i>										
Glucose to gluconate (without cofactor)	+	+	+	—	—	—	+	d	—	+
2-Ketogluconate to 2,5-diketogluconate	+	—	—	—	+	+	—	—	—	—
Glycerol dehydrogenase induced by glycerol, not hydroxyacetone	+	—	—	+	+	+	—	—	+	—
<i>Acid produced from:</i>										
Adonitol	d	+	+	+	—	—	v	—	—	+
L-Arabinose	—	—	+	+	+	+	+	+	+	+
D-Melibiose	—	—	+	+	+	+	+	+	+	+
myo-Inositol	d	—	+	+	+	+	+	d	+	d
D-Raffinose	—	—	+	+	+	+	d	+	+	+
L-Rhamnose	—	—	+	v	—	—	+	—	—	—
D-Sorbitol	+	—	+	+	+	+	+	d	+	—
Sucrose	+	+	+	—	+	+	d	+	+	+
D-Xylose	—	d	+	d	+	+	+	+	+	+
Growth at 5°C	—	+	+	+	+	+	+	+	+	—
Growth at 37°C	+	+	+	+	+	+	+	d	+	+
Growth at 40°C	+	+	—	+	—	—	+	—	—	d
<i>Growth in NaCl:</i>										
7% (w/v)	+	+	+	+	d	d	+	d	d	+
8.5% (w/v)	d	—	—	—	—	—	+	—	—	+
10% (w/v)	—	—	—	—	—	—	—	—	—	d
Tetrathionate reduced	d	—	—	+	+	+	—	—	+	—
Gas from glucose	—	—	—	d	+	+	—	d	+	—
β-Xylosidase	—	—	d	+	—	—	+	d	—	+
Tween 40 hydrolysis	+	+	+	+	+	+	+	+	+	+
Tween 60 hydrolysis	+	+	+	+	+	+	d	+	+	+
Chitin hydrolysis	+	d	+	—	d	d	—	+	d	—
Methyl red test	—	—	—	+	d	d	+	d	d	—
L-Histidine deaminase	—	—	—	—	—	—	—	—	—	—
Tryptophan deaminase	—	—	—	—	—	—	—	—	—	—
β-glucuronidase	—	—	—	—	—	—	—	—	—	—
Esculin hydrolysis	+	+	+	+	+	+	+	+	d	+

^aData from Grimont and Grimont (1995) and Gavini et al. (1979). For symbols see standard definitions.

TABLE BXII.γ.268. Identification of *S. marcescens* biogroups and biovars^a

Characteristics	Biogroups																	
	A1		A2		A6	A3				A4		A5	A8			TCT	TC	TT
	a	b	a	b	a	a	b	c	d	a	b		a	b	c			
Prodigiosin production	+	+	+	+	+	—	—	—	—	—	—	—	—	—	—	—	—	
<i>Growth on.</i> ^b																		
meso-Erythritol	+	+	+	+	+	+	+	+	+	+	+	+	+	—	—	—	—	
Benzoate	+	+	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
Quinate and/or 4-hydroxybenzoate	—	—	—	—	+	—	—	—	—	+	—	+	+	+	+	—	—	
3-Hydroxybenzoate	—	—	—	—	—	+	+	—	—	—	—	—	—	+	—	—	—	
Trigonelline	—	—	—	—	—	—	+	—	+	—	—	+	+	+	+	+	—	
D-Malate/ <i>m</i> -tartrate	+	—	—	+	d	—	—	d	—	+	+	+	d	—	—	+	—	
Gentisate	—	—	+	+	+	+	+	d	—	+	+	—	d	+	d	—	—	
Tetrathionate reduction	+	+	+	+	+	+	+	+	+	—	—	+	+	+	+	+	+	

^aFor symbols see standard definitions.

^bCarbon source utilization test.

7. *Serratia odorifera* Grimont, Grimont, Richard, Davis, Steigerwalt and Brenner 1978a, 461^{AL}
o.do.ri.fe'ra. M.L. fem. adj. *odorifera* bringing odors, fragrant.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is not produced. Cultures give off a musty, potato-like odor.

Physiological and nutritional characteristics are pre-

sented in Tables BXII.γ.266 and BXII.γ.267. Two biovars can be recognized (Table BXII.γ.272).

The capsular antigen reacts with *Klebsiella* antisera K4 or K68.

Rare opportunistic pathogen. Occasionally isolated from plants or food.

The mol % G + C of the DNA is: 54.6 (T_m).

TABLE BXII.γ.269. Correspondence between serovars and biogroups in *Serratia marcescens*

Biogroup ^a	O:H serovars ^b
A1	5:2; 5:3; 5:13; 5:23; 10:6; 10:13; 28:2
A2/6	5:23; 6,14:2; 6,14:3; 6,14:8; 6,14:9; 6,14:10; 6,14:13; 8:3; 13:5
A3	3:5; 3:11; 4:5; 4:18; 5:6; 5:15; 6,14:5; 6,14:6; <i>6,14:16</i> ; 6,14:20; <i>9:9</i> ; <i>9:11</i> ; <i>9:15</i> ; 9:17; 12:5; 12:9; <i>12:10</i> ; 12:11; <i>12:15</i> ; <i>12:16</i> ; 12A:17; <i>12:18</i> ; 12:20; <i>12:26</i> ; 13:11; <i>13:17</i> ; 15:3; 15:5; 15:8; 15:9; 17:4; 18:21; <i>18:26</i> ; <i>22:11</i> ; <i>23:19</i> ; 26:20
A4	1:1; 1:4; 2:1; 2:8; 3:1; 4:1; 4:4; 5:1; 5:6; 5:8; 5:24; 9:1; 13:1; 13:11; 13:13
A5/8	2:4; 3:12; 3,21:12; 4:12; 5:4; 6,14:4; 6,14:12; <i>8:4</i> ; 8:12; 15:12; 21:12; 25:12
TCT	<i>1:7</i> ; 2:7; <i>4:7</i> ; 5:7; 5:19; 7:7; 7:23; 10:9; 11:4; 13:7; 13:12; 16:19; <i>18:9</i> ; 18:16; <i>18:19</i> ; 19:14; <i>19:19</i> ; 24:6; 27:—
TC	10:8; 20:12

^aBiogroup A1 is composed of biotypes A1a and A1b; A2/6 of A2a, A2b, A6; A3 of A3a, A3b, A3c, A3d; A4 of A4a, A4b; A5/8 of A5, A8a, A8b, and A8.

^bSerovars for which exceptions to the correspondence occur are in italics.

TABLE BXII.γ.270. Identification of *Serratia entomophila* biotypes^a

Characteristics	Biotype	
	1b	2
<i>Growth on:</i> ^b		
D-Arabitol	+	—
L-Arabitol	—	+
D-Malate	—	d
D-Sorbitol	—	d
Quinate	+	d
D-Xylose	—	+

^aFor symbols see standard definitions.

^bThe type strain corresponds to biotype 1.

Type strain: ATCC 33077, CIP 79-1, DSM 4582, ICPB 3995, NCTC 11214.

GenBank accession number (16S rRNA): AJ233432.

8. ***Serratia plymuthica*** (Lehmann and Neumann 1896) Breed, Murray and Hitchens 1948, 481^{AL} (*Bacterium plymuthicum* (sic) Lehmann and Neumann 1896, 264.) *ply.mu'thi.ca*. M.L. adj. *plymuthica* pertaining to Plymouth, UK.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is produced by most strains.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

Most *S. plymuthica* strains studied were isolated from freshwater. Very rarely found in human sputum. No human infection reported.

The mol% G + C of the DNA is: 53.5–56.5 (*T_m*).

Type strain: ATCC 183, CIP 103239, DSM 4540, JCM 1244.

GenBank accession number (16S rRNA): AJ233433.

S. plymuthica is cited on the Approved Lists of Bacterial Names (Skerman et al., 1980) as *Serratia plymuthica* (Dyar 1895) Bergey, Harrison, Breed, Hammer and Huntoon 1923, 88. This is incorrect, for reasons discussed by Grimont et al. (1977b).

9. ***Serratia proteamaculans*** (Paine and Stansfield 1919) Grimont, Grimont and Starr 1978b, 503^{AL} (*Pseudomonas proteamaculans* Paine and Stansfield 1919, 38.)

pro.te.a.ma.cu'lans. M.L. n. *Protea* a plant generic name; L. v. *maculo* to spot; M.L. part. adj. *proteamaculans* spotting *Protea*.

S. proteamaculans and *S. liquefaciens* were thought to be synonymous based on DNA relatedness (Grimont et al., 1978b). However, subsequent observation of significant thermal instability of reassociated DNA fragments supported the separation of both species (Grimont et al., 1982b).

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is not produced.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

S. proteamaculans is found in the natural environment (plants, wild rodents, insects, and water) but exceptionally from human clinical specimens).

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 19323, CIP 103236, DSM 4543, NCPPB 245.

According to J.-P. Euzeby (<http://www.bacterio.cict.fr/corrections2.html>), the spelling should have been *proteimaculans*. However, since the name appeared in the Approved Lists, no correction is allowed.

- a. ***Serratia proteamaculans* subsp. *proteamaculans*** (Paine and Stansfield 1919) Grimont, Grimont and Starr 1978b, 503^{AL} (*Pseudomonas proteamaculans* Paine and Stansfield 1919, 38.)

The subspecies includes biovars C1c, EB, and RB. The type strain belongs to biotype C1c. Differential characteristics are given in Table BXII.γ.271.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 19323, CIP 103236, DSM 4543, NCPPB 245

GenBank accession number (16S rRNA): AJ233434.

- b. ***Serratia proteamaculans* subsp. *quinovora*** Grimont, Grimont and Irino 1983b, 438^{VP} (Effective publication: Grimont, Grimont and Irino 1982a, 71.) *qui.no'vo.ra*. M.L. *quinata*, from Spanish *quina* quinine; and L. v. *vor*o to devour; M.L. fem. adj. *quinovora* quinate devouring.

This subspecies corresponds to biotype RQ. Quinate utilization was observed for all strains of this subspecies in a minimal agar when an unwashed inoculum was used. In Biotype-100 strips, no growth is observed on quinate when Biotype Medium 1 is used, some strains grow from quinate when Biotype Medium 2 (containing more growth factors) is used.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 33765, CIP 81-95, DSM 4597.

GenBank accession number (16S rRNA): AJ233435.

10. ***Serratia rubidaea*** (Stapp 1940) Ewing, Davis, Fife and Lessel 1973, 224^{AL} (*"Bacterium rubidaeaum"* Stapp 1940, 259; *Serratia marinorubra* Zobell and Upham 1944, 255.) *ru.bi'dae.a*. L. *Rubus idaeus* raspberry, contracted and made to agree in gender with *Serratia*.

The cell morphology and colonial morphology are as given for the genus. Prodigiosin is produced by most strains.

Physiological and nutritional characteristics are presented in Tables BXII.γ.266 and BXII.γ.267.

Three biotypes (B1, B2, B3) correspond to subspecies “*S. rubidaea* subsp. *burdigalensis*”, “*S. rubidaea* subsp. *rubidaea*”, and “*S. rubidaea* subsp. *colindalensis*” (Grimont et al., manuscript in preparation) (Table BXII.γ.273).

S. rubidaea strains are rarely isolated, either in the natural environment or in human patients. May be found in ripe coconuts (Grimont et al., 1981c).

The mol% G + C of the DNA is: 53.5–58.5 (T_m).

Type strain: Ewing 2199-72, ATCC 27593, CIP 103234, DSM 4480, JCM 1240.

GenBank accession number (16S rRNA): AB004751, AJ233436.

TABLE BXII.γ.271. Identification of species in the *Serratia liquefaciens* complex^a

Characteristics	<i>S. liquefaciens</i>	<i>S. proteamaculans</i> subsp. <i>proteamaculans</i>			<i>S. proteamaculans</i> subsp. <i>quinovora</i>	<i>S. grimesii</i>	
	C1ab	C1c ^b	EB	RB	RQ	C1d ^c	ADC
<i>Carbon source utilization test:</i> ^d							
<i>trans</i> -Aconitate	—	+	+	d	+	—	+
Benzoate	—	—	d	d	—	+	—
<i>m</i> -Erythritol	—	—	+	—	—	—	—
Gentisate	—	—	—	+	—	—	—
D-Malate	+	—	—	—	d	v	v
L-Rhamnose	—	—	—	+	d	—	—
<i>m</i> -Tartrate	+	—	—	—	d	—	—
Arginine decarboxylase	—	—	—	—	—	+	+
Esculin hydrolysed	+	+	+	+	—	+	+

^aFor symbols see standard definitions.

^bThe type strain of *S. proteamaculans* (ATCC 19323) corresponds to biotype C1c.

^cThe type strain of *S. grimesii* (ATCC 14460) corresponds to biotype C1d.

^dBiotype-100 strips.

TABLE BXII.γ.272. Identification of *Serratia odorifera* biotypes^a

Characteristics	Biotype	
	1 ^b	2
<i>Growth on:</i>		
<i>m</i> -Erythritol	–	+
L-Fucose	d	+
D-Raffinose	+	–
Sucrose	+	–
D-Tartrate	+	–
Ornithine	+	–
decarboxylase	+	–
Acid from sucrose	+	–
Acid from raffinose	+	– ^c

^aFor symbols see standard definitions.

^bThe type strain corresponds to biotype 1.

^cSome strains positive in 3–7 days.

TABLE BXII.γ.273. Identification of *Serratia rubidaea* biotypes^a

Characteristics	Biotype ^b		
	B1	B2	B3
<i>Growth on:</i>			
Histamine	d	–	d
D-Melezitose	–	+	+
D-Tartrate	+	–	d
Tricarballoylate	–	d	–
Voges–Proskauer (O’Meara)	+	–	d
Lysine decarboxylase	+	+	–
Malonate (Leifson)	+	+	–

^aFor symbols see standard definitions.

^bThe biotypes were shown to correspond to subspecies designated as “*S. rubidaea* subsp. *burdigalensis*” (B1), “*S. rubidaea* subsp. *rubidaea*” (B2), and “*S. rubidaea* subsp. *colindalensis*” (B3).

Genus XXXV. *Shigella* Castellani and Chalmers 1919, 936^{AL}

NANCY A. STROCKBINE AND ANTHONY T. MAURELLI

Shi.gel' la. M.L. dim. *-ella* ending; M.L. fem. n. *Shigella* named after K. Shiga, the Japanese bacteriologist who first discovered the dysentery bacillus.

Straight rods, 1–3 × 0.7–1.0 μm, that conform to the general definition of the family *Enterobacteriaceae* and contain the enterobacterial common antigen. Gram negative. Nonmotile. Non-pigmented. Facultatively anaerobic, having both a respiratory and a fermentative type of metabolism. Catalase positive (with exceptions in *Shigella dysenteriae*). Oxidase negative. Chemoor-ganotrophic. **Ferment sugars without gas production** (a few exceptions produce gas). **Salicin, adonitol, and myo-inositol are not fermented**. Strains of *Shigella sonnei* ferment lactose and sucrose

upon extended incubation; however, other species do not utilize these substances in conventional medium. **Do not utilize citrate, malonate, or sodium acetate** (with exceptions in *Shigella flexneri* for sodium acetate) as a sole carbon source. **Do not grow in KCN or produce H₂S. Do not decarboxylate lysine**. Reduce nitrates to nitrites. Intestinal pathogens of humans and other primates, causing bacillary dysentery. Based on 16S rDNA sequencing, shigellae belong in the *Gamma*proteobacteria.

The mol% G + C of the DNA is: 49–53 (Laskin and Lechevalier, 1981).

Type species: *Shigella dysenteriae* (Shiga 1898) Castellani and Chalmers 1919, 935, Epit. Spec. Cons. Opinion 11 of the Jud. Comm. 1954a, 149 (*Bacillus dysenteriae* Shiga 1898, 817.)

FURTHER DESCRIPTIVE INFORMATION

Phylogenetic and systematic treatment Scientific evidence accumulated to date strongly supports the view that *Shigella* species are biotypes/pathotypes or clones of *Escherichia coli*. As early as 1957, Luria and Burrous (1957) showed that *E. coli* and the majority of *Shigella* species are in the same fertility system and suggested that genetic recombination between these organisms could play a role in the evolution of some *Shigella* serotypes and, more generally, in the evolution of natural populations of these bacteria. In the early 1970s, Brenner et al. (1972a, 1973a) published findings from DNA–DNA reassociation studies that provided the first direct insights into the evolutionary relationships between shigellae and *E. coli*. These authors found that, with the exception of *S. boydii* 13 (see discussion under Taxonomic comments), *Shigella* species were as related to *E. coli* (>75% nucleotide similarity) as they were to each other.

Given the variability that occurs within the defined *E. coli* species, the question then is whether the four species of *Shigella* represent the evolution of new species from a common ancestor or whether these organisms are clones of *E. coli*. Findings from multilocus enzyme electrophoresis (MLEE) analyses strongly support the conclusion that shigellae are more appropriately regarded as pathotypes or clones of *E. coli*. Principal component analysis of 302 electrophoretic types (ETs) from more than 1600 human and animal isolates of *E. coli* and 123 strains of the four species of *Shigella* showed that the *Shigella* strains clustered within two of the three overlapping groups of *E. coli* strains detected, rather than forming a group apart from those comprised by *E. coli*, as would be expected of a distinct genus (Ochman et al., 1983). Furthermore, with the exception of *S. boydii* 13, strains representing all four species of *Shigella* did not cluster in groups corresponding to each species. A dendrogram of the 23 ETs from *Shigella* constructed by average linkage clustering revealed two major clusters each containing strains of three or more species; one group consisted of ETs representing *S. flexneri*, *S. dysenteriae*, and *S. boydii* strains and the other consisted of ETs representing strains from all four species, with *S. sonnei* being most distantly related (Ochman et al., 1983).

MLEE findings from the study by Pupo et al. (1997) also showed that strains from the four species of *Shigella* clustered in two separate groups, with one group containing strains from the majority of *Shigella* species and the other group containing only *S. flexneri* strains. With the exception of one *S. flexneri* serotype, the *Shigella* species appeared as clones within the *E. coli* species. Nucleotide sequence analysis of the housekeeping gene maltose dehydrogenase (*mdh*) (Pupo et al., 1997) and analysis of restriction fragment length polymorphisms of rDNA (ribotyping) (Roland et al., 1998) also yielded largely similar clustering of shigellae, but showed somewhat different distributions of the *Shigella* spp. among the *E. coli* phylogenetic groups.

A careful analysis of 16S ribosomal DNA sequences by Cilia et al. (1996) also placed shigellae and *E. coli* in the same phylogenetic group, supporting the concept that *Shigella* species and *E. coli* are five nomenclatures that represent a single genomospecies. In fact, the variation of the 16S ribosomal RNA genes in the seven *rrn* operons of a single strain of *E. coli* shared a clade

with *S. dysenteriae*, *S. flexneri*, and *S. sonnei* sequences. There were informative base alleles that were shared by both *E. coli* and *Shigella* species.

Recent findings reported by Pupo et al. (2000) from the examination of nucleotide sequence similarity at eight housekeeping genes in four regions around the chromosome of *Shigella* and selected nonpathogenic *E. coli* strains (ECOR set) revealed three clusters of closely related strains, each including exclusively *Shigella* strains but from more than one species. *S. sonnei* and three *S. dysenteriae* strains from *S. dysenteriae* serotypes 1, 8, and 10 fell outside the three main clusters but were well within the population structure of *E. coli*. The presence of these seven different groups (clusters 1–3, *S. sonnei*, *S. dysenteriae* serotypes 1, 8, and 10) within *E. coli* suggests that the *Shigella* phenotype has arisen as many as seven times from ancestors that were *E. coli*. Pupo et al. (2000) propose that the many independent origins of the *Shigella* phenotype are an example of convergent evolution. Phenotypes that may have arisen by convergent evolution include the loss of catabolic pathways involving lactose fermentation, mucate utilization, lysine decarboxylation, the loss of motility, and the acquisition of the invasion plasmid (Pupo et al., 2000).

It has been suggested that *E. coli* arose from a common ancestor with *Salmonella* and diverged largely as the result of the acquisition of multiple genomic fragments (Lawrence and Ochman, 1998). It appears that *Shigella* differs from *E. coli* largely by the loss of genomic fragments. Maurelli et al. (1998) showed that shigellae lack a large fragment of the genome relative to *E. coli* K-12. It is interesting to note that the missing fragment included the *cadA* gene discussed in more detail below. The same pattern has been noted with another genomic fragment that derives from a cryptic prophage in *E. coli* K-12 and includes the *ompT* gene (Nakata et al., 1993). This gene encodes a surface protease that when introduced into *Shigella* inhibited the ability of the organism to spread among host cells. Another fascinating deletion occurs in *S. sonnei*. A large portion of the O-antigen gene cluster has been excised, leaving remnants of genes normally found at the ends of the cluster fused together (Lai et al., 1998). The clone carries a plasmid that has O-antigen genes that apparently arose in *Plesiomonas shigelloides* (Houng and Venkatesan, 1998; Chida et al., 2000). It has been postulated that the acquisition of the new O-antigen was a prerequisite for *S. sonnei* emerging in its modern niche and that this might have occurred about 10,000 years ago (Lai et al., 1998).

In contrast to the concept of species being defined as a group having a monophyletic origin, a species may be defined as a group of organisms sharing genetic drift (Hey, unpublished studies). By this latter definition, *Shigella* species might be classified independently of *E. coli*. Factors that affect the sharing of genetic drift among bacteria are natural selection and recombinant exchange of DNA. Shigellae are considered to be host-adapted to primates and thus have a narrower niche than *E. coli*. It is difficult to conclusively define species of bacteria based on recombination-exchange partners. Islands of DNA that clearly came from other species can frequently be found in bacterial genomes. However, it does appear that the frequency of recombination is proportional to the closeness of the species. Matsutani and Ohtsubo (1993) surveyed the distribution of five insertion elements (IS1, IS600, IS629, IS630, and IS640) originally found in *S. sonnei*. In general, the elements were found in high copy numbers in *S. sonnei*, *S. dysenteriae*, *S. flexneri*, and *S. boydii* (often more than 20 copies per genome), in moderate numbers in *E. coli*, and occa-

sionally, albeit rarely, in other *Enterobacteriaceae*. Thus, it may be that *Shigella* species are in the process of drifting apart from *E. coli*. The alternative hypothesis is that the insertion elements are more readily duplicated within clones than they exist in rather than across clones. However, the conclusion is the same since it would mean that the elements are more ancient in shigellae and this is an example of genetic drift that is not shared with *E. coli*.

If history were to be rewritten with the privilege of having our current understanding of the pathogenesis and diversity extent within *E. coli*, a very different nomenclature for *Shigella* would undoubtedly be adopted. Although phylogenetically it would be better to treat shigellae as pathotypes of *E. coli*, the members of the genus *Shigella* continue to be divided for historical and medical purposes into four species or subgroups: *S. dysenteriae* (subgroup A), *S. flexneri* (subgroup B), *S. boydii* (subgroup C), and *S. sonnei* (subgroup D). The useful information communicated about the disease caused by these organisms (shigellosis) through the genus epithet *Shigella* has preserved and will likely continue to preserve the current nomenclature until such time when there is a greater need to communicate the natural relationships that exist between these strains.

Biochemical characteristics of the genus are listed in Table BXII.γ.274. The biochemical reactions that are useful for differentiating the four species of *Shigella* are listed in Table BXII.γ.275. The ability to ferment D-mannitol is a particularly important differential biochemical trait for classifying the shigellae. Although exceptional strains occur, members of subgroup A are unable to ferment D-mannitol, while members of the other subgroups (B, C, and D) are able to ferment this sugar. Members of the genus *Shigella* have distinctive antigenic structures by which they are divided into a variety of serotypes and subserotypes. It is the combination of biochemical traits, particularly the D-mannitol phenotype, and antigenic properties that have guided the classification of members of this genus. The current schema for classifying *Shigella*, which recognizes 49 serotypes and subserotypes, is shown in Table BXII.γ.276.

Cell morphology Shigellae are nonmotile, Gram-negative rods with rounded ends. They are typically 1–3 μm in length and 0.7–1 μm in diameter. Fig. BXII.γ.209 shows negatively stained preparations of *S. dysenteriae* 1 and *S. flexneri* 2a. The bacterial cells in this figure are nonflagellated, which is typical of the members belonging to this genus. Under certain culture conditions, however, some cells in a culture have been observed to express flagella. See the section below on fine structure for a discussion of exceptions to the presence of flagella.

Cell wall composition The cell walls of shigellae resemble those of other Gram-negative bacteria in structure, composition, and endotoxic activity. When serologically characterizing living (unheated) isolates that are biochemically consistent with *Shigella*, it is important to be aware that living bacterial cells of certain strains of *Shigella* can produce a substance that will block their agglutination with O antisera. The masking effects of this substance, originally referred to as a K antigen of the B variety or B antigen, can be inactivated by heat. Treatment of nonagglutinable cultures at 100°C for 1 h restores their ability to be agglutinated by O antisera. Blocking substances also occur in certain *E. coli* strains and were shown by Ørskov et al. (1977) to be chemically and antigenically indistinguishable from the O antigens present in the cell-wall lipopolysaccharides of these bacteria. Blocking substances (B antigens) are no longer recognized as distinct antigens; however, their presence in unheated cells remains important from a practical standpoint.

TABLE BXII.γ.274. Characteristics of the genus *Shigella*^a

Test or Substrate	Result
H ₂ S (triple sugar iron agar)	—
Urea hydrolysis	—
Indole production	d ^b
Methyl red	+
Voges-Proskauer	—
Citrate (Simmons')	—
Citrate (Christenson's)	—
Growth in KCN	—
Motility at 37°C	—
Gelatin hydrolysis at 22°C	—
Lysine decarboxylase	—
Arginine dihydrolase	d
Ornithine decarboxylase	d ^c
Phenylalanine deaminase	—
D-Glucose:	
Acid	+
Gas	— ^d
Acid from:	
Adonitol	—
α-Methyl-D-glucoside	—
L-Arabinose	d
Cellobiose	—
Dulcitol	d
Erythritol	—
Glycerol	d
myo-Inositol	—
Lactose	d ^e
Maltose	d
D-Mannitol	d ^f
Raffinose	d
L-Rhamnose	d
Salicin	—
D-Sorbitol	d
Sucrose	d ^g
Trehalose	d
D-Xylose	d
Malonate (sodium) utilization	—
Mucate (sodium) utilization	—
Acetate (sodium) utilization	—
Esculin hydrolysis	—
β-Galactosidase (ONPG) ^h	d ⁱ
Nitrate to nitrite	+
Oxidase	—

^aFor symbols see standard definitions. Adapted from findings reported by Ewing (1971) based on the examination of 5166 cultures representative of all four species. Unless specified otherwise, results are for 48 h incubation at 36° ± 1°C.

^bSome strains of some serotypes of *S. dysenteriae*, *S. flexneri*, and *S. boydii* produce indole while strains of other serotypes are always negative. *S. sonnei* is always negative.

^cStrains of *S. boydii* 13 and *S. sonnei* are positive.

^dSome biotypes of *S. flexneri* 6 are positive; positive strains of *S. boydii* 13 and 14 have been described.

^eStrains of *S. sonnei* are usually positive after several days of incubation; positive strains of *S. flexneri* 2a, *S. boydii* 9, and *S. boydii* 15 have been described. Some strains of *S. dysenteriae* 1 ferment lactose slowly; however, all are positive for β-galactosidase.

^fStrains of *S. dysenteriae* are negative; negative biotypes of *S. flexneri* 4a ("S. *rabaulensis*", "S. *rio*") and *S. flexneri* 6 (Newcastle biotype) occur; negative biotypes of *S. sonnei* occur rarely.

^gStrains of *S. sonnei* are usually positive after several days of incubation.

^hONPG, o-nitrophenyl-β-D-galactopyranoside.

ⁱStrains of *S. dysenteriae* 1 and *S. sonnei* are positive; some positive strains of *S. flexneri* 2a and *S. boydii* 9 have been described.

Fine structure

Flagella By standard methods (radial spread from the site of inoculation in medium containing 0.3–0.4% agar), shigellae are nonmotile. Over the past decade, various investigators have shown that *Shigella* strains have the genes involved in producing flagella and, under certain conditions, can express these genes.

TABLE BXII.γ.275. Distinguishing biochemical reactions of the four species of *Shigella*^a

Test	<i>S. dysenteriae</i>	<i>S. boydii</i>	<i>S. flexneri</i>	<i>S. sonnei</i>
Indole production	d	d	d	—
Arginine dihydrolase	—	d	—	—
Ornithine decarboxylase	—	—	—	+
<i>Acid production from:</i>				
Lactose	—	—	—	(+) ^b
Sucrose	—	—	—	(+)
D-Mannitol	—	+	+	+
Dulcitol	d	d	—	—
D-Sorbitol	d	d	d	—
Raffinose	—	—	d	(+)
D-Xylose	—	d	—	—
Melibiose	—	d	d	—
β-Galactosidase (ONPG) ^c	d	d	—	+

^aFor symbols see standard definitions. Results are for 48 h incubation at 36° ± 1°C under conditions described by Ewing (1971).

^b(+) = 75% or more of strains gave a positive reaction after 3 or more days.

^cONPG, *o*-nitrophenyl-β-D-galactopyranoside.

Direct evidence for the existence of flagellum-related sequences in *Shigella* was reported by Tominaga et al. (1994), who identified and characterized cryptic genes for flagellin (*fliC*) in *S. flexneri* and *S. sonnei*. The cloned *fliC* gene from *S. flexneri* produced normal-type flagella when introduced into a strain of *E. coli* deleted for *fliC*, while the cloned *fliC* gene from *S. sonnei* produced curly-type flagella in this host strain. Findings from comparisons of nucleotide sequences of the *fliC* genes from *S. flexneri* and *S. sonnei* with those from *E. coli* and *Salmonella* serotype Typhimurium revealed various regions of high similarity between these bacteria. Similarities at and near the 5' and 3' constant regions of the *S. flexneri* and *E. coli fliC* genes suggested the *fliC* gene from *S. flexneri* is an *E. coli* type gene, while similarity observed between the *fliC* gene of *S. sonnei* at the operator and 3' constant region of *Salmonella* serotype Typhimurium and the downstream sequence of the *E. coli fliC* gene suggest the *S. sonnei* gene has undergone horizontal transfer and recombination (Tominaga et al., 1994). Coimbra et al. (2001) recently examined restriction fragment polymorphisms in the *fliC* genes of representative strains from each of the serotypes within the four species of *Shigella*. Seventeen F types, defined by *fliC* patterns sharing most of their bands, were observed among the 120 strains tested. Although O antigenic relationships between certain *E. coli* strains and shigellae are well documented, none of the F types observed among the tested *Shigella* strains matched the 62 F types previously described for *E. coli* (Machado et al., 2000). This method is a promising tool for the confirmation of atypical isolates as shigellae.

Prompted by reports that shigellae possessed the genes for flagella, Girón (1995) reexamined them for flagella by electron microscopy. He observed that strains of all four *Shigella* species, when cultivated under certain conditions, produced flagella. Motility was observed only by using low-concentration motility agar (0.175–0.2% agar), and genetic and environmental factors involved in the regulation of flagella are not yet identified. Expression of flagellated cells within a culture ranged from 1:300 to 1:1000, with flagellated cells typically producing one polar flagellum or occasionally 2–3 flagella in a semipolar topology. When expressed, the flagellum of *Shigella* is approximately 10 µm long and 12–14 nm in diameter and typically emanates from one pole of the bacterium. Putative flagellins of 33–38 kDa, which share immunologic similarities with *E. coli*, *Salmonella* spp., and

Proteus mirabilis flagellins, have also been identified (Girón, 1995).

It should be emphasized, however, that shigellae are non-motile under standard assay conditions. Al Mamun et al. (1997) showed that members of the four *Shigella* species became non-motile by undergoing various kinds of stable mutations in different flagellar genes. Findings from these investigators suggest that the loss of flagella in *S. boydii* is attributable to a defect or deletion in the *fliF* operon, while the primary cause for non-motility in *S. sonnei* is a deletion in the *fliD* gene. In *S. dysenteriae* and *S. flexneri*, IS1 insertions in the *fliH* master operon may be the primary cause of motility loss (Al Mamun et al., 1996).

Fimbrial adhesins In initial studies, *S. flexneri* was the only species of *Shigella* that was found to produce fimbriae (Duguid and Gillies, 1957). The fimbriae of *S. flexneri* were similar to those produced by *E. coli*; their width was uniformly about 0.01 µm and their length was between 0.3 and 2.0 µm. In these early studies, *S. flexneri* strains were found to readily undergo a reversible mutation between a fimbriate, hemagglutinating phase and a non-fimbriate, nonhemagglutinating phase. Findings by Snellings et al. (1997) confirmed the presence of fimbriae (type 1) on strains of *S. flexneri* and demonstrated their presence on recent clinical isolates of *S. boydii* and *S. dysenteriae*. Analysis of sequences upstream of the type 1 fimbriae subunit gene, *fimA*, showed that random phase variation between fimbriated and afimbriated states in *Shigella* was accompanied by the genomic rearrangement associated with phase variation in *E. coli* (Snellings et al., 1997). *Shigella* appears to switch between phases at a much lower frequency than *E. coli*, and this in combination with inappropriate growth conditions could partially explain the difficulty researchers have had demonstrating fimbriae in *Shigella*. The role fimbriae play in the pathogenesis of shigellae is not known. Some investigators have speculated that these structures may facilitate the formation of pellicles at the water–air interface in aquatic environments and thereby promote the survival of bacteria between outbreaks of disease (Snellings et al., 1997). Studies have also demonstrated the presence of type 3 and type 4 fimbrial adhesions and of afimbrial adhesions in *Shigella* strains (Qadri et al., 1989; Utsunomiya et al., 1992).

Another surface structure of note with respect to *Shigella* is the thin aggregative fimbria called curli. Curli have been demonstrated in a wide variety of *E. coli* and *Salmonella* spp. and are believed to mediate formation of biofilms or bacterial attachment to host intestinal cells. Rather than their presence, it is their complete absence from all four species of *Shigella*, as well as from enteroinvasive *E. coli* strains, that is notable here (Sakellaris et al., 2000). Shigellae are unable to produce curli because of insertions or deletions in the curli locus (*csg*); enteroinvasive *E. coli* strains are similarly affected or have another uncharacterized lesion(s) resulting in the lack of curli expression (Sakellaris et al., 2000). Since expression of curli occurs in a variety of *E. coli* strains, the widespread loss of curli from *Shigella* and enteroinvasive *E. coli* strains, which have a very similar mechanism of pathogenesis to *Shigella*, may represent a pathoadaptive mutation (Sakellaris et al., 2000). It is possible that during the divergence of *Shigella* from *E. coli*, the expression of curli in the new virulence niche was a selective disadvantage for *Shigella*.

Colonial or cultural characteristics On agar medium, colonies of *Shigella* strains can appear smooth and glistening or rough and dry. The degree of smoothness displayed depends in large part on the extent of polymerization of the O-antigen on the

TABLE BXII.γ-276. Serologic schema, antigenic formulae, and earlier designations of *Shigella* species

Subgroup and Species	Serotype/Subserotype	Antigenic Formula ^a	Main earlier designations or synonyms
Subgroup A			
<i>S. dysenteriae</i>	1		" <i>Bacterium shigae</i> ", " <i>S. shigae</i> " (Shiga, 1898)
	2		" <i>S. ambigua</i> ", " <i>S. schmitzii</i> " (Schmitz, 1917)
	3		" <i>S. largei</i> " Q771 (Large and Sankaran, 1934), <i>S. arabinotarda</i> A
	4		" <i>S. largei</i> " Q1167 (Large and Sankaran, 1934), <i>S. arabinotarda</i> B
	5		" <i>S. largei</i> " Q1030 (Large and Sankaran, 1934)
	6		" <i>S. largei</i> " Q454 (Large and Sankaran, 1934)
	7		" <i>S. largei</i> " Q902 (Large and Sankaran, 1934)
	8		Serotype 599-52 (Ewing et al., 1952b)
	9		Serotype 58 (Cox and Wallace, 1948)
	10		Serotype 2050-52 (Ewing, 1953)
	11		Serotype 3873-50 (Ewing and Hucks, 1952)
	12		Serotype 3341-55 (Ewing et al., 1958)
	13		Serotype 19809-73 (Shmilovitz et al., 1985)
	14		Serotype E22383 (Gross et al., 1989)
	15		Serotype E23507 (Gross et al., 1989)
Subgroup B			
<i>S. flexneri</i>	1a	I:4	V (Andrewes and Inman, 1919)
	1b	I:(4),6	VZ (Andrewes and Inman, 1919)
	2a	II:3,4	W (Andrewes and Inman, 1919)
	2b	II:7,8	WX (Andrewes and Inman, 1919)
	3a	III:(3,4), 6,7,8	Z (Andrewes and Inman, 1919)
	3b	III:(3,4),6	
	4a	IV:(3,4)	103 (Boyd, 1931)
	4b	IV:6	103Z
	4c	IV:7,8	(Pryamukhina and Khomenko, 1988)
	5a	V:(3,4)	P119 and P119X (Boyd, 1932a, b)
	5b	V:7,8	(Petrovskaya and Khomenko, 1979)
	6	VI:4	Boyd 88 (Boyd, 1931) Manchester bacillus, Newcastle bacillus, " <i>S. newcastle</i> "
	X	—:7,8,	X (Andrewes and Inman, 1919)
	Y	—:3,4	Y (Andrewes and Inman, 1919)
Subgroup C			
<i>S. boydii</i>	1		170 (Boyd, 1932a, b; Ewing, 1949)
	2		P288 (Boyd, 1932a, b; Ewing, 1949)
	3		D.1 (Boyd, 1932a, b; Ewing, 1949)
	4		P274 (Boyd, 1932a, b; Ewing, 1949)
	5		P143 (Boyd, 1938; Ewing, 1949)
	6		D19 (Boyd, 1932a, b; Ewing, 1949)
	7		Type T, Lavington I, " <i>S. etousae</i> " (Ewing, 1946)
	8		Serotype 112 (Cox and Wallace, 1948)
	9		Serotype 1296/7 (Boyd, 1946; Ewing et al., 1951)
	10		Serotype 430 (Ewing and Taylor, 1951) and D15 (Szturm et al., 1950)
	11		Serotype 34 (Ewing and Taylor, 1951)
	12		Serotype 123 (and "M") (Ewing and Hucks, 1952)
	13		Serotype 425 (Ewing and Hucks, 1952)
	14		Serotype 2770-51 (Ewing and Hucks, 1952)
	15		Serotype 703 (Ewing et al., 1952a)
	16		Serotype 2710-54 (Ewing et al., 1958)
	17		Serotype 3615-53 (Ewing et al., 1958)
	18		Serotype E10163 (1344-78) (Gross et al., 1980)
	19		Serotype E16553 (Gross et al., 1982)
Subgroup D			
<i>S. sonnei</i>			Sonne-Duval, Sonne III, Kruse E, " <i>S. ceylonensis</i> " A (Duval, 1904; Sonne, 1915)

^aThe serologic subdivision of serotypes is done for only *S. flexneri*. The roman numerals designate the "type" or serotype-specific antigen and the Arabic numerals designate group antigens that are shared among members of the different types. Different subserotypes are defined by the combination of group antigens they possess. Group antigens shown in parentheses may be expressed in varying amounts (undetectable to strongly positive) by members of certain subserotypes. For the reactions of *S. flexneri* in typing antisera see Ewing (1986c). For *S. flexneri* serotype 3, only two subserotypes are currently recognized; strains previously designated as subserotype 3c are now identified as subserotype 3b [III:(3,4), 6] (Petrovskaya and Khomenko, 1979; Brenner, 1984b).

lipopolysaccharide (LPS) molecule. For *S. sonnei*, colonies of freshly isolated strains with a smooth appearance are termed form I and are usually virulent. Form I colonies are genetically unstable and dissociate to rough-appearing colonies termed form II, which lack 2-amino-deoxy-L-altruonic acid from their LPS O-repeating unit (Kontrohr, 1977). This form variation in *S. sonnei*

is associated with the loss of the 180–220 kilobase pairs (kbp) invasion plasmid. Colonial variation, which also correlates with the presence of the invasion plasmid, has been described for colonies growing on agar containing the dye Congo red. Smooth colonies that appear red on this agar are virulent, while colorless colonies are avirulent (Maurelli et al., 1984a). The dye interacts

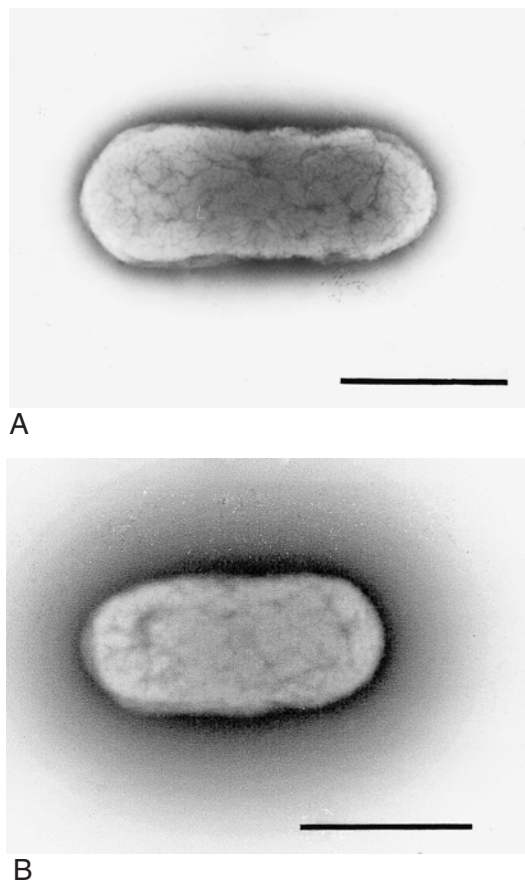


FIGURE BXII.γ.209. Electron micrograph of (A) *S. dysenteriae* 1 strain A5468 and (B) *S. flexneri* 2a strain 3342-87 prepared by negatively staining in 0.5% uranyl acetate. Bar = 1000 nm. (Micrograph courtesy of Charles D. Humphrey, CDC.)

with outer membranes and outer membrane proteins, but not with lipopolysaccharides. Only smooth colonies are scored for virulence using this dye because rough colonies, which are avirulent, can nonspecifically bind the dye, presumably by exposure of non-virulence-associated outer membrane proteins.

Nutrition and growth conditions The shigellae are aerobes and facultative anaerobes. Their optimal growth temperature is about 37°C. They generally grow less rapidly than most strains of *E. coli* and other members of the family *Enterobacteriaceae*. The average interval between cell divisions of *S. dysenteriae* in milk was reported to be 23 min compared with 12.5 min for *E. coli* (Sinclair, 1972). Ahmed et al. (1988) studied the nutritional requirements of 375 clinical isolates from the four species of *Shigella* for growth in a minimal medium containing glucose, ammonium sulfate, and inorganic salts and observed that most isolates would not grow in this medium. They found that virtually all isolates that failed to grow in the minimal medium could grow if the medium was supplemented with methionine, nicotinic acid, and tryptophan. Methionine and tryptophan appeared to be an obligatory requirement for *S. dysenteriae* 1 strains, while the combination of nicotinic acid and tryptophan was required for *S. dysenteriae* 2 strains.

In contrast to *Salmonella*, *Shigella* species and certain isolates of *E. coli* are acid-resistant (Gordon and Small, 1993; Small et al., 1994). Stationary phase cells of *S. flexneri* have the ability to sur-

vive for several hours at pH 2.5, which is hypothesized to contribute to their low infective dose by allowing them to safely pass through the stomach before colonizing the intestinal tract. The ability to survive exposure to acid is a complex phenotype, which depends on growth phase, media, and possibly a variety of proteins located at different subcellular locations. At least three genes, *hdeA*, *rpoS*, and *gadC*, play a role in mediating the acid-resistance phenotype under various conditions (Waterman and Small, 1996).

Metabolism and metabolic pathways Like *E. coli*, shigellae are chemoorganotrophic and have both a respiratory and a fermentative type of metabolism. They are generally less metabolically active than typical *E. coli* strains. They produce pyruvate from the fermentation of glucose and other sugars and convert it primarily by mixed acid (formic acid) fermentation to formic acid, acetic acid, and ethanol. In contrast to *E. coli* strains, shigellae do not produce gas from the fermentation of sugars (anaerogenic), because they lack the complex hydrogenlyase system (formic hydrogenlyase) that splits formic acid into equal amounts of CO₂ and H₂. Exceptions include some strains of *S. flexneri* serotype 6 and *S. boydii* serotypes 13 and 14. In addition to the traits described above, members of the genus *Shigella* do not hydrolyze urea and do not produce detectable amounts of acetoin (acetyl methyl carbinol) or 2,3-buteneglycol (Voges-Proskauer test negative). They also do not produce phenylalanine deaminase and do not produce acid from α-D-methylglucoside, erythritol, or esculin. They are uniformly negative for lysine decarboxylase. The pathogenic and evolutionary significance of this negative phenotype is addressed below in the pathogenicity section.

Shigella strains vary in their ability to metabolize glycerol. A study characterizing the glycerol dehydrogenase activity among *Shigella* species showed that the type of glycerol dehydrogenase produced could be used to help differentiate *Shigella* species from *E. coli* (Bouvet et al., 1995b). In this study, no strain of *Shigella*, with the exception of the Manchester biotype of *S. flexneri* serotype 6, was found to produce an NAD⁺-linked glycerol dehydrogenase, glyDH-II, while the majority of *E. coli* strains contained this enzyme.

Genetics Extensive genetic homology between *Shigella* and *E. coli* has permitted the use of classical genetic techniques, such as conjugal transfer and integration of chromosomal material between *E. coli* and *Shigella* (Luria and Burrous, 1957; Falkow et al., 1963; Formal et al., 1970, 1971; Sansonetti et al., 1983) and the generation of R plasmids (Timmis et al., 1985), to construct maps of the *Shigella* chromosome based on the known map for *E. coli* K-12. A *NotI* restriction map of the chromosome of *S. flexneri* 2a, with the assignment of nine virulence-associated loci identified by Tn5 insertions, was constructed by Okada et al. (1991). Genes that play a role in pathogenesis have received extensive study and were reviewed by Parsot and Sansonetti (1996), and are highlighted below in the pathogenicity section. Genes involved in O-antigen synthesis have also received considerable attention and are reviewed by Brahmabhatt et al. (1992) and discussed below in the antigenic structure section.

The work of Brenner et al. (1972a, 1973a) clearly established the genetic basis for the close biochemical and serologic relatedness between *E. coli* and *Shigella* that had been appreciated for many years. Their findings from DNA reassociation reactions followed by thermal elution chromatography on hydroxyapatite showed that, with the exception of *S. boydii* 13 strains, strains of

Shigella from all four species were as related to each other (sharing 80% or more of their nucleotide sequences) as they were to strains of *E. coli* (Brenner et al., 1973a). Estimates of genome size by I-CeuI macrorestriction analysis revealed that the chromosome sizes of all four *Shigella* species were in a similar range to those observed for *E. coli* (Shu et al., 2000). By this analysis, the genome sizes for the type strains of the four species of *Shigella* were 4.415×10^6 bp for *S. dysenteriae*, 4.792×10^6 bp for *S. flexneri*, 4.645×10^6 bp for *S. boydii*, and 4.501×10^6 bp for *S. sonnei*, while those for *E. coli* were 4.653×10^6 bp and 4.816×10^6 bp for two *E. coli* strains (Shu et al., 2000). The genome sequences of two different isolates of *S. flexneri* 2a have been published and are deposited in GenBank under accession numbers AE014073 and AE005674 (Jin et al. (2002); Wei et al., 2003).

Published ribosomal RNA sequences for the four species of *Shigella* and of *E. coli* are nearly identical, with more than 99% homology (Shu et al., 2000). Like *E. coli*, *Shigella* strains have seven ribosomal RNA operons as determined by restriction analysis with I-CeuI, an enzyme that specifically cuts a 26-bp site in the 23S rDNA sequence in *rrn* operons (Shu et al., 2000). The mapping of seven *E. coli* genes, each known to reside on a different I-CeuI fragment in *E. coli* K-12, revealed that some chromosomal rearrangements involving the fragments corresponding to fragments D and E of *E. coli* K-12 took place in *S. dysenteriae* and *S. flexneri*. Available findings did not permit a determination of the mechanism (translocation versus inversion of fragments) for these rearrangements. Hybridization patterns of *S. boydii* and *S. sonnei* strains were similar to those observed for *E. coli* strains, while the patterns for the *S. dysenteriae* and *S. flexneri* strains were each distinct.

Shigellae can carry a variety of insertion sequences (IS elements) including IS1 (Nyman et al., 1981; Kharat and Mahadevan, 2000); iso-IS1 (Ohtsubo et al., 1981); IS 2 (Soldati and Piffaretti, 1991); IS5 (Schoner and Schoner, 1981); IS200 (Gibert et al., 1990); IS 630 (Houng and Venkatesan, 1998); and IS911 (Prere et al., 1990). IS elements comprise over 6% of the sequenced genome of *S. flexneri* 2a; 314 IS elements were identified in strain 301 and 284 IS elements in strain 2457T (Jin et al. (2002); Wei et al., 2003). Some of these, such as IS 1 and iso-IS1, are present in large numbers in the chromosome of *Shigella* (50 to >150 copies per chromosome, respectively) and have been associated with negative phenotypes. For example, the inability of *S. sonnei* to utilize the β -glucoside salicin has been shown to involve the insertional inactivation of the gene *bglB*, encoding phospho- β -glucosidase B, by a novel IS (Kharat and Mahadevan, 2000). In contrast to adverse effects of IS elements on some genes, the IS 630 element appears to be required for the stable expression of form I antigen in *S. sonnei* (Houng and Venkatesan, 1998).

Plasmids, phages and phage typing, bacteriocins Shigellae carry a variety of plasmids. The most notable is the large (180–220 kbp) invasion plasmid, which belongs to the RepFIIA family of replicons (Silva et al., 1988) and is present in all virulent strains of *Shigella*. This plasmid carries genes that play an essential role in these organisms' ability to cause invasive disease and appears to contain regions from different origins (mol% G + C values ranging from 30% for *virF* to 49% for *sepA*). The complete DNA sequence of the virulence plasmid of *S. flexneri* has been determined (Buchrieser et al., 2000; Venkatesan et al., 2001). The organization and origin of the genes encoded on the plasmid

are discussed in these articles. In some serotypes, plasmids carry genes involved in synthesis of the O antigen (Kopecko et al., 1980; Watanabe and Timmis, 1984). Antibiotic resistance plasmids are also common among the shigellae. Because of the variety of plasmids carried by *Shigella*, plasmid profile analysis has been used widely to discriminate between strains during outbreak investigations. The use of this technique has declined in recent years in favor of molecular methods that examine differences in total cellular DNA and are less influenced by environmental or host pressures that can result in loss or acquisition of plasmids.

Shigellae have also been reported to carry Hsd (host specificity for DNA) plasmids. Strains of *S. flexneri* and *S. sonnei* carrying plasmid-encoded isoschizomers of either *EcoRII* or *NciI* (Lee et al., 1997) and a strain of *S. boydii* serotype 13 carrying a plasmid-encoded isoschizomer of *NruI* (Mise et al., 1986) have been described.

Shigellae are susceptible to a variety of bacteriophages, and this property has permitted the development of several phage typing systems (Thomen and Frobisher, 1945; Hammarström, 1947, 1949; Slopek et al., 1968, 1973; Lazlo et al., 1973; Pruneda and Farmer, 1977). A detailed description of the methods and the different bacteriophages used for typing *Shigella* was reported by Bergan (1979). Systems described to date, which typically use a panel of 10–20 bacteriophages to detect different patterns of lysis on a test strain, have focused on discriminating between strains within a serogroup or serotype. Discrimination between strains is generally good; for example, the method described by Lazlo et al. (1973) identified 90 types among more than 4000 strains of *S. flexneri* tested, and that of Pruneda and Farmer (1977) discriminated 87 different types among 265 strains of *S. sonnei*. Control of assay conditions, particularly the media and the smoothness of the cultures, is important for obtaining reproducible results. While the majority of strains can be characterized by phage typing, common phage types exist and it may be necessary to apply additional typing methods to achieve adequate discrimination.

Genes carried by some bacteriophages in shigellae have been identified. Among the serotypes of *S. flexneri*, bacteriophages have been demonstrated to carry genes, such as acetyltransferase, glucosyltransferase, or novel O-antigen polymerase genes, that play a role in determining the type and group determinants of the O antigens (Gemski et al., 1975; Lindberg et al., 1978; Clark et al., 1991a; Huan et al., 1997a, b; Mavris et al., 1997; Guan et al., 1999). Because many bacteriophages use O-antigenic polysaccharide chains as receptors for adsorption and infection, the modification of the host bacterial O antigens by the incoming phages is hypothesized to be a protective mechanism for excluding homologous phages from entering and may also help lysogenized cells evade preexisting host immunity.

Findings reported by McDonough and Butters (1999) provide evidence, in the sequences surrounding the Shiga toxin gene, suggesting that *S. dysenteriae* 1 was once lysogenized by a Shiga toxin-encoding lambdoid prophage, which became defective as a result of deletions after IS element insertions and rearrangements. These authors also provided evidence that Shiga toxin-encoding phages from *E. coli* can be stably introduced into *Shigella*. Other than *S. dysenteriae* 1, Shiga toxin-producing strains of *Shigella* are extremely rare.

Antigenic structure The different serotypes are distinguished by antigenic determinants that reside in the O antigen, which is part of the LPS molecule. The LPS molecule consists of three

parts: lipid A, which is made up of sugars and fatty acids and anchors the LPS molecule in the outer membrane; the core, which is made up of a single sequence of heptoses and hexoses and links the lipid A to the O antigen; and the O-antigen chain, which is composed of a repetitive sequence of hexoses and extends from the surface of the bacterium. The genetics of O antigen biosynthesis have been extensively studied and are reviewed by Schnaitman and Klena (1993) and Whitfield (1995). O antigens may contain from 10 to as many as 30 repeats of an oligosaccharide unit (O unit) each composed of three to six sugars. The diversity of O antigens is a result of varying combinations of sugars in the O unit, the type of chemical linkages between them, and the presence of nonsugar moieties such as O-acetyl residues or amino acids. Six to nineteen genes may be involved in the synthesis of the O antigen, and these are typically located together on the chromosome in a region (10 kbp or greater) called the *rfb* cluster. Depending on the sugars making up the O unit, serotypes may have completely different sets of genes. For certain shigellae, such as *S. dysenteriae* 1, *S. sonnei*, and *S. flexneri* 1–5, one or more genes for the synthesis of the O unit are carried on plasmids or lysogenic bacteriophages. Like *E. coli* and other Gram-negative bacteria, *Shigella* have a conserved 39-bp sequence, designated JUMPstart for “just upstream of many polysaccharide-associated gene starts,” located in the noncoding region upstream of the *rfb* cluster (Hobbs and Reeves, 1994). This region has been used to develop a molecular method for serotyping *Shigella* isolates (Coimbra et al., 1999).

The O antigens of most serotypes of *Shigella* are identical or partially related to the O antigens of *E. coli* (Ewing, 1953, 1986c). At least 13 identical and numerous reciprocal serologic relationships exist between *Shigella* and *E. coli* (Ewing, 1986c).

With the acceptance of provisional serotypes E22383 and E23507 as new serotypes (Ansaruzzaman et al., 1995), *S. dysenteriae* contains 15 serotypes, each with a distinctive antigen by which it can be recognized; there are few cross-reactions, either within the species or with other species that present difficulties for serotyping strains. Of the different serotypes within this species, the O antigen of *S. dysenteriae* 1 has been the most extensively studied because of efforts to develop LPS-based antidysentery vaccines. Genes involved in the synthesis of its O antigen have been identified, and their structure and function were reviewed by Schnaitman and Klena (1993). Eight of these genes are located together in the chromosome and two genes (*rfb* and *rfe*) lie outside the chromosomal cluster; *rfb* is carried on a 9-kbp multicopy plasmid.

S. flexneri contains eight serotypes and nine subserotypes. The serotypes are antigenically related, but each has a qualitatively distinct major (type) antigen; the group antigens are shared by other members of the species. Because of the important intra-group relations, highly absorbed sera are needed for the detailed serotyping of *S. flexneri*. The immunochemical and genetic bases of the complex antigenic structures of the species have been summarized by Petrovskaya and Bondarenko (1977). The O antigens of all serotypes, except *S. flexneri* 6, contain group antigens 3, 4 as a main primary structure. The type-specific antigens I, II, IV, and V and the group antigens 7, 8 are all the result of phage conversion of the 3, 4 antigens resulting in the incorporation of α -glycosyl and/or glucosyl groups to the common O-repeating unit of the LPS molecule. Type-specific antigen III and group antigen 6 differ from the above antigens in that they contain acetyl groups. Nevertheless, these antigens are also formed as a result of phage conversion of the 3, 4 antigens. The genes in-

involved in synthesis of the common O-repeating unit are located on the chromosome in the *rfb* cluster.

The classification of *S. flexneri* 6 as a serotype of *S. flexneri* was questioned by Petrovskaya and Bondarenko (1977), who proposed that it be transferred to *S. boydii* because the immunochemical structure and genetics of its O antigen were more consistent with those found among serotypes of *S. boydii* than among serotypes 1–5, Y and X of *S. flexneri*. In 1984, the International Committee on Systematic Bacteriology Subcommittee on the Taxonomy of *Enterobacteriaceae* considered this proposal and upheld the 1973 recommendation of the Subcommittee Working Group on *Shigella* to reject reclassification of *S. flexneri* 6 as a serotype of *S. boydii* on the grounds that it would have no practical value and would cause confusion by altering a well-accepted classification system (Brenner, 1984b).

S. boydii contains 19 serotypes and each has a qualitatively distinct antigen; there may be some cross-reactions with antisera to other *Shigella* species, but these seldom interfere with diagnosis. Serotypes 10 and 11 share a major antigen, although each possesses a specific antigen.

S. sonnei contains only one serotype but may exist in one of two forms or “phases”: form I (smooth colonial appearance) and form II (rough colonial appearance). Each form has a distinctive antigen, and antiserum containing agglutinins for both phases should be used for identification. Form I cells, which are virulent and carry the 180-kbp invasion plasmid, are typically isolated from acutely ill individuals. These cells can rapidly and irreversibly switch to form II cells, which are avirulent, when the invasion plasmid is lost. Form II cells may be isolated from ill individuals; however, they are more frequently isolated from individuals during convalescence and toward the end of an outbreak.

The O antigen unit of form I cells is structurally unique compared with those from other shigellae or *E. coli*, and it is chemically identical to that found in serotype 17 of *Plesiomonas shigelloides* (Rauss et al., 1970; Kenne et al., 1980). With the exception of two (*wzz* and *wbg*), the genes responsible for expression of the form I O antigen are located on the invasion plasmid and are nearly identical to those present in *P. shigelloides* (Shepherd et al., 2000). Factors influencing the loss of the plasmid carrying these genes are not well understood; however, Hough and Venkatesan (1998) propose that an insertion element present in the plasmid may be necessary for the stable expression of the form I antigen. These authors discovered the insertion element IS 630 within the O antigen gene cluster of all virulent strains of *S. sonnei* and observed that recombinant strains of *E. coli* carrying the cloned *S. sonnei* O antigen genes could stably express the form I antigen only when the IS 630 element was present.

While the form I O antigen expressed by *S. sonnei* is encoded by the plasmid-borne O antigen genes from *P. shigelloides* (Shepherd et al., 2000), findings by Lai et al. (1998) show that *S. sonnei* once had a typical chromosomal O antigen gene cluster. The remnant O antigen gene cluster apparently arose from a deletion due to homologous recombination between *manB* genes present in the adjacent O antigen and colanic acid gene clusters. The remnant chromosomal O antigen genes are not functional.

In addition to the recognized serotypes of shigellae, a number of provisional *Shigella* serotypes have been described. These may be added to the serotyping schema in the future, but in the meantime they remain *sub judice*, and antisera for their identification are usually available only at a very few reference laboratories. Provisional serotypes under consideration at present include serotypes E670/74 (Gross et al., 1989), 3162-96 (Kuijper

et al., 1997), 93-119 (Matsushita et al., 1997), 96-204 (Matsushita et al., 1998), and 96-265 (Frank Rogers, National Laboratory for Enteric Pathogens, Health Canada, Winnipeg, Canada; personal communication), which are biochemically consistent with members of subgroup A; serotypes Y394 (Wehler and Carlin, 1988), 88-893 (Matsushita et al., 1992a), 89-141 (Matsushita et al., 1992b), which are biochemically and antigenically consistent with members of subgroup B; and serotypes 1621-54 (Ewing et al., 1958), E28938 (Gross et al., 1989), and 99-4528 (Frank Rogers, National Laboratory for Enteric Pathogens, Health Canada, Winnipeg, Canada; personal communication), which are biochemically consistent with members of subgroup C.

Pathogenicity Shigellae are pathogens of humans and other primates. Although there have been occasional reports of infections in dogs, other animals are resistant to infection. Laboratory animals such as mice, rabbits, and guinea pigs may be infected orally but only following starvation and treatment with gastric antacids and antiperistaltic agents.

Shigellosis (bacillary dysentery) is transmitted orally through contaminated food and water or by direct fecal-oral spread. A review of the literature estimated the annual number of cases throughout the world to be 164.7 million, of which 163.2 million were in developing countries (with 1.1 million deaths). A total of 69% of all episodes and 61% of all deaths attributable to shigellosis involved children less than 5 years of age (Kotloff et al., 1999). Studies of volunteers have demonstrated that ingestion of as few as 200 organisms is sufficient to cause dysentery (DuPont et al., 1989). The incubation period is from 1 to 7 days with symptoms commonly manifesting on day 3. Shigellosis often (but not always) begins with a watery diarrhea that precedes the characteristic dysentery symptoms. The diarrhea phase probably results from production of enterotoxins by the bacteria as they transit through the small intestine.

The lesions of bacillary dysentery are usually restricted to the rectum and large intestine, but in severe cases part of the terminal ileum may be affected. *Shigella* penetrates into the epithelial cells lining the colon, multiplies within these cells, and spreads from cell to cell through the mucosa. The foci of infected cells coalesce to form abscesses. Small volume bloody and mucoid stools contain dead cells along with mucus and large numbers of bacteria. Typically there is acute inflammation with ulceration of the epithelium, and the presence of polymorphonuclear leukocytes in the stools is consistent with the inflammatory nature of shigellosis. Common clinical signs include fever, severe abdominal pain, and cramping. The most severe forms of shigellosis are caused by strains of *S. dysenteriae* 1, which also produces a potent cytotoxin (Shiga toxin) that has been shown to play a role in the severity of the illness it causes (Fontaine et al., 1988). Shiga toxin-producing strains can cause hemolytic uremic syndrome. *S. sonnei* strains cause milder forms of the disease, while *S. flexneri* and *S. boydii* strains can cause either severe or mild illness. Bacillary dysentery is a self-limiting disease. The organisms rarely spread deeper than the lamina propria, and bloodstream involvement is uncommon.

Treatment of shigellosis with appropriate antimicrobial therapy results in decreased duration of both symptoms and excretion of the organism (Haltalin et al., 1967; Salam and Bennish, 1991); however, empirical antimicrobial therapy for this illness has become complicated by the increasing incidence of multiple drug resistance among *Shigella* strains. Of historical note, the first observation of multiple, transferable drug resistance in bacteria

was in *Shigella* in Japan (Ochiai et al., 1959). Over the years, *Shigella* species have become increasingly resistant to most of the widely used antibiotics (Bennish et al., 1992). Multiple antibiotic resistance in *S. dysenteriae* 1 is so widespread that it is uniformly susceptible only to the fluoroquinolones (Sack et al., 1997). Antimicrobial resistance surveys of *Shigella* species conducted during the 1990s in developed and developing countries showed that multiple drug resistance exists in strains of all species (Keusch and Bennish, 1998; Mates et al., 2000; Replogle et al., 2000). The resistance patterns varied by species and by country, highlighting the need for monitoring of local antimicrobial resistance patterns to develop appropriate empirical therapy. Because *Shigella* infections are typically self-limiting and because antibiotic resistance frequently develops after treatment, some have proposed that antimicrobial therapy be reserved for the most severely ill patients, immunocompromised patients, and patients for whom eliminating carriage is a public health priority (Weissman et al., 1973; Replogle et al., 2000). Others feel that, because the infection is generally transmitted from person to person and the infected person represents the major reservoir of infection, each patient with a positive stool culture or with known bacillary dysentery should be treated to prevent the spread of infection (DuPont, 2000).

The hallmarks of *Shigella* pathogenicity are induction of diarrhea, the ability to invade eucaryotic cells, multiplication inside these cells, and spread from cell to cell. The rabbit ileal loop assay has been used as an experimental model to demonstrate production of at least two different enterotoxins by *S. flexneri*, which may play a role in the diarrhea caused by the organism (Fasano et al., 1995; Nataro et al., 1995). The invasive properties of *Shigella* have been demonstrated in experimental infection of monkeys (LaBrec et al., 1964) as well as in the Serény test, which measures the ability of the bacteria to invade and produce keratoconjunctivitis in the guinea pig eye (Serény, 1957). Invasion can also be demonstrated in mammalian cells in tissue culture (LaBrec et al., 1964; Ogawa et al., 1967; Day et al., 1981). The ability of the bacteria to spread from cell to cell after invasion is measured by the plaque assay (Oaks et al., 1985). While shigellae are killed after being taken up by polymorphonuclear leukocytes (Mandic-Mulec et al., 1997), they induce apoptosis in macrophages and kill these cells after uptake (Zychlinsky et al., 1992). *S. dysenteriae* serotype 1 produces Shiga toxin, a potent inhibitor of eucaryotic protein synthesis (O'Brien et al., 1992). Shiga-like toxins are also produced by enterohemorrhagic strains of *E. coli* (O'Brien et al., 1984).

Shigella strains invade the intestinal mucosal surface via a pathogen-directed phagocytic process that actively involves elements of the host cytoskeleton (Tran Van Nhieu et al., 2000). Unlike most other intracellular bacterial pathogens, however, *Shigella* strains rapidly lyse the endocytic vacuole upon entry and are released free into the host cell cytoplasm where they replicate. Shigellae are actively motile during growth inside the host cell. This motility is unusual in that it is not driven by bacterial-based flagella. Rather a bacterial protein, expressed in the outer membrane at one pole of the bacterium, catalyzes the polymerization of host cell actin filaments (Makino et al., 1986; Bernardini et al., 1989). Formation of actin tracks literally propels the bacterium through the cytoplasm as actin monomers polymerize into long filaments from one pole of the bacterium. This motility is required for cell-to-cell spread of *Shigella*. Mutant strains of *Shigella* that fail to display this intracellular motility produce a greatly attenuated form of disease in the monkey model, thus confirm-

ing this phenotype as a critical hallmark of *Shigella* pathogenesis (Sansone et al., 1991).

Virulence in *Shigella* spp. is dependent on expression of a group of genes encoded on a large plasmid. The role of this plasmid in *Shigella* virulence was first demonstrated in *S. flexneri* and *S. sonnei* (Sansone et al., 1981, 1982). Subsequently, all virulent species of *Shigella* and enteroinvasive strains of *E. coli* were shown to carry a 180–220-kbp plasmid that shares substantial DNA homology with these prototype virulence plasmids (Sansone et al., 1985). Introduction of these plasmids into plasmid-cured strains of *Shigella* or laboratory strains of *E. coli* K-12 imparts on the bacteria the ability to invade mammalian cells in tissue culture. Thus, the *Shigella* virulence plasmid encodes all the genes required for the invasive phenotype of *Shigella*. The DNA sequence of the virulence plasmid of *S. flexneri* 5a has been determined and deposited in GenBank under accession numbers AF348706, AL391753, and AF386526 (Buchrieser et al., 2000; Venkatesan et al., 2001; Jin et al. (2002)).

A 37-kbp region of the plasmid contains the minimal sequence needed for invasion (Maurelli et al., 1985; Baudry et al., 1987). The genes in this region are roughly grouped into two clusters transcribed from opposite strands. They encode the Ipa proteins, required for inducing uptake of the bacteria by mammalian cells, and Mxi/Spa proteins of a dedicated type III secretion apparatus for transport of the Ipa proteins outside the bacteria. Type III secretion systems are specialized transport systems found in animal and plant pathogens and are responsible for mediating interactions between the bacterial pathogen and its eucaryotic host (Cornelis and Van Gijsegem, 2000). The Mxi and Spa proteins of the *Shigella* type III secretion system show strong homologies with the analogous proteins of the type III secretion systems of *Salmonella* Typhimurium, enteropathogenic and enterohemorrhagic *E. coli*, and *Yersinia* spp. The *Shigella ipa* genes (*ipaBCDA*) encode the secreted effector molecules that mediate the bacteria-induced phagocytic (invasion) event. The Ipa proteins (invasion plasmid antigens) are the immunodominant antigens recognized by convalescent-phase sera from shigellosis patients and challenged monkeys (Maurelli et al., 1985; Oaks et al., 1986). Unlike the genes of the virulence plasmid that are involved in invasion and postinvasion steps, the chromosomal genes associated with virulence of *Shigella* appear to mostly encode modulators of pathogenicity.

Of interest from the point of view of bacterial evolution and pathogenesis are the genes that are present in nonpathogenic strains of *E. coli* but are missing from the chromosome of *Shigella*. A cryptic prophage in the *E. coli* K-12 chromosome that is missing in shigellae encodes an outer membrane protease, OmpT. When the *ompT* gene is introduced into *Shigella* by conjugation, the strain remains invasive but loses the ability to spread from cell to cell. This phenotype is due to OmpT protease degradation of IcsA, the bacterial protein responsible for actin polymerization and motility (Nakata et al., 1993). Another example of a missing genetic locus in the *Shigella* genome is *cadA*, the gene for lysine decarboxylase. Although lysine decarboxylase activity is present in >85% of *E. coli* strains, it is absent in all strains of *Shigella* spp. and enteroinvasive *E. coli*. When *cadA* is reintroduced into *S. flexneri*, the decarboxylation of lysine generates cadaverine. The *Shigella* enterotoxins are inhibited by cadaverine (Maurelli et al., 1998). Thus, these genes can be considered antivirulence genes and the missing genetic locus, a black hole. The example of *Shigella* illustrates another way that pathogens evolve from their nonpathogenic commensal relatives by both acquiring genes (e.g., the virulence plasmid) that contribute to virulence and

deleting genes that are incompatible with expression of these new virulence traits.

Growth temperature is the critical stimulus for regulation of virulence in *Shigella*. Wild-type *Shigella* strains grown at 37°C display all the attributes of virulence, whereas the same strains grown at 30°C are avirulent in the Serény test and fail to invade tissue culture cells (Maurelli et al., 1984b). The loss of virulence is reversible and the organisms regain full virulence (invasion and ability to provoke keratoconjunctivitis) after the growth temperature is shifted to 37°C. Paradoxically, growth of *Shigella* at 37°C in the laboratory leads to spontaneous deletion or loss of the virulence plasmid. Evidence suggests that expression of virulence genes *in vitro* is not well tolerated by the bacteria and leads to selection of variants that no longer express these genes either due to mutations in regulatory genes or deletion of plasmid sequences (Schuch and Maurelli, 1997). Strains grown at 30°C retain the virulence plasmid intact.

Despite many years of effort, an effective vaccine against shigellosis still has not been developed. Formulations have included killed whole cells, live attenuated strains, subunit vaccines (conjugate and proteosome), and ribosomal vaccines. The World Health Organization has placed *Shigella* vaccine development on its list of priorities for the Global Program for Vaccines and Immunization (WHO, 1997).

Ecology Shigellae are host-adapted to humans and subhuman primates, where they typically inhabit the intestinal tract. *Shigella* infections in captive primates are not uncommon; however, two groups of investigators (Takasaka et al., 1964; Carpenter and Cooke, 1965) found no evidence that wild primates living without human contact are infected. During the acute stage of the illness, shigellae are typically shed in large numbers in the feces. The number of organisms dramatically declines during recovery, and in some individuals a carrier state may develop in which organisms may persist in the feces for several weeks after the symptoms have subsided. The spread of *Shigella* infection is typically from person to person by fecal–oral transmission or from contaminated food or water. In situations without good sewage disposal and poor hygiene conditions, flies may act as vectors (Levine and Levine, 1991).

ENRICHMENT AND ISOLATION PROCEDURES

Food and water The minimum infecting dose of shigellae is small (10–100 organisms), and occurrence of the organisms in food, milk, and water may be significant even when only a small number of organisms are present. There are no reliable and effective enrichment methods, however, and the true incidence of *Shigella* contamination of foodstuffs cannot be accurately determined. The GN (Gram-negative) broth of Hajna (1955) may be useful for enrichment of *Shigella*, and it is recommended that the investigation of foodstuffs include an enrichment step using this medium. Subsequent steps in the isolation of *Shigella* from foods should follow the procedure recommended for fecal specimens.

Fecal specimens Freshly passed stools should be examined, although, if this is not possible, fecal swabs or rectal swabs showing marked fecal staining may be used. The specimens should be collected during the acute stage of the disease and before any chemotherapy is started. Specimens should be examined as soon after collection as possible. If the specimen includes blood and mucus, these should be included in the portion examined.

Although GN broth and Selenite broth are commonly used, no enrichment medium for *Shigella* consistently improves the

recovery rate beyond that obtained by direct plating alone. Because some strains of *Shigella* grow poorly on inhibitory media, both a relatively noninhibitory selective medium (e.g., MacConkey) and an inhibitory medium (e.g., xylose desoxycholate, deoxycholate citrate agar, or Hektoen enteric agar) should be used for isolation. Salmonella–Shigella (SS) agar is not recommended because it inhibits the growth of some strains of *S. dysenteriae* 1. Instructions for preparation of these media are given by Ewing (1986d) and Atlas (1997). Specimens are streaked onto the chosen media, and after overnight incubation at 35–37°C, lactose- or xylose-nonfermenting colonies are selected for further examination. Even when stool specimens from acute dysentery are examined, there may be only a scanty growth of *Shigella*. Suspicious colonies should be tested by biochemical and serologic methods (direct slide agglutination or coagglutination of antibody-coated latex beads) to confirm the identification of shigellae. For *S. flexneri*, which is the most difficult to prepare typing antisera against because of its shared group antigens, a panel of monoclonal antibodies has been developed for typing (Carlin et al., 1989). In some reference laboratories, molecular methods, such as DNA hybridization (Boileau et al., 1984), PCR (Frankel et al., 1990; Sethabutr et al., 1993; Yavzori et al., 1994; Houn et al., 1997; Villalobo and Torres, 1998; Sethabutr et al., 2000), enzyme-linked immunosorbent assays (Floderus et al., 1995; Pal et al., 1997), and immunomagnetic capture assays (Islam et al., 1993), targeting O antigens or invasion-associated genes/proteins are being used to detect shigellae and enteroinvasive *E. coli*.

Laboratory surveillance for *Shigella* infections is conducted in many countries and has been valuable for investigating outbreaks of shigellosis and for guiding vaccine development efforts. Serotyping has been and continues to be the primary method for discriminating between strains. During outbreaks or for highly prevalent serotypes such as *S. sonnei*, isolates are often subjected to a variety of additional phenotypic or molecular typing methods. Virtually all the secondary typing methods are predicated upon knowledge of the isolate's serotype. Other methods commonly used for differentiating between *Shigella* strains include colicin typing and phage typing. The colicin typing scheme for *S. sonnei* described by Abbot and Shannon (1958) has been used widely and distinguishes 14 types using 15 indicator strains (see Procedures for Testing Special Characters). Phage-typing schemes have also been used by many investigators. While a number of schemes have been described for *S. flexneri* and *S. sonnei*, only a few have been reported for *S. dysenteriae* and *S. boydii* (reviewed by Bergan, 1979). A variety of molecular subtyping methods have been applied to *Shigella*, which include ribotyping (Hinojosa-Ahumada et al., 1991; Faruque et al., 1992; Nastasi et al., 1993), IS element typing (Soldati and Piffaretti, 1991), enterobacterial repetitive intergenic consensus (ERIC) sequence-based PCR (Liu et al., 1995a), randomly amplified polymorphic DNA (Bando et al., 1998), and macrorestriction endonuclease analysis with pulsed-field gel electrophoresis (PFGE) (Soldati and Piffaretti, 1991; Liu et al., 1995a; Litwin et al., 1997). Because of its discriminatory potential, PFGE is the method preferred by many investigators for molecular subtyping studies, particularly for epidemiologic investigations of *S. sonnei* infections. This method proved invaluable in the investigation of a protracted outbreak of *S. sonnei* infections among members of traditionally observant Jewish communities in North America (Sobel et al., 1998) and was critical in an investigation of geographically unrelated cases of *S. sonnei* infection, identifying parsley from a single farm as the vehicle of infection (Centers for Disease Control, 1999).

MAINTENANCE PROCEDURES

Shigella cultures are maintained best frozen in liquid nitrogen. Liquid culture medium, such as trypticase soy broth, containing 10–20% glycerol is commonly used as a freezing medium. Cultures of *Shigella* may also be lyophilized or maintained on blood agar base or Dorset egg medium (stabbed slants or deeps) at room temperature; however, rough and degraded variants frequently arise under these conditions.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

For colicin typing of *S. sonnei*, the organism under investigation is inoculated heavily in a broad streak across a blood agar plate and incubated at 37°C for 24 h. The bacterial growth is then removed from the agar by scraping with a glass slide, and the organisms remaining are killed with chloroform. The 15 indicator strains are streaked onto the plate at right angles to the original line of growth. After further incubation for 8–12 h, the patterns of inhibition of growth of the indicator strains can be examined and compared with a key. It is important that controls be included in every batch of tests.

DIFFERENTIATION OF THE GENUS *SHIGELLA* FROM OTHER GENERA

The biochemical identification of *Shigella* is complicated by the similarity of some strains of other genera. In particular, strains of *Hafnia alvei*, *Providencia* spp., *Aeromonas* spp., and atypical *E. coli* can cause difficulties. Advances in our knowledge of the genetics of microorganisms have allowed the development of new approaches for identifying and detecting bacteria. Molecular strategies have been developed to facilitate microbial identification. Spierings et al. (1993) described a PCR assay targeting DNA sequences that encode the hypervariable surface-exposed regions of the outer membrane protein PhoE, which shows a high degree of species specificity and can differentiate strains of *Shigella* and *E. coli* from strains of other enteric bacteria.

Nonlactose-fermenting or anaerogenic strains of *E. coli* are commonly quite difficult to differentiate from shigellae. Of particular interest are members of the so-called Alkalescens Dispar (A-D) group, which contains nonmotile, anaerogenic biotypes of *E. coli*. These are best differentiated from *Shigella* by means of the Christensen's citrate and lysine decarboxylase tests, in which *Shigella* is always negative. The members of the A-D group were divided into eight serogroups based on their O antigens (Frantzen, 1950), although most of these are identical with or closely related to *E. coli* antigens.

TAXONOMIC COMMENTS

The genus *Shigella* consists of nonmotile organisms that conform to the definition of the family *Enterobacteriaceae* and have the biochemical and antigenic properties described above. Circumscription of the genus is problematic because of the close genetic relationship that exists between *Shigella* spp. and *E. coli*. There are no unique traits exclusively associated with *Shigella*. The negative phenotypes identified with *Shigella* spp. can occur in *E. coli*, particularly among the nonmotile, anaerogenic biotypes of *E. coli*, while positive phenotypes seen among typical *E. coli* isolates, such as lactose or sucrose fermentation and gas production from the fermentation of glucose, can occasionally occur in *Shigella*. Some *E. coli* strains (enteroinvasive *E. coli*) can also cause a dysentery-like illness, which is mediated by similar plasmid-encoded virulence genes as those possessed by *Shigella*.

Although pathogenicity (i.e., the ability to cause dysentery)

was not endorsed by the *Enterobacteriaceae* Subcommittee of the International Committee on Bacteriological Nomenclature for the classification of *Enterobacteriaceae* (Carpenter, 1963), it has influenced and continues to influence the classification of *Shigella* by virtue of the fact that only strains that are capable of causing dysentery receive consideration for addition to this group. Strains that do not cause dysentery but yield biochemical reactions consistent with those produced by *Shigella* are not considered for inclusion in this genus. The preservation of this association between bacteria classified as *Shigella* and a distinct form of diarrheal illness (dysentery) has sustained a functionally useful nomenclature that might otherwise have yielded over the years to persuasive findings that reveal a sufficiently close relationship between *Shigella* and *E. coli* to consider them the same species (Brenner et al., 1972a, 1973a).

The names adopted by Pupo et al. (2000), (*Escherichia coli* clone Dysenteriae, *Escherichia coli* clone Flexneri, *Escherichia coli* clone Boydii, and *Escherichia coli* clone Sonnei), reflect the well-documented natural relationships that exist between *Shigella* and *E. coli* and have merit from a phylogenetic perspective; however, their widespread adoption will happen only when it is important to users to communicate these relationships through the bacterial names. Until that time, the medically useful link between the genus epithet *Shigella* and shigellosis, the term for the disease caused by these bacteria, will sustain the current nomenclature.

Findings from numerous studies raise questions about the

inclusion of *S. boydii* 13 strains in the genus *Shigella*. The DNA-DNA reassociation findings reported by Brenner et al. (1982c) show that *S. boydii* 13 strains were related to each other, but clearly separable from other *Shigella* and *E. coli* (average 65% relatedness). Findings from MLEE (Ochman et al., 1983; Pupo et al., 1997), nucleotide sequence analysis of housekeeping genes (Pupo et al., 2000), ribotyping (Rolland et al., 1998), and esterase electrophoretic polymorphisms (Goullet and Picard, 1987) also show a distant relationship between *S. boydii* 13 strains and other shigellae.

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *SHIGELLA*

Biochemical characteristics useful for differentiating the species of *Shigella* are listed in Table BXII.γ.275.

List of species of the genus *Shigella*

1. ***Shigella dysenteriae*** (Shiga 1898) Castellani and Chalmers 1919, 935, Erit. Spec. Cons. Opinion 11 of the Jud. Comm. 1954a, 149^{AL} (*Bacillus dysenteriae* Shiga 1898, 817.)* *dys.en.te'ri.ae*. Gr. n. *dysenteria* dysentery; M.L. gen. n. *dysenteriae* of dysentery.

Colonies of serotype 1 often have a pinkish tinge on Leifson's deoxycholate citrate agar. Catalase is not produced by serotype 1, but is usually produced by strains of other serotypes. D-mannitol is not fermented. Dulcitol is fermented by strains of serotype 5. Gas from the fermentation of sugars is not produced. Indole is not produced by serotype 1, but is always produced by strains of serotype 2; strains of other serotypes vary in indole production. All the serotypes have, at one time or another, been known by other designations, and these are shown in Table BXII.γ.276. Members are intestinal pathogens of humans and subhuman primates, causing bacillary dysentery. Humans are the primary reservoir. A long-term carrier state occurs in a small percentage of cases (Levine et al., 1973). Serotype 1 causes more severe disease than other serotypes and produces a potent protein exotoxin (Shiga toxin), which plays a role in the severity of the disease attributable to this serotype and in the development of hemolytic uremic syndrome in some individuals. Large epidemics in developing countries are commonly caused by serotype 1. Disease caused by other serotypes may be mild or severe.

The mol% G + C of the DNA is: 53 (chemical analysis; Laskin and Lechevalier (1981)).

Type strain: ATCC 13313, CIP 57.28, NCTC 4837.

GenBank accession number (16S rRNA): X96966.

Additional Remarks: Neotype strain ATCC 13313 was designated by Judicial Commission, 1963.

2. ***Shigella boydii*** Ewing 1949, 634, Erit. Spec. Cons. Opinion 11 of the Jud. Comm. 1954a, 149^{AL}* *boy'di.i*. M.L. gen. n. *boydii* of Boyd; named after Sir John Boyd, a British bacteriologist.

Catalase is produced. D-mannitol is fermented. Dulcitol is usually fermented by serotypes 2, 3, 4, 6, and 10, but this may be delayed. D-xylose fermentation is variable. Indole may or may not be produced. Gas-producing biotypes of *S. boydii* serotype 13 (Rowe et al., 1975) and serotype 14 (Carpenter, 1961) have been described. Members are intestinal pathogens of humans and subhuman primates, causing bacillary dysentery. Disease may be mild or severe. Humans are the primary reservoir. A long-term carrier state occurs in a small percentage of cases (Levine et al., 1973).

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 8700, CIP 82.50, DSM 7532, NCTC 12985.

3. ***Shigella flexneri*** Castellani and Chalmers 1919, 937, Erit.

*Editorial Note: This species is also known as subgroup A.

*Editorial Note: This species is also known as subgroup C.

Spec. Cons. Opinion 11 of the Jud. Comm. 1954a, 149^{AL}* *flex'neri*. M.L. gen. n. *flexneri* of Flexner, named after Simon Flexner, an American bacteriologist.

Catalase is produced. D-mannitol is fermented, except by biotype Newcastle, serotype 6 and a D-mannitol negative, D-xylose positive biotype of serotype 4a (sometimes known as "*S. rabaulensis*"). Dulcitol is fermented by certain biotypes of serotype 6 (see Table BXII.γ.275), some of which produce gas from fermentable sugars. Indole is not produced by serotype 6; in other serotypes indole production is variable. Members are intestinal pathogens of humans and subhuman primates, causing bacillary dysentery. Humans are the primary reservoir. Disease may be mild or severe. A long-term carrier state occurs in a small percentage of cases (Levine et al., 1973). Infection in individuals with the HLA-B27 histocompatibility antigen can be complicated by Reiter chronic arthritis syndrome.

The mol% G + C of the DNA is: 49 (chemical analysis) (Laskin and Lechevalier, 1981) and 50.9 (nucleotide sequence analysis) (Jin et al., 2002) and (Wei et al., 2003).

Type strain: ATCC 29903, CIP 82.48, DSM 4782.

GenBank accession number (16S rRNA): X96963; complete genome: AE005673 and AE014073.

4. *Shigella sonnei* (Levine 1920) Weldin 1927, 182, Epit. Spec.

Cons. Opinion 11 of the Jud. Comm. 1954a, 149^{AL} (*Bacterium sonnei* Levine 1920, 31.)*
son'nei. M.L. gen. n. *sonnei* of Sonne, named after Carl Sonne, a Danish bacteriologist.

On deoxycholate citrate agar colonies are at first colorless, but after a few days show bright pink papillae consisting of lactose-fermenting cells. On MacConkey agar, phase I colonies are indistinguishable from colonies of other shigellas, but phase II colonies are larger, flatter, and more translucent, and have an irregular edge. On subculture, phase I colonies produce both phase I and phase II colonies, but phase II colonies give rise to phase II colonies only. D-mannitol is fermented rapidly, lactose and sucrose more slowly. Some strains may ferment D-xylose. Catalase is produced. Indole is not produced. Ornithine is decarboxylated; arginine may be decarboxylated. Members are intestinal pathogens of humans and subhuman primates, causing bacillary dysentery. Humans are the primary reservoir. A long-term carrier state occurs in a small percentage of cases (Levine et al., 1973). Illness is typically milder than that caused by members of the other *Shigella* subgroups.

The mol% G + C of the DNA is: 51 (CsCl buoyant density).

Type strain: ATCC 29930, CIP 82.49, DSM 5570, NCTC 12984.

Other Organisms

S. boydii 13 strains were first described in 1952 by Ewing and Hucks (1952) under the provisional designation "serotype 425". Six years later, serotype 425 was added to the *Shigella* scheme as *S. boydii* 13 (Ewing et al., 1958). The initial description of the serotype included 10 strains: eight from cases and two from carriers isolated from persons from Italy, Belgian Congo, Egypt, and the United States. Except for differences in their ability to grow on ammonium salts glucose agar, the strains were biochemically indistinguishable from each other and most closely resembled shigellae of subgroup C. In 1973, Brenner et al. (1973a) reported evidence from DNA-DNA reassociation studies of *Shigella* species challenging the placement of *S. boydii* 13 strains in the genus *Shigella*. These investigators found that the *S. boydii* 13 strains they examined were highly interrelated but averaged only about 65% relatedness to other shigellae or members of the genus *Escherichia*. In a subsequent study, Brenner et al. (1982c) published findings from additional *S. boydii* 13 strains that extended and confirmed the earlier observation that *S. boydii* 13 strains,

both anaerogenic and aerogenic varieties, are separable from other shigellae and *E. coli* based on DNA relatedness. The DNA-DNA reassociation evidence clearly demonstrates that *S. boydii* 13 strains represent a new, as yet unnamed, species. More recent findings from MLEE (Ochman et al., 1983; Pupo et al., 1997), nucleotide sequence analysis of housekeeping genes (Pupo et al., 2000), ribotyping (Rolland et al., 1998), and esterase electrophoretic polymorphisms (Goullet and Picard, 1987) are consistent with a distant relationship between *S. boydii* 13 strains and the other shigellae. At present, *S. boydii* 13 strains are primarily isolated by clinical laboratories and routinely keyed out as belonging to the genus *Shigella*. Because practical and logical factors are important in considering changes to existing systems of taxonomy and nomenclature (Brenner et al., 1973a), the renaming or reclassification of these bacteria is dependent on the development of methods that are useful for their identification in clinical laboratories.

Genus XXXVI. *Sodalis* Dale and Maudlin 1999, 273^{VP}

COLIN DALE, SERAP AKSOY, SUSAN C. WELBURN, IAN MAUDLIN AND AHARON OREN

so.da'lis. L. masc. n. *sodalis* a companion.

Cells rod-shaped. Nonmotile. Gram negative. Endospore formation not observed. **Microaerophilic.** Chemoorganotrophic. Axenic growth occurs in media with enzymatically digested proteins serving as carbon and nitrogen sources. **Certain carbohydrates are used with the production of acids.** The optimum tem-

perature for growth is 25°C, with little or no growth occurring at temperatures >30°C. Catalase and oxidase negative. **Found as secondary intracellular symbiotic bacteria in insects.**

The mol% G + C of the DNA is: 53.5.

Type species: *Sodalis glossinidius* Dale and Maudlin 1999, 273.

*Editorial Note: This species is also known as subgroup B; formerly known as "*S. paradysenteriae*" Flexner.

*Editorial Note: This species is also known as subgroup D.

FURTHER DESCRIPTIVE INFORMATION

The genus *Sodalis* consists of maternally transmitted intracellular symbiotic bacteria inhabiting the hemolymph and other organs of insects. Most intracellular symbiotic bacteria of insects are fastidious, and have thus far proved refractory to conventional culture techniques. *Sodalis glossinidius*, a multitissue secondary intracellular symbiont of the tsetse fly (*Glossina morsitans*), was the first true intracellular symbiont from insects to be cultivated *in vitro*. Cultivation was first achieved through the use of a mosquito (*Aedes albopictus*) feeder cell culture system. Subsequently, the organism was isolated in pure culture on a semidefined solid medium under microaerobic conditions. Therefore, *Sodalis* is the first example of the successful cultivation of an insect secondary intracellular symbiont on an agar-based medium. Phylogenetic characterization of *Sodalis* strains from different tsetse species revealed no differences based on their 16S rDNA sequences (Chen et al., 1999b). Other insects related to the tsetse fly carry bacterial symbionts with 16S rDNA sequences very similar to the 16S rDNA of *Sodalis*, but these symbionts are still awaiting isolation.

Sodalis glossinidius, the only species of the genus described thus far, is microaerophilic and can be cultivated only on solid media under a reduced-oxygen atmosphere. An atmosphere of 5% oxygen and 95% carbon dioxide is optimal for growth. Alternatively, aerotolerance-enhancing supplements such as catalase or fresh horse blood as a source of catalase activity can be added to the plates to enable growth in air.

The biochemical capacities of *Sodalis glossinidius* are rather restricted. Among the few carbohydrates used for growth the most efficient are N-acetyl-D-glucosamine and raffinose, which are metabolized with massive acid production. Glucose, glycol chitosan, mannitol, and sorbitol are metabolized to a lesser extent.

Sodalis glossinidius has a chromosome of about 2 Mb and about 134 kb plasmid DNA (Akman et al., 2001). Its chromosomal DNA is subject to extensive adenine and cytosine methylation. Hybridization of *Sodalis* DNA to *Escherichia coli* macroarrays revealed the presence of about 1800 orthologs, indicating that *Sodalis* has retained many of the capabilities of free-living bacteria. Based on this analysis, it appears that *Sodalis* has retained many genes involved in transcription, translation, regulation, nucleic acid, and amino acid biosynthetic pathways. However, it is possible that *Sodalis* might have lost genes in carbon compound catabolism, central intermediary metabolism, and fatty acid and phospholipid metabolism, which could account for the organism's restricted biochemical capacities.

To enter into insect cells, *Sodalis* uses a type III secretion system invasion gene (*invC*), very similar to virulence determi-

nants previously identified in the genera *Salmonella* and *Shigella* (Dale et al., 2001).

ENRICHMENT AND ISOLATION PROCEDURES

Sodalis glossinidius has been isolated from the hemolymph of *Glossina morsitans morsitans* in coculture with the mosquito *Aedes albopictus* cell line C6/36 (Welburn et al., 1987). Pure culture isolation was achieved through the use of solid-phase culture on serum-free Mitsuhashi-Maramorosch basal medium (Igarashi, 1978) with 1% Bacto-agar (Difco) under a microaerobic atmosphere (O₂/CO₂; 5:95). For growth under an air atmosphere, aerotolerance-enhancing supplements such as catalase or fresh horse blood cells (as source of catalase activity) should be added.

MAINTENANCE PROCEDURES

Sodalis glossinidius can be maintained by coculture in *Aedes albopictus* C6/36 cells at 25°C in liquid Mitsuhashi-Maramorosch medium (Igarashi, 1978) supplemented with 20% (v/v) heat-inactivated fetal calf serum by passaging the cells every 10 d (Welburn et al., 1987). For long-time storage bacterial cells are frozen in Mitsuhashi-Maramorosch medium supplemented with 20% (v/v) heat-inactivated fetal calf serum and 15% (v/v) glycerol.

TAXONOMIC COMMENTS

Sodalis was classified in the family *Enterobacteriaceae* based on phenotypic tests and its 16S rDNA sequence. As a microaerophilic insect symbiont, it differs sufficiently from the other described genera within the *Enterobacteriaceae* to warrant classification as a separate genus. Different species and subspecies of tsetse flies harbor intracellular symbionts having almost identical 16S rDNA sequences (Beard et al., 1993; Aksoy et al., 1997). In addition, other insects, including flour and rice weevils and certain psyllids also harbor intracellular symbionts with closely related 16S rDNA sequences (Thao et al., 2000a).

Sodalis harbors type III secretion system invasion genes that are phylogenetically related to those found previously in *Salmonella enterica* and *Shigella flexneri*.

FURTHER READING

- Akman, L., R.V.M. Rio, C.B. Beard and S. Aksoy. 2001. Genome size determination and coding capacity of *Sodalis glossinidius*, an enteric symbiont of tsetse flies, as revealed by hybridization to *Escherichia coli* gene arrays. *J. Bacteriol.* 183: 4517–4525.
- Dale, C. and I. Maudlin. 1999. *Sodalis* gen. nov. and *Sodalis glossinidius* sp. nov., a microaerophilic secondary endosymbiont of the tsetse fly *Glossina morsitans morsitans*. *Int. J. Syst. Bacteriol.* 49: 267–275.
- Dale, C., S.A. Young, D.T. Haydon and S.C. Welburn. 2001. The insect endosymbiont *Sodalis glossinidius* utilizes a type III secretion system for cell invasion. *Proc. Natl. Acad. Sci. U.S.A.* 98: 1883–1888.

List of species of the genus *Sodalis*

1. ***Sodalis glossinidius*** Dale and Maudlin 1999, 273.^{VP}
glos.si.ni' di.us. N.L. adj. *glossinidius* of the genus *Glossinia*.

The characteristics are as described for the genus, with the following additional information. The cells are 1.0–1.5 × 2.0–12.0 μm. When grown intracellularly in *Aedes albopictus* cell culture, the organisms appear as pleomorphic rods. Temperature range for growth: 18–28°C. Colonies on Mitsuhashi-Maramorosch agar (Igarashi, 1978) are shiny, off-white, and concave, with entire edges.

Negative for DNase, gelatinase, urease, nitrate reductase,

indole production, hippurate hydrolysis, arginine dihydrolyase, lysine decarboxylase, ornithine decarboxylase, phenylalanine decarboxylase, and starch hydrolysis. Produces α-galactosidase and β-N-acetylglucosaminidase. Does not produce α-fucosidase, β-galactosidase, α-glucosidase, β-glucosidase, β-glucuronidase, α-mannosidase, and β-xylosidase. Utilization of N-acetyl-D-glucosamine and raffinose accompanied by a high level of acid production. Utilizes glucose, glycol chitosan, mannitol, and sorbitol with accompanying weak acid production. No increase in growth is detected

when any of the following carbon sources are incorporated in the media: acetic acid, adonitol, δ -aminovaleric acid, L-arabinose, *n*-butanol, citric acid, dulcitol, ethanol, fructose, fumaric acid, galactose, glycerol, glycolic acid, histamine, *p*-hydroxybenzoic acid, α -ketoglutaric acid, lactose, D,L-malic acid, maltose, mannose, melibiose, methyl- α -D-glucopyranoside, *myo*-inositol, *n*-propanol, pyruvic acid, quinic acid, rhamnose, ribose, saccharic acid, salicin, sarcosine, sorbose, succinic acid, sucrose, starch, trehalose, or xylose.

Contains a number of large extrachromosomal DNA elements.

Found as secondary intracellular symbionts in the midgut, fat body, and hemolymph of the tsetse fly *Glossina morsitans morsitans*.

The mol% G + C of the DNA is: 53.5 (T_m).

Type strain: M1, NCIMB 13495.

GenBank accession number (16S rRNA): M99060.

Genus XXXVII. *Tatumella* Hollis, Hickman and Fanning 1982, 267^{VP} (Effective publication: Hollis, Hickman, Fanning, Farmer, Weaver and Brenner 1981, 86) (Group EF-9 Hollis, Hickman, Fanning, Brenner and Weaver 1980)

J.J. FARMER III

Ta.tum.ell la. M.L. dim. neut. -ella ending; M.L. fem. n. *Tatumella* named to honor Harvey Tatum, an American bacteriologist who made many contributions to our understanding of the classification and identification of fermentative and nonfermentative bacteria of medical importance.

Small rod-shaped cells 0.6–0.8 $\times$ 0.9–3 μ m, conforming to the general definition of the family *Enterobacteriaceae*. Contains the enterobacterial common antigen. Gram negative. **Nonmotile at 36°C; over half the strains are motile by means of polar, subpolar, or lateral flagella when grown at 25°C.** Facultatively anaerobic. Catalase positive (weak and slow). Oxidase negative. Nonpigmented. **Stock cultures often die within a few weeks on laboratory media. Biochemically more active at 25°C than at 36°C.** Ferment, rather than oxidize, D-glucose; without the formation of visible gas. Reduce nitrate to nitrite. **Very inactive biochemically; positive tests only for Voges–Proskauer (Coblentz method), phenylalanine deaminase, and fermentation of sucrose, trehalose, and D-mannose. Negative for most tests:** indole production, methyl red, Voges–Proskauer (O'Meara method), citrate utilization (Simmons), H₂S production (TSI), urea hydrolysis, lysine decarboxylase, arginine dihydrolase, ornithine decarboxylase, growth in the presence of cyanide (KCN test), malonate utilization, esculin hydrolysis, ONPG, gelatin hydrolysis (22°C), lipase (corn oil), DNase, gas production during fermentation, and the fermentation of lactose, D-mannitol, dulcitol, adonitol, *myo*-inositol, D-sorbitol, L-arabinose, raffinose, L-rhamnose, maltose, D-xylose, cellobiose, α -methyl-D-glucoside, erythritol, D-arabitol, glycerol, and mucate. Have very large zones of inhibition around antibiotics; susceptible to colistin, nalidixic acid, sulfadiazine, gentamicin, streptomycin, kanamycin, tetracycline, chloramphenicol, carbenicillin, ampicillin, and cephalothin (disk diffusion method on Mueller–Hinton agar). **Large zone of inhibition around a penicillin G (10 U) disk, in contrast to most other *Enterobacteriaceae*.**

Isolated from human clinical specimens, mainly from the **respiratory tract** where clinical significance is questionable. Also isolated from blood cultures suggesting clinical significance. Probably **a rare opportunistic pathogen or colonizer of humans.**

The mol% G + C of the DNA is: 53–54.

Type species: *Tatumella ptyseos* Hollis, Hickman and Fanning 1982, 267 (Effective publication: Hollis, Hickman, Fanning, Farmer, Weaver and Brenner 1981, 86.)

FURTHER DESCRIPTIVE INFORMATION

Literature Since *Tatumella* was described in 1981, there have been only a few reports in the literature. In addition to the genus

and species names, “EF-9” should also be included as a search term in computerized literature searches, although it is rarely used today.

The *Tatumella* literature reports include reviews and taxonomic studies of new *Enterobacteriaceae* (Gilchrist, 1995; Aleksic and Bockemühl, 1999), and surveys or comparisons of the family *Enterobacteriaceae* for evolutionary relatedness based on 16S rRNA sequencing (Spröer et al., 1999), the enterobacterial common antigen (Ramia et al., 1982), catalase production (Chester and Moskowitz, 1987), metabolic pathways (Bouvet and Grimont, 1987a; Bouvet et al., 1989), polyamines (Hamana, 1996), and siderophores (Rabsch and Winkelmann, 1991).

Sources, strains, collections Strains of *Tatumella* have been isolated from a variety of human clinical specimens, but not from other sources. The original collection of Hollis et al. (1981) included isolates from body sites that are normally sterile such as blood (three isolates) and urine (one), but 38 were from throat, pharynx, tracheal aspirate, or sputum. One was from feces.

Clinical significance of human isolates The clinical significance and the ability of *T. ptyseos* to actually cause infections should be carefully evaluated in cases that yield this organism. Systematic study and good case reports are needed. Tan et al. (1989) reported a case of presumed sepsis in a neonate with jaundice who recovered uneventfully after treatment with antibiotics. This report and the three blood isolates described by Hollis et al. (1981) suggest that *T. ptyseos* should at least be considered as a rare opportunistic pathogen.

Isolation, identification, culture preservation, phenotypic characterization Strains of *Tatumella* are not difficult to grow, but may grow more slowly than typical *Enterobacteriaceae*. Stock cultures may die within a few weeks on agar or in semisolid stock culture media. This characteristic is very unusual, since most cultures of *Enterobacteriaceae* can be kept almost indefinitely in sealed tubes kept at room temperature. *Tatumella* cultures frozen in 5% rabbit blood and stored at -40°C to -60°C have remained viable after storage for up to 14 years. This latter method or storage in liquid nitrogen (or perhaps freeze-drying) should be suitable for long-term preservation. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for

biochemical testing, identification, computer programs, and antibiotic susceptibility. On MacConkey agar at 24 h, all strains grew and produced colorless (lactose negative) colonies.

Biochemical reactions and differentiation from other *Enterobacteriaceae* Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* gives the results for *Tatumella* in 47 biochemical tests normally used for identification (Farmer, 1999). The most striking feature of *T. ptyseos* strains is their biochemical inactivity. They are also more active biochemically at 25°C than 36°C. The flagellation of *Tatumella* is also unusual. Strains are nonmotile at 36°C, but 66% are motile at 25°C. No flagella are produced on most cells, but those seen are polar, subpolar, or lateral rather than peritrichous. In contrast to most other *Enterobacteriaceae*, cultures of *T. ptyseos* have large zones of inhibition around 10 U penicillin G disks (range of 15–36 mm, mean of 24 mm, standard deviation of 4.6 mm). There are no genus- or species-specific tests or sequences for the identification of *Tatumella*. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the section on *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae*. Several computer programs greatly facilitate analyzing the results.

TAXONOMIC COMMENTS

History The name *Tatumella* was coined by Hollis et al. (1981) for a group of organisms that had previously been known as “Group EF-9”, a name coined and used by the Special Bacteriology Section at the Centers for Disease Control (Hollis et al., 1980). Group EF-9 had been known for many years, but its taxonomic position had not been studied.

Circumscription of *T. ptyseos* (Hollis et al., 1981) Twenty-seven “suspect” strains of Group EF-9 were studied by DNA–DNA hybridization, biochemical reactions, and antibiotic susceptibility. Twenty of the strains were very highly related to the type strain (89% or greater with ΔT_m values of 0–0.5) by DNA–DNA hybridization (60°C, hydroxyapatite method with ^{32}P). A group of six strains was less related (80–90% with ΔT_m values of 5.4–7.8), but was included in the species *T. ptyseos*. These strains may eventually be classified as one or more distinct biogroups. One strain (H34) was considerably less related (66% related at 60°C, which dropped to 44% at 75°C), and was not included in *T. ptyseos*. This probably represents a second species of *Tatumella*. By DNA hy-

bridization, other taxa in the family *Enterobacteriaceae* were related by 7–38%, including 25–30% relatedness of *Escherichia*, the type genus of the family. Based on the DNA–DNA hybridization and phenotypic data, Group EF-9 was proposed as a new genus, *Tatumella*, with *T. ptyseos* as the only species (Hollis et al., 1981).

Phylogeny based on 16S rRNA sequencing A 16S rRNA tree that includes *T. ptyseos* can be found in Fig. BXII.γ.198 in the chapter on the genus *Budvicia* in this *Manual*. The 16S rRNA sequencing data agree with DNA–DNA hybridization and phenotypic data that *Tatumella* is a distinct genus of *Enterobacteriaceae*. This tree generally agrees with the tree published by Spröer et al. (1999), in that *Tatumella* is on a distinct branch. However, the two trees differ in the placement of the closest neighbors and other organisms (see also Fig. BXII.γ.189 in the chapter on the family *Enterobacteriaceae*).

Nomenclatural problem: the irregular author citation in the names *Tatumella* Hollis, Hickman and Fanning 1982, 267^{VP} and *Tatumella ptyseos* Hollis, Hickman and Fanning 1982, 267^{VP} The author citation for the names *Tatumella* and *Tatumella ptyseos* has traditionally been written as “Hollis, Hickman and Fanning 1982, 267^{VP}” rather than the expected citation of “Hollis, Hickman, Fanning, Farmer, Weaver and Brenner 1982, 267^{VP}”. The original paper (Hollis et al., 1981) had six authors rather than three, but on page 86 the authors stated: “We propose the following citations of the genus and species, which recognizes the greater contribution of three of the authors: *Tatumella* Hollis, Hickman and Fanning; *Tatumella ptyseos* Hollis, Hickman and Fanning.”

This irregular author citation continued through correspondence with the editor of the *International Journal of Systematic Bacteriology* that requested validation of the genus and species name. In Validation List 8, the request for a “three-author citation” was granted and was without comment since only three authors, “Hollis, Hickman and Fanning”, were listed under the column heading “Author(s)” for both names. This unusual citation appears to conflict with the rules of the Bacteriological Code (International Code of Nomenclature of Bacteria) for citing authors of new taxa. Although this is a small point, it is one that can lead to instability in nomenclature. The problem could be addressed by amending the Bacteriological Code, or through an Opinion of the Judicial Commission.

FURTHER READING

Hollis, D.G., F.W. Hickman, G.R. Fanning, J.J. Farmer, III, R.E. Weaver and D.J. Brenner. 1981. *Tatumella ptyseos* gen. nov., sp. nov., a member of the family *Enterobacteriaceae* found in clinical specimens. *J. Clin. Microbiol.* 14: 79–88.

List of species of the genus *Tatumella*

1. ***Tatumella ptyseos*** Hollis, Hickman and Fanning 1982, 267^{VP} (Effective publication: Hollis, Hickman, Fanning, Farmer, Weaver and Brenner 1981, 86.)
pty' se.os. Gr. n. *ptyseos* a spitting (or less literally from sputum, the most common source of clinical isolates).

The characteristics are as given for the genus and the results for 47 biochemical tests are given in Table BXII.γ.193 of the chapter on the family *Enterobacteriaceae*. Isolated from human clinical specimens, primarily from the respiratory tract; should probably be considered a rare opportunistic pathogen.

The mol% G + C of the DNA is: 53–54 (Bd).

Type strain: H36, ATCC 33301, CDC D6168, CDC 9591-78, DSM 5000.

GenBank accession number (16S rRNA): AJ233437.

Additional Remarks: Hollis et al. (1981) include 25 other strains in the species; however, six of these were less related to the type strain than the others (see previous discussion). One additional strain (very highly related to the type strain) was also deposited in the American Type Culture Collection (ATCC 33302 [H3 9558-78 = A7744]), from human sputum, Connecticut, USA.

Genus XXXVIII. *Trabulsiella* McWhorter, Haddock, Nocon, Steigerwalt, Brenner, Aleksic, Bockemühl and Farmer 1992, 327^{VP} (Effective publication: McWhorter, Haddock, Nocon, Steigerwalt, Brenner, Aleksic, Bockemühl and Farmer 1991, 1482) (Enteric Group 90 Murlin, Brenner, Steigerwalt and Farmer 1988)

JOHN A. LINDQUIST AND J.J. FARMER III

Tra.bul.si.el'la. M.L. dim. ending -ella; M.L. fem. n. *Trabulsiella* named to honor L.R. Trabulsi, a Brazilian bacteriologist who did many important studies on the enteric pathogens of the family *Enterobacteriaceae*.

Rod-shaped cells, conforming to the general definition of the family *Enterobacteriaceae*. Gram negative. Motile. Facultatively anaerobic. Catalase positive. Oxidase negative. Ferment, rather than oxidize, D-glucose and other carbohydrates and produce visible gas during fermentation. Reduce nitrate to nitrite. **Positive for methyl red, citrate utilization (Simmons), H₂S production (TSI), lysine decarboxylase, ornithine decarboxylase, motility at 36°C, growth in the presence of cyanide (KCN test), esculin hydrolysis, ONPG, tyrosine hydrolysis, and the fermentation of D-mannitol, salicin, D-sorbitol, L-arabinose, L-rhamnose, maltose, D-xylose, trehalose, cellobiose, esculin, mucate, D-mannose, and D-galactose. Arginine dihydrolase is positive, but delayed.** Negative for Voges-Proskauer, urea hydrolysis, phenylalanine deaminase, malonate utilization, gelatin hydrolysis (22°C), lipase (corn oil), DNase, and the fermentation of sucrose, dulcitol, adonitol, myo-inositol, raffinose, α-methyl-D-glucoside, erythritol, melibiose, and D-arabitol. **Biochemically similar to, and can be confused with, *Salmonella* in routine screening tests. Do not completely type in *Salmonella* O or H antisera, and are negative in the *Salmonella*-specific test: lysis by bacteriophage O1, MUCAP (4-methylumbelliferyl caprylate), and a commercial gene probe (Gene-Trak Systems).** Susceptible to colistin, nalidixic acid, gentamicin, streptomycin, kanamycin, tetracycline, chloramphenicol, and carbenicillin; **resistant to penicillin, ampicillin, and cephalothin**; variable resistance to sulfadiazine and carbenicillin (disk diffusion method on Mueller-Hinton agar).

Ecological niches are not known, but have been isolated from environmental samples and human feces. However, there is **no evidence that *Trabulsiella* causes diarrhea or intestinal infections.** Apparently a **very rarely isolated genus of *Enterobacteriaceae*.**

The mol% G + C of the DNA is: not determined.

Type species: *Trabulsiella guamensis* McWhorter, Haddock, Nocon, Steigerwalt, Brenner, Aleksic, Bockemühl and Farmer 1992, 327 (Effective publication: McWhorter, Haddock, Nocon, Steigerwalt, Brenner, Aleksic, Bockemühl and Farmer 1991, 1483.)

FURTHER DESCRIPTIVE INFORMATION

Literature Since *Trabulsiella* was described in 1991, there have been only a few reports in the literature. In addition to the genus and species names, "Enteric Group 90" should also be included as a search term in computerized literature searches, although it is rarely used today.

Sources Six strains of *Trabulsiella* were isolated from vacuum cleaner dust on the island of Guam, three were from human stool specimens (one from New York, USA, and two from Germany), one was from wheat flour from Oregon, USA, and two were from environmental material (not further specified) in Malaysia. Many of these isolates were originally isolated and studied as "suspect *Salmonella* cultures".

Isolation, identification, culture preservation, phenotypic characterization Strains of *Trabulsiella* are not difficult to grow,

and are typical *Enterobacteriaceae* in most respects. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility. Strains of *Trabulsiella* will often appear as typical *Salmonella* strains on some enteric plating and screening media; however, they do not type completely in *Salmonella* O and H antisera.

Biochemical reactions and differentiation from other *Enterobacteriaceae* Table BXII.γ.193 in the section on the family *Enterobacteriaceae* gives the results for *Trabulsiella* in 47 biochemical tests normally used for identification (Farmer, 1999). There are no genus- or species-specific tests or sequences for the identification of *Trabulsiella*. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the chapter on the family *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 of that chapter. Several computer programs greatly facilitate analyzing the results. *Trabulsiella* strains are phenotypically most like *Salmonella* subgroups 4 and 5, which are rarely isolated from human clinical specimens.

TAXONOMIC COMMENTS

In 1985 the vernacular name Enteric Group 90 was coined for a group of strains that had been studied at the *Salmonella* Laboratory at the Centers for Disease Control and Prevention. From 1985 to 1988 eight strains were characterized that were phenotypically distinct from all of the described organisms in the family *Enterobacteriaceae*. They had been submitted as "biochemically *Salmonella*, but will not serotype". Since the strains were biochemically very close to *Salmonella*, particularly to *Salmonella* DNA hybridization groups 4 and 5, they were originally suspected as being a new DNA hybridization group in the genus *Salmonella*. However, by DNA-DNA hybridization the strains were very highly related to each other, but were more distantly related to other organisms in the family *Enterobacteriaceae*. Strain ATCC 49490 (CDC 370-85) that eventually became the type strain of *Trabulsiella guamensis* was 41% related by DNA-DNA hybridization to *Salmonella* serotype Typhimurium and to a strain of *Salmonella* DNA hybridization group 3b. Less relatedness was found to *Kluyvera ascorbata* (39%), *Shigella flexneri* (38%), *Klebsiella terrigena* (38%), and strain 6003-71 of the *Enterobacter agglomerans* complex (38%). Other *Enterobacteriaceae* were 6-37% related. Because strains of Enteric Group 90 were distinct by both DNA-DNA hybridization and phenotype, a new genus *Trabulsiella* with a single species *Trabulsiella guamensis* was proposed (McWhorter et al., 1991). *Trabulsiella* and *Trabulsiella guamensis* gained standing in nomenclature in 1992 when they appeared on Validation List 41 (McWhorter et al., 1992).

FURTHER READING

McWhorter, A.C., R.L. Haddock, F.A. Nocon, A.G. Steigerwalt, D.J. Brenner, S. Aleksic, J. Bockemuhl and J.J. Farmer, III. 1991. *Trabulsiella*

guamensis, a new genus new species of the family *Enterobacteriaceae* that resembles *Salmonella* subgroups 4 and 5. J. Clin. Microbiol. 29: 1480–1485.

List of species of the genus *Trabulsiella*

1. *Trabulsiella guamensis* McWhorter, Haddock, Nocon, Steigerwalt, Brenner, Aleksic, Bockemühl and Farmer 1992, 327^{VP} (Effective publication: McWhorter, Haddock, Nocon, Steigerwalt, Brenner, Aleksic, Bockemühl and Farmer 1991, 1483.)

guam.en'sis. M.L. adj. *guamensis* pertaining to Guam, the largest island of the Micronesian group of the Pacific Ocean, where the first strains were isolated.

The characteristics are as given for the genus. Isolated from environmental samples, food, and human feces. However, there is no evidence that *Trabulsiella guamensis* can cause diarrhea or intestinal infections. Apparently it is a very rarely isolated species of *Enterobacteriaceae*, whose importance in microbiology may be more as a nuisance, because of its biochemical similarity to *Salmonella* and possible misidentification as *Salmonella*. The type strain was isolated from vacuum cleaner contents on the island of Guam obtained during a survey to isolate *Salmonella* from environmental reservoirs, and to look at the role of reptile feces as a source of *Salmonella* infections. Vacuum cleaner contents were a convenient sample of "indoor dirt".

In the addendum section of their paper, McWhorter et al. (1991) described two different biochemical patterns given by *T. guamensis* strains, but they did not name them as biogroups. The eight strains from the United States and

Guam were negative for indole production, gelatin hydrolysis (film method, 36°C), and esculin hydrolysis (within 2 d), but the four strains from Germany and Malaysia were positive. Strain 2421-87 (feces, Germany) was the only one of the original eight strains studied that had this latter pattern, but it was 98% related to the type strain by DNA–DNA hybridization at 60°C and the value did not drop when the temperature was raised to 75°C. Thus, the available data indicate that the two groups of strains should be considered as biogroups of *T. guamensis*.

The mol% G + C of the DNA is: not determined.

Type strain: ATCC 49490, CDC 0370-85.

Additional Remarks: The American Type Culture Collection includes four other strains of *Trabulsiella guamensis* including strain ATCC 49493 (CDC 2421-87, from human feces, Germany) that is from the indole-positive biogroup described above.

The biochemical and geographic differences of the two groups of strains may reflect evolutionary divergence that should be investigated with methods other than DNA–DNA hybridization that measure relatedness. J.J. Farmer (unpublished data) recently found a group of three strains in the collection of the Enteric Reference Laboratory, CDC that were very similar to *Trabulsiella*, but were H₂S negative. These could represent a second species of *Trabulsiella*.

Genus XXXIX. *Wigglesworthia* Aksoy 1995b, 849^{VP}

SERAP AKSOY

Wigg.les.worth'i.a. M.L. fem. n. *Wigglesworthia* named after the parasitologist W.B. Wigglesworth.

Obligate intracellular, bacteriome-tissue associated, primary endosymbionts of tsetse flies. In the adult fly, the bacteria lie free in the cytoplasm of differentiated epithelial cells, bacteriocytes, which form the U-shaped bacteriome organ (previously referred to as mycetome) in the anterior midgut (Figs. BXII.γ.210A, B). The endosymbionts are rod shaped (2–4 × 8–10 μm) with a typical Gram-negative cell wall and inner membrane structure, lack flagella (Fig. BXII.γ.210C). The cells appear to divide by binary fission. The bacteria are obligate intracellular organisms, as their experimental elimination from tsetse by procaryotic-specific antibiotic treatment, by lysozyme, or with symbiont-specific antibodies provided in the blood meal results in retarded growth of the insect and a decrease in egg production, causing loss of the ability of these aposymbiotic hosts to reproduce (Hill et al., 1973; Nogge, 1976, 1978, 1980). In addition to *Wigglesworthia*, tsetse harbors a secondary (S)-symbiont in midgut tissue. The specific function(s) of *Wigglesworthia* and S-symbionts are not known; however, since the ability to reproduce can be partially restored when the aposymbiotic tsetse receive a blood meal that is supplemented with B-complex vitamins (thiamine, pantothenic acid, pyridoxine, folic acid, and biotin), it is suggested that the gut-tissue associated endosymbionts may play a role in metabolism that involves these compounds (Nogge, 1981). The recently sequenced genome of *Wigglesworthia* shows the presence of var-

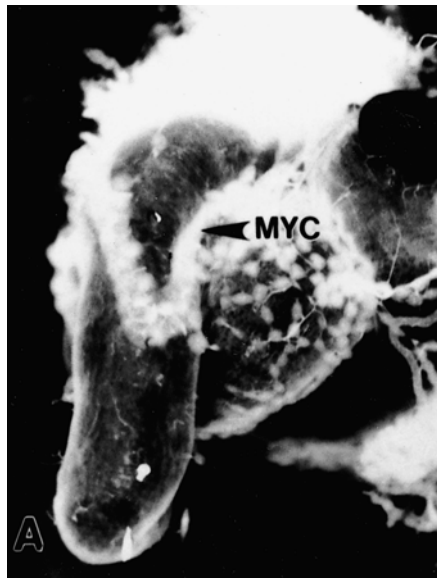
ious biosynthetic pathways for vitamin metabolites (Akman et al., 2002). The 16S rDNA-based phylogeny of *Wigglesworthia* characterized from eight different tsetse species displays concordance with their insect host species, indicating an ancient association for this bacterium with an ancestral tsetse that has subsequently radiated along with the host insect species (Chen et al., 1999b). The bacterium is not present in the ovaries, but may be transmitted to the intrauterine larvae of tsetse via its milk-gland secretions.

Type species: *Wigglesworthia glossinidia* Aksoy 1995b, 849.

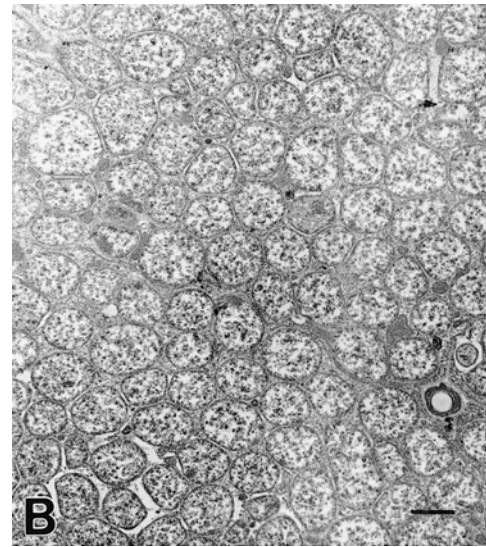
FURTHER DESCRIPTIVE INFORMATION

Various insects that rely on a single food source (blood or phloem) have established symbiotic associations with microorganisms in order to supplement their limited diets. Stuhlman found bacteria in the intestines of tsetse (Stuhlmann, 1907) and Roubaud described a mycetome (bacteriome)-like organ in midgut (Roubaud, 1919) that was confirmed by Wigglesworth and Buxton (Wigglesworth, 1929; Buxton, 1955). In ultrastructural studies, Reinhardt et al. (1972) described the large bacteroid found in tsetse mycetomes, and Pinnock and Hess (1974) distinguished these organisms from the smaller S-symbionts found in midgut tissue.

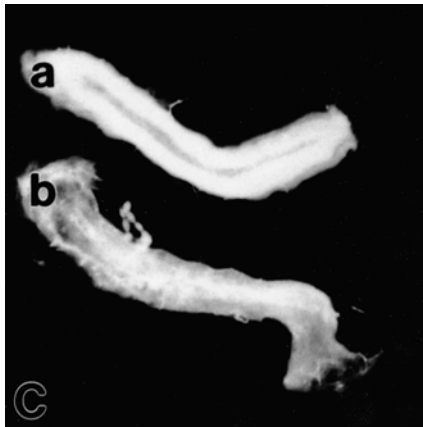
The well-defined bacteriome structure in tsetse guts can be



A



B



C

FIGURE BXII.γ.210. Bacteriome structure and *Wigglesworthia* in bacteriocytes. *A*, The arrow points at the U-shaped bacteriome (previously called mycetome, MYC) structure in the anterior midgut region of *G. m. morsitans*. (Reproduced with permission from S. Aksoy et al., *Insect Molecular Biology*, 4: 15–22, 1995, ©Blackwell Science Ltd., Oxford.) *B*, Electron micrograph of *Wigglesworthia* showing the packed endosymbionts in the bacteriocytes of *G. morsitans*. The endosymbionts are surrounded by a double-membrane ($\times 5000$; bar = 4 μm). *C*, Dissected bacteriomes: (*a*) from a female maintained on a regular blood meal, and (*b*) from female that has received an antibiotic containing blood meal for one week. The bacteriome structure in the presence of antibiotics has disintegrated. (Reproduced with permission from S. Aksoy et al., *Insect Molecular Biology*, 4: 15–22, 1995, ©Blackwell Science Ltd., Oxford.)

dissected and transiently maintained in tissue culture (Wink, 1979). Gene expression studies *in vivo* and *in vitro* have shown one of the major gene products of *Wigglesworthia* to be a molecular-chaperonin, the *groEL* homolog of *Escherichia coli* (Aksoy, 1995a). The overexpression of stress-related proteins has also been reported for the bacteriome endosymbionts of aphids (Kakada and Ishikawa, 1991) and may be due to the intracellular nature of these microorganisms. PCR-based experiments show that *Wigglesworthia* is not present in ovary tissue (O'Neill et al., 1993), thus ruling out a transovarial transmission route; however, previous microscopy studies have reported the presence of microorganisms with similar morphology in milk-gland tissue of tsetse (Ma and Denlinger, 1974). Thus, milk-gland secretions may provide a route for the maternal transmission of *Wigglesworthia* to the intrauterine larvae of tsetse.

The S-symbionts of tsetse harbored in midgut-tissue and *Wolbachia* harbored in ovary tissue are morphologically different from *Wigglesworthia*. They are short rods (1–2 μm in length), and inside host cells they are surrounded by host membranes. There is a clear lytic zone around the cells reminiscent of true *Rickettsiaceae* to which the genus *Wolbachia* is related. Since phylogenetic

analysis of tsetse S-symbionts has shown them to be enteric microorganisms, it is not clear what this lytic zone signifies (Reinhardt et al., 1972). The organization and copy number of the 16S rDNA gene in *Wigglesworthia* is different from *Buchnera* (the phylogenetically closely related P-endosymbionts of aphids), tsetse S-symbionts, and *E. coli*. The free-living bacterium *E. coli* and tsetse S-symbionts have multiple copies of rDNA operons encoding the 16S and 23S gene products. *Wigglesworthia* was shown to have a single operon encoding its 16S rDNA by Southern blot analysis (Aksoy, 1995a). The genome sequence analysis now indicates that there are two copies of the rDNA operon in *Wigglesworthia*, unlike the single operon found in *Buchnera* (Akman et al., 2002). In *Buchnera*, the 16S and 23S rDNA genes are not genetically linked, while in *Wigglesworthia* they are found to be transcriptionally linked (Aksoy, 1995a; Akman et al., 2002).

TAXONOMIC COMMENTS

While it has not been possible to maintain the bacteriome-tissue symbionts *in vitro*, recent PCR-based amplification techniques and the use of nucleic acid sequences in phylogenetic recon-

structions have provided additional insight into the relationships among these intracellular bacteria. Their characterization from different species of aphids, whiteflies, carpenter ants, and tsetse have shown that they each form unique lineages within the *Gammaproteobacteria*, while the organisms from mealybugs are in the *Betaproteobacteria* and those associated with cockroaches are members of flavibacteria. Their phylogeny in each case has been found to parallel the phylogeny of their host insect species, indicating the ancient and obligatory nature of these symbiotic relationships (Baumann et al., 1998). The phylogeny of *Wigglesworthia* was analyzed from different flies representing the four species groups of genus *Glossina* (*morsitans*, *palpalis*, *machadomyia*, and *fusca*). For this analysis, the 16S rDNA sequence that is present as a single-copy gene was used (Aksoy et al., 1995; Chen et al., 1999b). The

phylogenetic relationships of the tsetse host species were also analyzed by using an insect gene, the internal transcribed spacer-2 region of rDNA (ITS-2). These sequences were then used to generate phylogenetic alignments by maximum parsimony and two representative trees are shown in Fig. BXII.γ.211: for *Wigglesworthia* (Fig. BXII.γ.211b) and for the tsetse host species (Fig. BXII.γ.211a). The alignment of *Wigglesworthia* included the tsetse midgut S-symbiont clade and the S-symbionts of the weevil, *Sitophilus zeamais*; *Escherichia coli* was chosen as the out group. All trees generated could distinguish the tsetse symbionts, *Wigglesworthia*, and midgut S-symbionts, as distinct lineages within the *Gammaproteobacteria*. While within the S-symbiont clade no differences were observed in their 16S rDNA sequence, *Wigglesworthia* analyzed from the *fusca* group (*G. brevipalpis*) was found

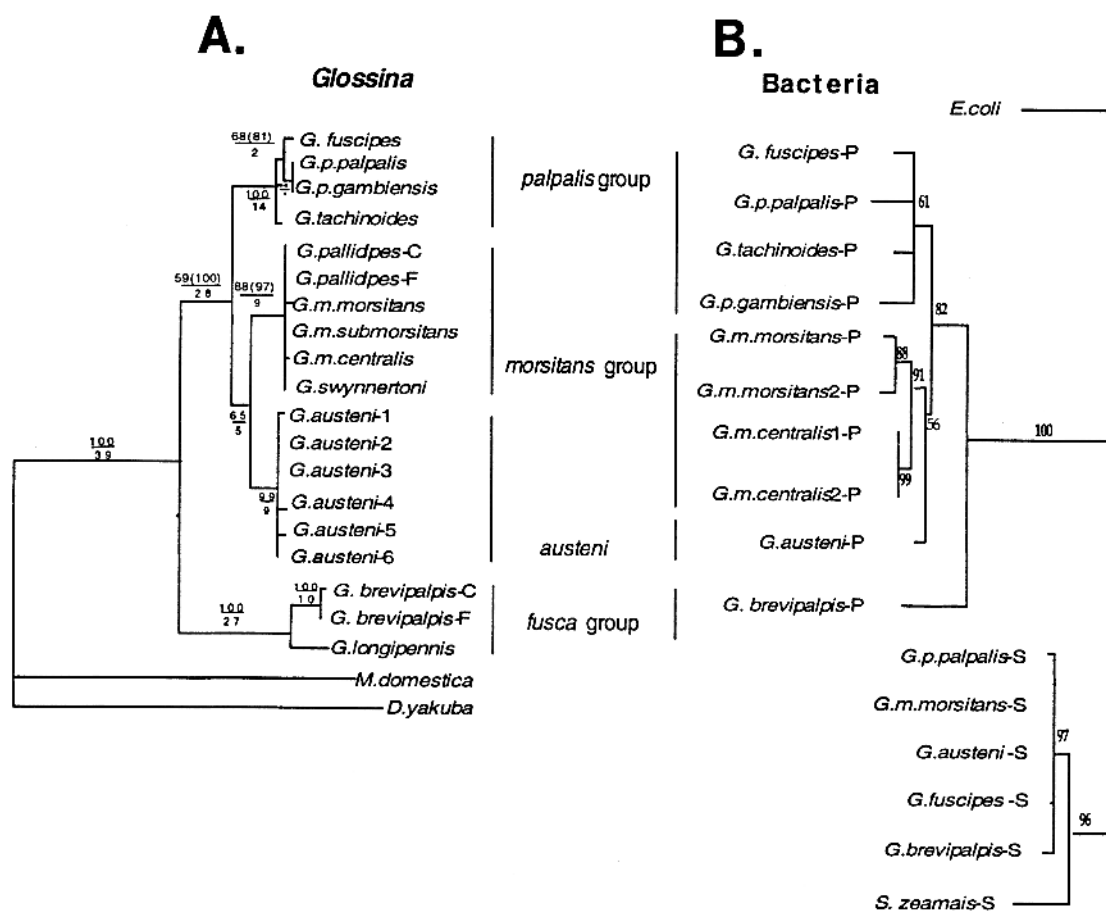


FIGURE BXII.γ.211. The phylogenetic placement of *Glossina* species and their symbionts. (Reproduced with permission from X. Chen et al., Journal of Molecular Evolution, 48: 49–58, 1999, ©Springer-Verlag.) A, The phylogenetic tree constructed by maximum parsimony using the ITS-2 sequence. Tree length = 292, CI = 0.897. The bootstrap confidence values (300 replications) are presented at the nodes; the values in parentheses are bootstrap values where gaps have been included as weighted characters; numbers below denote branch length values. *D. yakuba* and *M. domestica* sequences are used as out groups. The grouping of the *palpalis*, *morsitans*, *austeni*, and *fusca* species are indicated. The origination of the *G. austeni* flies are as follows: 1, colony maintained in Seibersdorf tsetse laboratories originating from Tanzania; 2, field collected from Zanzibar; 3, field collected from Shimba Hills (Kenya); 4, colony established from Shimba Hills (Kenya); 5 and 6, field collected from Hell's Gate Park in South Africa. C denotes flies analyzed from colonies. F denotes field isolates. *G. pallidipes*-C is from the colony at International Livestock Research Institute (ILRI) originating from Shimba Hills area of Kenya. *G. pallidipes*-F was a field collected fly from Nguruman region of Kenya. *G. brevipalpis*-C is from the colony at ILRI established from Kenya. *G. brevipalpis*-F is a field sample from South Africa. B, Phylogenetic analysis of *Wigglesworthia* and midgut S-symbionts based on their 16S rDNA sequence analysis. P denotes primary symbiont *Wigglesworthia*, and S denotes the secondary symbionts analyzed from the corresponding species of *Glossina*. *E. coli* was used as the out group. The analysis included 893 sites, of which 205 were variable and 121 were informative. One representative tree is shown (tree length 216, CI = 0.843, bootstrap values for 500 replications are shown at the nodes). *G. m. centralis*-1 was obtained from the colony maintained at ILRI in Kenya, while *G. m. centralis*-2 is from the colony maintained at Alberta and has originated from Zambia. The GenBank accession numbers for the 16S rDNA loci of *Wigglesworthia* from *G. p. gambiensis*, *G. m. morsitans*-2, *G. m. centralis*-1, *G. m. centralis*-2, *G. austeni*, and *G. fuscipes* are AF022875–AF022880, respectively.

to occupy the deepest branch. The *Wigglesworthia* sequences analyzed from the *morsitans* and *palpalis* group species formed sister taxa. The placement of *Wigglesworthia* from *G. austeni* (*machadomyia*), however, was less certain, since equally robust trees could be generated that differed only in the placement of *G. austeni* with respect to the other two subgenera. The low bootstrap-analysis confidence value (56) at the node where *G. austeni* symbiont forms a sister group with the *morsitans* species demonstrates this uncertainty. The ITS-2 sequence-based alignment (Fig.

BXII.γ.211a) displayed similar phylogenetic relationships among the tsetse host species; the fusca flies were found to be the most divergent, while the *palpalis*, *morsitans*, and *machadomyia* subgenera were of more recent origin, and again the evolutionary relatedness of *machadomyia* with respect to the *palpalis* and *morsitans* subgenera could not be resolved with high confidence. The comparative analysis of the two phylogenies indicates their strong similarity, arguing for concordance of *Wigglesworthia* with their host tsetse species.

List of species of the genus *Wigglesworthia*

1. ***Wigglesworthia glossinidia*** Aksoy 1995b, 849^{VP}
glos.si.nid' i.a. M.L. fem. adj. *glossinidia* referring to *Glossina*, the genus of tsetse flies.

The characteristics are as described for the genus and as depicted in Fig. BXII.γ.210.

The mol% G + C of the DNA is: not known.

Type strain: endosymbiont of *Glossina morsitans morsitans*.

GenBank accession number (16S rRNA): L37339.

Genus XL. ***Xenorhabdus*** Thomas and Poinar 1979, 354^{AL} emend. Thomas and Poinar 1983, 878

RAYMOND J. AKHURST AND NOËL E. BOEMARE

Xe.no.rhab'dus. Gr. n. *xenos* unwanted guest (less literally, pathogen); Gr. fem. n. *Rhabdus* rod; M.L. masc. n. *Xenorhabdus* pathogenic rod-shaped bacterium.

Asporogenous, rod-shaped cells 0.3–2 × 2–10 μm and occasionally with filaments 15–50 μm in length. Spheroplasts, averaging 2.6 μm in diameter, appear in the last third of exponential growth. **Proteinaceous crystalline inclusions** develop in a large proportion of cells in stationary phase cultures. Gram negative. Motile by means of peritrichous flagella. Swarming may occur on 0.6–1.2% agar. Facultatively anaerobic, with both respiratory and fermentative types of metabolism. **Optimum temperature usually ~28°C or less;** a few strains grow at 40°C. Acid (no gas) production from glucose; ferment some other carbohydrates but poorly. **No catalase activity; nitrate is not reduced to nitrite.** Negative for most tests used to differentiate *Enterobacteriaceae*. Lipase detected with Tween 20 and egg yolk agar; most strains lipolytic on Tween 40, 60, 80, and/or 85. Positive for deoxyribonuclease and protease. **Phase shift occurs to varying degrees in stationary phase cultures,** giving rise to phase II, which lacks dye adsorption, antibiotic production, protein inclusions, and some other characteristics of the phase I isolated from the natural environment. **Only known from the intestinal lumen of entomopathogenic nematodes of the family Steinernematidae** and insects infected by these nematodes.

The mol% G + C of the DNA is: 43–50 (Bd).

Type species: ***Xenorhabdus nematophila*** (Poinar and Thomas 1965) Thomas and Poinar 1979, 355 (*Achromobacter nematophilus* Poinar and Thomas 1965, 249.)

FURTHER DESCRIPTIVE INFORMATION

Analyses of 16S rDNA sequences show that *Xenorhabdus* is most closely related to *Photorhabdus*, a genus composed largely of bacteria symbiotically associated with entomopathogenic nematodes of the family *Heterorhabditidae*. The next nearest phylogenetic neighbors are *Proteus vulgaris* and *Arsenophonus nasoniae* (Suzuki et al., 1996; Brunel et al., 1997; Liu et al. 1997a; Szállás et al., 1997).

Xenorhabdus species are insect pathogenic bacteria that occur naturally in the intestinal vesicle of nonfeeding infective stage

entomopathogenic nematodes of the family *Steinernematidae* (Bovien, 1937; Poinar and Leutenegger, 1968; Bird and Akhurst, 1983). After invading an insect host, the nematode commences development, releasing *Xenorhabdus* into the nutrient-rich hemolymph. The bacteria proliferate, killing the insect host and producing suitable nutrient conditions for growth and reproduction of the nematodes as well as an array of antibiotics and bacteriocins to minimize competition. As the nutrient source becomes depleted, the immature nematodes develop into the infective stage that will transport *Xenorhabdus* to a new nutrient source. The association between each *Xenorhabdus* species and nematode is specific, with each nematode only naturally associated with a single *Xenorhabdus* species. However, *X. bovienii*, *X. poinarii*, and *X. beddingii* are each associated with more than one *Steinernema* species (Akhurst, 1983a, 1986b; Fischer-Le Saux et al., 1998).

A highly significant feature of bacteria of the genus *Xenorhabdus* is a phase variation that occurs in stationary phase (Akhurst, 1980). Only phase I *Xenorhabdus* have been detected in nature, but under *in vitro* conditions a variable proportion undergoes a profound change affecting colony and cell morphologies, motility, endo- and exo-enzymes, including respiratory enzymes, and secondary metabolites (Akhurst, 1980; Boemare and Akhurst, 1988; Smigielski et al., 1994; Givaudan et al. 1995; Fodor et al. 1997; Table BXII.γ.277). The timing and extent of phase change is largely unpredictable, but the rate of change from phase I to phase II is generally greater than the reverse. Phase I *Xenorhabdus* provide and protect essential nutrients for the nematodes by killing and metabolizing the insect host and producing a range of antimicrobial agents. Although phase II variants may also kill the insect host, they are less effective in providing growth conditions for the nematodes (Akhurst, 1980, 1982a), and have never been found associated with naturally occurring nematodes; the role of phase II is uncertain.

Xenorhabdus cells are rod-shaped cells that are highly variable in size, ranging from 0.3 × 2 μm to 2 × 10 μm with occasional

filaments 15–50 µm in length. Varying proportions of the cells occur as spheroplasts (diameter ~2.6 µm) in stationary phase cultures. Proteinaceous inclusion bodies, commonly two morphs, are evident in many of the rod-shaped phase I cells in stationary phase (Fig. BXII.γ.212). The glycocalyx surrounding *Xenorhabdus*

cells is irregular in thickness, with a mean depth of 142 nm in phase I *X. nematophila** and 49 nm in phase II (Brehélin et al.,

*Editorial Note: *Xenorhabdus nematophila* was originally named *Xenorhabdus nematophilus*; the name was corrected by Euzéby and Boemare (2000).

TABLE BXII.γ.277. Differential characteristics for phase I and II of *Xenorhabdus* and *Photorhabdus* species^a

Characteristic ^b	1. <i>X. nematophila</i>		2. <i>X. beddingii</i>		3. <i>X. bovienii</i>		4. <i>X. japonica</i>		5. <i>X. poinarii</i>	
	Phase I	Phase II	Phase I	Phase II	Phase I	Phase II	Phase I	Phase II	Phase I	Phase II
<i>Colonial properties:</i>										
Morphology ^c	g	t	g	t	g	t	g	nr	g	t
Dye adsorption	+	–	+	–	+	–	+	–	+	–
Pigmentation ^d	ow	ow	lb	ow	y	ow	yb	nr	b	ow
<i>Ultrastructural elements and cytological properties:</i>										
Protoplasmic inclusions	+	–	+	–	+	–	nr	nr	+	–
<i>Enzymatic activities:</i>										
Antibiotics	+	+w	+	–	+	–	+	–	d	d
Lecithinase (egg yolk agar)	d	–	+	–	[+]w	–	+	–	–	–
Phospholipase (lecithin agar)	+	dw	+	–	+	–	nr	nr	[–]	–
Lipolysis (Tween 80)	[–]	+	+	+	+	+	–	–	+	+
Simmons citrate	+	+	+	+	–	+	nr	nr	d	+
Phenylalanine deaminase	–	d	–	–	–	–	+	–	–	–
Gelatin hydrolysis	+	d	+	+	+	+	+	–	+	+

^aAll tests were done at 28° ± 1°C for *Xenorhabdus* and *Photorhabdus*. Data compiled from the following references: Akhurst (1980, 1982a, 1986a), Boemare and Akhurst (1988); Brehélin et al. (1993); Nishimura et al. (1994).

^bSymbols: +, 90–100% of strains are positive; [+], 76–89% are positive; d, 26–75% are positive; [–], 11–25% are positive; –, 0–10% are positive; w, weak reaction.

^cg, granular; t, translucent; nr, not reported.

^dow, off-white; y, yellow; b, brown; lb, light brown; yb, yellowish brown.

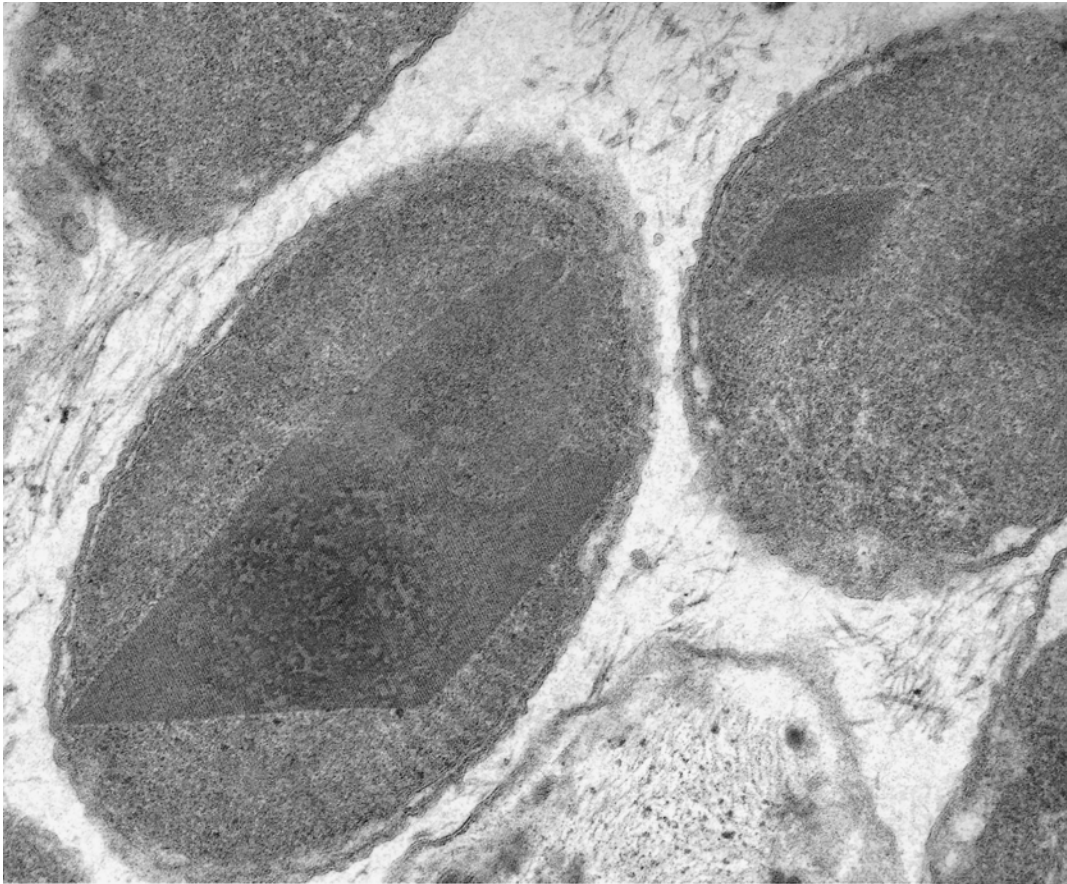


FIGURE BXII.γ.212. Protein inclusions in *X. nematophila*, showing a rhomboidal crystalline inclusion and an ovoid inclusion body.

1993). Phase I, but not phase II, cells have a peritrichous array of fimbriae with diameter of 6.4 nm, morphologically similar to the type I fimbriae of *E. coli* (Binnington and Brooks, 1993; Moureaux et al., 1995).

Phase I *Xenorhabdus* such as those isolated from the nematode associates or insects infected by the nematodes form convex, circular colonies with slightly irregular margins and a diameter of 1.5–2 mm after 4 d at 28°C; they also have a slightly granular appearance and, in some species, are pigmented (yellow, brown). These colonies adsorb dyes, such as bromothymol blue and neutral red, taking on intense coloration. Phase II variants generally form similar colonies but are flatter and wider (diameter 2.5–3.5 mm after 4 d at 28°C) and with lesser pigmentation; these colonies adsorb dyes only very weakly and no coloration is evident in 4-d colonies.

The major cellular fatty acids of *Xenorhabdus* are C_{16:0}, C_{16:1}, C_{18:1}, and C_{17:0 cyc}, and the respiratory quinone system is ubiquinone-8 (Suzuki et al., 1990).

Xenorhabdus species are easily grown *in vitro* on a range of complex liquid and solid media and in minimal media supplemented with nicotinic acid, *p*-aminobenzoic acid, serine, tyrosine, and/or proline (Grimont et al., 1984b); nutrient agar is suitable for all strains. They are mesophilic; most grow between 15 and 30°C but strains growing at 4°C or at 40°C have been isolated.

On the basis of 16S rDNA sequence data, *Xenorhabdus* can be distinguished from its nearest phylogenetic neighbor, *Photorhabdus* (Fig. BXII.γ.213), by the sequence TTCCG at positions 208–211 (*E. coli* numbering) of the 16S rDNA where *Photorhabdus* has a longer version (TGAAAG) (Szállás et al., 1997). DNA–DNA hybridization demonstrated the existence of a number of undescribed species in the genus (Fig. BXII.γ.214). Data have been recorded for some of the strains that would be assigned to new *Xenorhabdus* species (Boemare and Akhurst, 1988) but too few to warrant a decision on their taxonomic status; there are no 16S rDNA data for any of these strains.

Conjugation of plasmids from *E. coli* has been applied successfully for *X. nematophila* and *X. bovienii* (Smigielski and Akhurst, 1992; Francis et al. 1993; Forst and Nealson, 1996). In contrast, transformation of *Xenorhabdus* has not been generally successful. Although Xu et al. (1989) reported the transformation of the type strain of *X. nematophila* with a broad host-range vector, they and other workers have been unsuccessful with other strains of *X. nematophila* and other *Xenorhabdus* spp. The increase in transformation efficiency after the vector had been passed through *X. nematophila* suggested the presence of a restriction modification system. This suggestion was further supported by the demonstration that *Xenorhabdus* DNA is not easily digested by several common restriction enzymes (Akhurst et al., 1992). Endonuclease activity has subsequently been detected in a range of *Xenorhabdus* spp. (Akhurst et al., 1992).

Hypovirulent and avirulent mutants have been produced in *X. nematophila* by chemical and transposon mutagenesis (Xu et al., 1991; Dunphy, 1994). Francis et al. (1993) developed a transposon mutagenesis system for *X. bovienii* by constitutively expressing *lamB* on the surface of the bacterium, allowing them to be infected with lambda particles carrying the Tn10 transposon. This process produced various dye-binding, lipase, protease, hemolytic, and DNase mutants.

Couche et al. (1987) demonstrated the presence of one or two small plasmids (3.6–12 kb) in some, but not all, *X. nematophila* and *X. bovienii* strains. The plasmid profiles did not differ between phases I and II. Smigielski and Akhurst (1994) reported two

megaplasms (71.8 and 118.5 kb) as well as two additional plasmids (6.5 and 17 kb) in the A24 strain of *X. nematophila*. They also demonstrated that all strains of *X. nematophila*, *X. bovienii*, *X. beddingii*, and *X. poinarii* contained megaplasms (48 to >680 kb) and that there were no differences in megaplasms profile between phases I and II.

The presence of lysogenic phage in and production of bacteriocins by *X. nematophila* was demonstrated by Boemare et al. (1992). Lysis of both phases of *X. nematophila*, *X. bovienii*, and *X. beddingii* in response to mitomycin C or heat shock released complete and partial phages. In addition to inhibiting nonhost *Xenorhabdus*, the bacteriocins inhibited *P. luminescens*, *Proteus vulgaris*, and *Morganella morganii* but no other Gram-negative or Gram-positive bacteria tested, indicating that the bacteriocins act against closely related genera in contrast to the antibiotics produced by phase I variants that have a wide spectrum of activity. The *X. nematophila* bacteriocin xenorhabdicolin was shown to consist of two major protein bands, corresponding to the sheath and core, and five minor bands; the phage head capsid had one major and two minor subunits (Thaler et al., 1995).

Antibiograms scored after 3 d at 28°C show that *Xenorhabdus* spp. are inhibited by streptomycin, neomycin, gentamicin, tetracycline, kanamycin, and colistin, but not penicillin. Most strains are resistant to ampicillin and cephaloridine and, to a lesser extent, furazolidone, whereas resistance to chloramphenicol is limited. Resistance to streptomycin, tetracycline, and kanamycin after selection has been demonstrated for *X. nematophila*.

Xenorhabdus spp. are insect pathogens only when delivered into the insect hemocoel, either by their nematode symbiont or by injection; they are not pathogenic when applied topically. Most are highly toxic for larvae of the greater wax moth, *Galleria mellonella*, with an LD₅₀ of <20 cells (Akhurst and Dunphy, 1993). *X. poinarii* has very little pathogenicity for *G. mellonella*, (LD₅₀ = 5000 cells) when injected alone, although it is highly pathogenic when co-injected with axenic *Steinernema glaseri*, its natural host (Akhurst, 1986b). *X. nematophila* produces an orally active toxin complex of the same family as those produced by *Photorhabdus*, *Serratia entomophila*, and *Yersinia pestis* (Glare and Hurst, 2002) and a smaller single unit toxin that is toxic by injection (East et al., 1998). Their toxicity varies between insects, with *X. nematophila* having an LD₅₀ of about 500 for *Hyalophora cecropia* caterpillars (Götz et al., 1981) and no toxicity to maggots of the genus *Chironomus* (Götz and Boman, 1985). *Xenorhabdus* spp. do not have any demonstrable effects on vertebrates (Poinar and Thomas, 1967; Poinar et al., 1982; Obendorf et al. 1983).

Xenorhabdus species have been found only in the intestinal tract of infective-stage nematodes of the genus *Steinernema* (syn. *Neoaplectana*) and in insects killed by these nematodes. There is a high degree of specificity in these associations, with each nematode species being naturally associated with only one *Xenorhabdus* species (Akhurst, 1983a; Table BXII.γ.278). This specificity is determined by the ability of the nematode to retain the bacterium in its intestinal vesicle (Bird and Akhurst, 1983; Akhurst and Boemare, 1990). Although there have been reports of other bacteria being associated with *Steinernema* (Lysenko and Weiser, 1974; Mracek, 1977; Aguilera et al., 1993), closer examinations have always demonstrated that only *Xenorhabdus* have a specific association (Akhurst, 1982b; Bonifassi et al., 1999).

The infective stage nematodes act as vectors, transporting the bacteria into the insect via natural orifices (mouth, anus, spiracles) and then into the hemocoel. The infective (dauer) nematode, a third-stage juvenile with closed mouth and anus, recom-



FIGURE BXII.γ.213. Phylogenetic tree showing the relationship of the genus *Xenorhabdus* to members of the *Gammaproteobacteria*. Note that since the publication of this figure, *Chromatium vinosum* has been transferred to the genus *Allochromatium* as *Allochromatium vinosum* (Imhoff et al., 1998b) and that *Vibrio marinus* has been transferred to the genus *Moritella* as *Moritella marina* (Urakawa et al., 1998, 1999b); in addition, the spelling of the species name *japonicus* has been corrected to *japonica* and that of *nematophilus* to *nematophila* (Euzéby and Boemare, 2000). (Reprinted with permission from T. Suzuki et al., *Journal of Basic Microbiology*, 5: 351–354, 1996 ©Wiley-VCH Verlag, Berlin.)

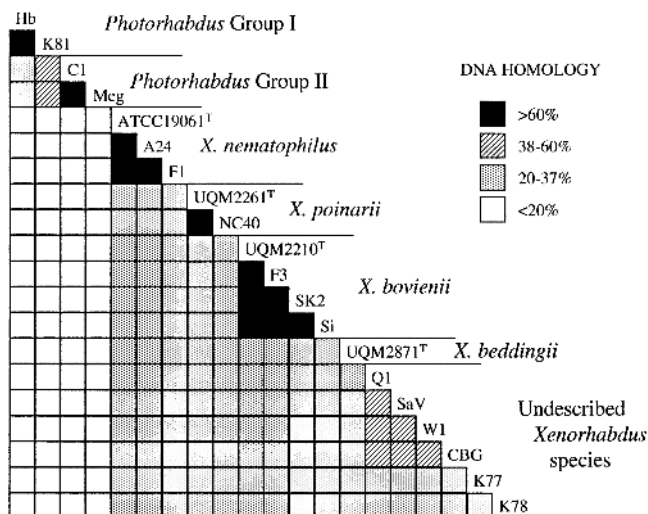


FIGURE BXII.γ.214. DNA relatedness of 20 strains of bacterial symbionts of entomopathogenic nematodes (modified from Boemare et al., 1993), indicating that strains representing a number of undescribed species have been isolated. In addition, the spelling of the species name *nematophilus* has since been corrected to *nematophila*.

mences development in the hemocoel, releasing its symbiotic bacterium and an inhibitor of the inducible antibacterial enzymes (Poinar and Himsforth, 1967; Götz et al., 1981). As *Xenorhabdus* multiplies it provides essential nutrients for nematode maturation and reproduction (Poinar and Thomas, 1966) and produces antibiotics (Paul et al., 1981; Akhurst, 1982a; McInerney et al. 1991a, b; Li et al., 1996).

Xenorhabdus alternates between a nutrient rich (insect) and nutrient poor (nematode) existence. Forst et al. (1997) hypothesized that phase II may be induced by the nematode gut conditions and better adapted to the nutrient poor conditions of the intestinal vesicle of the nonfeeding nematode. However, this hypothesis does not account for the fact that the bacteria isolated from field-collected infective stage juveniles are inevitably phase I. Smigielski et al. (1994) found differences in the activity of respiratory enzymes of phases I and II that indicate the greater potential of phase II to survive in soil environments than phase I. The lack of a record of isolation of *Xenorhabdus* directly from soil may be due to its slow growth, lack of a suitable selective medium, and/or the difficulty of identifying phase II, which has few positive characters in most standard tests for identifying bacteria. However, Morgan et al. (1997) found that *X. nematophila* declined very quickly in river water and soil, becoming undetectable after 2 d and 7 d, respectively. Although the phase status

TABLE BXII.γ-278. Biochemical characterization of the genus *Xenorhabdus*^a

Characteristic ^b	1. <i>X. nematophila</i>	2. <i>X. beddingii</i>	3. <i>X. bovienii</i>	4. <i>X. japonica</i>	5. <i>X. poinarii</i>
Mol% G + C of DNA	43–48	45.5–50	44–47	45.9	42.6–49
<i>Isolated from:</i>					
<i>S. carpocapsae</i>	+	—	—	—	—
<i>S. feltiae</i> , <i>S. intermedium</i> , <i>S. affine</i> , <i>S. kraussei</i>	—	—	+	—	—
<i>S. glaseri</i> , <i>S. cubanum</i>	—	—	—	—	+
Unidentified <i>Steinernema</i> spp. (<i>Steinernema longicaudum</i> ?)	—	+	—	—	—
<i>S. kushidai</i>	—	—	—	+	—
Pathogenicity for lepidopteran insects	+	+	+	—	—
Motility	d	d	d	d	d
Peritrichous flagella	+	+	+	+	+
Protoplasmic inclusions	d	d	d	nr	d
Bioluminescence	—	—	—	—	—
Pigmentation ^c	ow	lb	y	yb	br
Dyes adsorption	d	d	d	d	d
Antimicrobial production	d	d	d	d	d
Oxidation-fermentation	F	F	F	F	F
Catalase	—	—	—	—	—
Nitrate reduced to nitrite	—	—	[—]w	—	—
Oxidase, Kovac's	—	—	—	—	—
KCN, growth in	—	—	—	—	—
Indole production	—	—	—	—	—
Methyl red	—	—	—	—	—
Voges-Proskauer	—	—	—	—	—
Simmons citrate	+	+	+	—	+
Hydrogen sulfide production	—	—	—	—	—
ONPG (β-galactosidase)	—	—	—	—	—
Esculin hydrolysis	—	+	—	—	d
Urease, Christensen's	—	—	—	—	—
Phenylalanine deaminase	d	—	[—]	dw	[—]
Tryptophan deaminase	—	+w	+	—	—
Lysine decarboxylase	—	—	—	—	—
Ornithine decarboxylase	—	—	—	—	—
Arginine dihydrolase	—	—	—	—	—
D-Glucose, acid production	+	+	+	+	+
D-Glucose, gas production	—	—	—	—	—
<i>Acid production from:</i>					
D-Adonitol	—	—	—	—	—
L-Arabinose	—	—	—	—	—
Cellobiose	—	—	—	—	—
Dulcitol	—	—	—	—	—
Fructose	+	+	+w	+	+
Glycerol	+w	+	+	+	+
N-Acetylglucosamine	+	+	+	+	+
myo-Inositol	+w	—	dw	—	—
Lactose	—	—	—	—	—
Maltose	+	+	+	+	+
D-Mannitol	—	—	—	—	—
D-Mannose	+	+	+	+	+
Melibiose	—	—	—	—	—
α-Methyl-D-glucoside	—	—	—	—	—
Raffinose	—	—	—	—	—
L-Rhamnose	—	—	—	—	—
Ribose	—	+	+	—	—
Salicin	—	+	—	—	—
D-Sorbitol	—	—	—	—	—
Sucrose	—	—	—	—	—
Trehalose	+	+	+w	+	+
D-Xylose	—	—	—	—	—
<i>Utilization of:</i>					
Diaminobutane	—	—	[+]	—	—
L-Fucose	—	—	—	—	—
DL-Glycerate	+	+	+	—	[—]
L-Histidine	—	+	+	—	[+]
myo-Inositol	+w	—	dw	—	—
DL-Lactate	+	+	+	+	+
D-Mannitol	—	—	—	—	—
Ribose	—	+	[+]	—	—
L-Tyrosine	—	+	[+]	—	—
Gelatin hydrolysis, Kohn's	+	+	+	d	+
Lecithinase (egg yolk agar)	d	d	d	d	—
Lipase (Tween 80)	d	+w	+	—	+w
DNase	+	+	+	+	+w

^aAll tests were done at 28° ± 1°C for *Xenorhabdus*. Data from Akhurst (1986a,b); Boemare and Akhurst (1988); Yamanaka et al. (1992); Brehélin et al. (1993); Nishimura et al. (1994); Fischer-Le Saux et al. (1998).

^bSymbols: +, 90–100% of strains are positive; [+], 76–89% are positive; d, 26–75% are positive; [—], 11–25% are positive; —, 0–10% are positive; w, weak reaction; nr, not reported; F, fermentative.

^cow, off-white; y, yellow; br, brown; lb, light brown; yb, yellowish brown.

of *X. nematophila* used in that study was not specified, it seems likely that they used phase I. At this time, there is no satisfactory explanation of the ecological role of phase II.

ENRICHMENT AND ISOLATION PROCEDURES

Xenorhabdus spp. generally grow well at 25–28°C on nutrient or similar agar (e.g., Luria-Bertani, trypticase soy), but best growth has been achieved with yeast extract-salts (YS; Dye, 1968) and medium X (Götz et al., 1981) agars. *Xenorhabdus* can be isolated from the infective-stage nematodes by the hanging drop technique or by maceration (Poinar and Thomas, 1966; Akhurst, 1980). For both methods, the infective juveniles must first be surface sterilized; this is readily achieved by immersing a small number of live infective-stage nematodes (<100), free of debris, in 0.1% merthiolate for 1 h at room temperature and then rinsing thoroughly in several changes of sterile water. In the hanging-drop technique, individual surface-sterilized infective juveniles are transferred to a drop of aseptically collected insect hemolymph on a coverslip that is then inverted over a cavity to prevent desiccation and incubated at 25°C until the nematodes commence development (1–3 d). At this time, they void their symbiotic bacteria, which can be subcultured from the hemolymph onto an agar medium (e.g., nutrient agar) 1 d later. A more rapid method involves the maceration of 50–100 surface-sterilized infective juveniles in a nutrient broth by means of a tissue homogenizer. The macerate should be plated onto an agar medium immediately (10–100 µl aliquots) and incubated at 28°C for 3 d. The inclusion of suitable controls to confirm that the surface-sterilization procedure has been effective is essential for both methods. *Xenorhabdus* can also be isolated by the less labor-intensive method of collecting hemolymph from an insect (e.g., *G. mellonella*) larva within 24 h of its death after infection by *Steinernema*. With this last method, bacteria other than *Xenorhabdus* may also be isolated; these bacteria may be carried into the host on the exterior of the nematodes or may be picked up into the hemolymph from the insect cuticle. Contamination by other bacteria can be minimized by burying the insect in clean, damp sand, adding a small number of nematodes to the surface of the sand and incubating at 20–25°C until one to five nematodes infect the insect. This last method is better suited to reisolation of a *Xenorhabdus* strain rather than to determining the identity of the bacteria specifically associated with a nematode species.

MAINTENANCE PROCEDURES

Storage in 20% glycerol at –70°C or below is very useful for long-term maintenance of *Xenorhabdus* cultures; they do not store well at –20°C. Short-term storage (less than 1 month) is best conducted at 10–15°C because survival on agar or in broth at 4°C is very poor, and when cultures are maintained at temperatures in excess of 15°C there is a significant risk of a proportion of the culture undergoing phase change.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

All tests for phenotypic characterization of *Xenorhabdus* spp. should be conducted at 28°C.

Dye adsorption in most *Xenorhabdus* species can be tested on nutrient agar containing 0.0025% (w/v) bromothymol blue and 0.004% (w/v) triphenyltetrazolium chloride (NBTA; Akhurst, 1980). Dye-adsorbing phase I colonies will appear dark blue, while nonadsorbent phase II colonies will be red. As *X. poinarii* does not adsorb bromothymol blue, dye adsorption in this species can be assessed on MacConkey agar; adsorbing colonies are dark

red (Akhurst, 1986b). Congo red (Francis et al., 1993) and some other dyes can also be used for most *Xenorhabdus* spp.

Insect pathogenicity is tested by injection of 100, 1000, and 10,000 cells (total count) from a 24-h broth culture into final instar *G. mellonella* larvae. The injected larvae should be placed on dry filter paper in Petri dishes and incubated at 25°C for 3 d. Most *Xenorhabdus* spp. will kill >50% at a dosage of 100 cells; all will kill >50% at 10,000 cells. The use of a less susceptible host (e.g., *Manduca sexta*) has enabled detection of differences in pathogenicity between the two phases of a strain (Volgyi et al., 1998).

Antibiotic production is tested by spot inoculating *Xenorhabdus* spp. onto nutrient agar and incubating at 28°C for 3 d. At this time the bacteria are killed by exposure to chloroform vapor for ~1 h. After the chloroform has evaporated from the agar, semi-solid nutrient agar (0.5%) inoculated with a suitable indicator organism (*Micrococcus luteus* or another Gram-positive species) is poured to form a thin layer. After incubation at 28°C overnight, a halo of inhibition around antibiotic-producing colonies will be evident.

DIFFERENTIATION OF THE GENUS *XENORHABDUS* FROM OTHER GENERA

Xenorhabdus spp. are easily distinguished from other *Enterobacteriaceae* by the absence of catalase in the former and, with the exception of *Photorhabdus*, by their inability to reduce nitrate. The key characteristics for differentiating *Xenorhabdus* from its most closely related genera are shown in Table BXII.γ.279.

TAXONOMIC COMMENTS

The first symbiotic bacterium isolated from entomopathogenic nematodes was described as a new species *Achromobacter nematophilus* (Poinar and Thomas, 1965). With the rejection of the genus *Achromobacter* (Hendrie et al., 1974), this new species could not be accommodated within any existing genus. Thomas and Poinar (1979) described a new genus, *Xenorhabdus*, to accommodate the bacterial symbionts of entomopathogenic nematodes as two species, *X. nematophila*, symbionts of the family *Steinernematidae*, and “*X. luminescens*”, associated with the *Heterorhabditidae*. These bacteria have low DNA–DNA relatedness (4%) to the type species of the type genus of the family *Enterobacteriaceae* (Farmer, 1984b) and lack nitrate reductase, which is positive for all other genera in the family. However, they do have the enterobacterial common antigen (Ramia et al., 1982), and phylogenetic analyses based on 16S rDNA (Suzuki et al., 1996; Brunel et al., 1997; Liu et al., 1997a; Szállás et al., 1997) confirm their relatedness to the *Enterobacteriaceae*.

From a phenotypic study of bacterial symbionts of the *Steinernematidae* and *Heterorhabditidae*, five groups were recognized

TABLE BXII.γ.279. Characteristics differentiating *Xenorhabdus*, *Photorhabdus*, and *Proteus*^a

Characteristic	<i>Xenorhabdus</i>	<i>Photorhabdus</i>	<i>Proteus</i>
Bioluminescence	–	+	–
Catalase	–	+	+
Annular hemolysis	–	d	–
Urea hydrolysis	–	d	+
Indole	–	d	d
H ₂ S production	–	–	[+]
Nitrate reductase	–	–	+
Acid from mannose	+	+	–

^aFor symbols see standard definitions.

within the genus and the subdivision of *X. nematophila* into subspecies proposed (Akhurst, 1983b). A more comprehensive phenotypic study (Boemare and Akhurst, 1988) led to the elevation of the subspecies to species status, as *X. nematophila*, *X. bovienii*, *X. poinarii*, and *X. beddingii* (Akhurst and Boemare, 1988). *Xenorhabdus japonica*¹, symbiotically associated with *Steinernema kushidai*, was described later (Nishimura et al., 1994). DNA–DNA hybridization (Suzuki et al., 1990; Akhurst et al., 1996) and 16S rDNA (Suzuki et al., 1996; Brunel et al., 1997; Liu et al., 1997a; Szállás et al., 1997) analyses validated the inclusion of these five species in, and the exclusion of *P. luminescens* from, the genus.

DNA–DNA hybridization analysis indicates that there are more than the five *Xenorhabdus* species described to date (Fig. BXII.γ.214). Too few isolates of the other “species” have been characterized phenotypically to allow description of new species.

List of species of the genus *Xenorhabdus*

1. ***Xenorhabdus nematophila*** (Poinar and Thomas 1965) Thomas and Poinar 1979, 355^{AL} (*Achromobacter nematophilus* Poinar and Thomas 1965, 249.)

ne.ma.to'phi.la. Modern entomological term n. *nematode*; Gr. adj. *philus* loving or having affinity for; M.L. adj. *nematophila* nematode-loving.

The species characteristics are listed in Tables BXII.γ.277 and BXII.γ.278. No isolates known to grow at temperatures in excess of 34°C. Neither phase is pigmented. Most isolates sensitive to furazolidone.

Only found associated with one species of nematode, *Steinernema carpocapsae*, but distributed globally.

The mol% G + C of the DNA is: 43–48 (Bd).

Type strain: ATCC19061, DSM 3370.

GenBank accession number (16S rRNA): D78009, X82251.

2. ***Xenorhabdus beddingii*** (Akhurst 1986a) Akhurst and Boemare 1993, 864^{VP} (Effective publication: Akhurst and Boemare 1988, 1844) (*Xenorhabdus nematophilus* subsp. *beddingii* Akhurst 1986a, 456.)

bed.din'gi.i. M.L. gen. n. *beddingii* of Bedding, named for R.A. Bedding, who made significant contributions to the development of *Xenorhabdus*/*Steinernema* associations for insect pest control.

The species characteristics are listed in Tables BXII.γ.277 and BXII.γ.278. All isolates grow at 34°C, some at 38°C. Hydrolyses esculin. Inhibited by cephaloridine and ampicillin. The brown pigmentation is not strong. Phase I species is highly unstable, producing the very stable phase II.

Associated with two undescribed species of *Steinernema* from Australia, one of which may be *Steinernema longicaudum*, which was described from China.

The mol% G + C of the DNA is: 45.5–50 (Bd, HPLC).

Type strain: UQM 2871 (phase I of strain Q58), ATCC 49542, DSM 4764.

GenBank accession number (16S rRNA): D78006, X82254.

3. ***Xenorhabdus bovienii*** (Akhurst 1983b) Akhurst and Boemare 1993, 864^{VP} (Effective publication: Akhurst and Boemare 1988, 1843) (*Xenorhabdus nematophilus* subsp. *bovienii* Akhurst 1983b, 45.)

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bo.vi.en'i.i. M.L. gen. n. *bovienii* of Bovien, named for P. Bovien who first reported the presence of bacteria in the intestinal vesicle of a *Steinernema* species.

The species characteristics are listed in Tables BXII.γ.277 and BXII.γ.278. No growth at 34°C; some strains will grow at 5°C. Resistant to carbenicillin.

Associated with several species of entomopathogenic nematode (*Steinernema feltiae*, *Steinernema intermedium*, *Steinernema kraussei*, *Steinernema affine*) in temperate regions.

The mol% G + C of the DNA is: 44.3 (HPLC)–46.9 (Bd).

Type strain: UQM2210 (phase I of strain T228), ATCC 35271, DSM 4766.

GenBank accession number (16S rRNA): X82254, D78007.

4. ***Xenorhabdus japonica*** Nishimura, Hagiwara, Suzuki and Yamanaka 1995, 619^{VP} (Effective publication: Nishimura, Hagiwara, Suzuki and Yamanaka 1994, 209.)

ja.po'ni.ca. M.L. adj. *japonica* of Japan.

The species characteristics are listed in Tables BXII.γ.277 and BXII.γ.278. Does not grow at 37°C. Arginine dihydrolyase activity detected in phase II. Pigmentation is yellowish brown.

Only known to be associated with *Steinernema kushidai* in Japan.

The mol% G + C of the DNA is: 45.9 (HPLC).

Type strain: IAM 14265.

GenBank accession number (16S rRNA): D78008, Z76739.

5. ***Xenorhabdus poinarii*** (Akhurst 1983b) Akhurst and Boemare 1993, 864^{VP} (Effective publication: Akhurst and Boemare 1988, 1843) (*Xenorhabdus nematophilus* subsp. *poinarii* Akhurst 1983b, 45.)

poi.nar'i.i. M.L. gen. n. *poinarii*, of Poinar, named for G.O. Poinar, Jr., who made major contributions to the understanding of entomopathogenic nematode/bacterial interactions.

The species characteristics are listed in Tables BXII.γ.277 and BXII.γ.278. This is the most heat tolerant *Xenorhabdus*, with all strains growing at 36°C and some at 40°C. The intensity of pigmentation in phase I varies from light to reddish brown. Antibiotic production by phase I and phase II is variable. Associated with *Steinernema glaseri* and *Steinernema cubanum* in the USA and Caribbean. The nematode/bacterium association does not appear to be as strong for

1. Euzéby and Boemare (2000) noted that *-rhabdus* has a feminine root and so species names for genera ending in *-rhabdus* should also be feminine.

this species as for other *Xenorhabdus*; its symbiosis may be more primitive. It is not pathogenic for greater wax moth (*Galleria mellonella*) larvae unless associated with its nematode partner.

The mol% G + C of the DNA is: 42.6 (HPLC)–49 (Bd).
Type strain: UQM 2216 (phase I of strain G), ATCC 35272, DSM 4768.
GenBank accession number (16S rRNA): D78010, X82253

Genus XLI. *Yersinia* Van Loghem 1944, 15^{AL}

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Yer.sī' ni.a. M.L. fem. n. *Yersinia* named for the French bacteriologist A.J.E. Yersin, who first isolated the causal organism of plague in 1894.

Straight rods to coccobacilli, 0.5–0.8 × 1–3 µm. Endospores are not formed. Capsules are not present, but an envelope occurs in *Y. pestis* strains grown at 37°C and in cells from clinical sources. Gram negative, **nonmotile at 37°C, but motile with peritrichous flagella when grown below 30°C, except for *Y. pestis*, which is always nonmotile.** Growth occurs on ordinary bacteriological media. Colonies of yersiniae are translucent to opaque, 0.1–1.0 mm in diameter after 24 h. Optimum growth temperature, 28–29°C. Facultatively anaerobic, having both a respiratory and a fermentative type of metabolism. Oxidase negative. Catalase positive. Nitrate is reduced to nitrite with a few exceptions in specific biovars. D-glucose and other carbohydrates are fermented with **acid production but little or no gas. Phenotypic characteristics are often temperature dependent, and usually more characteristics are expressed by cultures incubated at 25–29°C than at 35–37°C.** The enterobacterial common antigen is present in all species investigated. Widely distributed in nature with some species adapted to specific animal hosts and humans. Several species are pathogenic for humans and animals including *Y. pestis*, the causative agent of plague. A significant cause of food-borne and water-borne disease. Belongs to the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 46–50.

Type species: *Yersinia pestis* (Lehmann and Neumann 1896) Van Loghem 1944, 15 (*Bacterium pestis* Lehmann and Neumann 1896, 194; *Yersinia pseudotuberculosis* subsp. *pestis* Bercovier, Mollaret, Alonso, Brault, Fanning, Steigerwalt and Brenner 1981a, 383.)

FURTHER DESCRIPTIVE INFORMATION

Cells of *Yersinia* species are small, coccoid-shaped Gram-negative bacilli that resemble cells of *Pasteurellaceae* rather than cells of *Enterobacteriaceae*. Pleomorphism occurs depending on the type of medium used and the temperature of incubation. Rods, coccobacilli, and small chains of four to five elements (especially in liquid media) can occur. Gram, Giemsa, or Wayson-stained smears reveal a more pronounced tendency to bipolar staining in *Y. pestis* than in the other species. Spores or specific inclusions are not formed. No definite capsule occurs, but *Y. pestis* displays a carbohydrate-protein envelope termed capsular antigen or fraction I (F-1) when cultured at 37°C (Burrows, 1963), or in direct smears taken from infected hosts (mice, guinea pigs, humans). L forms have been described for *Y. enterocolitica* (Pease, 1979).

Yersinia species are nonmotile after growth at 37°C but motile at 22–29°C, except *Y. pestis*, which is nonmotile irrespective of incubation temperatures. Fresh isolates of *Y. enterocolitica* and *Y. pseudotuberculosis* may require a few subcultures to express their motility. Motile cells have 2–15 peritrichous flagella characterized by a long wavelength (Nilehn, 1969).

Yersinias do not differ from other *Enterobacteriaceae* in their

fine structure and overall cell wall composition. Lipopolysaccharides (O antigens) have been isolated and characterized (Davies, 1958; Rische et al., 1973). The whole-cell lipid composition of all *Yersinia* species investigated exhibits a pattern shared with other *Enterobacteriaceae* (Tornabene, 1973; Jantzen and Lassen, 1980). *Y. pestis* lipopolysaccharide has core components in common with other *Enterobacteriaceae* but lacks extended O-group side chains (Perry and Fetherston, 1997).

Placement of the genus *Yersinia* within the *Enterobacteriaceae* family is supported by both biochemical and DNA–DNA relatedness studies. Further studies (Ahmad et al., 1990) comparing rDNA sequence analysis to clustering based on aromatic amino acid synthesis placed the genus *Yersinia* in one of three “enteroclusters” along with *Cedecea*, *Edwardsiella*, *Hafnia*, *Khuyvera*, *Proteus*, *Providencia*, and *Morganella*. Sequence analysis of the 16S rRNA gene performed by Ibrahim and colleagues (1993) of representatives of 10 of the 11 *Yersinia* species reveal that yersiniae form a coherent cluster within the *Gammaproteobacteria* with sequence similarities ranging from 96.9–99.8%. Within the *Gammaproteobacteria* the closest relative is *Hafnia alvei* (96.5% sequence similarity).

Intragenetically, phylogenetic analysis disclosed five sublines, with *Y. enterocolitica*, *Y. rohdei*, and *Y. ruckeri* forming separate sublines. A separate subline was formed by *Y. pestis*, *Y. pseudotuberculosis*, and *Y. kristensenii*, while *Y. mollaretii*, *Y. intermedia*, *Y. bercovieri*, *Y. aldovae*, and *Y. kristensenii* formed a fifth subline.

Yersinia species grow on nutrient agar without enrichment. A small colony diameter differentiates yersiniae from all other *Enterobacteriaceae*. After incubation for 24–30 h at 30°C or 37°C, *Y. pestis* forms minute colonies (0.1 mm) that can be discerned only with difficulty by the naked eye. After 48 h their diameter increases to 1.0–1.5 mm. The colonies are slightly opaque, butyrous, smooth, round, and have somewhat irregular edges. The use of enriched media (serum, blood, yeast extract) does not dramatically improve growth, and after 48 h the colony sizes are similar to those found on nutrient agar. All other *Yersinia* species grown on nutrient agar at 25–37°C produce visible colonies in 24 h. The colonies reach a diameter of 1.0–1.5 mm after 24–30 h, and 2.0–3.0 mm after 48 h. After 18 h they are translucent, smooth, and round with irregular edges, but after 48 h the centers become elevated and the edges become more regular, producing a “Chinese hat” shape. *Y. pestis* colonies may appear slightly mucoid. When cultured for 48 h, all *Yersinia* species dissociate into small (0.5 mm) and large (2 mm) colonies. This phenomenon appears to depend on the medium used (Bercovier et al., 1979).

Growth is moderate in liquid media: incubation of *Yersinia* for 48 h will yield the same turbidity that occurs in 18 h with other *Enterobacteriaceae*. When grown in nutrient broth, clumps of *Y.*

pestis cells adhere to the side of the tube, which slowly forms a deposit at the bottom of the tube while the supernatant remains relatively clear; this is followed by the appearance of a pellicle, which in turn disintegrates to form flocculent masses and a larger deposit. This phenomenon is attenuated in peptone water. *Y. pseudotuberculosis* occasionally grows in a manner similar to that of *Y. pestis*. All other *Yersinia* species give uniform turbidity in nutrient broth and in peptone water.

Y. pestis and *Y. pseudotuberculosis* give variable growth responses on MacConkey agar. All other species grow well on this medium, with colonies reaching a size similar to that observed on nutrient agar. On Salmonella-Shigella agar incubated at 25°C, *Y. pestis* hardly grows at all, whereas all the other species produce pinpoint colonies in 24–30 h. When incubated on this medium at 37°C, *Y. enterocolitica* is only partially inhibited, whereas all other species are severely inhibited (Nilehn, 1969; Bottone, 1977; Bercovier et al., 1979).

All *Yersinia* species except *Y. pestis* can grow at 25°C on synthetic mineral-salt media with various carbohydrates as the energy source (Burrows and Gillett, 1966; Bercovier et al., 1979). *Y. pestis* requires L-methionine, L-isoleucine, L-valine, and L-phenylalanine, and or L-threonine for growth. When incubated at 37°C on synthetic mineral-salt media all *Yersinia* species become auxotrophic, and the addition of biotin and thiamine, and pantothenate and glutamic acid for *Y. pestis*, is necessary to promote growth (Burrows and Gillett, 1966; Brubaker, 1991). The growth of *Y. pestis* on such media is further enhanced by the addition of a reducing agent, and by incubation in a CO₂-enriched atmosphere (Brubaker, 1991). Virulent strains of *Y. pestis* require Ca²⁺ or ATP for growth at 37°C but not at 25°C (Zahorchak et al., 1979). This temperature-dependent requirement for Ca²⁺, mediated by the Lcr plasmid, has also been described for virulent strains of *Y. pseudotuberculosis* and *Y. enterocolitica*.

The growth range is 4–42°C, with an optimum temperature of 28–29°C. *Y. pestis* and *Y. pseudotuberculosis* tolerate a pH range of 5.0–9.6; other *Yersinia* species can grow in a pH range of 4.0–10.0. The optimum pH for all species is 7.2–7.4. *Yersinia* species can grow in peptone water without the addition of NaCl. *Y. pestis* and *Y. pseudotuberculosis* tolerate up to 3.5% NaCl, and the other species can tolerate up to 5% NaCl. *Y. pseudotuberculosis*, but not *Y. enterocolitica*, which lacks tellurite reductase, is the only species that grows well on media containing 0.06% tellurite (Brzin, 1968).

Yersinia species do not differ significantly from other *Enterobacteriaceae* in their general metabolism (Brubaker, 1991) (Table BXII.γ.280). They produce acid during fermentation of D-glucose. *Y. enterocolitica*, *Y. frederiksenii*, and *Y. intermedia* produce acetoin (positive Voges-Proskauer test) when incubated at 28°C, whereas this characteristic is variable for *Y. ruckeri* and is absent in *Y. pestis* and *Y. pseudotuberculosis*. No *Yersinia* species produces acetoin at 37°C (Table BXII.γ.281).

The main physiological and biochemical characteristics of the various *Yersinia* species are given in Table BXII.γ.282. *Yersinia* ferment carbohydrates usually without gas production, a characteristic that is constant for *Y. pestis* and *Y. pseudotuberculosis*, but other species may produce a few bubbles after 2–3 days at 28°C. Because the optimum growth temperature of yersiniae is 28–29°C, some biochemical activities are often temperature-dependent (cellobiose and raffinose fermentation, ornithine decarboxylase, ONPG (*o*-nitrophenyl-β-D-galactopyranoside) hydrolysis, indole production, and the Voges-Proskauer reaction) and are more constantly expressed at 28°C rather than at 37°C. All species

except *Y. intermedia* reduce nitrate to nitrite by a type B nitrate reductase; *Y. intermedia* strains have either a type A nitrate reductase, like most *Enterobacteriaceae*, or a type B reductase. ONPG activity of yersiniae does not correspond to a true β-galactosidase, but only to an ONPG-ase (Le Minor and Coynault, 1976). In addition to the characteristics given in Tables BXII.γ.281 and BXII.γ.282, *Yersinia* species are able to attack polypectate in 5–7 days and starch in 3–7 days. Yersiniae are neither hemolytic nor proteolytic, except *Y. ruckeri*, which liquefies gelatin, and some strains of *Y. pestis*, which have fibrinolytic and coagulase activity linked to the production of Pesticin 1. Plasminogen activator coagulase activity and pesticinogeny are plasmid (pPCD1) encoded. Lecithinase activity in *Y. enterocolitica* is strain-dependent. *Y. pseudotuberculosis*, *Y. enterocolitica*, and *Y. ruckeri* strains have a lipase that is active on corn oil, but only *Y. intermedia*, *Y. frederiksenii*, and *Y. enterocolitica* biovar I express a lipase-esterase that is active on Tween 80.

Transformation of auxotrophic strains of *Y. enterocolitica* by prototrophic strains using the Juní-Janik technique has been reported (Callahan and Koroma, 1979). F lac⁺ episomes from *E. coli* have been transferred to *Y. pestis* (Martin and Jacob, 1962), to *Y. pseudotuberculosis* (Lawton et al., 1968b), and to *Y. enterocolitica* (Cornelis and Colson, 1975), but usually with a low frequency (10⁻⁴–10⁻⁶). This has allowed chromosomal mapping of *Y. pseudotuberculosis* (Lawton and Stull, 1971; McMahon, 1973). Gene transfer by conjugation between *Y. pseudotuberculosis* and *Y. pestis* has also been demonstrated (Lawton et al., 1968a).

R factors have been transferred to *Y. pestis* and *Y. pseudotuberculosis* (Ginoza and Matney, 1963) and to *Y. enterocolitica* (Knapp and Lebek, 1967). Wild strains of *Yersinia* carrying R plasmids (Cornelis et al., 1973; Kanazawa and Ikemura, 1979) appear to be rare. This could be explained, at least for *Y. enterocolitica*, by the presence of a restriction-modification system (Cornelis and Colson, 1975). A self-transmissible plasmid coding for lactose fermentation has been described in *Y. enterocolitica* (Cornelis et al., 1976). This plasmid is freely transmissible between strains of *Y. enterocolitica* and *E. coli*.

Other plasmids related to various virulence tests (Ca²⁺ dependency, autoagglutination, lethality for mice and gerbils, Sereny test) have been demonstrated in *Y. pestis* (Ferber and Brubaker, 1981; Brubaker, 1991), *Y. pseudotuberculosis* (Gemski et al., 1980b), and *Y. enterocolitica* (Gemski et al., 1980a; Zink et al., 1980). These plasmids of 40–48 MDa molecular weight constitute a family of related plasmids (Ben-Gurion and Shafferman, 1981; Portnoy et al., 1981). *Y. pestis* and *Y. pseudotuberculosis* have never been found to be lysogenic, whereas of 1252 strains of *Y. enterocolitica* studied, 86.4% were lysogenic when grown at 25°C but not at 37°C (Nicolle et al., 1973). Phages active on *Y. pestis* and *Y. pseudotuberculosis* have been described (Gunnison et al., 1951; Girard, 1953), but they are not host-specific and are used only for presumptive bacteriological diagnosis. Coliphages T2, T3, and T7 are also active on *Y. pseudotuberculosis* and *Y. pestis* (Hertman, 1964). A phage typing system, useful in epidemiological investigations has been developed for *Y. enterocolitica* (Nicolle et al., 1973): strains of *Y. enterocolitica* serogroup O:3 are associated with phagovar VIII in Europe, IXa in the Republic of South Africa, and IXb in Canada and the United States (Bottone, 1997).

Strains of *Y. pestis* produce a bacteriocin active on *Y. pseudotuberculosis* (Ben-Gurion and Hertman, 1958). This was named Pesticin I by Brubaker and Surgalla (1962) after they detected a second bacteriocin (Pesticin 11) that was produced by *Y. pestis* and *Y. pseudotuberculosis*. Pesticin I has a narrow host range, being

TABLE BXII.γ.280. Differentiation of *Yersinia* from other genera^a

Test	<i>Yersinia</i>	<i>Citrobacter</i>	<i>Enterobacter</i>	<i>Escherichia</i>	<i>Hafnia</i>	<i>Klebsiella</i>	<i>Proteus</i>	<i>Salmonella</i>
Voges-Proskauer	—	—	+ ^b	—	[+]	d	d	—
Citrate (Simmons)	—	+	+	[—]	—	d	d	+
H ₂ S production	—	d	—	—	—	—	d	+
Phenylalanine deaminase	—	—	—	—	—	—	+	—
Lysine decarboxylase	— ^c	—	[—] ^d	[+]	+	[+]	—	+
Motility, 37°C	—	+	+ ^b	[+]	[+]	—	+	+
Motility, 25°C	+ ^e	+	+ ^b	[+]	[+]	—	+	+
KCN, growth	— ^c	d	+ ^f	— ^g	+	+	+	+ ^h
Malonate utilization	—	d	+ ⁱ	d	d	+ ^j	—	d
D-Glucose, gas	— or W	+	+ ^k	+	+	[+]	[+]	+
L-Arabinose, acid	+ ^l	+	+	+	+	+	—	+
D-Mannitol, acid	+ ^m	+	+	+ ⁿ	+	+	—	+
Mucate, acid	—	+	d	d	—	+ ^o	—	+ ^p

^aFor symbols see standard definitions; W, weak.^bExcept *E. asburiae*.^cExcept some strains of *Y. ruckeri*.^d*E. aerogenes* and *E. gergoviae* are positive.^eExcept *Y. pestis* and some strains of *Y. ruckeri*.^fExcept *E. gergoviae* and some strains of *E. agglomerans*.^gExcept *E. hermannii* and a few strains of *E. vulneris*.^hExcept *S. bongori* and some strains of *S. choleraesuis* subsp. *houtenae*.ⁱExcept *E. asburiae* and *E. sakazakii*.^jExcept *K. ozaenae*.^kExcept *E. agglomerans*.^lExcept *Y. ruckeri* and some strains of *Y. aldovae*, *Y. kristensenii*, and *Y. pseudotuberculosis*.^mExcept some strains of *Y. aldovae*.ⁿExcept *E. blattae*.^oExcept *K. ozaenae* and *K. rhinoscleromatis*.^pExcept *S. choleraesuis* subsp. *diarizonae* and *S. choleraesuis* subsp. *houtenae*.

active against a few strains of *E. coli*, serogroup IA and IB *Y. pseudotuberculosis* (Burrows, 1963), and serogroup O:8 *Y. enterocolitica* (Brubaker, 1991). *Y. pestis* strains that produce Pesticin I also elaborate a fibrinolytic factor and a coagulase (Brubaker, 1972). A bacteriocin-like activity associated with the presence of phage tails has been described in *Y. enterocolitica* (Nicolle et al., 1973). *Y. intermedia* produces a bacteriocin-like substance at 25°C but not at 37°C that is active on certain strains of *Y. enterocolitica*, *Y. intermedia*, *Y. frederiksenii*, and *Y. kristensenii* (Bottone et al., 1979). Cafferkey et al. (1989) have shown that bacteriocin production by "avirulent" strains of serogroup O:3 occurs at both 25°C and at 37°C and is active against isolates of serogroups O:3, and single strains tested of O:8 and O:9 irrespective of the presence of the virulence plasmid.

The antigenic structure of *Yersinia* species is complex, but antigens are shared by *Y. pestis*, *Y. pseudotuberculosis*, and *Y. enterocolitica*. The common enterobacterial antigen has been found in these species (Le Minor et al., 1972; Maeland and Digranes, 1975). The fraction I envelope antigen (FI) of *Y. pestis* is best produced when cultures are incubated at 37°C on protein-rich media (Fox and Higuchi, 1958). This antigen, a large gel-like capsule or envelope, is heat labile (10 min at 100°C), water soluble, and dissociates during growth into a glycoprotein (FIA) and a carbohydrate-free protein (FIB). Passive hemagglutination with FI antigen is used for serologic surveys in plague foci. The presence of this antigen has also been demonstrated in *Y. pseudotuberculosis* (Quan et al., 1965). V and W antigens expressed by virulent strains of *Y. pestis* cultivated at 37°C appear to be related to the presence of a 45 MDa plasmid (Ben-Gurion and Shaffer, 1981; Ferber and Brubaker, 1981). Production of plasmid-

mediated V and W antigens has also been described in *Y. pseudotuberculosis* (Gemski et al., 1980b) and in *Y. enterocolitica* (Gemski et al., 1980a). The somatic antigen of *Y. pestis* is rough (R antigen) and therefore no serogroups have been described in this species. This R antigen is also present in *Y. pseudotuberculosis* (Thal and Knapp, 1971). In addition, *Y. pestis* and *Y. pseudotuberculosis* share at least 11 of 18 antigens (Lawton et al., 1960). *Y. pestis* and *Y. enterocolitica* express common protein antigens (Barber and Eylan, 1976). The antigenic scheme for *Y. pseudotuberculosis* (Thal and Knapp, 1971) comprises eight main thermostable serogroups (I to VIII) with nine subtypes (IA, IB; IIA, IIB, IIC; III; IVA, IVB; VA, VB; VI; VII; VIII), and five thermolabile flagellar H antigens (a–e). Antigenic relationships have been demonstrated between *Y. pseudotuberculosis* (serogroups 11, IV, IVA, and VI) and *Salmonella* serogroups B and D, *E. coli* serogroups O17, O55, and O77, and *Enterobacter cloacae* (Knapp, 1968; Mair and Fox, 1973).

Wauters et al. (1972) initially described 34 different O antigen and 20 H antigen serogroups in *Y. enterocolitica*. This classification included some serogroups defined by strains belonging to *Y. intermedia* (O:17) and *Y. kristensenii* (O:11, O:12, O:28). Wauters et al. (1991) subsequently expanded the O and H antigenic schema to include approximately 60 serogroups. Serogroup expansion has also shown sharing of antigens between pathogenic and nonpathogenic *Y. enterocolitica* and *Y. bercovieri* (O:8), *Y. frederiksenii* (O:3), and *Y. mollaretii* (O:3). Nevertheless, serogrouping in conjunction with biogrouping (Table BXII.γ.283), is a useful epidemiological marker. Cross-reactions occur between *Y. enterocolitica* serogroup O:9 and *Brucella* species (Hurvell and Lindberg, 1973; Corbel, 1975).

Yersinia species are susceptible *in vitro* to the following anti-

TABLE BXII.γ.281. Reactions of *Yersinia* spp. at 25–28°C and 37°C^a

Test	1. <i>Y. pestis</i>	2. <i>Y. aldovae</i>	3. <i>Y. bercovieri</i>	4. <i>Y. enterocolitica</i>	5. <i>Y. frederiksenii</i>	6. <i>Y. intermedia</i>	7. <i>Y. kristensenii</i>	8. <i>Y. mollaretii</i>	9. <i>Y. pseudotuberculosis</i>	10. <i>Y. rohdei</i>	11. <i>Y. ruckeri</i>
<i>Voges-Proskauer:</i>											
37°C	–	–	–	–	–	–	–	–	–	–	–
25–28°C	–	+	–	+	+	+	–	–	–	–	–
<i>Citrate, Simmons:</i>											
37°C	–	–	–	–	[–] ^b	–	–	–	–	–	–
25–28°C	–	d	–	–	d	–	–	–	–	[+] ^c	–
<i>Urease:</i>											
37°C	–	[+]	d	[+]	[+]	[+]	[+]	[–]	+	d	–
25–28°C	–	+	+	+	+	+	+	+	+	d	–
<i>Ornithine decarboxylase:</i>											
37°C	–	d	[+]	+	+	+	+	[+]	–	[–]	+
25–28°C	–	+	+	+	+	+	+	+	–	[+]	+
<i>Motility:</i>											
37°C	–	–	–	–	–	–	–	–	–	–	–
25–28°C	–	+	+	+	+	+	+	+	+	+	[+]
<i>Glycerol, acid:</i>											
37°C	d	–	–	+	[+]	d	d	[–]	d	d	d
25–28°C	d	+	[+]	+	+	+	+	+	+	[+]	d
<i>myo-Inositol, acid:</i>											
37°C	–	–	–	d	[–]	[–]	[–]	–	–	–	–
25–28°C	–	+	d	d	–	[+]	d	d	–	–	–
<i>Melibiose, acid:</i>											
37°C	[–]	–	–	–	–	[+]	–	–	d	d	–
25–28°C	d	–	–	–	–	+	–	–	+	d	–
<i>Raffinose, acid:</i>											
37°C	–	–	–	–	d	d	–	–	[–]	d	–
25–28°C	–	–	–	–	–	+	–	–	[–]	d	–
<i>L-Rhamnose, acid:</i>											
37°C	–	–	–	–	+	+	–	–	d	–	–
25–28°C	–	+	–	–	+	+	–	–	+	–	–
<i>Salicin, acid:</i>											
37°C	d	–	[–]	[–]	+	+	[–]	[–]	[–]	–	–
25–28°C	[+]	+	–	[–]	+	+	–	[–]	d	–	–
<i>D-Xylose, acid:</i>											
37°C	+	d	+	d	+	+	[+]	d	+	d	–
25–28°C	+	+	+	d	+	+	+	+	+	+	–
<i>Mucate, acid:</i>											
37°C	–	–	–	–	–	–	–	–	–	–	–
25–28°C	–	d	+	–	[–]	d	–	+	–	–	–
<i>Esculin hydrolysis:</i>											
37°C	d	–	[–]	[–]	[+]	[–]	–	–	+	–	–
25–28°C	+	+	[+]	d	+	+	–	[–]	+	–	–

^aFor symbols see standard definitions.^b[–], 11–25% positive.^c[+], 26–75% positive.

microbial agents: tetracycline, chloramphenicol, aminoglycosides (amikacin, streptomycin, gentamicin, kanamycin, and neomycin), sulfonamides (alone or in combination with trimethoprim), imipenem, aztreonam, and fluoroquinolones (Hoogkamp-Korstanje, 1987; Bonacorsi et al., 1994). They are variably susceptible to colistin and are resistant to erythromycin and novobiocin. *Y. pestis* and *Y. pseudotuberculosis* are usually susceptible to β-lactam antibiotics, but their susceptibility to penicillin is in the range of susceptible to intermediate. Resistance to ampicillin (Borowski and Zaremba, 1973) and to streptomycin (Kanazawa and Ike-mura, 1979) has been described for *Y. pseudotuberculosis*, *Y. enterocolitica*, and *Y. intermedia* (Bottone, 1977). *Y. frederiksenii* and *Y. kristensenii* are resistant to penicillin and slightly susceptible or resistant to other β-lactam antibiotics (ampicillin, carbenicillin, cephalothin) (Bercovier et al., 1979). The level of resistance is strain dependent (Zaremba and Aldová, 1979) and temperature dependent (Chester and Stotzky, 1976). Antibiotic susceptibility patterns of *Y. enterocolitica* are serogroup specific. *Y. enterocolitica* strains produce both a constitutive β-lactamase (active on am-

picillin, carbenicillin, penicillin, and cephalosporins) and an inducible β-lactamase active only on cephalosporins and penicillin (Cornelis and Abraham, 1975). *Y. enterocolitica* strains that are resistant to tetracycline, chloramphenicol, streptomycin, and kanamycin have been reported (Zaremba and Aldová, 1979). Newer β-lactam antibiotics such as ceftriaxone, ceftazidime, cefotaxime, and moxalactam have excellent activity against *Y. enterocolitica*, as do imipenem and aztreonam (Hornstein et al., 1985).

Y. pestis is the causative agent of plague. Plague is primarily a disease of wild rodents. *Y. pestis* is transmitted among wild rodents by flea bites or ingestion of contaminated animal tissues (Butler, 1983). In fleas, the bacterium multiplies and blocks the esophagus and the pharynx. The fleas regurgitate the bacteria when they take their next blood meal and transmit *Y. pestis* to humans if no other hosts are available. Infective flea bites produce the typical bubonic form of plague within 2–8 d. *Y. pestis* multiplies intracellularly in macrophages and extracellularly and proceeds through the lymphatic system. The lymph nodes near the flea

TABLE BXII.γ.282. Differential characteristics of the species of the genus *Yersinia*

Test ^a	1. <i>Y. pestis</i>	2. <i>Y. aldovae</i>	3. <i>Y. bercovieri</i>	4. <i>Y. enterocolitica</i>	5. <i>Y. frederiksenii</i>	6. <i>Y. intermedia</i>	7. <i>Y. kristensenii</i>	8. <i>Y. mollaretii</i>	9. <i>Y. pseudotuberculosis</i>	10. <i>Y. rohdei</i>	11. <i>Y. ruckeri</i>
Pathogenic	+	(+) ^b	(+)	+	(+)	(+)	(+)	(+)	+	(+)	(+)
Mol% G + C content	46	46	50	48.5 ± 0.5	48	48.5 ± 0.5	48.5 ± 0.5	50–51	46.5	48.7–49.4	48.0 ± 0.5
Motility (22°C)	– ^c	–	–	+	–	+	–	–	+	–	–
Nitrate reductase	V ^{d,e}	+	+	+	+	+	+	+	+	+	+
Urease	– ^c	–	d	+	+	+	+	–	+	d	–
Simmons citrate	–	d	–	–	V	+ 22°C or –	–	–	–	+	–
Ornithine decarboxylase	–	d	+	+	+	+	+	+	–	+	+
Acetylmethyl-carbinol (22°C)	–	+	+	+	+	+	–	+	–	+	+
B-Galactosidase	d	–	–	+	+	+	d	–	d	d	d
Indole	–	–	–	V	+	+	V	–	–	–	–
H ₂ S production	–	–	–	–	–	–	–	–	–	–	–
Aesculin hydrolysis	+	–	–	V	+	+	–	–	+	–	–
Gelatin	–	–	–	–	–	–	–	–	–	–	–
Methyl red	+ ^e	+ ^d	V	V	+	+	+	+	+	+	+
Glucose	+	+	+	+	+	+	+	+	+	+	+
Lactose	–	–	–	–	d	d	–	d	–	–	–
Sucrose	–	–	+	+	+	+	–	+	–	+	–
Glycerol	V ^d	–	–	+	+	d	d	–	V	d	d
α-Methylglucoside	–	–	–	–	–	+ (22°C)	–	–	–	–	–
Cellobiose	–	d	+	+	+	+	+	+	– ^c	–	–
Melibiose	V ^d	–	–	–	–	22°C ^e	– ^c	–	+	d	–
Mucate	–	d	+	–	V	V	–	+	–	–	–
Raffinose	–	–	–	–	d	d	–	–	–	d	–
L-Rhamnose	–	–	–	–	d	d	–	–	+ ^c	–	–
Trehalose	+	+	+	V	V	+	–	+	+	+	+
Sorbose	–	NR ^f	NR	V	+	+	+	NR	–	NR	NR
Sorbitol	d	d	+	+	+	+	+	+	– ^c	+	d
Arabinose	+	+	+	+	+	+	+	+	+	+	–
Maltose	V ^d	–	+	+	+	+	+	d	+	–	+
D-Xylose	+	d	+	+	+	+	+	d	+	d	–
Mannitol	+	+	+	+	+	+	+	+	+	+	+
Pyrazinamidase	–	–	V	V	+	+	+	–	–	+	+
<i>Usual habitat:</i>											
Animals				+	+		+		+		+
Birds									+		
Dogs										+	
Fish		+			+	+					+
Fleas	+										
Mammals	+										
Rodents	+					+				+	
Humans	+	+	+	+		+		+	+	+	+
Environment				+			+				
Soil			+				+				
Sewage						+					
Water		+			+	+				+	+
Food					+			+			+
Vegetables			+					+			

^aAll *Yersinia* spp. are fermentative; negative for H₂S, oxidase, arginine dihydrolase, phenylalanine deaminase, malonate, dulcitol, DNase (25°C), and pigmentation; all are positive for mannitol and mannose. Motile species have peritrichous flagella, and motility is more pronounced at temperatures less than or equal to 30°C. Except for delayed reactions in some *Y. ruckeri*, all species are negative for gelatin hydrolase and lysine decarboxylase.

^b(+), Opportunistic pathogens.

^cKey test to differentiate species.

^dKey test to differentiate biovars.

^eV, Strain instability or variable.

^fNR, Not reported.

bite are the first to become inflamed, enlarged, and painful, and constitute the bubo. As the evolution of the infection is usually rapid with massive growth of *Y. pestis* in the blood, characteristic lesions are not found in the spleen or liver at autopsy. Untreated,

the disease evolves in 5–10 d to profound septicemia in which *Y. pestis* may be seen in peripheral blood smears (Mann et al., 1984). Secondary pneumonia may result from hematogenous spread. From the latter, primary pneumonic plague can spread

TABLE BXII.γ.283. Differentiation of biogroups of *Yersinia enterocolitica*^a

Test	Biogroup Reaction					
	1A	1B ^b	2	3	4	5
Lipase activity	+	+	—	—	—	—
Salicin (acid production in 24 h)	+	—	—	—	—	—
Esculin hydrolysis (24 h)	±	—	—	—	—	—
Xylose (acid production)	+	+	+	+	—	V
Trehalose (acid production)	+	+	+	+	+	—
Indole production	+	+	V	—	—	—
Ornithine decarboxylase	+	+	+	+	+	+
Voges-Proskauer test	+	+	+	+	+	+
Pyrroline oxidase activity	+	—	—	—	—	—
Sorbose (acid production)	+	+	+	+	+	—
Inositol (acid production)	+	+	+	+	+	+
Nitrate reduction	+	+	+	+	+	—

^aSymbols: +, positive; —, negative; (+), delayed positive; V, variable.

^bBiogroup 1B is comprised mainly of strains isolated in the United States.

from person to person or from animals to humans by means of droplets. Both bubonic and pneumonic plague have been transmitted to humans by the domestic cat—bubonic through a scratch and bite from an infected cat (Thornton et al., 1975; Weniger et al., 1984), and pneumonic through face-to-face exposure to a cat with pneumonic plague (Werner et al., 1984; Doll et al., 1994). In this clinical form, death generally occurs in less than 4 d. Pestis minor cases, in which the bacteria remain self-limited in buboes followed by self-cures have been described in endemic plague areas (Pollitzer, 1954).

Virulence of *Y. pestis* is associated with several factors (Surgalla et al., 1968; Perry and Fetherston, 1997). Among these are synthesis of the fraction I (FI) gel-like antiphagocytic capsule or surface antigen, VW antigens (associated with Ca^{2+} dependency for growth on magnesium oxalate medium, autoagglutination in broth cultures at 37°C, pigment production (incorporation of Congo red dye or hemin into cell surfaces, which results in dark greenish-brown or red colonies at 26°C but not at 37°C), presence of siderophore (yersiniabactin) iron-acquisition system, pH6 antigen, a fibrillar adhesin surface structure that may facilitate entry into macrophages (Straley, 1993) and is synthesized in host macrophage lysosomes (Lindler and Tall, 1993), and serum resistance necessary for growth in blood and transmission between insect vector and mammalian hosts. Pesticin production (Pst plasmid) is necessary for dissemination of *Y. pestis* from peripheral sites of inoculation; Pst is a 9.5-kb plasmid that also encodes plasminogen activator, a fibrinolysin, coagulase activity, which may be limited to rabbit plasma (Brubaker, 1991), and murine toxin, a 6-adrenergic antagonist that is highly lethal for mice and rats, causing circulatory collapse. LD₅₀ dose for mice inoculated with strains expressing the aforementioned virulence factors is 1–10 organisms. Avirulent strains of *Y. pestis* never produce VW antigens except in the case of the vaccine strain EV76, whose attenuated virulence has resulted from a mutation in its iron metabolism. Virulent strains and the EV76 strain harbor a 45-MDa plasmid. In contrast to the VW antigens, the lack of any of the other virulence factors does not completely abolish the virulence of *Y. pestis* strains.

Y. pseudotuberculosis is responsible for epizootics in nearly all animal species, especially in rodents and birds. Animals are usually contaminated by the oral route and, after 1–2 weeks of incubation, the bacteria are found in mesenteric lymph nodes. The main symptoms are mesenteric adenitis and chronic diarrhea. Infection evolves either in self-cure, or in fatal septicemia. *Y.*

pseudotuberculosis is an intracellular parasite and, like *Y. pestis*, reaches the lymphatic system. At autopsy, caseous lesions are found in Peyer's patches, mesenteric lymph nodes, the spleen, and the liver. Humans orally contaminated by *Y. pseudotuberculosis* develop either a mesenteric adenitis, which simulates acute appendicitis, or, in the compromised host, a severe septicemia. *Y. pestis* and *Y. pseudotuberculosis* share a common 70-kb plasmid low calcium response (LCR) that encodes for calcium dependency and four outer membrane proteins, and the V and W antigens. *Y. pseudotuberculosis* chromosomally controlled virulence factors include the outer membrane protein invasins, which promotes host cell penetration (Isberg and Falkow, 1985).

Y. enterocolitica has been recognized as pathogenic for chinchillas, hares, monkeys, and humans. The pathogenicity for animals is similar to that of *Y. pseudotuberculosis*. In humans the most common presentation is acute enteritis in children. Concurrent mesenteric lymphadenitis and terminal ileitis simulating appendicitis may also be present. In young adults, acute terminal ileitis and mesenteric lymphadenitis appear to be more common. The extent of gastrointestinal tract pathology depends on the serogroup of the invading strain and age and underlying status of the host. In adults secondary clinical forms of infection include arthritis and erythema nodosum. Septicemia with metastatic abscesses is a rarer complication and is usually associated with immunosuppression or occurs in the setting of iron overload in individuals receiving the iron chelator desferrioxamine (Robins-Browne and Pipic, 1985). Septicemia and profound shock has been associated with transfusion of *Y. enterocolitica* contaminated blood (reviewed by Bottone, 1999). Secondary immunologically mediated sequelae include arthritis and erythema nodosum. Arthritis is closely associated with the presence of the histocompatibility antigen HLA-27 (Dequeker et al., 1980), especially among Scandinavians, and *Y. enterocolitica* serogroup 0:3 biotype 4, phage type 8, and serogroup 0:9 infections. Production of a heat-stable enterotoxin (ST) resembling *E. coli* ST (Okamoto et al., 1981) has been demonstrated *in vitro* (Pai and Mors, 1978), but its role in pathogenicity is not clear: *Y. enterocolitica* strains do not produce ST when incubated *in vitro* at temperature above 30°C, and direct proof of ST production *in vivo* has not been reported.

Virulent plasmid-containing strains of *Y. pestis*, *Y. pseudotuberculosis*, and *Y. enterocolitica* rapidly become avirulent when grown at 37°C, which results in the loss of the virulence plasmid. Cross-immunity among these three species has been demonstrated

(Thal, 1973; Alonso et al., 1978). Human chemoprophylaxis with streptomycin (drug of choice), tetracycline, and chloramphenicol; vaccination; and the spreading of insecticides and rodenticides are the suggested measures for controlling plague. For persons exposed to *Y. pestis*, tetracycline, doxycycline, and trimethoprim-sulfamethoxazole are recommended (Poland, 1989).

The pathogenicity of *Y. intermedia*, *Y. kristensenii*, *Y. frederiksenii*, *Y. bercovieri*, and *Y. mollaretii* in humans and animals is not clearly established. They appear more like opportunistic pathogens than true pathogens (Bercovier et al., 1978; Wauters et al., 1988; Bottone, 1997). ST-producing strains of these three species have been described (Kapperud, 1980), but their clinical significance is still unknown. *Y. ruckeri* is a fish pathogen responsible for red mouth disease, especially in rainbow trout. An inflammation of the mouth and the throat is the main characteristic of the disease, which is enzootic (Rucker, 1966). The bacterium is usually isolated from the kidneys of fish undergoing a systemic infection.

Epidemiology The geographical distribution of *Y. pestis* is widespread, and the organism has been isolated from all continents except Australia and Antarctica. Plague is enzootic in Africa (Central, East, and South Africa), in North and South America, certain regions of Asia (Southeast Asia, Mongolia), and the former USSR, India, and southwestern United States and the Pacific coastal area (Perry and Fetherston, 1997). Between epidemics, *Y. pestis* remains localized in enzootic or maintenance hosts (Butler, 1983) and has been isolated from more than a hundred different naturally infected species of rodents, but rarely from predatory animals (carnivores and birds, the latter being resistant to the infection). The spread of plague is usually accomplished by the epizootic cycle of rodents to fleas and fleas to rodents. The reservoir for *Y. pestis* is soil contaminated by infected dead fleas and rodents in which the microorganism survives for months in deep rodent burrows.

Rodents coming from noninfected areas become infected when they dig burrows in previously contaminated areas (Mollaret et al., 1963). This cycle constitutes "sylvatic plague". When urban rodents come in contact with rural rodents, *Y. pestis* can spread between rodents and to humans through flea bites. The epidemiology of plague is thus linked to the ecology of both fleas and rodents.

Y. pseudotuberculosis is distributed worldwide. It has been found in numerous animal species, especially rodents and birds, in soil, and in humans (Wetzler, 1970). In Japan, cats and dogs have been associated with human cases (Fukushima et al., 1989). Wild animals, which are often asymptomatic carriers, are considered the reservoir of the bacteria. Humans and animals are contaminated orally either by direct contact with sick or asymptomatic animals or through food contaminated by the excretions of these animals. The incidence of *Y. pseudotuberculosis* infection varies with the season and is highest during colder months. *Yersinia* species multiply at 4°C and therefore have a selective advantage over other bacteria at low temperatures; this explains why *Y. pseudotuberculosis*, *Y. enterocolitica*, *Y. frederiksenii*, and *Y. kristensenii* are more frequently isolated from the environment during cold rather than hot seasons. Human and animal infections follow this seasonal distribution as well.

Y. enterocolitica has been isolated from a wide variety of sources (live and inanimate) in every country in which it has been sought and probably has a worldwide distribution (Mollaret et al., 1979). As shown in Table BXII.γ.284, biovar 1A strains are ubiquitous, having been found in a wide range of animal and environmental

sources (including foods), whereas other biovars or serogroups are frequently associated with a specific host (Bercovier et al., 1980a): biovar 5 strains have been isolated mainly from hares in Europe; biovar 4, serogroup 0:3 strains are responsible for most human gastrointestinal infections in Europe, Canada, and the Republic of South Africa, and recently in the United States as well (Bottone et al., 1987; Ostroff, 1995). Serogroup 0:8, heretofore the most frequently isolated *Y. enterocolitica* strain in the United States, has decreased in frequency in the United States, and is only sporadically reported in other parts of the world (Ostroff, 1995). Serogroup 0:9 strains are the second most common *Y. enterocolitica* isolates in Europe and Japan.

Y. intermedia and *Y. frederiksenii* have been identified in Europe, the United States, Australia and New Zealand, Israel, and Japan. These two species have been isolated mainly from fresh water and foods and only rarely from nonirrigated soil, humans or animals other than fish (Kapperud, 1977; Bercovier et al., 1978; Brenner et al., 1980a; Ursing et al., 1980a). *Y. kristensenii* has been found in Europe, the United States, Japan, and Australia. Strains of this species have been isolated mainly from soil, foods, and asymptomatic animals; isolates from other environmental sources and from human infection are rare (Bottone and Robin, 1977; Bercovier et al., 1980b). *Y. bercovieri* has been recovered from terrestrial ecosystems (Bercovier et al., 1978), while *Y. mollaretii* has been isolated from human stool specimens, most raw vegetables, soil, and drinking water (Fukushima, 1985; Kaneko and Maruyama, 1987). Evidence of human disease potential for either *Y. bercovieri* or *Y. mollaretii* are lacking.

Y. ruckeri, which is taxonomically tentative in the genus *Yersinia*, has been encountered mainly in the United States and Canada as a natural component of fresh water ecosystems. *Y. ruckeri* causes enteric red mouth disease of salmon and fish when fish are exposed to large numbers of bacteria (Ross et al., 1966). The disease is usually enzootic and occasionally epizootic in fish hatcheries especially under conditions of stress or poor environmental factors (Stevenson, 1997). Five serovars and two biotypes of *Y. ruckeri* have been identified. Serovar 1 strains are usually D-sorbitol positive, while serovar 2 isolates are usually D-sorbitol negative (Stevenson and Daly, 1982; Stevenson and Airdrie, 1984). A 36-MDa transferable plasmid encoding resistance to tetracycline and sulfonamides has been identified in some *Y. ruckeri* strains (DeGrandis and Stevenson, 1985).

ENRICHMENT AND ISOLATION PROCEDURES

Isolation of *Yersinia* strains from noncontaminated normally sterile samples (blood, lymph nodes) can be performed by using blood agar or nutrient agar incubated for 48 h at 28°C, or 24 h at 37°C followed by 24 h at room temperature. The isolation of *Y. pestis* from contaminated samples requires inoculation (subcutaneously or percutaneously) of animals (guinea pigs, mice, or rats). The organism can be recovered postmortem from the spleen, liver, or lymph nodes of the inoculated animals. All other *Yersinia* species can be isolated from clinical or food samples by inoculating standard or special selective media (reviewed by Bottone, 1992), of which cefsulodin-irgasan-novobiocin (CIN) agar (Schiemann, 1979), and virulent *Yersinia enterocolitica* (VYE) agar formulated by Fukushima (1987) seem the most useful. The latter medium was developed because of the growth inhibition of many *Y. bercovieri* and *Y. pseudotuberculosis* strains on CIN agar (Fukushima and Gomyoda, 1986). All these media should preferably be incubated for 48 h at 28–29°C or for 24 h at 37°C followed by 48–72 h at room temperature. Recovery of *Yersinia*

TABLE BXII.γ.284. Correlation of biogroup and serogroup with ecologic and geographic distribution

Biogroup	Associated with human infections	Serogroup(s)	Ecologic distribution
1B	+	O:8, O:4, O:13a, 13b, O:18, O:20, O:21	Environment, pigs (O:8), mainly in the U.S.A.
2	+	O:9, O:5, 27	Pigs, Europe (O:9), United States (O:5, 27), Japan (O:5, 27)
3	+	O:1, 2, 3, O:5, 27	Chinchilla, pigs (O:5, 27)
4	+	O:3	Pigs, Europe, U.S.
5	+	O:2,3	Hare
1A ^a	—	O:5, O:6, 30, O:7, 8, O:18, O:46, nontypeable	Environment, pigs, food, water animal and human feces, U.S.

^a*Y. enterocolitica* isolates comprising biogroup 1A may be opportunistic pathogens in patients with underlying disorders.

strains from contaminated samples can be improved by various cold enrichment techniques (van Pee and Straiger, 1979; Lee et al., 1980).

MAINTENANCE PROCEDURES

Stab inoculations of *Yersinia* strains in conventional stock culture media stored in the dark at room temperature or at 4°C provide living cultures for 10 years or more, if the tubes are tightly sealed. Lyophilization and deep-freeze storage in 10% glycerol are suitable preservation techniques. To keep a strain fully virulent, it should never be subcultured at 37°C, but always at 25–28°C.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Methods to test tetrathionate reduction, tellurite reduction, and the type of nitrate reductase have been described or referenced by Bercovier et al. (1979). The Ca²⁺-dependency of virulent *Yersinia* strains is evaluated on magnesium oxalate medium¹ (Higuchi and Smith, 1961) as follows. Inoculate 0.1 ml of a bacterial suspension (10⁵ cells/ml) onto two plates: one is incubated at 37°C, the other at 26°C. Check colony numbers on the two plates after 2–3 d. Colonies growing at 26°C but not at 37°C are Ca²⁺-dependent. A fully virulent strain should give confluent growth at 26°C, whereas only 10–100 colonies should appear at 37°C. The autoagglutination test (Laird and Cavanaugh, 1980) to detect virulent *Yersinia* strains is performed by inoculating 10 or more isolated colonies, individually into a pair of tubes (13 × 100 mm) containing 2 ml of RPMI-1640 medium containing 10% fetal calf serum and 25 ml HEPES buffer (N-2-hydroxyethylpiperazine-N'-2-ethanesulfonic acid). One tube is incubated at 37°C, the other at 26°C. After incubation at 26°C for 18 h, virulent isolates give a uniform turbidity; at 37°C a layer of agglutinated bacteria appears at the bottom of the tube and the overlying supernatant remains clear. Avirulent strains give uniform turbidity at both 26°C and 37°C, and rough strains show spontaneous agglutination at both temperatures.

DIFFERENTIATION OF THE GENUS *YERSINIA* FROM OTHER GENERA

Characteristics useful for differentiating *Yersinia* from other physiologically similar genera are listed in Table BXII.γ.280, and in Tables BXII.γ.193, BXII.γ.194, and BXII.γ.196 in the chapter on the family *Enterobacteriaceae*.

1. Magnesium oxalate medium consists of blood agar base (BBL, or any other manufacturer if the Ca²⁺ content of the base is low), 40.0 g; distilled water, 830 ml. Sterilize at 121°C for 15 min and cool to 45°C. From stock solution sterilized by filtration, aseptically add the following ingredients: MgCl₂ solution (23.8 g/l), 80 ml; sodium oxalate solution (33.5 g/l), 80 ml; and glucose solution (180.1 g/l), 10 ml.

TAXONOMIC COMMENTS

The genus *Yersinia* was proposed by van Loghem (1944) in order to separate *Y. pestis* and *Y. pseudotuberculosis* (formerly in the genus *Pasteurella*) from *Pasteurella* species *sensu stricto* (i.e., *P. multocida*, etc.), from which they differ in their negative oxidase reaction and in their DNA base composition. The genus *Yersinia* belongs to the family *Enterobacteriaceae*. *E. coli* tDNA (i.e., the genes coding for transfer RNA) and *Y. pestis* tDNA are 63% related (Brenner et al., 1976), a value similar to that found for *E. coli* tDNA and *Hafnia alvei* tDNA. All *Yersinia* species express the common enterobacterial antigen. Their physiological characteristics and their fatty acid content are similar to those of all *Enterobacteriaceae* species. The mol% G + C range of *Yersinia* species is 46–50 and is consistent with that for *Enterobacteriaceae* species.

The genus *Yersinia* currently consists of 11 different species. Based on DNA–DNA hybridization studies, all of these species are more closely related to each other than to any other *Enterobacteriaceae* species (Brenner et al., 1980e; Bercovier et al., 1984; Aleksic et al., 1987; Wauters et al., 1988). The genus *Yersinia* can be considered a very homogeneous taxon.

DNA relatedness among *Yersinia* species is 40% or higher, except for *Y. ruckeri*, which is at most 38% related to other *Yersinia* species. DNAs of *Y. ruckeri* strains have been shown to be 30% related to *Serratia* species (Ewing et al., 1978). *Y. ruckeri* was included in *Yersinia* because its mol% G + C of 48 is closer to that of *Yersinia* species than to that of *Serratia* species. Because the phenotypic characteristics of *Y. ruckeri* are very different from those of other *Yersinia* species (Tables BXII.γ.281 and BXII.γ.282), it might constitute a new genus by itself. Phylogenetic studies would be helpful in clarifying this problem.

Strains of *Y. enterocolitica* belonging to the five different biovars (Table BXII.γ.283), including the metabolically inactive biovar 5 strains, constitute a homogeneous genomospecies (Bercovier et al., 1980a). The strains originally described as *Y. enterocolitica*-like organisms or atypical *Y. enterocolitica* have been separated into three different species: *Y. intermedia* (Brenner et al., 1980a), *Y. frederiksenii* (Ursing et al., 1980a), and *Y. kristensenii* (Bercovier et al., 1980c). *Y. frederiksenii* consists of three genetic groups based on DNA–DNA hybridization (Ursing et al., 1980a). For practical reasons, because there are no phenotypic differences among the three genetic groups, only one species, *Y. frederiksenii*, has been proposed for these rhamnose-positive strains. More study of phenotypic characteristics is needed to separate the three genetic groups.

The DNAs of *Y. pestis* strains, regardless of biovar, and of *Y. pseudotuberculosis* are 90% or more interrelated. This explains the antigenic and biochemical similarities of the two species. Based on DNA data, Bercovier et al. (1980b) proposed that the two

species constitute a single species, divided into two subspecies: *Y. pseudotuberculosis* subsp. *pseudotuberculosis* and *Y. pseudotuberculosis* subsp. *pestis*. Presently, however, the two species remain distinct, largely because of fear of omitting the subspecies epithet subsp. *pestis*, which may present a potential hazard for laboratory-acquired infections (Williams, 1983).

Ursing et al. (1980b) have shown, based on DNA and physiological data, that *Y. philomiragia* (Jensen et al., 1969) is not related to the genus *Yersinia*, and furthermore, that it is not a member of the family *Enterobacteriaceae*. This species has now been assigned to the genus *Francisella* as *F. philomiragia*.

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *YERSINIA*

Characteristics useful in differentiating the various species of *Yersinia* are listed in Table BXII.γ.282.

List of species of the genus *Yersinia*

- Yersinia pestis*** (Lehmann and Neumann 1896) Van Loghem 1944, 15^{AL} (*Bacterium pestis* Lehmann and Neumann 1896, 194; *Yersinia pseudotuberculosis* subsp. *pestis* Bercovier, Mollaret, Alonso, Brault, Fanning, Steigerwalt and Brenner 1981a, 383.)
pes' tis. L. n. *pestis* plague, pestilence.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. Three biogroups have been described in relation to the geographical distribution of the organism: (a) biogroup *antiqua* produces acid aerobically from glycerol, reduces nitrate to nitrite, does not ferment melibiose, and is found in Central Asia and Central Africa; (b) biogroup *medievalis* produces acid from both glycerol and melibiose, but does not reduce nitrate to nitrite; it is found in Iran and the former USSR; and (c) biogroup *orientalis* (synonym: *oceanic*) does not produce acid from either glycerol or melibiose but reduces nitrate to nitrite and is distributed worldwide. Some rare atypical strains positive in their reactions for urease and L-rhamnose have been reported. *Y. pestis* is the causative agent of plague. The disease can be reproduced experimentally in mice, rats, guinea pigs, and monkeys.

The mol% G + C of the DNA is: 46 (T_m).

Type strain: ATCC 19428, NCTC 5923.

- Yersinia aldovae*** Bercovier, Steigerwalt, Guiyoule, Huntley-Carter, and Brenner 1984, 171^{VP}
al.do'vae. M.L. gen. n. *aldovae* in honor of Eva Aldova, the Czechoslovakian microbiologist who first isolated the bacterium.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. The species is composed of isolates previously referred to as *Y. enterocolitica*-like, group X2 (Bercovier et al., 1980a). Two biovars exist—one typically sucrose negative, and one sucrose pos-

itive. Differentiation of *Y. aldovae* from urease-positive *Hafnia alvei* may be achieved by production of gas by *H. alvei* from D-glucose at 36°C and susceptibility to *Hafnia* specific bacteriophage (Guinée and Valkenburg, 1968). *Y. aldovae* has been isolated from aquatic ecosystems—drinking water, river water, fish, and rarely soil. *Y. aldovae* has not been implicated in animal infections and has not been isolated from humans.

The mol% G + C of the DNA is: 48 (T_m).

Type strain: ATCC 35236, CNY 6065.

GenBank accession number (16S rRNA): X75277.

- Yersinia bercovieri*** Wauters, Janssens, Steigerwalt, and Brenner 1988, 428^{VP}
ber.co.vi.e'ri. M.L. n. *bercovieri* in honor of Herve Bercovier, who first described biogroups 3A and 3B and has made outstanding contributions to the taxonomy and ecology of yersiniae.

Formerly called *Y. enterocolitica* biogroup 3b (Bercovier et al., 1978). Later transferred to *Y. enterocolitica* biogroup 6 (Wauters et al., 1987) because of negative Voges-Proskauer reaction. *Y. bercovieri* must be distinguished from rarely occurring pathogenic strains of serogroup O:3 that are Voges-Proskauer-negative variants of biogroup 3 (Kaneko and Maruyama, 1987). The species conforms to the definition of the family *Enterobacteriaceae* and the genus *Yersinia*. The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. *Y. bercovieri* strains have been isolated from human stool specimens, animals, raw vegetables, soil, and water. There is no evidence of pathogenicity for humans.

The mol% G + C of the DNA is: 50 (T_m).

Type strain: ATCC 43970, CDC 2475-87.

- Yersinia enterocolitica*** (Schleifstein and Coleman 1943)

Frederiksen 1964, 104^{AL} (*Bacterium enterocoliticum* Schleifstein and Coleman 1943, 56.)
en.ter.o.co.li'ti.ca. Gr. n. *enteron* intestine; Gr. n. *colon* of the colon; Gr. suff. *iticos* pertaining to; M.L. fem. adj. *enterocolitica* pertaining to the intestine and colon.

The characteristics are as described for the genus and listed in Tables BXII.γ.281 and BXII.γ.282. The Voges–Proskauer test is usually positive at 22–28°C and negative at 37°C. Biovars of *Yersinia enterocolitica* are listed in Table BXII.γ.283 and, like phagovars and serogroups, are useful epidemiological tools. Rare atypical strains that are either positive for their reactions on Simmons' citrate and for acid production from lactose and raffinose (due to a metabolic plasmid) or negative for urease activity have been reported. When incubated at 20°C, *Y. enterocolitica* strains produce a broad-spectrum mannose-resistant hemagglutinin that is lost at 37°C (MacLagan and Old, 1980). *Y. enterocolitica* causes diarrhea, terminal ileitis, mesenteric lymphadenitis, autoimmune arthritis, abscesses, and septicemia in humans. The disease can be reproduced experimentally in mice, gerbils, and monkeys. Other human infections are listed in Table BXII.γ.285. The species has been isolated from a wide variety of sources in the environment (live and inanimate) including foods and from healthy humans and animals, especially the pig.

The mol% G + C of the DNA is: 48.5 ± 1.5 (*T_m*, Bd).

Type strain: 161, ATCC 9610, CIP 80-27, DSM 4780.

GenBank accession number (16S rRNA): M59292.

Additional Remarks: The type strain belongs to biovar I B, serogroup 0:8, and phagovar X.

5. ***Yersinia frederiksenii*** Ursing, Brenner, Bercovier, Fanning, Steigerwalt, Brault and Mollaret 1981, 217^{VP} (Effective publication: Ursing, Brenner, Bercovier, Fanning, Steigerwalt, Brault and Mollaret 1980a, 213.)
fred.er.ik.sen'i.i. M.L. gen. n. *frederiksenii* of Frederiksen; named after the Danish microbiologist Wilhelm Frederiksen, who made a substantial contribution to the study of the genus *Yersinia*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. This species is composed of three different genomospecies. One group

is positive for β-xylosidase and citrate (Simmons), and the type strain belongs to this group. The other two groups are variable or negative for these tests. More phenotypic studies are needed to differentiate the three groups. Some strains are able to ferment raffinose and lactose when they harbor a metabolic plasmid. *Y. frederiksenii* has been isolated mainly from fresh water sources, fish, foods, and occasionally from healthy or sick humans and animals.

The mol% G + C of the DNA is: 48 (*T_m*).

Type strain: 6175, CIP 80-29.

6. ***Yersinia intermedia*** Brenner, Bercovier, Ursing, Alonso, Steigerwalt, Fanning, Carter and Mollaret 1981, 217^{VP} (Effective publication: Brenner, Bercovier, Ursing, Alonso, Steigerwalt, Fanning, Carter and Mollaret 1980a, 207.)
in.ter.me'di.a. L. fem. adj. *intermedia* intermediate; here it implies that biochemical reactions of this species seem midway between *Y. enterocolitica* and *Y. pseudotuberculosis*.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. Media with a high bile salt content (0.8%) are inhibitory, especially when incubated at 37°C. Some biochemical characteristics (citrate utilization; cellobiose, L-rhamnose, and raffinose fermentation) are always expressed at 25–28°C but are inconstant at 37°C. Either a type A or a type B nitrate reductase is present. Eight biovars have been described (Brenner et al., 1980a) based on the fermentation of melibiose, L-rhamnose, α-methyl-D-glucoside, and raffinose, and on the utilization of citrate (Simmons). Of the strains studied, 96% are positive for at least four of these five tests. *Y. intermedia* has been isolated mainly from fresh water sources, fish, foods, and occasionally from both sick and healthy humans (Bottone et al., 1974; Punsalang et al., 1987).

The mol% G + C of the DNA is: 48.5 ± 0.5 (*T_m*, Bd).

Type strain: 3953, Bottone 48, Chester 48, ATCC 29909, CIP 80-28.

7. ***Yersinia kristensenii*** Bercovier, Ursing, Brenner, Steigerwalt, Fanning, Carter and Mollaret 1981b, 217^{VP} (Effective publication: Bercovier, Ursing, Brenner, Steigerwalt, Fanning, Carter and Mollaret 1980c, 219.)
kris.ten.se'ni.i. M.L. gen. n. *kristensenii* of Kristensen, named after the Danish microbiologist Martin Kristensen, who first isolated this organism.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. Growth is delayed (7 d) when cultures are incubated at 41°C and even at 37°C for some isolates. Some strains utilize citrate (Simmons) after 7-d incubation at 25°C. Most strains produce a "musty" or "cabbage-like" odor when grown on nutrient agar. Some strains produce an enterotoxin (ST) when incubated at 22°C and also at 37°C (Kapperud, 1980). *Y. kristensenii* strains have been isolated mainly from environmental sources such as soil, fresh water, foods, and rarely from healthy human and animal carriers. *Y. kristensenii* may occasionally be associated with human enteritis (Bottone and Robin, 1977), and *Y. kristensenii* isolates may be lethal to mice pretreated with iron dextrin (Robins-Browne et al., 1991).

The mol% G + C of the DNA is: 48.5 ± 0.5 (*T_m*, Bd).

Type strain: 105, ATCC 33638, CIP 80-30.

TABLE BXII.γ.285. Spectrum of *Yersinia enterocolitica* infections

Type of infection	Manifestation/population
Gastrointestinal	Enterocolitis, predominantly in young children; concomitant bacteremia may also be present in infants Pseudoappendicitis syndrome (children older than 5 years; adults), acute mesenteric lymphadenitis; terminal ileitis
Septicemia	Especially in immunosuppressed individuals and those in iron overload or being treated with deferrioxamine; transfusion related (usually leads to septic shock syndrome)
Metastatic	Focal abscesses: liver, kidney, spleen, lung; cutaneous manifestations: cellulitis, pyomyositis, pustules, and bullous lesions; pneumonia, cavitory pneumonia; meningitis; panophthalmitis; endocarditis, infected mycotic aneurysm; osteomyelitis
Postinfection sequelae	Arthritis (associated with HLA-B-27), myocarditis, glomerulonephritis, erythema nodosum

8. ***Yersinia mollaretii*** Wauters, Janssens, Steigerwalt, and Brenner 1988, 427^{VP}
mol.la.re'ti.i. M.L. gen. n. *mollaretii* in honor of Henri H. Mollaret, head of the National *Yersinia* Center at the Pasteur Institute in Paris, for his years of study on the classification and epidemiology in the genus *Yersinia*.

Formerly called *Y. enterocolitica* biogroup 3A (Bercovier et al., 1978). Later transferred to *Y. enterocolitica* biogroup 6 (Wauters et al., 1987) because of negative Voges-Proskauer reaction (typical serotype 0:3 strains are Voges-Proskauer positive). *Y. mollaretii* must be distinguished from rarely occurring pathogenic strains of serogroup 0:3 that are Voges-Proskauer-negative variants of biogroup 3 (Kaneko and Maruyama, 1987). The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. The species conforms to the definition of the family *Enterobacteriaceae* and the genus *Yersinia*. *Y. mollaretii* strains have been isolated from human stool specimens, meat, raw vegetables, soil, and drinking water, and lack virulence markers. There is no evidence of pathogenicity for humans.

The mol% G + C of the DNA is: 50–51 (T_m).

Type strain: ATCC 43969, CDC 2465-87.

9. ***Yersinia pseudotuberculosis*** (Pfeiffer 1889) Smith and Thal 1965, 220^{AL} (*Bacillus pseudotuberculosis* Pfeiffer 1889, 5; *Yersinia pseudotuberculosis* subsp. *pseudotuberculosis* (Pfeiffer 1889) Smith and Thal 1965, 220.)
pseu.do.tu.ber.cu.lo'sis. Gr. adj. *pseudes* false; M.L. fem. n. *tuberculosis* tuberculosis; M.L. gen. n. *pseudotuberculosis* of false tuberculosis.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. Some freshly isolated strains may require subculturing before expressing their motility. Strains belonging to serogroup IV are citrate positive (Simmons) and malonate positive. Up to 5% of *Y. pseudotuberculosis* strains have been reported to produce acid from adonitol. Some strains, mostly of serogroup III, produce an exotoxin that differs from the *Y. pestis* toxin. Its biological activity is not well defined. *Y. pseudotuberculosis* is a human and animal pathogen responsible for mesenteric lymphadenitis, diarrhea, and septicemia. The disease can be reproduced experimentally in guinea pigs challenged per os and in mice.

The mol% G + C of the DNA is: 46.5 (T_m).

Type strain: ATCC 29833, DSM 8992, NCTC 10275.

Additional Remarks: The type strain belongs to serogroup I.

10. ***Yersinia rohdei*** Aleksic, Steigerwalt. Bockemühl, Huntley-Carter, and Brenner 1987, 330^{VP}
roh'de.i. M.L. gen. n. *rohdei* of Rohde; named in honor of the late Rolf Rohde, who founded the National Reference Center for *Salmonella* in Hamburg, Federal Republic of Germany, and who made many significant contributions to the diagnostic and serological identification of *Enterobacteriaceae*, especially *Salmonella*.

The characteristics are as described for the genus and listed in Tables BXII.γ.281 and BXII.γ.282. Two biovars are recognized based on positive (biovar 1) or negative (biovar 2) reactions for both raffinose and melibiose. *Y. rohdei* strains were isolated from the feces of dogs and humans and from surface water. The clinical significance of *Y. rohdei* as a diarrheal agent in humans and animals is uncertain.

The mol% G + C of the DNA is: 49.1 (T_m).

Type strain: H271-36/78, ATCC 43380.

Additional Remarks: The type strain was isolated from dog feces in Germany in 1978.

11. ***Yersinia ruckeri*** Ewing, Ross, Brenner and Fanning 1978, 37^{AL}
ru'cker.i. M.L. gen. n. *ruckeri* of Rucker; named after R.R. Rucker, who studied the red mouth disease and its etiological agents.

The characteristics are as described for the genus and as listed in Tables BXII.γ.281 and BXII.γ.282. The cells are $1 \times 2\text{--}3 \mu\text{m}$. Filaments can be seen in old cultures (48 h at 22°C). Colonies on nutrient agar are smooth, circular, and slightly raised. Growth is delayed or inhibited on *Salmonella*-*Shigella* agar incubated at 37°C but not at 22°C. Corn oil is hydrolyzed when the test is performed at 22°C but not at 37°C. *Y. ruckeri* is one of the agents responsible for red mouth disease in rainbow trout. The disease can be transmitted experimentally from fish to fish. The organism was initially isolated only in North America, but reports of enteric red mouth disease caused by *Y. ruckeri* have also assumed increasing importance in fish farms in Europe (Home et al., 1984). First isolated in 1959 from a dead muskrat found in a marshy area at the Bear River Migratory Bird Refuge in northern Utah. Other strains have been isolated from water in the same area.

The mol% G + C of the DNA is: 48 ± 0.5 (Bd).

Type strain: ATCC 29473.

GenBank accession number (16S rRNA): X75275.

Genus XLII. *Yokenella* Kosako, Sakazaki, and Yoshizaki 1985, 224^{VP} (Effective publication: Kosako, Sakazaki, and Yoshizaki 1984, 124)

J.J. FARMER III AND FRANCES W. BRENNER

Yoken.el'la. dim. ending *-ella*; Japanese abbreviation "Yoken" that stands for the National Institute of Health, Tokyo, Japan; M.L. fem. n. *Yokenella* pertaining to Yoken, the National Institute of Health, Tokyo, Japan where the new group of organisms was recognized and studied.

Rod-shaped cells, conforming to the general definition of the family *Enterobacteriaceae*. Contain the enterobacterial common antigen. Gram negative. Motile, with peritrichous flagella. Facultatively anaerobic. Catalase positive (weak). Oxidase negative.

Ferment, rather than oxidize, D-glucose and other carbohydrates. Reduce nitrate to nitrite.

Positive for methyl red, citrate utilization (Simmons), lysine decarboxylase, ornithine decarboxylase, motility at 36°C, and

growth in the presence of cyanide (KCN test), β -galactosidase (ONPG test), tetrathionate reductase, and the fermentation of D-mannitol, L-arabinose, L-rhamnose, maltose, D-xylose, trehalose, cellobiose, melibiose, D-mannose, D-galactose, L-fucose, and D-gluconate. Produce visible gas during fermentation.

Negative for indole production, Voges-Proskauer, H_2S production (TSI), urea hydrolysis, phenylalanine deaminase, arginine dihydrolase, malonate utilization, gelatin hydrolysis (22°C), lipase (corn oil and Tween 80), DNase, esculin hydrolysis, pectinase, chitinase, β -xylosidase, growth at 4°C, and the fermentation of lactose, sucrose, dulcitol, salicin, adonitol, *myo*-inositol, D-sorbitol, raffinose, α -methyl-D-glucoside, erythritol, D-arabitol, glycerol, and mucate.

Susceptible to nalidixic acid, sulfadiazine, gentamicin, kanamycin, tetracycline, and chloramphenicol (disk diffusion method on Mueller-Hinton agar); usually resistant to colistin, penicillin, ampicillin, carbenicillin, and cephalothin; variable susceptibility to streptomycin.

Ecological niches are not known, but has been isolated from human clinical specimens, insects, and water. Probably an opportunistic pathogen for humans. A rarely isolated member of the *Enterobacteriaceae*.

The mol% G + C of the DNA is: 58–59.3 (T_m).

Type species: *Yokenella regensburgei* Kosako, Sakazaki, and Yoshizaki 1985, 224 (Effective publication: Kosako, Sakazaki, and Yoshizaki 1984, 124.)

FURTHER DESCRIPTIVE INFORMATION

Different names for the same organism It is essential for the reader to understand that three vernacular names ("Hafnia" species 3, Enteric Group 45, and NIH biogroup 9) and one scientific name (*Koserella trabulsii*) have been used in the literature for the organism that is known today as *Yokenella regensburgei*. This is explained in the section Taxonomic Comments. Until recently, the names *Koserella* and *K. trabulsii* appeared in the literature, so we have inserted these names in parentheses if they were used in the original article.

Literature Since *Yokenella* and *Koserella* were described in 1984–85, there have been several reports in the literature. In addition to the two genus and two species names, "Hafnia species 3", "Enteric Group 45", and "NIH biogroup 9" should also be included in computerized literature searches, although these names are rarely used today.

Sources, strains, collections Strains of *Y. regensburgei* have been isolated from a variety of human clinical specimens and other sources. The original isolates of Kosako et al. (1984) were from human clinical specimens; urine (three isolates), wound (one), and abscess (one), and six isolates from insect intestine. The original collection of Hickman-Brenner et al. (1985a) included the type strain for *K. trabulsii* isolated from a wrist wound of a 28-year-old man. The other isolates were from human wounds (four), sputum (two), throat (one), stool (one), knee fluid (one), well water (one), and unknown (one). Since that original report, the CDC has received 10 additional human strains (three wound, two blood, two sputum, one urine, one lung biopsy, and one unknown). The first blood isolate was from a 35-year-old woman who had a bone marrow transplant. Group G streptococci and *Klebsiella pneumoniae* were also isolated. No further clinical information was given for the remaining strains.

Isolates from other sources Six of the original strains reported by Kosako et al. (1984) were originally isolated by F. Haas

(Germany) from the intestine of *Heteroptera* (*Pyrrhocoris apterus*). Cassel-Beraud and Richard (1988) studied stools of 88 insect eating bats (*Chaerephon pumila*), which yielded many different bacterial strains. Strains of *K. trabulsii* were identified among the 20 different species identified, most of which were *Enterobacteriaceae*.

Clinical significance Whether *Yokenella* strains can actually cause infections is still being evaluated. Systematic study and additional case reports are needed. Abbott and Janda (1994a) reported two cases that yielded *Y. regensburgei*; one was a 74-year-old man with a "septic knee", and the other was a 35-year-old immunocompromised woman without overt signs of sepsis but with a positive blood culture. Yague et al. (1989) described two cases (*K. trabulsii*) from Spain. The first case involved an isolate from the stool of a 4-year-old boy. *Campylobacter jejuni* was also isolated and was considered to be the causative agent. The second case involved an isolate from the urine of an 88-year-old man with fever and respiratory problems. The colony count was over 100,000 CFU/ml, but there was no evidence of a urinary tract infection. *K. trabulsii* was not considered significant in either case.

Other literature The literature includes reviews and taxonomic studies of new *Enterobacteriaceae* (Richard, 1989; Gilchrist, 1995; Pokhil, 1996; Aleksic and Bockemühl, 1999), and surveys or comparisons of the family *Enterobacteriaceae* for acid phosphatases (Thaller et al., 1995) and polyamines (Hamana, 1996).

Isolation, identification, culture preservation, phenotypic characterization Strain of *Y. regensburgei* are not difficult to grow, and are typical *Enterobacteriaceae* in most respects. See the chapter on the family *Enterobacteriaceae* for general information on growth, plating media, working and frozen stock cultures, media and methods for biochemical testing, identification, computer programs, and antibiotic susceptibility.

Biochemical reactions and differentiation from other *Enterobacteriaceae* Table BXII.γ.193 in the chapter on the family *Enterobacteriaceae* gives the results for *Y. regensburgei* in 47 biochemical tests normally used for identification (Farmer, 1999). There are no genus- or species-specific tests or sequences for the identification. The best approach is to do a complete set of phenotypic tests (growth on plating media, biochemical tests, antibiogram, etc.) and compare the results with all the other named organisms in the family *Enterobacteriaceae*. This approach is described in more detail in the chapter on the family *Enterobacteriaceae*. The biochemical results of a test strain can be compared to all the organisms listed in Table BXII.γ.193 in that section. Several computer programs greatly facilitate analyzing the results. Biochemically, *Yokenella regensburgei* is closest to *Hafnia alvei*. These two organisms can be differentiated because *Y. regensburgei* is Voges-Proskauer and glycerol negative; citrate (Simmons), cellobiose, and melibiose positive; and resistant to colistin and to the *Hafnia*-specific bacteriophage of Guinée and Valkenburg (1968). It also gives a weak, delayed, catalase reaction. These are in contrast to the reactions of *Hafnia alvei*.

TAXONOMIC COMMENTS

Today there is general agreement on the use of the names *Yokenella* and *Yokenella regensburgei*. However, the names *Koserella* and *Koserella trabulsii* also appear in the literature. Table BXII.γ.192 in the chapter on the family *Enterobacteriaceae* summarizes how these organisms were discovered, named, and how *Yokenella regensburgei* eventually displaced *Koserella trabulsii*.

Agreement to use *Yokenella* rather than *Koserella* Although the CDC workers did not agree with the manner in which Kosako and Sakazaki (1991) proceeded in trying to solve the nomenclatural problem of priority, they decided to use *Yokenella* and *Y. regensburgi* instead of *Koserella* and *K. trabulsii*. In McWhorter et al. (1991), one of us (JJF) wrote: "We now acknowledge that *Y. regensburgi* has priority over *K. trabulsii* because *Y. regensburgi* was published first and gained standing in nomenclature in the same issue of the *International Journal of Systematic Bacteriology* as *K. trabulsii*. In a nomenclatural sense, *K. trabulsii* should now be considered a 'junior subjective synonym' of *Y. regensburgi* and thus an illegitimate name. We will now use the name *Y. regensburgi* instead of the name *K. trabulsii*."

McWhorter et al., 1991 conceded that only quirks in the validation process had given priority to *Koserella* over *Yokenella*, and agreed that since the names *Yokenella* and *Yokenella regensburgi* were published first and should have appeared first in a validation list, they should be used. Publications, databases, computer programs, and biochemical charts of CDC have reflected this use of *Yokenella* rather than *Koserella*. For example, compare Table 1 of Farmer (1999) that used *K. trabulsii* with Table 1 of Farmer (1995) that used *Y. regensburgi*. This informal acceptance resolved the issue as far as usage; however, only an Opinion of the Judicial Commission in the future can resolve the nomenclatural problem described below.

Nomenclatural problem — do *Yokenella* and *Y. regensburgi* have priority over *Koserella* and *K. trabulsii*? Under the rules of the Bacteriological Code, the answer to this question is unclear. Kosako and Sakazaki (1991) argued clearly and logically for their priority, but this was clearly an opinion of Kosako and Sakazaki, which has no judicial standing under the Bacteriological Code. Unfortunately, they did not end the title of their paper with "Request for an Opinion" as previously agreed to resolve the problem of priority (Kosako et al., 1987). The rules of the Bacteriological Code are interpreted differently by different people, and their application in unique circumstances is also a matter of interpretation. Thus, the question of priority remains as only a small nomenclatural problem because the matter of usage was settled in the early 1990s.

FURTHER READING

- Abbott, S.L. and J.M. Janda. 1994. Isolation of *Yokenella regensburgi* (*Koserella trabulsii*) from a patient with transient bacteremia and from a patient with a septic knee. *J. Clin. Microbiol.* 32: 2854–2855.
- Hickman-Brenner, F.W., G.P. Huntley-Carter, G.R. Fanning, D.J. Brenner and J.J. Farmer. 1985. *Koserella trabulsii*, a new genus and species of *Enterobacteriaceae* formerly known as enteric group-45. *J. Clin. Microbiol.* 21: 39–42.
- Kosako, Y., R. Sakazaki and E. Yoshizaki. 1984. *Yokenella regensburgi* gen. nov., sp. nov.: a new genus and species in the family *Enterobacteriaceae*. *Japan J. Med. Sci. Biol.* 37: 117–124.

List of species of the genus *Yokenella*

1. ***Yokenella regensburgi*** Kosako, Sakazaki, and Yoshizaki 1985, 224^{VP} (Effective publication: Kosako, Sakazaki, and Yoshizaki 1984, 124.)
re.gens.bur'ge.i. M.L. gen. *regensburgi* pertaining to Regensburg, Germany, where the type strain of *Y. regensburgi* and five other strains in the original paper (Kosako et al., 1984) were isolated from insects.

The characteristics are as given for the genus. Isolated from human clinical specimens (blood, urine, feces, throat and sputum, wounds, abscess, and others), insect intestines, and well water. Its clinical significance is not known, but if it is pathogenic for humans it is probably a very weak pathogen. However, it is also possible that it is colonizing rather than infecting body sites that are normally not sterile. There

is no evidence that it can cause diarrhea or intestinal infections. It should be considered a rarely isolated species of *Enterobacteriaceae* that is probably an opportunistic pathogen for humans. The type strains was isolated by F. Haas from the intestine of an insect (*Pyrrhocoris apterus*) in Regensburg, Germany.

The mol% G + C of the DNA is: 58–59.3 (T_m).

Type strain: ATCC 49455, CDC 3020-86, NIH 725-83, JCM 2403.

Additional Remarks: The American Type Culture Collection includes the five strains of *Y. regensburgi* that were originally deposited by Hickman-Brenner et al. (1985a) as *Koserella trabulsii*. Four were from human clinical specimens and one was from well water.

Order XIV. *Pasteurellales* ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Pas.teu.rel.la'les. M.L. fem. n. *Pasteurella* type genus of the order; *-ales* suffix to denote order; M.L. fem. n. *Pasteurellales* the *Pasteurella* order.

The order *Pasteurellales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the order contains the family *Pasteurellaceae*.

Description is the same as for the family *Pasteurellaceae*.

Type genus: ***Pasteurella*** Trevisan 1887, 94 (Nom. Cons. Opin. 13, Jud. Comm. 1954b, 153.)

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